

Quantum Cryptography with Continuous Variables



Arpan Akash Ray

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PROEFSCHRIFT

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To those who came before me
known and unknown
whose struggles gave me the freedom to learn
and the privilege of education.

*“The scientist does not study nature because it is useful;
he studies it because he delights in it.”*

— Jules Henri Poincaré

Acknowledgements

No voyage is ever successful without the many sailors who keep the ship moving, often unseen beneath the deck. This is especially true of a journey as immense as a PhD.

Who, after all, am I? I exist in this world not only as a collection of atoms, but more so as a bundle of memories in the minds of other individuals. A collection of atoms may write a thesis; it may even be awarded a doctoral degree. Yet neither this book nor the piece of parchment bearing my name can, by itself, make an impact on the lives of individuals. That impact exists only in the memories of the people I have had the good fortune to meet along the way. My true doctorate, therefore, is not merely the degree I receive, but the mark this journey has allowed me to leave on others.

For that, I would like to thank the many people who sailed with me.

It must begin with Boris. As did this journey, more than four years ago. I did not know what to expect from this voyage. I had heard tales from many PhD students about what a gruelling undertaking it could be. But my stars must have aligned remarkably well, as I found a supervisor such as yourself. You never made me feel that you were my superior merely because you were my supervisor; instead, you were always a caring adviser. And that led me to fully explore my ideas (even if they were incorrect) and new directions of research (even if they may turn out to be futile) and voice my opinions with confidence (even if some were a bit stupid), knowing that you would not shoot me down without hearing me out.

You were also a true mentor. Not everyone could talk about everything with their supervisor, but I know that I could always shoot you an email about anything and expect a prompt reply. You were also a friend. When I first arrived here from a natural-science institute, I felt quite isolated in this computer science department. The only person with whom I could really talk about science was you. But because you made sure that you were almost always available, I never truly felt that isolation. We did a lot of physics and math together, interspersed with funny anecdotes and the occasional scream of your Reindeer. I must say some of the most fun I had was talking science to you. As you would say after many long afternoons and plenty of cups of coffee, “time passes quickly when you are having fun”.

I am truly grateful to have a supervisor and mentor like you. And I thank you for all you have done for me. Thank you especially, for making this PhD more about the spirit of discovery than about a degree.

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As many would already know, when I am not doing physics, I am doing music. Music has always played a big role in my life, and I met a lot of very talented musicians (turned friends) in Europe. From Stichting SPIC MACAY, I would thank Akshay, Ravi and Aananth for the jamming sessions and the friendship. Thank you Prashant Ji, for sharing your home and your music with me. Thank you Shasank; you are of course much more than a friend, and I will never forget playing together under the trees in the misty morning of Mohali. Thank you Siddharth Ji, for sharing your knowledge with me; I am immensely grateful for everything I have learnt from you. Thank you, Sandip Da, also for sharing your love of music with me as well. I have had the wonderful opportunity to perform in concerts with many of the people mentioned above, as well as with several others. With every performance, I learnt so much from all of you. Thank you, therefore, to each of you, and to the many others whom I may have missed, for inspiring me through the brilliance of your art.

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*Arpan Akash Ray
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अभिस्वीकृति

कोई भी समुद्री यात्रा उन अनेक नाविकों के बिना सफल नहीं हो सकती जो जहाज़ को आगे बढ़ाते रहते हैं, भले ही उनका काम अक्सर दिखाई न दे। डॉक्टरेट जैसी विराट यात्रा के लिए यह बात और भी सच है।

आखिर मैं कौन हूँ? मैं इस दुनिया में केवल परमाणुओं का एक समूह नहीं हूँ; मैं उन लोगों की स्मृतियों में कहीं ज़्यादा जीवित हूँ जिनसे मुझे इस यात्रा के दौरान मिलने का अवसर मिला। परमाणुओं का एक समूह शोधप्रबंध लिख सकता है और उसे डॉक्टरेट की उपाधि भी मिल सकती है। लेकिन न तो यह पुस्तक, और न ही मेरे नाम वाला वह प्रमाणपत्र, अपने आप में किसी व्यक्ति के जीवन पर प्रभाव डाल सकता है। वह प्रभाव केवल उन लोगों की स्मृतियों में जीवित रहता है जिनसे इस यात्रा के दौरान मिलने का सौभाग्य मुझे मिला। इसलिए मेरी असली डॉक्टरेट केवल वह उपाधि नहीं है जो मुझे मिलेगी, बल्कि वह छाप है जिसे इस यात्रा ने मुझे दूसरों के जीवन पर छोड़ने का अवसर दिया।

इसी कारण मैं उन सभी लोगों को धन्यवाद देना चाहता हूँ जिन्होंने इस यात्रा में मेरा साथ दिया।

इसकी शुरुआत बोरिस से ही होनी चाहिए, ठीक वैसे ही जैसे चार वर्ष से भी अधिक पहले इस यात्रा की शुरुआत हुई थी। मुझे नहीं पता था कि इस सफ़र से क्या अपेक्षा करनी चाहिए। मैंने अनेक डॉक्टरेट विद्यार्थियों से सुना था कि यह यात्रा कितनी कठिन और थका देने वाली हो सकती है। लेकिन मेरे सितारे सचमुच बहुत मेहरबान रहे होंगे कि मुझे आप जैसा शोध-निर्देशक मिला। आपने मुझे कभी यह महसूस नहीं कराया कि केवल मेरे सुपरवाइज़र होने के कारण आप मुझसे ऊपर हैं; आप हमेशा एक परवाह करने वाले सलाहकार रहे। इसी वजह से मैं अपने विचारों को खुलकर परख सका, भले ही वे ग़लत हों; शोध की नई दिशाओं में जा सका, भले ही वे आगे चलकर निष्फल साबित हों; और अपनी राय आत्मविश्वास से रख सका, भले ही उनमें से कुछ थोड़ी मूर्खतापूर्ण रही हों। मुझे हमेशा भरोसा था कि आप मेरी बात पूरी सुने बिना मुझे खारिज नहीं करेंगे।

आप एक सच्चे मार्गदर्शक भी थे। हर कोई अपने सुपरवाइज़र से हर विषय पर बात नहीं कर सकता, लेकिन मुझे हमेशा पता था कि मैं आपको किसी भी बात के बारे में ईमेल कर सकता हूँ और शीघ्र उत्तर की उम्मीद कर सकता हूँ। आप एक मित्र भी थे। जब मैं एक प्राकृतिक-विज्ञान संस्थान से यहाँ आया था, तब इस कंप्यूटर विज्ञान विभाग में मुझे काफ़ी अलग महसूस होता था। विज्ञान के बारे में खुलकर बात करने के लिए मेरे पास केवल आप ही थे। लेकिन आपने लगभग हमेशा उपलब्ध रहने की कोशिश की, इसलिए मैंने उस अलगाव को कभी सचमुच महसूस नहीं किया। हमने साथ में बहुत-सी भौतिकी और गणित की, जिनके बीच मज़ेदार किस्से और कभी-कभार आपके रेनडियर की चीख भी सुनाई देती थी। मुझे कहना ही होगा कि विज्ञान के बारे में आपसे बातचीत करना इस डॉक्टरेट के सबसे आनंददायक अनुभवों में से एक था। जैसा कि आप कई लम्बी दोपहरों और न जाने कितने कप कॉफ़ी के बाद कहा करते थे, “जब मज़ा आ रहा हो, तो समय जल्दी बीत जाता है।”

आप जैसा सुपरवाइज़र और मार्गदर्शक पाकर मैं सचमुच अत्यन्त कृतज्ञ हूँ। आपने मेरे लिए जो कुछ भी किया, उसके लिए आपका धन्यवाद। विशेष रूप से इस डॉक्टरेट को केवल एक उपाधि प्राप्त करने की प्रक्रिया न बनाकर खोज की भावना से भरी यात्रा बनाने के लिए धन्यवाद।

मुझे बेहद बुद्धिमान और मज़ेदार लोगों के साथ कार्यालय साझा करने का सौभाग्य भी मिला। टोन, मेरी अंतहीन बातें सुनने और मुझे गणित की बारीकियों को गंभीरता से लेना सिखाने के लिए धन्यवाद। कोपेनहेगन में एक

अपार्टमेंट साझा करने के बाद—क्या यात्रा थी वह!—हमें जल्दी ही समझ आ गया कि हममें सचमुच बहुत कुछ समान है। हम बड़ी सहजता से भौतिकी से मज़ाक-मस्ती और वहाँ से गणित तक पहुँच जाते थे। मेरे अपरंपरागत हास्यबोध और कॉफी के प्रति आपकी दीवानगी के बीच हमने एक ऐसा संबंध बना लिया जिसे मैं बहुत लम्बे समय तक सँजोकर रखूँगा। मुझे दौड़ने की गंभीरता से लेने के लिए प्रेरित करने और एक अत्यन्त कठोर प्रशिक्षक बनने के लिए धन्यवाद। आपके साथ कदम मिलाने की कोशिश मैं कभी नहीं भूलूँगा। जब भी आपने हमारे लिए पियानो बजाया, उसके लिए भी धन्यवाद। आपका संगीत सुनना हमेशा आनंददायक था।

डोमिनिक, मेरी बातें सुनने और मुझे क्लासिकल कंप्यूटर सुरक्षा से परिचित कराने के लिए आपका भी धन्यवाद। हमने तीन वर्षों तक एक ही कार्यालय साझा किया और मुझे लगता है कि अंत तक हम लगभग हर सम्भव विषय पर बात कर चुके थे। हमने जो मज़ाक किए, उन्हें मैं कभी नहीं भूलूँगा। कार्यालय के भीतर और बाहर साथ बिताए समय की अनेक सुखद स्मृतियाँ मेरे पास हैं।

बशीर, यद्यपि हम एक-दूसरे से बहुत अधिक नहीं मिलते थे, लेकिन जब भी मिलते थे, मुझे भौतिकी के दार्शनिक आधार, समाज, अर्थशास्त्र और इनके बीच की हर चीज़ पर चर्चा करना बहुत अच्छा लगता था। बाड बूकेलो में हुई हमारी लम्बी चर्चाएँ भी मेरे लिए बहुत मूल्यवान हैं। किसी भी चर्चा में आप जिस स्पष्टता और समझ की गहराई के साथ अपनी बात रखते थे, उससे मैं हमेशा बहुत प्रभावित होता था। मैं आशा करता हूँ कि मैं आपसे तर्क करने की कला का कुछ हिस्सा सीख पाया हूँ।

अपने कार्यालय से बाहर निकलने पर मुझे दो अन्य कार्यालयों में भी मित्र मिल गए।

योरिट, कभी-कभार की सैर, बारबेक्यू, मुफ्त खाना हासिल करने के अभियानों और मेरे संदेश छापने के बीच हमने बहुत-से मज़ेदार पल साथ बिताए। मत्तेओ, यह कहना गलत नहीं होगा कि आप पूरे ग्रुप के जीवन में, और मेरे जीवन में भी, ढेर सारी हँसी लेकर आए। कृपया मुझे अपनी दिलचस्प रील्स भेजना बन्द न करें। लियोन, मुझे ज़िन्दगी को ऑप्टिमाइज़ करना सिखाने के लिए धन्यवाद। मैं आपके लिए ढेरों शुभकामनाएँ और हर मामले में सबसे ऑप्टिमल परिणाम की कामना करता हूँ। कून, मेरा मित्र बनने और सम्भवतः ग्रुप के सबसे अच्छा इंसान होने के लिए धन्यवाद। चार वर्षों के बाद भी मुझे आपका उच्च उच्चारण समझने में कठिनाई होती है। गेलारेह, शुरुआत में मुझे लगा था कि आप बहुत गम्भीर व्यक्ति हैं, लेकिन यह जानकर मुझे राहत मिली कि आपका हास्यबोध हम सब जैसा ही है। हमें सवारी देने के लिए gaadi खरीदने पर विशेष धन्यवाद। एदुआर्दो, हमारे मज़ाक सहने और बार-बार यह याद दिलाने के लिए धन्यवाद कि “अब हम बच्चे नहीं रहे।”

कुछ ऐसे लोग भी हैं जिन्हें याद किया जाना चाहिए, भले ही उनके कार्यालय थोड़े दूर थे। टॉम, साथ में हुई सभी मज़ेदार खाना पकाने की बैठकों के लिए धन्यवाद। मुझे यह कहते हुए भी प्रसन्नता है कि मैं पूकी की खराबों का गर्वित मालिक हूँ।

नितीश, ऐसा मित्र होने के लिए धन्यवाद जिसे मैं किसी भी समय सहायता के लिए पुकार सकता था। कोपेनहेगन की हमारी यात्रा के दौरान मुझे पहली बार आपके अस्तित्व का पता चला और मैं बहुत आभारी हूँ कि वहाँ आपसे परिचित होने का अवसर मिला। उस मुलाकात ने आपके, टोन और मेरे बीच एक गहरी मित्रता को जन्म दिया। चूँकि इस कंप्यूटर साइंस विभाग में हम कुछ हद तक अलग हैं, इसलिए आपके साथ क्वांटम की बातें करना हमेशा बहुत मज़ेदार था। और जैसा कि आप जानते हैं, मैं हमेशा आपकी कहानियों की प्रतीक्षा में रहता हूँ। मुझे आशा है कि आपका अफ्रीकी रेस्तराँ जल्द खुलेगा, क्योंकि मैं आपके सबसे अच्छे ग्राहकों में से एक बनने वाला हूँ!

मुझे अपने सहयोगियों और मित्रों, लोरेंज़, रेने, फिलिप और फ्लोरियन, का भी धन्यवाद करना चाहिए। दिमित्री

और यूसुफ के साथ भी मेरी कुछ बेहद रोचक चर्चाएँ हुई, और आशा है कि वे जल्द ही मेरे सहयोगी भी बनेंगे।

मैं अपनी समिति के सदस्यों का अत्यन्त आभारी हूँ, जिन्होंने इस शोधप्रबंध को पढ़ने और इसका मूल्यांकन करने के लिए समय निकाला। एलेक्स, एंथनी, चिगो, क्रिस, एलेनी और प्रभंजन, आप सभी का धन्यवाद। मेरे पूरे डॉक्टरेट के दौरान मुझे व्यावहारिक फोटोनिक्स के अनेक पहलुओं से परिचित कराने के लिए चिगो को विशेष धन्यवाद।

मिरयम, इन सभी वर्षों में आपकी सहायता के लिए विशेष धन्यवाद, और योलांडे, अपनी हँसी से गलियारों को हमेशा जीवंत बनाए रखने के लिए आपका धन्यवाद।

स्टैश, रॉय, मिकेले, सौम्या, मार्टिन, पावलो, इंग्रिड, बर्नार्डो, एलेसेंड्रो, क्रिस्टोफ़र, सैंड्रो, निकोला, सावियो, कादिर, जोआओ और सुभा, हँसी-मज़ाक और चर्चाओं के लिए आप सभी का धन्यवाद। यदि मैं किसी और का नाम लेना भूल गया हूँ, तो मुझे क्षमा करें। मेरे साथ अपने जीवन के मूल्यवान क्षण साझा करने के लिए आपका बहुत-बहुत धन्यवाद। मैं इन स्मृतियों को बहुत लम्बे समय तक सँजोकर रखूँगा।

यदि मैं उन लोगों को धन्यवाद न दूँ जिन्होंने काम के बाहर मेरे जीवन को छुआ, तो ये आभार अधूरे रह जाएंगे।

मार्टिन, मैं स्वयं को भाग्यशाली मानता हूँ कि आइंडहोवन में अपने पहले सप्ताह के दौरान मैं आपसे बिल्कुल संयोग से मिला। छोटे जर्मन कस्बों में भारतीय रेस्तराँ ढूँढने से लेकर आपके घर दावतें उड़ाने तक, आपके और टेटी के साथ समय बिताना हमेशा बहुत सुखद होता है। मुझे अपने परिवार का हिस्सा महसूस कराने के लिए धन्यवाद।

अखिल और देवांगी, हमने अपनी अनेक यात्राओं में बहुत मज़ा किया और मुझे विश्वास है कि भविष्य में भी हम साथ में ट्रिपिंग करते रहेंगे। रोशन और राजेश, आप दोनों बहुत लम्बे समय से मेरे सबसे घनिष्ठ मित्र रहे हैं। यदि मैं हमारे साथ बिताए हर पल को गिनाना शुरू कर दूँ, तो यह अनुच्छेद शोधप्रबंध से भी लम्बा हो जाएगा। बहुत-से लोग शायद इसे न समझें, लेकिन डी3 एक ऐसा समूह है जिसमें मुझे कभी यह सिद्ध नहीं करना पड़ा कि मैं उसका सदस्य हूँ।

ऋतिक, एक सच्चा मित्र होने के लिए धन्यवाद। हमारे बीच भी कुछ उतार-चढ़ाव आए, लेकिन हमने सब कुछ सुलझा लिया—मेरे और गिट के विपरीत—और किसी तरह अब हमारी मित्रता को चौदह वर्ष हो चुके हैं और यह गिनती अभी जारी है। मुझे लगता है कि मुझे हमेशा गेम खेलने के लिए परेशान करते रहने पर भी आपका धन्यवाद करना चाहिए। इतने वर्षों के बाद भी मुझे विश्वास नहीं होता कि अब आप मुझे आमने-सामने परेशान कर सकते हैं।

বুবলি দি, সেই বড় বোন হয়ে থাকার জন্য তোমাকে ধন্যবাদ, যাঁর আমার বিশেষ প্রয়োজন ছিল। তোমার মতো একজন নিজের মানুষ পেয়ে আমি খুবই আনন্দিত, এবং তুমি যে কাজ করো, তার জন্য আমি ততটাই গর্বিত। আমার জীবনের ভরসার স্তম্ভ হয়ে থাকার জন্য ধন্যবাদ!

विन्सेंट, मुझे याद नहीं कि हम पहली बार क्यूकेडी समर स्कूल में मिले थे या आइंडहोवन में, लेकिन मुझे बहुत प्रसन्नता है कि हम मिले। यह एक विचित्र संयोग है कि हमारे रास्ते कितनी बार एक-दूसरे से टकराए और हमारे जीवन किस तरह आपस में जुड़ते गए। मुझे आशा है कि आने वाले वर्षों में भी यह सिलसिला जारी रहेगा।

अमांदिनी, सम्भवतः आप आइंडहोवन में मेरी पहली मित्र थीं। भले ही अब आप बहुत दूर रहती हैं, लेकिन मुझे बहुत खुशी है कि मैं आपसे मिला। मुझे आज भी काम के बाद डे योंग पार्क में हमारी सैर याद है, और यह भी कि

आप किसी न किसी तरह अपने जीवन में हमेशा इतने सारे रोमांच पैदा कर लेती थीं। अपनी कहानियाँ मेरे साथ साझा करने और सबसे बढ़कर मेरी मित्र होने के लिए धन्यवाद।

क्विटिन, हम ने एक-दूसरे की मेज़ बानी करते हुए और अपनी मासिक बैठकों के दौरान बहुत मज़ा किया। कॉम्प्लेक्सिटी थ्योरी की बारी कियों के बारे में बातने के लिए आपका धन्यवाद।

जैसा कि बहुत-से लोग पहले से जानते हैं, जब मैं भौतिकी नहीं कर रहा होता, तब संगीत कर रहा होता हूँ। संगीत मेरे जीवन का बहुत बड़ा हिस्सा रहा और यूरोप में मुझे अनेक अत्यन्त प्रतिभाशाली संगीतकारों से मिलने का अवसर मिला, जो बाद में मेरे मित्र बन गए। स्पिक मैके से अक्षय, रवि और आनंत को जैमिंग सत्रों और मित्रता के लिए धन्यवाद देना चाहूँगा।

प्रशांत जी, अपना घर और अपना संगीत मेरे साथ साझा करने के लिए धन्यवाद। शशांक, आप निःसन्देह केवल एक मित्र से कहीं अधिक हैं। मोहाली की धुंध भरी सुबह में पेड़ों के नीचे आ पके साथ जुगलबंदी मैं कभी नहीं भूलूँगा। सिद्धार्थ जी, मुझे आपने शिष्य के रूप में ग्रहण करने के लिए धन्यवाद; आपसे सीखी हर चीज़ के लिए मैं अत्यन्त कृतज्ञ हूँ। सन्दीप दा, সংগীতের প্রতি তোমার ভালোবাসা আমার সঙ্গে ভাগ করে নেওয়ার জন্য তোমাকেও ধন্যবাদ।

ऊपर उल्लिखित अनेक लोगों एवं और भी कुछ अन्य लोगों के साथ संगीत कार्यक्रमों में प्रस्तुति देने का अद्भुत अवसर मुझे मिला। हर प्रस्तुति में मैंने आप सभी से बहुत कुछ सीखा। इसलिए आप में से प्रत्येक को, और उन अनेक अन्य लोगों को जिनका उल्लेख करना मुझ से छूट गया हो, अपनी कला की प्रतिभा से मुझे प्रेरित करने के लिए धन्यवाद।

নিঃসন্দেহে, আমার বাবা-মায়ের অবিরাম সহযোগিতা ছাড়া এর কোনোটিই সম্ভব হতো না। আমার প্রতি তোমার যে বিশ্বাস আর ভালোবাসা, তার জন্য ধন্যবাদ। জেঠি, তোমার ভালোবাসা আর আশীর্বাদ এর জন্য ধন্যবাদ।

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आइंथोवेन, सितम्बर २०२६

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1

Introduction

Nature isn't classical, dammit.

— Richard P. Feynman

On a voyage to chart quantum cryptography, there is perhaps no better beginning than the celebrated quantum key distribution protocol.

At the most basic level, quantum key distribution (QKD) is a method by which two distant parties, conventionally called Alice and Bob, generate a shared cryptographic key. The goal is to create a random string of bits that they both know and that an adversary does not. These distant parties are connected by a quantum communication channel, through which physical quantum systems are transmitted, and a classical communication channel, through which they exchange ordinary messages according to the protocol. The quantum part produces correlated data. The classical part is then used to check whether the data are sufficiently reliable, to correct any differences between Alice's and Bob's strings, and finally to compress away any information that may have leaked to an adversary. If the protocol succeeds, the output is a shared *secret* key.

To truly grasp the gravity of this achievement, we must understand our adversary Eve. Eve is not merely a passive eavesdropper listening to a wire. She may control the channel in all possible ways. She may replace it, interrupt it, inject (quantum) information, store quantum information, perform measurements, and wait until the most favourable moment to use every piece of information she has gathered. In the most demanding formulations, even the internal functioning of the devices used by Alice and Bob is not trusted in the usual way.¹ If the laws of physics allow it, Eve can do it.

The footnotes dotted around this chapter cater for the more inquisitive reader and thus may usually be safely ignored.

¹This is the guiding idea behind device-independent quantum key distribution, where security is inferred

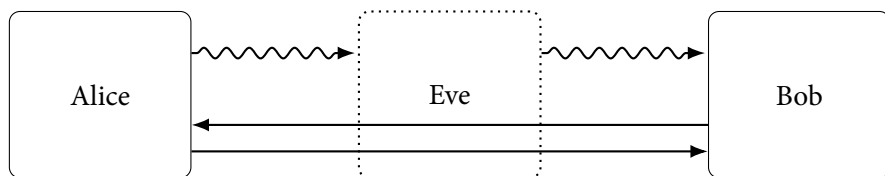


Figure 1.1: A schematic view of quantum key distribution. Alice and Bob use quantum communication (shown here with squiggly lines) together with *authenticated* classical communication (shown here with a straight line) to generate a shared key. The direction of the arrows shows the allowed direction of information transfer. In the case of the classical channel, Alice and Bob are allowed to send and receive messages. Eve may attack the quantum channel and listen to (but not modify) the classical communication, but a successful protocol ensures that her information about the final key is negligible.

And yet, at the end of the protocol, Alice and Bob may hold the same secret string (the *key*). At first sight, this appears almost disappointingly mundane. The final key is only a sequence of symbols. Nothing in its appearance reveals whether it is private or already known to someone else. Written on a page, it looks no different from any other random string. Secrecy is therefore not a property that can be read off from the key itself. Its meaning lies not in what the string is, but in what can be said about it.

Eve is all-powerful. Yet, to say that Alice and Bob share a *secret* key is to make a statement about Eve's ignorance.

This is the remarkable point. To call the key *secret* is to make a claim about its relation to another system or another possible holder of information. In quantum key distribution, this claim is made against Eve. Alice and Bob do not merely possess the same string. This string is, in a precise operational sense, absent from Eve's knowledge. From the data produced from their measurement devices, and from the laws of quantum mechanics, they can turn this relational statement into a theorem: Eve knows essentially nothing about their shared key, even though she is an entity with unbounded abilities.

This is the marvel of quantum cryptography. In classical computational cryptography, security is often obtained by assuming that some mathematical problem is too difficult for an adversary to solve within a realistic amount of time. Quantum cryptography begins from a different premise. It asks whether secrecy can be certified directly from the laws governing physical systems. The answer, initiated by the discovery of quantum key distribution [26, 69], is that quantum theory contains restrictions on information itself: unknown states cannot be copied perfectly, incompatible measurements cannot be predicted simultaneously, and sufficiently non-classical correlations cannot be freely shared with a third party². These restrictions can be converted, through a security proof, into

from observed correlations rather than from a detailed model of the devices themselves; see, for example, Refs. [2, 157].

²These principles enter different families of QKD proofs in different ways. In prepare-and-measure QKD,

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quantitative bounds on Eve's information.

But this is perhaps not the end of the marvel. The relation between quantum cryptography and fundamental physics may be more intricate than it first appears. The quantum key distribution protocol described above does not merely generate a key; it generates evidence about what lies outside the trusted laboratories, using only data produced within them. This idea is eerily similar to the black hole radiation decoding problem, where observers outside a black hole are able to obtain knowledge about the interior³.

The key is thus more than a practical resource. It is the residue of a confrontation between information and physics. Alice and Bob end with a string of numbers, but what they have really produced is a "certificate": a finite, operational statement about the limits of an unknown and powerful Eve. That such a statement can be made at all is the central wonder from which quantum cryptography begins.

1.1. Quantum physics

At the end of the nineteenth century, classical mechanics and classical electrodynamics had achieved an extraordinary range of successes, but they could not consistently account for the behaviour of microscopic systems. Black-body radiation, atomic spectra, the stability of atoms, and later the photoelectric effect all pointed towards the same conclusion: matter and radiation could not be understood by simply extending the concepts of classical physics to smaller scales. The first decisive step was Planck's introduction of energy quantisation in his analysis of black-body radiation [158]. Einstein then sharpened the physical meaning of this idea by proposing that light itself could behave as if its energy were carried in discrete quanta [66]. Bohr's model of the atom brought quantisation into the structure of matter, explaining the stability and spectral lines of hydrogen by postulating discrete atomic orbits and transitions [36]. These developments did not yet constitute quantum physics in its modern form, but they exposed the inadequacy of the classical picture.

The decisive change came in the mid-1920s. Heisenberg's matrix mechanics replaced the classical description of trajectories by algebraic relations between observable quantities [96]. Schrödinger's wave mechanics introduced a different but equivalent description

non-orthogonality and no-cloning capture the impossibility of a passive classical interception of the quantum signal [198, 58]; security proofs such as those of Lo-Chau and Shor-Preskill make this intuition precise within an unconditional security framework [130, 169]. In entropic security proofs, measurement incompatibility is converted into a lower bound on Eve's uncertainty about the raw key through entropic uncertainty relations [134, 29]. In entanglement-based and device-independent QKD, Bell nonlocality and the monogamy of non-classical correlations allow secrecy to be inferred from the observed statistics themselves [3, 157, 142].

³The link between these two ideas has been recently studied. However, the precise arguments are beyond the scope of this thesis. The interested reader is directed to Ref. [39].

in terms of wave functions and differential equations [167]. Born then gave the wave function its probabilistic interpretation, arguing that it does not directly describe a material wave in space but rather determines the probabilities of measurement outcomes [38]. In the formulations of Dirac and von Neumann, quantum physics acquired its abstract mathematical form: states are represented by vectors or density operators on Hilbert spaces, physical quantities by operators, and measurements by rules that assign probabilities to possible outcomes [61, 184]. A theory that began as a set of repairs to classical physics had become a new framework for describing physical reality.

This new framework also changed the meaning of a physical state. In classical mechanics, the state of a system is usually thought of as a point in phase space: if the position and momentum of every particle were known exactly, then the future evolution would be determined by the equations of motion. Probabilities enter only because an observer may be ignorant of the exact state. Quantum mechanics does not fit into this picture. Even when a quantum system is prepared as deterministically as the theory allows, the outcomes of later measurements may still be intrinsically probabilistic.

This distinction is crucial for quantum information. Information is always carried by physical systems, but quantum physics changes what kind of information a physical system can carry and what is accessible. To make such statements precise, one needs the mathematical language of quantum mechanics. In its modern formulation, a physical system is associated with a Hilbert space $\mathcal{H}$ (which is a linear vector space equipped with an inner product). A *pure* state is represented by a unit vector $|\psi\rangle$ in this space, while a general state is represented by a density operator (matrix⁴) ρ satisfying

$$\rho \geq 0, \quad \text{Tr}[\rho] = 1. \quad (1.1)$$

For the purposes of this thesis, we do not need a complete axiomatic reconstruction of quantum mechanics. Instead, we will isolate a few structural features of the theory that later become central in quantum information and quantum cryptography.

The first and most important distinction is that the probabilities observed in an experiment are not obtained by revealing a pre-existing classical value, but through the measurement postulate. For a measurement described by positive operators $\{M_x\}_x$ satisfying $\sum_x M_x = \mathbb{I}$, the probability of observing outcome x is

$$p(x) = \text{Tr}[M_x \rho], \text{ equivalently for pure states } p(x) = \text{Tr}[M_x |\psi\rangle\langle\psi|]. \quad (1.2)$$

This rule, historically rooted in Born's probabilistic interpretation [38], is one of the points where quantum mechanics departs most sharply from classical mechanics. The state encodes all probabilities that the theory assigns to possible measurements, but it does not

⁴All operators can be represented as matrices. The trace (Tr) of a matrix is the sum of its diagonal elements.

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generally allow one to regard all measurement outcomes as simultaneously pre-existing classical facts.

The second structural feature is superposition. If $|\psi_1\rangle$ and $|\psi_2\rangle$ are possible pure states of a system, then a normalised linear combination

$$|\psi\rangle = \alpha|\psi_1\rangle + \beta|\psi_2\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad (1.3)$$

is again a possible pure state⁵. This is not simply the statement that a system may be in one classical alternative or another. It means that the relative amplitudes and phases between alternatives can influence later measurement statistics. Superposition is therefore one of the basic reasons why quantum systems can encode and process information in ways that have no direct classical analogue [61, 152].

The third feature is incompatibility. In classical mechanics, one may in principle assign simultaneous values to all physical properties of a system. Quantum theory does not permit this in general. Observables represented by non-commuting operators cannot be treated as jointly sharp quantities. The canonical example is position and momentum, whose incompatibility lies behind Heisenberg's uncertainty principle [97]. This is the physical and mathematical origin of the fact that information encoded in one quantum basis cannot, in general, be read without consequences for information encoded in a complementary basis. Later, this feature becomes one of the conceptual roots of quantum cryptography: the attempt to extract information from an unknown quantum state is constrained by the measurement structure of the theory.

The fourth feature is composition. If two systems have Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, then the joint system is described on the tensor product space

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B. \quad (1.4)$$

This rule is responsible for one of the most radical consequences of the theory: entanglement. A pure state of two systems is called entangled if it cannot be written as a product

$$|\psi\rangle_{AB} = |\phi\rangle_A \otimes |\chi\rangle_B. \quad (1.5)$$

⁵This should be distinguished from a classical probabilistic mixture. The mixed state corresponding to “ $|\psi_1\rangle$ with probability $|\alpha|^2$ and $|\psi_2\rangle$ with probability $|\beta|^2$ ” is

$$\rho_{\text{mix}} = |\alpha|^2 |\psi_1\rangle\langle\psi_1| + |\beta|^2 |\psi_2\rangle\langle\psi_2|.$$

By contrast, the superposition state gives

$$\rho_{\text{sup}} = |\psi\rangle\langle\psi| = |\alpha|^2 |\psi_1\rangle\langle\psi_1| + |\beta|^2 |\psi_2\rangle\langle\psi_2| + \alpha\beta^* |\psi_1\rangle\langle\psi_2| + \alpha^*\beta |\psi_2\rangle\langle\psi_1|.$$

The last two terms are coherence terms. They have no analogue in a classical mixture and are responsible for interference effects.

A general state is entangled if it cannot be expressed as a probabilistic mixture of product states.⁶ For an entangled state, the joint system therefore possesses properties that cannot be reduced to independently specified states of its constituent parts. Einstein, Podolsky, and Rosen famously used such correlations to argue that the quantum-mechanical description of reality was incomplete [67]. Bell later showed that this debate was not merely philosophical: theories satisfying an appropriate notion of locality must obey inequalities that are violated by quantum-mechanical predictions [25]. Entanglement therefore reveals that the quantum description of a composite system cannot, in general, be reduced to a classical catalogue of local properties.

These principles also impose restrictions on information. An unknown quantum state cannot be copied perfectly [198, 58]. This is the infamous no-cloning theorem. Similarly, the information obtainable from a quantum system is limited by the measurement one performs and by the non-orthogonality of the states in question [100, 152]. Thus quantum physics is not only a physical theory of microscopic particles. It is also a theory about what can be known, copied, and shared.

This is the point at which quantum physics becomes relevant for cryptography. Classical information can be copied without changing it. Quantum information cannot. Classical correlations can, in principle, be distributed and duplicated freely. Quantum correlations obey monogamy-type restrictions that limit how strongly they can be shared across systems. Classical uncertainty may reflect ignorance about an underlying fact. Quantum uncertainty can arise because the theory itself does not assign simultaneous definite values to incompatible observables. These differences make quantum mechanics more than a new physical substrate for old information-processing tasks. They make it a new theory of information-bearing systems.

The aim of the next subsections is to place this physical theory next to the separate historical development of cryptography. Cryptography begins from a different set of questions: how can messages be protected, how should secrecy be defined, and what assumptions are needed to prove security? Quantum cryptography emerges when these questions are asked not only about mathematical symbols, but about physical systems governed by the laws described above.

⁶A bipartite state is called separable if it can be written as

$$\rho_{AB} = \sum_i p_i \rho_A^{(i)} \otimes \rho_B^{(i)}, \quad p_i \geq 0, \quad \sum_i p_i = 1.$$

Such a state may contain classical correlations, but it is not entangled. Any state that admits no such decomposition is entangled.

1.2. Cryptography

Cryptography is much older than its modern mathematical form. For most of its history, it was the art of secret writing: a collection of methods by which a message could be transformed so that only its intended recipient could recover its meaning. Historical accounts of cryptography are therefore often histories of a struggle between cipher-makers and code-breakers, where each new method of concealment is eventually met by a new method of attack [110].

A decisive conceptual step was the recognition that the security of a cryptographic system should not rely on the secrecy of the method itself. Kerckhoffs formulated this principle in the nineteenth century in the context of military cryptography: a system should remain secure even if everything about the system is public, except for the secret key [114]. This idea separates the public design of a cryptographic protocol from the private information that enables its use. In modern language, the algorithm may be known to everyone, while the key remains secret. This shift became foundational because it turned cryptography from an art of obscurity into a discipline in which systems can be openly criticised, standardised and proved secure under explicit assumptions.

The next major transformation came when cryptography was given an information-theoretic formulation. Shannon's theory of secrecy systems introduced a precise mathematical language for encryption, plaintexts, ciphertexts, keys, and adversarial information [168]. In this framework, the security of a cryptosystem could be formulated in terms of the statistical relationship between the plaintext and the ciphertext, rather than judged merely by how complicated the encryption procedure appeared. The strongest security notion within this framework is perfect secrecy. If M is the plaintext and C is the ciphertext, then perfect secrecy requires

$$\Pr[M = m \mid C = c] = \Pr[M = m] \quad (1.6)$$

for every message m and every ciphertext c . In words, observing the ciphertext does not change the adversary's probability distribution over the message. Shannon showed that the one-time pad achieves this level of secrecy when the key is uniformly random, at least as long as the message, used only once, and kept secret [168]. This is perfect because in this case secrecy is independent of the adversary's computational power.

However, information-theoretic security poses a central difficulty. Perfect secrecy requires a secret key that is as large as the message to be protected. The one-time pad therefore moves the problem rather than eliminating it: if Alice and Bob already share a long secret key, they can communicate securely; but how should they obtain such a key in the first place? This key-distribution problem is one of the fundamental tensions in cryptography⁷.

⁷The astute reader will note that this is also one of the points at which quantum cryptography entered naturally.

One way around this difficulty is to relax the requirement of security against an adversary with unlimited computational power. Cryptographic schemes had long relied implicitly on the practical difficulty of reversing an encryption procedure, but modern computational cryptography made this dependence explicit: security is based on the assumption that certain mathematical problems cannot be solved efficiently. The decisive development for key distribution was public-key cryptography. Diffie and Hellman showed that two parties could establish a shared secret over a public channel without first sharing a secret key, provided that the adversary could not efficiently solve the underlying mathematical problem [59]. Rivest, Shamir, and Adleman subsequently introduced RSA, a concrete public-key system whose security is related to the difficulty of factoring large integers [164]. Other foundational public-key systems are based on the discrete logarithm problem, most notably the ElGamal cryptosystem [70]. This line of thought also led to elliptic-curve cryptography, independently proposed by Miller and Koblitz, where security is based on the presumed hardness of the elliptic-curve discrete logarithm problem [146, 115].

In the paradigm of computational cryptography, secrecy is to be guaranteed against an adversary limited by feasible computation. This, of course, implied that words such as “hard” and “feasible” had to be given mathematical meaning. The early 1980s mark the point at which this viewpoint became systematic. Yao’s work on trapdoor functions clarified the abstract structure behind public-key cryptography: there should be operations that are easy to perform in one direction, hard to invert without additional secret information, and easy to invert for someone who possesses that secret information [201]. This connected cryptographic security directly to computational complexity. Blum and Micali then gave a complexity-theoretic account of pseudorandomness: a sequence need not be truly random in the information-theoretic sense if no efficient algorithm can predict its future bits from its past ones [34]. Goldwasser and Micali further sharpened the foundations of encryption by showing that security should be formulated against polynomially bounded adversaries under explicit hardness assumptions, and that randomness in encryption is not merely an implementation detail but a conceptual ingredient of security [83]. Around the same period, Dolev and Yao showed that communication protocols themselves could be treated as mathematical objects inside an adversarial model, where the attacker controls the network and protocols are analysed by what such an attacker can derive [62].

Together, these developments transformed cryptography into a formal mathematical discipline. A cryptographic claim had to specify the adversary, the resources available to that adversary, the task by which success is judged, and the assumption under which success is unlikely. Security thus became a theorem of the form: under these assumptions, no adversary of this kind can achieve this goal except with a controlled probability. Classical modern cryptography derives such impossibility statements mainly from mathematical hardness assumptions. Quantum cryptography, as we will see shortly, asks whether some of them can instead be derived from the laws governing physical systems.

This formal viewpoint is now extended to all areas of cryptography. Encryption is only one cryptographic task. Authentication asks whether a message really came from a claimed sender. Digital signatures provide a public way to verify authorship. Zero-knowledge proofs let one party convince another of the truth of a statement without revealing the underlying witness [84]. Secure computation lets parties jointly compute a function of their private inputs while revealing as little else as possible. Across these tasks, the same conceptual structure appears: one defines what an adversary is allowed to do, what it means for the protocol to be secure, and under which mathematical assumptions the security proof holds.

Thus, while quantum mechanics arose from physical crises in our understanding of matter and radiation, modern cryptography arose from the mathematical formalisation of secrecy, adversaries, and proof. The two fields begin from different directions. Quantum mechanics asks what physical systems can be and how they behave. Cryptography asks what information an adversary can possess and what can be certified despite that adversary. Once both are viewed through the language of information, their meeting becomes almost unavoidable. If information is always carried by physical systems, and if quantum mechanics changes the rules governing such systems, then it is natural to ask whether cryptographic security can be derived not only from mathematical hardness assumptions, but also from the laws of physics themselves.

1.3. Quantum cryptography

The preceding two sections describe two traditions that developed for very different reasons. Quantum mechanics emerged from the failure of classical physics to describe microscopic systems, and it led to a theory in which information-bearing physical states cannot always be copied, distinguished, or jointly measured. Cryptography, by contrast, emerged from the practical problem of secret communication, and later became a mathematical discipline in which security became provable through well-defined adversarial models. Quantum cryptography begins at the point where these two stories meet. If information is always represented by physical systems, and if quantum mechanics changes the rules governing such systems, then cryptographic security need not be based only on mathematical hardness assumptions. It may also be based on the laws of physics.

Historically, this idea first appeared not as quantum key distribution, but as quantum money. In Wiesner's proposal of conjugate coding, information was encoded into quantum states chosen from incompatible bases [192]. This ensured that the security of the object was tied to the fact that an adversary could not perfectly learn the encoded information without knowing which basis had been used, nor could the adversary produce arbitrary perfect copies of the underlying quantum states. In this sense, Wiesner's idea

already contained the conceptual seed of quantum cryptography: physical systems could be designed so that their very quantum structure limited what an adversary could do.

Bennett and Brassard transformed this insight into the first quantum key distribution protocol (now known as BB84 [26]). Their protocol addressed precisely the tension left open by Shannon's theory of perfect secrecy. The one-time pad provides information-theoretic secrecy, but only if Alice and Bob already share a secret key of sufficient length. As we discussed, classical public-key cryptography solved the distribution problem by relying on computational assumptions. Quantum key distribution offers a different route: Alice and Bob use quantum states, together with public classical communication for reconciliation⁸ of errors, to generate a shared random key whose secrecy can be certified against an adversary not limited by computational power.

This was the first major innovation in quantum cryptography. In classical cryptography, an intercepted signal can in principle be copied without leaving any trace. In quantum mechanics, the relevant signal states may be non-orthogonal, and measurements in the wrong basis cannot reveal all the encoded information without affecting the statistics observed later. The security intuition of BB84 is thus connected directly to the structural features discussed earlier: non-orthogonality, measurement incompatibility, and the impossibility of perfect copying. The later development of security proofs made this intuition precise. In particular, entanglement-based methods and error-correction/privacy-amplification arguments led to proofs of unconditional security for QKD [130, 169], while subsequent reviews systematised both the theoretical and practical aspects of the field [82, 166].

A second route was opened by Ekert's entanglement-based protocol [69]. Alice and Bob receive entangled particles and test a subset of their measurement results for a violation of a Bell inequality. The purpose of this test is cryptographic: a Bell violation rules out an explanation based solely on pre-existing local values shared between the devices and an adversary. Moreover, because non-classical correlations cannot be shared arbitrarily, sufficiently strong correlations between Alice and Bob limit the correlations that Eve can possess with their outcomes. The EPR⁹ correlations that had originally been used to question the completeness of quantum mechanics thus became a means of certifying secrecy. This principle was later developed into device-independent cryptography, where security is inferred from observed input-output correlations without requiring a complete model of the devices' internal operation.

⁸Reconciliation is the classical post-processing step in QKD in which Alice and Bob use authenticated public communication to correct discrepancies between their correlated raw strings. Since this public discussion may reveal some information to Eve, its leakage must be accounted for before privacy amplification is applied.

⁹The term EPR refers to the 1935 argument of Einstein, Podolsky, and Rosen, who used entangled continuous-variable states with perfectly correlated position and momentum observables to argue that quantum mechanics was incomplete [67]. Bohr's reply defended the completeness of the quantum description, and the debate later became one of the conceptual roots of modern entanglement theory [37].

Thus we observe how the object of the security proof changes accordingly. In Shannon's theory, (information-theoretic) secrecy is a statement about a joint probability distribution: the ciphertext must reveal no information about the plaintext. In computational cryptography, security is a statement about the inability of an efficient algorithm to distinguish or recover information under a hardness assumption. In QKD, the task of the security proof is to convert observed experimental data into a statement about this external quantum system.

Once quantum mechanics can be used to certify not only secrecy but also limits on an adversary's correlations with a physical system, it is natural to ask which other cryptographic tasks can be realised from the same physical restrictions. Wiesner's quantum money already suggested one such direction: the no-cloning principle can be used not only to hide information, but also to prevent duplication of a cryptographic object. This idea reappears in unclonable encryption. In ordinary encryption, an adversary who intercepts a ciphertext can copy it, store the copy indefinitely, and wait for side information or a later key leak. Unclonable encryption asks for a stronger property: the adversary should not be able to split the encrypted information into two usable copies that can later be independently decoded. Gottesman introduced this task as a way of extending the no-cloning property from unknown quantum states to classical messages encoded into quantum systems [86]. The point is again the same fusion of the previous two sections: a cryptographic requirement is formulated mathematically, but the reason it can be achieved comes from a physical restriction on information.

Another direction is position-based cryptography. Classical cryptography normally treats identity as something established by a key, password, certificate, or computational capability. Position-based cryptography lets geographical location itself serve as a credential. The natural task is position verification: a set of verifiers wish to check that a prover is located at a claimed point in space. Classically, this turns out to be impossible, because separated adversaries can coordinate their responses using pre-shared information and communication strategies [48]. Quantum mechanics changes the landscape but does not make the problem trivial. Quantum position verification and related quantum tagging proposals use quantum states together with relativistic signalling constraints to test whether a response could have been produced from the claimed location [113]. At the same time, entanglement gives adversaries new powers: with a sufficient amount of pre-shared entanglement, general position-verification schemes can be attacked through forms of instantaneous non-local quantum computation [43, 24]. Position verification therefore illustrates both sides of quantum cryptography at once. Quantum states can create new cryptographic possibilities, but entanglement can also create new attacks.

This broader development shows that quantum cryptography is a systematic study of which cryptographic statements can be derived from the physical structure of information.

QKD derives secrecy from the impossibility of an adversary being too correlated with the final key. Unclonable encryption derives protection from the impossibility of freely copying quantum-encoded information. Quantum position verification derives security of location by combining quantum information processing with spacetime constraints¹⁰. In each case, the central question is the same: what can Alice and Bob prove about an adversary when the adversary is limited only by the laws of physics?

1.4. Continuous-variable quantum systems

The discussion so far has described the conceptual meeting point between quantum mechanics and cryptography. Quantum mechanics supplies restrictions on information, while cryptography supplies the language in which such restrictions become statements about security claims against well-defined adversaries. To turn this meeting point into an actual protocol, however, one must choose which physical quantum systems carry the information.

There are two broad ways in which quantum information is commonly represented. In discrete-variable (DV) quantum information, the relevant degrees of freedom are described by a finite-dimensional state space. The simplest example is a qubit, whose state lies in a two-dimensional Hilbert space. Such a qubit may be implemented using the polarisation of a single photon, two time bins, two paths of an interferometer, the spin of an electron, or two energy levels of an atom. Mathematically, the elementary structure is finite-dimensional: a basis may be written as $\{|0\rangle, |1\rangle\}$, and a general pure qubit state has the form

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (1.7)$$

This setting has been central to the development of quantum information theory. It gives a direct quantum analogue of the classical bit and provides the natural language for many foundational protocols, including the original formulations of quantum key distribution.

Continuous-variable (CV) quantum information begins from a different physical object. Instead of restricting attention to a finite set of basis states, one uses observables with a continuous spectrum. The paradigmatic example is a *mode*¹¹ of the electromagnetic field.

¹⁰Position verification is not limited to an application of the no-cloning theorem. It can be achieved by sending only classical data, but the prover must have quantum capabilities. This is shown by a recent certified randomness (proof-of-quantumness)-based protocol [111].

¹¹A mode is an independent degree of freedom of the electromagnetic field, specified by a complete set of properties such as its frequency, propagation direction, and polarisation. After quantisation, each mode behaves mathematically like a harmonic oscillator, with field quadratures playing the role of position and momentum variables; see, for example [189]. We will delve into the mathematical structure more in Section 2.3.

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Such a mode can be described by two conjugate quadratures, usually denoted by $\hat{q}$ and $\hat{p}$, which satisfy the canonical commutation relation

$$[\hat{q}, \hat{p}] = i, \quad (1.8)$$

where we have chosen units in which $\hbar = 1$. These quadratures are the quantum analogues of the real and imaginary components of the field amplitude. They play, for optical modes, a role similar to position and momentum in ordinary quantum mechanics. A state of an optical mode may therefore be described by a wavefunction in the q quadrature, by its expansion in photon-number states, or by a phase-space representation such as the Wigner function [41, 189]. We will go into more mathematical details in the next chapter.

The mathematical differences between discrete and continuous variables form the vertebrae of this thesis. A qubit is described by a finite-dimensional Hilbert space. An optical mode, by contrast, is described by an infinite-dimensional Hilbert space. Its quadrature outcomes range continuously, and physically relevant states must often be constrained by energy rather than by dimension¹². This changes the technical structure of the theory. In finite dimensions, many statements can be phrased in terms of matrices of fixed size and probability distributions over finite alphabets. In the continuous-variable setting, one must deal with unbounded operators, continuous probability distributions, and finite-energy approximations.

At the same time, continuous-variable systems possess a powerful simplifying structure. Many experimentally natural states of light are *Gaussian* states¹³: vacuum states, coherent states, squeezed states, thermal states, and their multimode generalisations. These states are completely characterised by their first and second moments. This gives continuous-variable quantum information a phase-space language [41, 189]. It also makes the connection with optical experiments particularly direct, since coherent states, squeezed states, beam splitters, phase shifters, and measurements (homodyne/heterodyne)¹⁴ are efficiently described in this language.

From a practical point of view, this makes the continuous-variable setting particularly attractive. DV optical protocols often rely on preparing and detecting individual photons. Such technologies have advanced enormously, but true single-photon generation and detection remain experimentally demanding, especially when high efficiency, low dark counts and low noise are required [92, 68]. CV implementations, by contrast, can

¹²Although it is true that the dimension of a CV system is unbounded, certain approximations can cause finite dimensions to appear due to a physical bound on energy. This becomes relevant in Chapter 5.

¹³States which admit a Gaussian Wigner function are called Gaussian states. For details, see Section 2.3.3.

¹⁴A homodyne measurement measures only a single canonical quadrature direction (either q or p). A heterodyne measurement measures both directions simultaneously, but it effectively doubles the noise, as two separate measurements are made. This is, of course, expected due to the uncertainty principle discussed above.

leverage mature coherent optical communication technology and standard telecom components [189]. They often use coherent or squeezed states of light and measure field quadratures by homodyne or heterodyne detection, where the signal is compared with a strong local oscillator using standard photodiodes [41, 189].

This practical motivation also determines how continuous-variable cryptographic protocols must be studied. The motivation behind using CV systems is not merely to create a mathematically different version of quantum cryptography. The appeal of the CV setting is that it is close to the physics of existing optical communication infrastructure. But this also means that any useful CV protocol must be analysed under the imperfections that such infrastructure inevitably introduces.

This is especially important because loss and noise affect CV protocols in a different way from many DV optical protocols. In a single-photon picture, optical loss is often registered as a missing detection event. In a CV protocol, Bob typically still obtains a quadrature outcome, but the signal has been attenuated and contaminated by noise. Thus the operational question changes from asking only whether a photon arrived to analysing how much information remains in noisy continuous data. Security and performance in the CV case therefore must be viewed through the lens of loss, noise, limits of state preparation, detector inefficiency and finite-energy constraints. This introduces the need for statistical techniques which are not needed in DV quantum cryptography.

The current thesis is concerned with perfect (information-theoretic) security in a world of physical imperfection. The central question is how far the cryptographic power of quantum mechanics can be developed within continuous-variable systems while retaining a realistic view of the physical channel. The original examples of quantum cryptography were often formulated in DV language, but optical communication is naturally continuous-variable at the level of field quadratures. This raises a broad question: which cryptographic primitives can be built, analysed and proved secure when the information carriers are continuous-variable quantum states, and when loss and noise are treated as central features of the model rather than as afterthoughts?

1.5. Quantum cryptography with continuous variables

1.5.1. Research question I: Reconciliation for long-distance CVQKD

The first concrete setting in which the continuous-variable approach to quantum cryptography became practically compelling was quantum key distribution. Early CVQKD protocols showed that secret keys could be generated by encoding information into continuous optical degrees of freedom, rather than into single-photon qubits [160, 99, 46]. A particularly important step was the coherent-state protocol with Gaussian modulation, in

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which Alice draws random quadrature displacements, sends weak coherent states through an optical channel, and Bob measures field quadratures using homodyne detection [89]. Together with reverse reconciliation, this made it possible to extract keys even when the channel transmission is below one half [88]. In this form, CVQKD is very close in spirit to classical coherent optical communication: information is carried by field amplitudes, transmitted through fibre, and recovered by quadrature measurements.

After the quantum transmission stage, however, Alice and Bob do not yet share a secret key. They share correlated continuous data. In the Gaussian-modulated homodyne setting, Bob's outcome may be modelled as

$$Y = \sqrt{T}X + N, \quad (1.9)$$

where X is Alice's modulation variable, T is the channel transmission, and N contains shot noise and excess noise. The task of the classical post-processing is to convert many samples of X and Y into identical discrete strings while leaking as little information as possible to Eve. This step is called information reconciliation, and it is one of the central bottlenecks in long-distance CVQKD.

In DV QKD, optical loss often appears as a missing detection event, which can be discarded. In CVQKD, Bob always obtains a measurement outcome. Loss therefore does not create erasures in the same way; it weakens the correlation between Alice's and Bob's continuous data. At long distances, the transmission T becomes very small¹⁵, and the reconciliation problem becomes an extremely low signal-to-noise ratio decoding problem. If the signal-to-noise ratio is denoted by ε , then in the low-SNR regime the mutual information between Alice and Bob scales as

$$I(X; Y) = \frac{1}{2} \log_2(1 + \varepsilon) = \frac{\varepsilon}{2 \ln 2} + \mathcal{O}(\varepsilon^2). \quad (1.10)$$

Thus each optical pulse carries only a very small amount of useful shared information. To obtain a useful key, the reconciliation procedure must operate close to the Shannon limit while remaining fast enough for the optical repetition rate.

The security benchmark is given by the Devetak–Winter expression. For Gaussian modulation and one-way reverse reconciliation, the relevant ideal asymptotic key ratio (key bits generated per laser pulse) is governed by

$$\Delta I = I(X; Y) - I(E; Y), \quad (1.11)$$

where $I(E; Y)$ denotes Eve's information about Bob's variable [57]. In the long-distance, low-transmission regime, and in the limit of large modulation variance, this approaches

$$\Delta I \longrightarrow \frac{T}{2 \ln 2} + \mathcal{O}(T^2), \quad (1.12)$$

¹⁵For an optical fibre of length L and attenuation coefficient α measured in dB/km, the channel transmittance is $T(L) = 10^{-\alpha L/10}$. For standard telecom fibre, one typically takes $\alpha \simeq 0.2$ dB/km.

which we refer to as the low- T Devetak–Winter value. This is the natural target for reconciliation: a good post-processing scheme should convert the weak Gaussian correlations between Alice and Bob into secret bits at a ratio as close as possible to this benchmark.

A large amount of work has already gone into this problem. Existing reconciliation methods for CVQKD include multidimensional reconciliation, turbo codes, LDPC codes, polar codes, Raptor-like codes, and other low-rate coding constructions [20, 127, 131, 107, 145, 106, 91, 90, 108, 202, 200]. These methods have achieved impressive reconciliation efficiencies and have pushed CVQKD to increasingly long distances. Nevertheless, long-distance operation places pressure on more than nominal efficiency alone. The decoder must be implementable at high throughput, preferably in real time, and this becomes difficult for very low-rate iterative decoding schemes. Moreover, many practical schemes reject blocks that fail to decode. Such frame rejection is natural from an engineering point of view, but from a security point of view it is a form of post-selection: conditioning on accepted blocks may also condition Eve’s quantum state. Therefore, the achieved fraction of the Devetak–Winter value is only a meaningful secret-key benchmark when acceptance, leakage, and residual errors are accounted for consistently.

This leads to the first research question:

Can we design a reconciliation system for long-distance CVQKD that is parallelisable and produces secret-key rates closer to the Devetak–Winter value than existing methods?

1.5.2. Research question II: practical CV quantum position verification

Quantum key distribution answers one fundamental question in cryptography: can two distant parties generate a secret key in the presence of an adversary? Quantum position verification (QPV) asks a different question. Can the *location* of a party itself be used as a cryptographic credential? In a position-verification protocol, a set of verifiers wish to test whether a prover is located at a claimed position P . The prover should be accepted only if the responses received by the verifiers must have originated from that position, taking into account the finite speed of light. Thus the primitive QPV combines relativistic signalling constraints with the information-processing restrictions imposed by quantum mechanics.

The idea of position-based cryptography was introduced by Chandran, Goyal, Moriarty and Ostrovsky [48]. In a purely classical world, however, position verification is impossible in full generality. If the challenge information can be copied, then adversaries placed near the verifiers can duplicate and redistribute the relevant data quickly enough to simulate a prover at the claimed position. Quantum mechanics appears to change the situation because unknown quantum states cannot be copied perfectly [198]. This observation

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led to quantum position verification, also studied in the early literature under the name quantum tagging [112, 137, 138, 118]. In such protocols, the verifiers send classical and quantum challenges arranged so that an honest prover at P can combine them in time, while spatially separated adversaries cannot.

The adversarial model is central to the meaning of security in QPV. In the usual one-dimensional setting, two verifiers V_0 and V_1 are placed on opposite sides of the claimed position P . The adversaries, traditionally called Alice and Bob, are not located at P ; instead, they sit between the verifiers and the claimed position and try to make the verifiers accept. They may intercept quantum and classical messages, perform arbitrary local operations, and communicate with one another, but they are constrained by the same relativistic timing conditions as the honest prover. The strongest versions of the model also allow them to share entanglement before the protocol begins. This last resource is what makes QPV interesting.

Although the no-cloning theorem rules out the simple classical copying attack, a universal quantum attack is possible if the adversaries share sufficiently large amounts of entanglement [43, 24]. This attack is based on ideas related to instantaneous nonlocal quantum computation. However, the entanglement required by such attacks can grow with the input size, making them physically unrealistic in many regimes. This motivates the bounded-entanglement model: rather than asking for security against attackers with arbitrary entanglement, one asks whether a protocol can remain secure against attackers whose pre-shared quantum resources are limited. Much of modern QPV is shaped by this tension between unconditional impossibility and resource-bounded security [44, 170, 64, 33, 54, 5, 15, 19, 18].

For a communication protocol to be useful, it must tolerate imperfect channels. Loss and noise directly affect what the provers observe, and therefore directly affect whether honest and dishonest behaviour can be distinguished. In finite-dimensional optical protocols, photon loss is particularly delicate because an adversary may try to exploit no-detection events. This has led to increasing attention to loss- and error-tolerant QPV protocols [7, 72, 4]. The continuous-variable setting changes this picture. A coherent or squeezed optical state passing through a lossy and noisy channel does not simply produce a binary arrival-or-no-arrival event. Instead, attenuation reduces the signal and excess noise broadens the distribution of quadrature outcomes. Thus, in CV-QPV, the honest prover produces non-deterministic continuous data, and security means showing that any admissible attack produces a statistically distinguishable distribution.

This leads to the research question studied in Chapter 4:

Is it possible to achieve secure position verification using continuous-variable quantum states in the presence of realistic loss and noise, while also maintaining

security against attackers with access to a bounded amount of entanglement?

1.5.3. Research question III: continuous-variable unitary two-designs

As we discussed in Section 1.1, the state of a quantum system is represented by a vector, or more generally by a density operator, on a Hilbert space. The reversible transformations of such a system are described by unitary operators. These can be represented as matrices U satisfying

$$U^\dagger U = U U^\dagger = I. \quad (1.13)$$

A unitary in quantum theory represents an ideal reversible quantum operation: if the system is in state ρ , the operation transforms it as

$$\rho \mapsto U \rho U^\dagger. \quad (1.14)$$

Unitary operations preserve inner products, probabilities, and the total norm of the state. A recurring theme in quantum information theory is the usefulness of applying such operations at random. Many protocols and proof techniques become simpler if one can average over a uniformly random unitary, drawn from what is known as the *Haar* measure on the unitary group. One can think of Haar averaging as the ideal way of randomising a reversible quantum operation. Just as an ideal random number generator provides samples from a uniform distribution over numbers or bit strings, the Haar measure provides the mathematically ideal notion of a uniformly random unitary. Such averages provide a clean mathematical model of generic quantum dynamics and appear in tasks such as decoupling, benchmarking, tomography, and fidelity estimation [56]. In practice, however, Haar-random unitaries are not realistic physical objects. Sampling them exactly is computationally expensive in large dimensions, and implementing a generic Haar-random unitary is far beyond what is usually available in an experiment [144, 56]. This motivates the notion of a unitary design.

A unitary t -design is an ensemble of unitaries whose first t moments agree with those of the Haar measure. Mathematically, if μ is a probability distribution over unitaries on a d -dimensional Hilbert space, then the associated t -fold averaging (*twirling*) channel is

$$\mathcal{G}_\mu^{(t)}(X) := \int U^{\otimes t} X (U^\dagger)^{\otimes t} d\mu(U). \quad (1.15)$$

Here, $X \in \mathbb{C}^{\otimes 2t}$. The measure μ is an exact unitary t -design when this twirl agrees with the corresponding Haar twirl.

For many applications, one does not need all Haar moments. The case $t = 2$ is already powerful, because it captures the second-moment behaviour of Haar-random unitaries. This is why unitary two-designs appear naturally in randomised benchmarking, where

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averaging over a suitable gate set simplifies the description of noise, and in tomography and randomised measurement protocols, where sufficiently random measurement bases allow one to extract many properties of a quantum state efficiently [56, 135, 104]. The use case of decoupling also naturally fits in this picture. Decoupling is a randomisation mechanism that destroys correlations between a target system and a reference system by scrambling information across a very large system. Averaging over 2-designs facilitates this scrambling in an efficient manner [65]. An *approximate*¹⁶ two-design relaxes exact equality: it only requires the two-fold twirl to be close to the Haar twirl in a suitable norm. This relaxation is essential, since approximate designs often retain the useful averaging properties of Haar randomness while being much easier to construct and implement.

In finite-dimensional quantum information, unitary designs are now a mature tool. For example, the Clifford group forms a unitary two-design, and several efficient approximate constructions are known [56, 50, 95]. A particularly relevant construction for this thesis is the alternating-basis construction of Nakata *et al.*, where one alternates random phase unitaries diagonal in two complementary bases [149]. Repeated alternation contracts the part of the two-copy operator space that is not fixed by the Haar twirl, and the resulting ensemble approaches a unitary two-design. This gives a clear finite-dimensional recipe: rather than sampling a generic unitary, one can build pseudorandomness from simple phase operations in complementary bases.

The situation is much less clear for continuous-variable systems. A single bosonic mode is described by an infinite-dimensional Hilbert space, for instance $L^2(\mathbb{R})$, and there is no Haar (normalisable) probability measure on this space. Moreover, the most natural CV candidates do not solve the problem. Gaussian states and Gaussian unitaries are experimentally accessible and play in CV quantum information a role somewhat analogous to stabiliser states and Clifford operations in finite dimensions. However, Gaussian states do not form a continuous-variable unitary two-design, and in fact fail to approximate the required second-moment Haar behaviour in the full infinite-dimensional setting [35]. More generally, recent no-go results show that exact CV state and unitary t -designs do not exist, under natural infinite-dimensional extensions of the standard definition, for $t \geq 2$ [105]. Thus the direct question “is there an exact CV analogue of a finite-dimensional unitary two-design?” has a negative answer.

This negative answer does not end the story. It instead clarifies what the right question should be. Physical CV systems are never accessed with infinite energy and infinite resolution. Realistic states occupy an effectively finite region of phase space, and realistic measurements cannot distinguish arbitrarily fine structure. Therefore, rather than demanding an exact design on the full Hilbert space, one can ask for a regularised design: a construction that reproduces second-moment Haar behaviour on a finite-dimensional,

¹⁶The approximation is often quantified by a parameter ε .

physically motivated approximation of the CV system, while retaining a clear connection to quadrature physics.

This leads to the research question addressed in Chapter 5:

Can we construct a physically motivated ε -approximate 2-design for CV systems?

1.5.4. Research questions IV and V: continuous-variable unclonable encryption

An unknown quantum state cannot, in general, be copied perfectly. As we saw in the previous section, this principle has long been used as a cryptographic primitive: instead of relying only on computational assumptions, one can design protocols whose security follows from the physical structure of quantum theory. Unclonable encryption is another such primitive.

In ordinary encryption, the main goal is secrecy: an adversary should not learn the plaintext from the ciphertext. Unclonable encryption asks for something stronger. Here a classical message is encrypted into a quantum ciphertext, and security means that an adversary should not be able to split this ciphertext into two parts that can both later be decrypted once the key is revealed. Thus the ciphertext is not merely a cipher; it is also non-duplicable. This makes unclonable encryption a natural primitive for one-time uses of encrypted data, where copying and replaying a ciphertext must be made impossible.

The term *unclonable encryption* was introduced by Gottesman [86]. Later work refined the security definitions and connected the primitive to stronger cloning-style security games [122, 42]. In the modern framework, an adversary receives a challenge ciphertext and applies an arbitrary quantum operation that splits it into two systems. These systems are handed to two separated parties, Bob and Charlie, who are not allowed to communicate. After the secret key is revealed, both parties try to recover the plaintext. The scheme is *unclonable secure* if the probability that both parties recover the plaintext is small. A stronger notion is *unclonable indistinguishability*: instead of asking whether both parties can recover a uniformly random plaintext, one asks whether they can both distinguish which of two possible plaintexts was encrypted. This is closer in spirit to indistinguishability-based encryption security, but now with the additional requirement that two separated parties must both succeed.

Until recently, unclonable encryption has been studied mostly in DV quantum systems. This leaves open the natural question in the context of this thesis, namely, that of a continuous-variable-based protocol. However, transferring unclonable encryption from the DV setting to the continuous-variable setting is not a routine translation.

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The ciphertexts are now states of infinite-dimensional systems; measurements have continuous outcomes; infinite squeezing is practically not possible; and realistic channels introduce loss and excess noise. A useful CV unclonable-encryption protocol must therefore satisfy two competing requirements: it must be decryptable by the honest receiver using realistic measurements, while still preventing an adversary from producing two successful decryptions.

This motivates the first research question addressed in Chapter 6:

Is it possible to achieve unclonable encryption using continuous-variable quantum states in the presence of realistic loss and noise?

This first question is aimed at establishing unclonable security, but it does not by itself address the stronger notion of unclonable indistinguishability. In finite-dimensional quantum information, unclonable indistinguishability can be obtained from decoupling-based constructions [30]. The decoupling idea is that a sufficiently random unitary can wash out correlations between a protected system and an adversarial reference system. Since decoupling arguments depend only on second moments of the random unitary, Haar randomness can often be replaced by a unitary two-design or by an approximate two-design [174]. This connects directly to the previous subsection: once one has a physically motivated continuous-variable approximate two-design, it becomes possible to use it as the randomising operation in a CV encryption scheme.

This leads to the second research question:

Is it possible to design a continuous-variable unclonable encryption protocol which is also unclonable-indistinguishability-secure?

1.6. Summary and guide to the thesis

The aim of this thesis is to investigate how far continuous-variable quantum systems can be pushed as a platform for quantum cryptography. The guiding point of view is that continuous variables are both a different mathematical representation of quantum information and a practically motivated framework. The practical angle forces one to confront issues that are categorically different from finite-dimensional models, like finite-energy constraints, finite squeezing, and continuous measurement outcomes. The central objective running through the thesis is therefore to investigate and propose solutions to these problems, and to study if continuous-variable systems could be strong contenders for future quantum cryptographic deployments.

The thesis is arranged in the order in which one naturally develops continuous-variable quantum cryptography from established primitives towards newer and less explored ones.

We begin with continuous-variable quantum key distribution, the most mature part of the field. CVQKD already has a well-developed theoretical and experimental literature, but long-distance implementations remain limited by the difficulty of information reconciliation at a very low signal-to-noise ratio. Chapter 3 addresses this bottleneck by developing a random-codebook reconciliation method based on MAP score thresholding and conservative leakage accounting. The point of this chapter is to enable long-distance quantum key distribution by overhauling the classical (and cheaper) part of the protocol.

The next step is continuous-variable quantum position verification. Unlike CVQKD, CV-QPV had only appeared in the literature in a very limited form, and its security had not been systematically established. Chapter 4 studies this problem in two stages. First, it considers a simple coherent-state protocol and proves security against adversaries without pre-shared entanglement, while also analysing concrete attacks and realistic channel imperfections. Second, it enriches the protocol structure in order to obtain security against attackers with bounded pre-shared entanglement. In this way, the chapter turns CV-QPV from a briefly suggested possibility into a protocol family with full security proofs and an explicit noise analysis.

After this, the thesis moves from concrete communication protocols to a more structural question in CV quantum information: the construction of unitary designs. Unitary two-designs are a standard tool in finite-dimensional quantum information, but their continuous-variable analogue had remained open because the infinite-dimensional setting does not admit a straightforward Haar-random construction. Chapter 5 addresses this problem by introducing a regularised CV Hilbert space based on finite quadrature windows and finite resolution, and by constructing approximate unitary two-designs from random phase operations in conjugate quadrature bases. This chapter therefore provides a new technical tool for CV quantum information, while also preparing the ground for the final cryptographic application of the thesis.

The final part of the thesis concerns unclonable encryption. This is a much newer primitive than QKD, and before the work presented here no continuous-variable version had been investigated or proposed. Chapter 6 develops two complementary approaches. The first uses finitely squeezed CV states and proves unclonable security through a reduction to a continuous-variable monogamy-of-entanglement game. The same construction is then analysed under realistic loss, noise, and detector imperfections, showing that the protocol can operate in experimentally feasible regimes. The second approach uses the approximate CV two-designs developed in Chapter 5 and proves the stronger notion of unclonable indistinguishability through decoupling. Together, these results show that unclonable encryption can also be realised and proved secure in the continuous-variable setting, and again establish a protocol family.

The thesis can therefore be read as a progression from the established to the speculative.

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Chapter 3 starts from CVQKD, where the main challenge is to improve performance within an already mature framework. Chapter 4 then moves to CV-QPV, where the primitive had been suggested but needed concrete security proofs and noise analysis. Chapter 5 studies CV approximate unitary two-designs, a long-standing structural question whose resolution provides a new way of scrambling for CV systems. Finally, Chapter 6 applies these ideas to unclonable encryption, a new cryptographic direction in which CV systems had not previously been explored. Taken together, the chapters support the main conclusion of the thesis: continuous-variable systems do indeed provide a rich and practically relevant setting for quantum cryptography, but their security theory must be developed under the constraints of specific features of the CV domain.

The chapters of this thesis are based on the following works:

Paper	Chapter
Arpan Akash Ray and Boris Škorić, “Random coding for long-range continuous-variable QKD,” arXiv:2512.15990 [quant-ph], 2025. Under review.	Chapter 3
Rene Allerstorfer, Llorenç Escolà-Farràs, Arpan Akash Ray, Boris Škorić, Florian Speelman, and Philip Verduyn Lunel, “Security of a Continuous-Variable based Quantum Position Verification Protocol,” arXiv:2308.04166 [quant-ph], 2023. Under review.	Chapter 4
Rene Allerstorfer, Llorenç Escolà-Farràs, Arpan Akash Ray, Boris Škorić, and Florian Speelman, “Continuous-variable quantum position verification secure against entangled attackers,” <i>Physical Review A</i> 110 , 062605, 2024. DOI: 10.1103/PhysRevA.110.062605.	Chapter 4
Arpan Akash Ray and Boris Škorić, “Continuous-variable approximate unitary 2-design, with applications to unclonable encryption,” arXiv:2603.06393 [quant-ph], 2026. Under review.	Chapter 5, Chapter 6
Arpan Akash Ray and Boris Škorić, “Practical Unclonable Encryption with Continuous Variables,” arXiv:2503.02648 [quant-ph], 2025. Accepted for publication in <i>Physical Review A</i> .	Chapter 6

2

Mathematical Preliminaries

The enormous usefulness of mathematics in the natural sciences is something bordering on the mysterious and that there is no rational explanation for it.

— Eugene P. Wigner

This chapter collects the mathematical background used throughout the thesis. The goal is to present an overview of the main mathematical ideas from (quantum) information theory and continuous-variable quantum optics. We will assume that the reader is familiar with linear algebra, operator theory, probability and methods of statistics. We also assume a basic understanding of quantum mechanics and the density matrix formalism. We begin with primary notions from classical and quantum information theory, then introduce bosonic modes and their phase-space description, and finally discuss Gaussian states, Gaussian operations, measurements, and noisy channels.

Throughout the chapter we work in units where $\hbar = 1$. Logarithms are taken in base 2 unless explicitly stated otherwise. We use the quadrature convention in which the phase-space position and momentum operators are

$$\hat{x} = \frac{a + a^\dagger}{\sqrt{2}}, \quad \hat{p} = \frac{a - a^\dagger}{i\sqrt{2}}. \quad (2.1)$$

Together with $[a, a^\dagger] = 1$, this gives $[\hat{x}, \hat{p}] = i$. Moreover, if $|0\rangle$ denotes the vacuum state, defined by $a|0\rangle = 0$, then Eq. (2.1) implies

$$\langle 0|\hat{x}|0\rangle = \langle 0|\hat{p}|0\rangle = 0, \quad \langle 0|\hat{x}^2|0\rangle = \langle 0|\hat{p}^2|0\rangle = \frac{1}{2}. \quad (2.2)$$

Thus in our convention the vacuum has quadrature variance $1/2$ in each quadrature.¹

¹In the covariance-matrix convention introduced below, this means that the vacuum covariance matrix is the

2.1. Classical information theory

We begin with a few basic quantities from classical information theory. In this thesis they appear whenever classical data are generated from a quantum system, for example as the outcome of a homodyne or heterodyne measurement. Since these outcomes are random variables, the relevant question is not only what their probability distribution is, but also how much uncertainty they contain, how much one variable reveals about another, and how distinguishable two distributions are.

Let X be a discrete random variable with alphabet $\mathcal{X}$ and probability mass function p_X . The Shannon entropy of X is defined as

$$H(X) = - \sum_{x \in \mathcal{X}} p_X(x) \log p_X(x). \quad (2.3)$$

Entropy measures the uncertainty of a random variable in an operational sense: it quantifies the average number of bits needed to describe samples of X in an optimal compression scheme. A deterministic random variable has zero entropy, whereas a random variable that is spread over many possible outcomes has larger entropy. In particular, the uniform distribution on an alphabet of size $|\mathcal{X}|$ has entropy $\log |\mathcal{X}|$.

Let X and Y be jointly distributed. The conditional entropy of X given Y is defined by

$$H(X|Y) = H(XY) - H(Y), \quad (2.4)$$

where $H(XY)$ denotes the entropy of the joint random variable (X, Y) . This is the remaining uncertainty about X after Y is known. The mutual information between X and Y is defined as

$$I(X; Y) = H(X) - H(X|Y) = H(X) + H(Y) - H(XY). \quad (2.5)$$

It quantifies the information shared by the two random variables. $H(X)$ and $H(Y)$ represent the information contained in the two variables, while $I(X; Y)$ is their overlap: the part of the information about X that is also contained in Y . Equivalently, it is the reduction in the uncertainty of X obtained by learning Y . The expression is symmetric, so the same quantity also measures the reduction in the uncertainty of Y obtained by learning X . If X and Y are independent, then learning one provides no information about the other and $I(X; Y) = 0$. At the opposite extreme, if $Y = X$, then all the information in X is shared and $I(X; X) = H(X)$.

identity matrix.

2. Mathematical Preliminaries

Another useful quantity is the classical relative entropy, or Kullback–Leibler divergence. For two probability distributions p and q on the same alphabet, it is defined as

$$D(p\|q) = \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)}. \quad (2.6)$$

The relative entropy measures how different the distribution q is from the reference distribution p . More precisely, it is the average log-likelihood ratio when the data are generated according to p . It is non-negative and vanishes if and only if $p = q$, but it is not symmetric in its two arguments and therefore is not a distance. With the standard convention that $p(x) \log(p(x)/0) = +\infty$ for $p(x) > 0$, the definition implies that $D(p\|q) = +\infty$ whenever $\text{support}(p) \not\subseteq \text{support}(q)$.

A special case that appears frequently is the entropy of a binary random variable. If the distribution is $(p, 1 - p)$, then the binary entropy is

$$h_2(p) = -p \log p - (1 - p) \log(1 - p). \quad (2.7)$$

This function is maximal at $p = 1/2$, where the bit is completely random, and vanishes at $p = 0$ or $p = 1$, where the bit is deterministic.

Continuous-variable protocols involve real-valued random variables. Let X have a probability density function f_X . Then the continuous analogue of entropy is the differential entropy

$$h(X) = - \int_{\text{supp}(f_X)} f_X(x) \log f_X(x) dx. \quad (2.8)$$

Although this expression resembles the discrete entropy, its interpretation is more subtle. One way to interpret differential entropy is through typical sets.² For n independent copies of a continuous random variable X , the typical set has volume approximately $\exp(nh(X))$ [53]. Thus differential entropy is the logarithm, per sample, of the volume occupied by typical sequences, rather than the logarithm, per sample, of the number of typical discrete sequences.

This also explains why differential entropy can be negative. Unlike a probability mass function, a probability density is not itself a probability: only its integral over a region gives

²Let $X_1, \dots, X_n$ be independent copies of a random variable X with density f_X . For $\varepsilon > 0$, the ε -typical set is the set of sequences $x^n = (x_1, \dots, x_n)$ satisfying

$$\left| -\frac{1}{n} \log \left(\prod_{i=1}^n f_X(x_i) \right) - h(X) \right| \leq \varepsilon.$$

Equivalently, the joint density of a sequence in the typical set is approximately $\exp[-nh(X)]$. Under the usual regularity assumptions, the probability that $(X_1, \dots, X_n)$ lies in this set approaches one as n becomes large.

a probability. Therefore a density may take values larger than one when the distribution is concentrated in a region of volume smaller than one. In such a case the logarithmic volume of the typical set may be negative. This does not mean that the random variable has “negative uncertainty”. It only means that, in the chosen units, the distribution is concentrated in a small region of the real line.

Changing the units changes the numerical value of the differential entropy. Indeed, for $\alpha > 0$,

$$h(\alpha X) = h(X) + \log \alpha. \quad (2.9)$$

Thus differential entropy is not invariant under a change of scale. It is an important quantity for many continuous information-theoretic expressions, especially when comparing uncertainties or computing mutual informations between continuous random variables.

For a Gaussian random variable $X \sim \mathcal{N}(\mu, \sigma^2)$, the differential entropy is

$$h(X) = \frac{1}{2} \log(2\pi e \sigma^2). \quad (2.10)$$

This formula will be used repeatedly, because homodyne and heterodyne measurements of Gaussian quantum states produce Gaussian classical outcome distributions. It should be noted that the entropy of the measurement result is determined entirely by its variance.

2.2. Quantum theory and information

We now review the quantum-information-theoretic formalism used throughout the thesis. The emphasis in this section is mathematical rather than physical: a quantum system is described by a Hilbert space, its state by a density operator, its measurements by POVMs, and its evolutions by quantum channels. This is the standard language of quantum information theory. Later, in Section 2.3, we specialise this abstract formalism to continuous-variable systems by explaining how bosonic modes arise from canonical quantisation.

We first work in finite dimension, where the relevant objects can be treated as matrices and no domain issues arise. The same conceptual framework also applies to infinite-dimensional systems, although additional care is needed when defining unbounded operators and computing entropies. Those continuous-variable aspects will be addressed in the later sections.

A finite-dimensional quantum system is represented by a Hilbert space $\mathcal{H}$. A pure state is represented by a unit vector $|\psi\rangle \in \mathcal{H}$, where global phases have no physical significance. More generally, a possibly mixed state is represented by a density operator ρ on $\mathcal{H}$, satisfying

$$\rho \geq 0, \quad \text{Tr } \rho = 1. \quad (2.11)$$

2. Mathematical Preliminaries

We write $\mathcal{D}(\mathcal{H})$ for the set of density operators on $\mathcal{H}$. The density operator associated with a pure state $|\psi\rangle$ is $|\psi\rangle\langle\psi|$.

Mixed states may arise from classical uncertainty about which pure state was prepared. For example, if a preparation device emits the state $|\psi_z\rangle$ with probability p_z , then the resulting state is

$$\rho = \sum_z p_z |\psi_z\rangle\langle\psi_z|. \quad (2.12)$$

It can be seen that a single mixed state may admit many such decompositions. In particular, mixed states also arise naturally as reduced states of entangled systems (which we will show shortly). Thus density operators can provide a unified description of both classical and quantum correlations and should be considered more general [152, 194].

If a composite system consists of two subsystems A and B , the corresponding Hilbert space is

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B. \quad (2.13)$$

Given a bipartite state $\rho_{AB} \in \mathcal{D}(\mathcal{H}_A \otimes \mathcal{H}_B)$, the state accessible to an observer holding only subsystem A is the reduced state

$$\rho_A = \text{Tr}_B \rho_{AB}. \quad (2.14)$$

The partial trace is characterised by the identity (in this chapter, the identity matrix is denoted by $\mathbb{I}$)

$$\text{Tr}[M_A \rho_A] = \text{Tr}[(M_A \otimes \mathbb{I}_B) \rho_{AB}], \quad (2.15)$$

for every operator M_A on $\mathcal{H}_A$. This expresses the operational meaning of the reduced state: it gives exactly the same statistics for all measurements performed on subsystem A alone as the full state ρ_{AB} would give when the measurement is applied locally.

A pure bipartite state is a product state if it can be written as

$$|\psi\rangle_{AB} = |\phi\rangle_A \otimes |\eta\rangle_B. \quad (2.16)$$

If no such decomposition exists, the state is entangled. For mixed states, the corresponding notion is separability. A bipartite mixed state ρ_{AB} is separable if it can be written as

$$\rho_{AB} = \sum_z p_z \rho_A^z \otimes \rho_B^z, \quad (2.17)$$

where $\{p_z\}_z$ is a probability distribution and $\rho_A^z \in \mathcal{D}(\mathcal{H}_A)$, $\rho_B^z \in \mathcal{D}(\mathcal{H}_B)$ are local states. Such a state can be prepared using only local state preparations and shared classical randomness. A state that is not separable is entangled. Entanglement is therefore a form of correlation that cannot be explained as classical randomness over local quantum states.

We will often need to quantify how distinguishable two quantum states are. This is important in cryptography because security is usually formulated by comparing a real state produced by a protocol with an ideal state that would represent perfect security. If no physical measurement can distinguish these two states with significant advantage, then they are operationally close.

The trace distance between two states ρ and σ is

$$T(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1, \quad (2.18)$$

where $\|X\|_1 = \text{Tr} \sqrt{X^\dagger X}$ is the Schatten 1-norm. The trace distance has a direct operational interpretation: it is the maximum possible statistical distance between the outcome distributions obtained by measuring ρ and σ . Equivalently,

$$T(\rho, \sigma) = \max_{0 \leq M \leq \mathbb{I}} \text{Tr}[M(\rho - \sigma)]. \quad (2.19)$$

Thus trace distance measures the largest distinguishing advantage achievable by any binary quantum measurement. In particular, if one is given either ρ or σ with equal prior probability, the optimal probability of guessing the state correctly is

$$p_{\text{guess}} = \frac{1}{2} (1 + T(\rho, \sigma)). \quad (2.20)$$

This is the Helstrom discrimination bound [98].

Another important distinguishability measure is the fidelity. With the convention used in this thesis, the fidelity between two states ρ and σ is

$$F(\rho, \sigma) = \text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}}. \quad (2.21)$$

Fidelity measures overlap rather than distinguishability: it is large when the states are close and small when they are nearly orthogonal. For pure states, it reduces to

$$F(|\psi\rangle\langle\psi|, |\phi\rangle\langle\phi|) = |\langle\psi|\phi\rangle|. \quad (2.22)$$

Thus two identical pure states have fidelity one, while orthogonal pure states have fidelity zero. More generally, Uhlmann's theorem characterises the fidelity as the maximum overlap between purifications of the two states [177, 152, 194]. This makes it particularly useful when comparing states that are part of larger systems.

Trace distance and fidelity are related by the Fuchs–van de Graaf inequalities:

$$1 - F(\rho, \sigma) \leq T(\rho, \sigma) \leq \sqrt{1 - F(\rho, \sigma)^2}. \quad (2.23)$$

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These inequalities show that the two quantities define equivalent notions of closeness, even though they emphasise different operational viewpoints [76].

A related distance measure, used frequently in one-shot quantum information theory, is the purified distance

$$\mathcal{P}(\rho, \sigma) = \sqrt{1 - F(\rho, \sigma)^2}. \quad (2.24)$$

For pure states, this becomes

$$\mathcal{P}(|\psi\rangle\langle\psi|, |\phi\rangle\langle\phi|) = \sqrt{1 - |\langle\psi|\phi\rangle|^2}. \quad (2.25)$$

The purified distance is useful because it behaves well under purifications and quantum operations, and because it is closely connected to smoothing procedures in one-shot entropy theory [175]. Operationally, small purified distance again means that the two states cannot be reliably distinguished, but it is often technically more convenient than trace distance when optimising over nearby states.

2.2.1. Measurements and POVMs

Measurements are the bridge between the quantum description of a system and the classical data obtained from it. The starting point is the Born rule. If a measurement is described by orthogonal projectors $\{\Pi_y\}_{y \in \mathcal{Y}}$ satisfying

$$\Pi_y \Pi_{y'} = \delta_{y,y'} \Pi_y, \quad \sum_{y \in \mathcal{Y}} \Pi_y = \mathbb{I}, \quad (2.26)$$

then the probability of obtaining outcome y on a state ρ is

$$p(y) = \text{Tr}(\rho \Pi_y). \quad (2.27)$$

After obtaining outcome y , the corresponding post-measurement state is

$$\rho_y = \frac{\Pi_y \rho \Pi_y}{\text{Tr}(\rho \Pi_y)}. \quad (2.28)$$

This is the usual projective-measurement form of the Born rule.

For quantum information theory, however, projective measurements are not general enough. A measurement may involve coupling the system to an ancilla, performing a projective measurement on the larger system, and then discarding degrees of freedom. The effective measurement on the original system is then described by a positive operator-valued measure, or POVM. For a finite outcome set $\mathcal{Y}$, a POVM is a collection of positive semidefinite operators $\{M_y\}_{y \in \mathcal{Y}}$ satisfying

$$M_y \geq 0, \quad \sum_{y \in \mathcal{Y}} M_y = \mathbb{I}. \quad (2.29)$$

The probability of outcome y is

$$p(y) = \text{Tr}(\rho M_y). \quad (2.30)$$

2

Thus a POVM should be viewed as the most general way of assigning probabilities to classical measurement outcomes while remaining compatible with the Born rule [152, 194, 188].

The POVM elements determine the outcome probabilities, but they do not by themselves determine the post-measurement state. To describe both the classical outcome and the conditional quantum state, one uses measurement operators. Let a measurement outcome y be associated with an operator K_y satisfying

$$M_y = K_y^\dagger K_y. \quad (2.31)$$

The probability of outcome y is

$$p(y) = \text{Tr}(K_y \rho K_y^\dagger) = \text{Tr}(\rho M_y), \quad (2.32)$$

and the normalised post-measurement state conditioned on that outcome is

$$\rho_y = \frac{K_y \rho K_y^\dagger}{p(y)}. \quad (2.33)$$

More generally, an outcome y may correspond to several measurement operators $\{K_{y,\ell}\}_\ell$. In that case,

$$M_y = \sum_\ell K_{y,\ell}^\dagger K_{y,\ell}, \quad (2.34)$$

the probability of outcome y is

$$p(y) = \text{Tr} \left[\sum_\ell K_{y,\ell} \rho K_{y,\ell}^\dagger \right], \quad (2.35)$$

and the conditional post-measurement state is

$$\rho_y = \frac{\sum_\ell K_{y,\ell} \rho K_{y,\ell}^\dagger}{p(y)}. \quad (2.36)$$

The collection of completely positive maps

$$\mathcal{M}_y(\rho) = \sum_\ell K_{y,\ell} \rho K_{y,\ell}^\dagger \quad (2.37)$$

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is called a quantum instrument. It contains more information than the POVM: the POVM specifies only the measurement statistics, while the instrument also specifies how the quantum state is updated after the outcome is observed.

Projective measurements are recovered as the special case $K_y = \Pi_y$, so that $M_y = \Pi_y$. The homodyne and heterodyne measurements introduced later are continuous-outcome analogues of the same formalism. In those cases, sums over y are replaced by integrals over real measurement outcomes, and the POVM elements become operator-valued densities.

2.2.2. Unitary dynamics

Before introducing general quantum channels, we first recall the special case of closed-system dynamics. A closed quantum system is one that is not coupled to an external environment. Its evolution must preserve normalisation and transition probabilities. In the Hilbert-space formalism, such an evolution is represented by a unitary operator U satisfying

$$U^\dagger U = U U^\dagger = \mathbb{I}. \quad (2.38)$$

At the level of pure states, the corresponding transformation is

$$|\psi\rangle \mapsto U|\psi\rangle. \quad (2.39)$$

The unitarity condition ensures that inner products are preserved:

$$\langle\psi|\phi\rangle = \langle\psi|U^\dagger U|\phi\rangle. \quad (2.40)$$

Thus a unitary transformation preserves the overlap, and hence the distinguishability³, of pure states.

In the density-operator formalism, a unitary operator U defines a map

$$\mathcal{U} : \mathcal{D}(\mathcal{H}) \longrightarrow \mathcal{D}(\mathcal{H}), \quad \mathcal{U}(\rho) = U\rho U^\dagger. \quad (2.41)$$

This is the density-matrix form of closed-system evolution. This can be trivially shown. Suppose first that the input state is pure, $\rho = |\psi\rangle\langle\psi|$. Then $\mathcal{U}(|\psi\rangle\langle\psi|) = U|\psi\rangle\langle\psi|U^\dagger = (U|\psi\rangle)(U|\psi\rangle)^\dagger$. Thus the density-operator expression reproduces the usual pure-state rule. For a mixed state with ensemble representation $\rho = \sum_z p_z |\psi_z\rangle\langle\psi_z|$, linearity gives $\mathcal{U}(\rho) = \sum_z p_z U|\psi_z\rangle\langle\psi_z|U^\dagger$. Hence the same unitary is applied to every component of the mixture.

³For pure states, distinguishability is determined by their overlap (see discussion immediately above Section 2.2.1). If two pure states have inner product close to one, no measurement can reliably tell them apart; if they are orthogonal, there exists a measurement that distinguishes them perfectly. Since unitary evolution preserves inner products, it also preserves the optimal ability to distinguish pure states.

The map $\mathcal{U}$ maps density operators to density operators. Positivity is preserved because, for every $|\varphi\rangle \in \mathcal{H}$,

$$\langle\varphi|\mathcal{U}(\rho)|\varphi\rangle = \langle\varphi|U\rho U^\dagger|\varphi\rangle = \langle\chi|\rho|\chi\rangle \geq 0, \quad |\chi\rangle = U^\dagger|\varphi\rangle, \quad (2.42)$$

whenever $\rho \geq 0$. The trace is preserved because

$$\mathrm{Tr}[\mathcal{U}(\rho)] = \mathrm{Tr}[U\rho U^\dagger] = \mathrm{Tr}[\rho U^\dagger U] = \mathrm{Tr}[\rho]. \quad (2.43)$$

Thus $\mathcal{U}(\rho) \in \mathcal{D}(\mathcal{H})$ whenever $\rho \in \mathcal{D}(\mathcal{H})$.

Closed-system unitary dynamics is usually described in terms of a Hamiltonian. For the moment, it is enough to regard the Hamiltonian H as the self-adjoint operator that generates time evolution. Physically, it represents the energy observable of the system, and mathematically it determines the infinitesimal change of the state in time. Hamiltonians will be introduced more explicitly in Section 2.3, where we discuss harmonic oscillators and bosonic modes.

If the Hamiltonian is time-independent, the unitary operator at time t is

$$U(t) = \exp(-iHt). \quad (2.44)$$

This defines a one-parameter family of maps

$$\mathcal{U}_t(\rho) = U(t)\rho U(t)^\dagger. \quad (2.45)$$

If $\rho(t) = \mathcal{U}_t(\rho(0))$, then differentiating gives the von Neumann equation

$$\frac{d\rho(t)}{dt} = -i[H, \rho(t)]. \quad (2.46)$$

Thus, in closed-system quantum dynamics, the Hamiltonian plays the role of the generator of time evolution, just as the Hamiltonian function generates time evolution in classical mechanics⁴.

The same dynamics can be described in two equivalent pictures. In the Schrödinger picture, the state is transformed by the map $\mathcal{U}$, while observables are kept fixed. In the Heisenberg picture, the state is kept fixed and an observable M is transformed according to

$$M \mapsto U^\dagger M U. \quad (2.47)$$

⁴In classical Hamiltonian mechanics, the state of a system is specified by phase-space coordinates (q, p) and a Hamiltonian function $H(q, p)$, usually interpreted as the energy of the system. The Hamiltonian determines the equations of motion through Hamilton's equations, $\dot{q} = \partial H / \partial p$ and $\dot{p} = -\partial H / \partial q$. In this sense, H is the generator of time translations. The quantum Hamiltonian plays the analogous role, with commutators replacing Poisson brackets. We will see this process of *quantisation* in action in Section 2.3.

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The two descriptions give the same measurement statistics, since

$$\mathrm{Tr}[\mathcal{U}(\rho)M] = \mathrm{Tr}[U\rho U^\dagger M] = \mathrm{Tr}[\rho U^\dagger M U]. \quad (2.48)$$

For composite systems, unitary dynamics may act locally or globally. A local unitary on a bipartite system has the tensor product form $U_A \otimes U_B$, whereas a general unitary on $\mathcal{H}_A \otimes \mathcal{H}_B$ can create correlations and entanglement between the two subsystems.

2.2.3. Quantum channels and CPTP maps

Unitary evolution describes the dynamics of a closed quantum system. However, cryptographic protocols rarely involve only closed-system dynamics. A system may pass through a noisy channel, interact with an environment, be partly discarded, or be measured. We therefore need a more general description of physical transformations of quantum states. This is provided by completely positive trace-preserving maps, also called quantum channels.

A quantum operation from system A to system B is a linear map

$$\Phi : \mathcal{L}(\mathcal{H}_A) \longrightarrow \mathcal{L}(\mathcal{H}_B), \quad (2.49)$$

where $\mathcal{L}(\mathcal{H})$ denotes the space of linear operators on $\mathcal{H}$.

The two essential requirements for a deterministic quantum evolution are the following. First, if the input is a *valid* quantum state, then the output must again be a valid quantum state. Here “valid” means positive semidefinite and normalised, so that it gives non-negative probabilities summing to one for every measurement through the Born rule. Second, the same requirement must continue to hold if the system is only one part of a larger joint system. That is, if the input system A is correlated or entangled with an arbitrary reference system R , then applying the evolution to A alone must still produce a valid joint state on BR .

The first requirement gives positivity and trace preservation. Positivity means that

$$\rho \geq 0 \implies \Phi(\rho) \geq 0, \quad (2.50)$$

and trace preservation means that

$$\mathrm{Tr}[\Phi(\rho)] = \mathrm{Tr}[\rho], \quad (2.51)$$

for all states, or more generally for all trace-class operators⁵ ρ . Trace preservation ensures that normalised input states are mapped to normalised output states.

⁵In infinite-dimensional Hilbert spaces, density operators are required to be trace-class. An operator A is trace-class if its Schatten 1-norm is finite. This condition guarantees that quantities such as $\mathrm{Tr} A$ and expectation values $\mathrm{Tr}(\rho M)$ are well-defined for bounded observables M . In finite dimension every operator is trace-class, so this distinction does not arise.

Positivity alone is not sufficient. A physical operation may act on a subsystem that is entangled with an inaccessible reference system. If R is such a reference, then the extended map $\Phi \otimes \text{id}_R$ must also map positive operators to positive operators. This leads to the second requirement and motivates complete positivity:

$$X_{AR} \geq 0 \implies (\Phi \otimes \text{id}_R)(X_{AR}) \geq 0, \quad (2.52)$$

for every auxiliary Hilbert space $\mathcal{H}_R$. A completely positive trace-preserving map is called a CPTP map or quantum channel [152, 194, 188].

Every finite-dimensional CPTP map admits a Kraus representation [116]

$$\Phi(\rho) = \sum_k A_k \rho A_k^\dagger, \quad (2.53)$$

where the Kraus operators $A_k : \mathcal{H}_A \rightarrow \mathcal{H}_B$ satisfy

$$\sum_k A_k^\dagger A_k = \mathbb{I}_A. \quad (2.54)$$

The condition in Eq. (2.54) is exactly the trace-preserving condition. Indeed,

$$\text{Tr}[\Phi(\rho)] = \sum_k \text{Tr}[A_k \rho A_k^\dagger] = \text{Tr}\left[\rho \sum_k A_k^\dagger A_k\right] = \text{Tr}[\rho]. \quad (2.55)$$

The Kraus form makes clear why the map is completely positive: even after adding a reference system, the extended map has the form

$$(\Phi \otimes \text{id}_R)(X) = \sum_k (A_k \otimes \mathbb{I}_R) X (A_k^\dagger \otimes \mathbb{I}_R), \quad (2.56)$$

which preserves positivity.

Quantum channels also admit a Stinespring dilation [171]. This means that any channel can be represented as an isometry into a larger Hilbert space followed by discarding the environment:

$$\Phi(\rho_A) = \text{Tr}_E[V_E \rho_A V_E^\dagger], \quad V_E^\dagger V_E = \mathbb{I}. \quad (2.57)$$

Equivalently, one may regard the system as interacting unitarily with an environment initially prepared in some state, after which the environment is traced out. This gives the physical interpretation of a general quantum channel: it is the effective evolution of a subsystem of a larger closed system [152, 188].

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The measurements introduced in Section 2.2.1 also fit into this framework. Consider a measurement instrument with outcome y and Kraus operators $\{K_{y,\ell}\}_\ell$. The completely positive map associated with outcome y is

$$\mathcal{M}_y(\rho) = \sum_\ell K_{y,\ell} \rho K_{y,\ell}^\dagger. \quad (2.58)$$

Its trace gives the probability of observing outcome y :

$$p(y) = \text{Tr}[\mathcal{M}_y(\rho)]. \quad (2.59)$$

The conditional post-measurement state is then

$$\rho_y = \frac{\mathcal{M}_y(\rho)}{\text{Tr}[\mathcal{M}_y(\rho)]}. \quad (2.60)$$

The map $\mathcal{M}_y$ is completely positive but generally not trace-preserving, because its trace is the probability of that particular outcome.

If the measurement outcome is recorded in a classical register Y , then the whole measurement process is a CPTP map

$$\mathcal{M}(\rho) = \sum_y |y\rangle\langle y|_Y \otimes \mathcal{M}_y(\rho). \quad (2.61)$$

Thus measurements are not outside the channel formalism. The non-normalised outcome maps $\mathcal{M}_y$ describe individual branches of a measurement, while the full map that includes all outcomes and records the classical result is trace-preserving.

It is useful to distinguish CPTP maps from completely positive trace non-increasing maps. A completely positive map Ψ is trace non-increasing if

$$\text{Tr}[\Psi(\rho)] \leq \text{Tr}[\rho], \quad (2.62)$$

for all positive trace-class ρ . In Kraus form this corresponds to

$$\Psi(\rho) = \sum_k B_k \rho B_k^\dagger, \quad \sum_k B_k^\dagger B_k \leq \mathbb{I}. \quad (2.63)$$

Such maps describe probabilistic operations, postselection, or individual measurement outcomes. They become normalised states only after conditioning on the event that the operation succeeds.

Quantum channels include unitary evolution, adding or discarding an ancilla, partial trace, noisy evolution, and measurements with classical outcomes. In continuous-variable cryptography, optical loss, detector inefficiency, thermal noise, and excess noise are all modelled as quantum channels. Gaussian channels, introduced later, are the subclass of quantum channels that map Gaussian states to Gaussian states.

2.2.4. Quantum information theory

In the previous section we introduced density operators as the mathematical objects describing quantum states. We now explain how these objects become carriers of information. The guiding analogy is with classical probability distributions. A classical random variable X with distribution p_X stores information through the probabilities assigned to its possible values. In quantum theory, a state ρ plays an analogous role: it determines the probability distribution of every measurement that can be performed on the system through the Born rule. Thus operationally, a quantum state can be considered as a rule for assigning probabilities to all possible measurement outcomes.

This connection is especially transparent when the quantum state is diagonal in some orthonormal basis $\{|x\rangle\}_{x \in \mathcal{X}}$. Given a classical probability distribution p_X , we may encode it as the density operator

$$\rho_X = \sum_{x \in \mathcal{X}} p_X(x) |x\rangle\langle x|. \quad (2.64)$$

Measuring this state in the basis $\{|x\rangle\}$ reproduces the original probability distribution. In this sense, diagonal density matrices can be regarded as classical probability distributions⁶. General density matrices extend this picture by allowing superpositions and coherences between basis states.

This analogy also explains the form of the quantum entropy. In classical information theory, the Shannon entropy is the entropy of a probability distribution. For a quantum state ρ , the intrinsic probability distribution associated with the state is its spectrum. Indeed, every finite-dimensional density operator has a spectral decomposition

$$\rho = \sum_i \lambda_i |i\rangle\langle i|, \quad (2.65)$$

where $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$. The eigenvalues of ρ are therefore a probability distribution. The von Neumann entropy is defined as the Shannon entropy of this eigenvalue distribution:

$$S(\rho) = -\text{Tr}(\rho \log \rho). \quad (2.66)$$

Equivalently, using Eq. (2.65),

$$S(\rho) = -\sum_i \lambda_i \log \lambda_i. \quad (2.67)$$

⁶This observation can be turned into a resource-theoretic [49] distinction between classical and non-classical states, once a preferred basis has been chosen. States that are diagonal in this basis are called incoherent and behave like classical probability distributions, while states with off-diagonal elements possess quantum coherence [23]. The resource theory of coherence [173] studies which operations cannot create such off-diagonal terms and introduces coherence measures to quantify the amount of this non-classical resource. A concise review of the resource theory of coherence can be found in my master's thesis [161].

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Thus, when ρ is diagonal in a fixed classical basis, the von Neumann entropy reduces exactly to the Shannon entropy of the corresponding classical distribution. For a pure state, the spectrum is $(1, 0, \dots, 0)$, and therefore $S(\rho) = 0$. Mixed states have nonzero entropy because their spectra are spread over more than one eigenvalue [152, 194].

For a bipartite state ρ_{AB} , the reduced states are $\rho_A = \text{Tr}_B \rho_{AB}$, $\rho_B = \text{Tr}_A \rho_{AB}$. The quantum conditional entropy is defined by

$$S(A|B)_\rho = S(\rho_{AB}) - S(\rho_B), \quad (2.68)$$

in direct analogy with the classical expression $H(X|Y) = H(XY) - H(Y)$. It measures the uncertainty in system A when system B is available as side information. However, unlike its classical counterpart, the quantum conditional entropy can be negative. For example, if ρ_{AB} is a pure entangled state, then $S(\rho_{AB}) = 0$, while $S(\rho_B) > 0$, so $S(A|B)_\rho < 0$. This negativity is a signature of entanglement and has an operational interpretation in tasks such as state merging [101, 194].

The quantum mutual information of a bipartite state is

$$I(A; B)_\rho = S(\rho_A) + S(\rho_B) - S(\rho_{AB}). \quad (2.69)$$

Equivalently,

$$I(A; B)_\rho = S(\rho_A) - S(A|B)_\rho. \quad (2.70)$$

This quantity measures the total correlation between the two systems, including both classical and quantum correlations. If $\rho_{AB} = \rho_A \otimes \rho_B$, then $I(A; B)_\rho = 0$. Conversely, correlations between the two systems increase the mutual information.

A particularly important case is a classical–quantum state. Such a state describes a classical random variable correlated with a quantum system. If X is a classical register and B is a quantum system, then a classical–quantum state has the form

$$\rho_{XB} = \sum_{x \in \mathcal{X}} p_X(x) |x\rangle\langle x|_X \otimes \rho_B^x. \quad (2.71)$$

The states ρ_B^x represent the quantum side information held by system B when the classical value is x . The average state on B is

$$\rho_B = \sum_{x \in \mathcal{X}} p_X(x) \rho_B^x. \quad (2.72)$$

For such a state, the mutual information becomes

$$I(X; B)_\rho = S(\rho_B) - \sum_{x \in \mathcal{X}} p_X(x) S(\rho_B^x). \quad (2.73)$$

This is the Holevo information. It quantifies how much information about the classical variable X is contained in the quantum system B . It is central in communication and cryptography because it bounds the amount of classical information that can be extracted from a quantum ensemble by measurement [100, 152, 194].

Finally, we will use the quantum analogue of the classical relative entropy. For two states ρ and σ , the quantum relative entropy is

$$D(\rho\|\sigma) = \text{Tr}[\rho(\log \rho - \log \sigma)], \quad (2.74)$$

with $D(\rho\|\sigma) = \infty$. When ρ and σ are diagonal in the same basis, this reduces to the classical Kullback–Leibler divergence between their eigenvalue distributions. Thus quantum relative entropy extends classical relative entropy from probability distributions to density operators.

As in the classical case, relative entropy is non-negative and vanishes if and only if $\rho = \sigma$. It is not symmetric, and therefore it is not a distance. Its importance comes from its operational and structural properties: it measures distinguishability between states and is monotone under quantum channels. For every CPTP map $\mathcal{N}$,

$$D(\rho\|\sigma) \geq D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma)). \quad (2.75)$$

This data-processing inequality expresses the fact that applying a physical operation cannot increase the distinguishability of two states [178, 194].

2.3. Bosonic modes

We now describe in detail the basic continuous-variable quantum system: a bosonic *mode*. The word “mode” should be understood as one independent *harmonic* degree of freedom. Here “harmonic” means that the corresponding classical dynamics is that of an oscillator with a quadratic energy function, so that its motion is sinusoidal and characterised by a frequency. As discussed in Section 1.4, a single mode may correspond to one normal mode of the electromagnetic field⁷, or equivalently to one quantum harmonic oscillator. The continuous-variable structure enters because the classical phase space of such a mode is described by two real variables: a position-like coordinate and a momentum-like coordinate.

⁷For a field, a normal mode is an independent oscillation pattern of the field. In free space, such a mode may be labelled by a wave vector, frequency, and polarisation; in a cavity, it may instead be labelled by the allowed standing-wave pattern and polarisation. After decomposing the field into such modes, each mode behaves mathematically like an independent harmonic oscillator.

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To see where the mathematical structure comes from, recall first the classical harmonic oscillator. In suitable dimensionless variables, its phase-space coordinates are denoted by x and p , and its Hamiltonian is

$$H_{\text{cl}} = \frac{\omega}{2}(x^2 + p^2). \quad (2.76)$$

The variables x and p are canonical coordinates: they satisfy the Poisson bracket relation

$$\{x, p\} = 1. \quad (2.77)$$

Canonical quantisation replaces classical observables by operators and replaces Poisson brackets by commutators according to the rule⁸

$$\{\cdot, \cdot\} \longrightarrow \frac{1}{i}[\cdot, \cdot], \quad (2.78)$$

where we use units with $\hbar = 1$ and i is the imaginary number. Thus the classical variables x and p become operators $\hat{x}$ and $\hat{p}$ satisfying

$$[\hat{x}, \hat{p}] = i. \quad (2.79)$$

This is the canonical commutation relation. It is the starting point of the continuous-variable description. Mathematically, it is the reason why a continuous-variable quantum system cannot be described by a finite-dimensional Hilbert space: two finite-dimensional matrices cannot satisfy $[\hat{x}, \hat{p}] = i\mathbb{I}$ ($\mathbb{I}$ being the identity matrix)⁹.

The quantum Hamiltonian of a single bosonic mode is

$$\hat{H} = \frac{\omega}{2}(\hat{x}^2 + \hat{p}^2). \quad (2.80)$$

Here $\hat{H}$ is the energy observable of the mode. It also generates the unitary time evolution of the system: in the Heisenberg picture, an operator $\hat{O}$ evolves as

$$\hat{O}(t) = e^{i\hat{H}t}\hat{O}e^{-i\hat{H}t}. \quad (2.81)$$

⁸This rule can be thought of as a central quantisation prescription. Its motivation is the correspondence principle: in the classical theory, the Poisson bracket generates infinitesimal canonical transformations, while in quantum mechanics the commutator generates infinitesimal unitary transformations. Requiring the quantum theory to reproduce the classical equations of motion in the appropriate limit leads to the replacement $\{A, B\} \mapsto [\hat{A}, \hat{B}]/(i\hbar)$. See, for example, Dirac's formulation of canonical quantisation and standard modern treatments of quantum mechanics [61, 165].

⁹If $\hat{x}$ and $\hat{p}$ acted on a d -dimensional Hilbert space, then taking the trace of $[\hat{x}, \hat{p}] = i\mathbb{I}_d$ would give $0 = id$ (using the cyclicity of the trace of finite matrices), a contradiction. Hence an exact bosonic mode requires an infinite-dimensional Hilbert space.

It is convenient to introduce the annihilation and creation operators

$$a = \frac{\hat{x} + i\hat{p}}{\sqrt{2}}, \quad a^\dagger = \frac{\hat{x} - i\hat{p}}{\sqrt{2}}. \quad (2.82)$$

Equivalently,

$$\hat{x} = \frac{a + a^\dagger}{\sqrt{2}}, \quad \hat{p} = \frac{a - a^\dagger}{i\sqrt{2}}. \quad (2.83)$$

Using $[\hat{x}, \hat{p}] = i$, one obtains

$$[a, a^\dagger] = 1. \quad (2.84)$$

In terms of these operators, the Hamiltonian becomes

$$\hat{H} = \omega \left(a^\dagger a + \frac{1}{2} \right). \quad (2.85)$$

The operator

$$\hat{n} = a^\dagger a \quad (2.86)$$

is called the number operator. Its eigenstates satisfy

$$\hat{n}|n\rangle = n|n\rangle, \quad n = 0, 1, 2, \dots \quad (2.87)$$

The eigenstates $\{|n\rangle\}_{n=0}^\infty$ of the number operator form a basis called the Fock basis of the single-mode Hilbert space. In an optical mode, the integer n is interpreted as the photon number. The term “bosonic” refers to the fact that any number of identical excitations can occupy the same mode, so the occupation number n is unbounded.

The energy eigenvalues of a mode are

$$\hat{H}|n\rangle = \omega \left(n + \frac{1}{2} \right) |n\rangle. \quad (2.88)$$

This equally spaced energy spectrum is why the term “harmonic” appears in the quantum theory. Each application of $a^\dagger$ raises the excitation number by one, while each application of a lowers it by one.

The same harmonic structure appears in the time evolution. Using Eq. (2.85), the Heisenberg evolution of the annihilation and creation operators is

$$a(t) = e^{i\hat{H}t} a e^{-i\hat{H}t} = e^{-i\omega t} a, \quad a^\dagger(t) = e^{i\omega t} a^\dagger. \quad (2.89)$$

Thus the annihilation operator evolves by a phase oscillating at frequency ω . Consequently, the quadrature operators evolve as

$$\hat{x}(t) = \hat{x} \cos(\omega t) + \hat{p} \sin(\omega t), \quad (2.90)$$

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and

$$\hat{p}(t) = \hat{p} \cos(\omega t) - \hat{x} \sin(\omega t). \quad (2.91)$$

The quadratures therefore rotate sinusoidally in phase space. This shows how the bosonic degree of freedom is a harmonic oscillator.

A field mode should be understood in this same sense. After decomposing a free field into independent modes, each mode has its own annihilation operator a_k , frequency ω_k , and Hamiltonian contribution $\omega_k(a_k^\dagger a_k + 1/2)$. Each mode evolves independently as $a_k(t) = e^{-i\omega_k t} a_k$. Thus a mode is an independent oscillatory quantum degree of freedom with a definite frequency, and the single-mode theory above describes one such degree of freedom.

As discussed in Section 1.4, the operators $\hat{x}$ and $\hat{p}$ are called quadrature operators. They are the quantum analogues of the two canonical coordinates of the classical oscillator. In optical continuous-variable systems, they correspond to two field quadratures of a single electromagnetic mode.

For an m -mode system, each mode has its own pair of quadrature operators $(\hat{x}_j, \hat{p}_j)$. We collect them into the phase-space operator vector

$$\hat{r} = (\hat{x}_1, \hat{p}_1, \dots, \hat{x}_m, \hat{p}_m)^T. \quad (2.92)$$

The canonical commutation relations can then be written compactly as

$$[\hat{r}_j, \hat{r}_k] = i\Omega_{jk}, \quad (2.93)$$

where

$$\Omega = \bigoplus_{j=1}^m \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (2.94)$$

The matrix Ω is called the symplectic form. A reversible linear transformation of the quadrature operators must preserve the canonical commutation relations. The matrices satisfying this condition form the symplectic group. We will discuss symplectic transformations in Section 2.4.

2.3.1. Quadrature eigenstates and wave functions

The quadrature operators have continuous spectra. Formally, one introduces generalised eigenstates $|x\rangle$ of $\hat{x}$ and $|p\rangle$ of $\hat{p}$, satisfying

$$\hat{x}|x\rangle = x|x\rangle, \quad x \in \mathbb{R}, \quad (2.95)$$

and

$$\hat{p}|p\rangle = p|p\rangle, \quad p \in \mathbb{R}. \quad (2.96)$$

These are *not* normalisable physical states. They are nevertheless useful because they provide the continuous basis in which wave functions are written.

The generalised eigenstates are delta-normalised:

$$\langle x|x'\rangle = \delta(x - x'), \quad \langle p|p'\rangle = \delta(p - p'). \quad (2.97)$$

Their overlap is the Fourier kernel

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi}} e^{ixp}. \quad (2.98)$$

This relation expresses the fact that the x and p descriptions of a state are related by a Fourier transform. A pure single-mode state $|\psi\rangle$ can be represented in the x basis by the wave function

$$\psi(x) = \langle x|\psi\rangle. \quad (2.99)$$

In this representation, the quadrature operators act as

$$\langle x|\hat{x}|\psi\rangle = x\psi(x), \quad \langle x|\hat{p}|\psi\rangle = -i\frac{d}{dx}\psi(x). \quad (2.100)$$

One can check directly from this representation that $[\hat{x}, \hat{p}] = i$, as required by Eq. (2.79).

Thus the generalised eigenstates are useful for representing wave functions, measurement probability densities, and formal resolutions of the identity, but care should be taken not to regard them as physical states.

Physical continuous-variable states are always normalisable density operators. In particular, the Gaussian states to be introduced in Section 2.3.3, such as coherent, squeezed, thermal, and two-mode squeezed states, are physical states with well-defined phase-space distributions and finite moments, provided the squeezing and energy are finite. This distinction between *ideal* quadrature eigenstates and physical states will be important later when continuous variables are discretised or when homodyne measurements are treated as ideal measurements or when security statements are made under finite-energy assumptions.

2.3.2. Wigner representation

The density-operator formalism is the most general way to describe quantum states, but for continuous-variable systems it is often more useful to work in phase space. To formulate quantum mechanics in phase space, one must represent not only states, but also observables, measurement operators, Hamiltonians, and other operators by functions of the quadrature variables. The Weyl transform [191] provides this correspondence between operators on the Hilbert space and functions on phase space.

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For an m -mode system, we write $x, p \in \mathbb{R}^m$ and collect the phase-space coordinates into the vector

$$r = (x_1, p_1, \dots, x_m, p_m)^T \in \mathbb{R}^{2m}. \quad (2.101)$$

The Weyl transform

For an operator $\hat{O}$ on an m -mode system, its Weyl symbol is defined as

$$O_W(x, p) = 2^m \int_{\mathbb{R}^m} e^{2ip \cdot y} \langle x - y | \hat{O} | x + y \rangle d^m y, \quad (2.102)$$

where $p \cdot y$ denotes the Euclidean inner product. The Weyl transform replaces the position-basis matrix elements of an operator by a scalar-valued function on phase space.

The transformation is invertible for suitable classes of operators. From the symbol O_W one can recover the position-basis matrix elements through

$$\langle x - y | \hat{O} | x + y \rangle = \frac{1}{(2\pi)^m} \int_{\mathbb{R}^m} e^{-2ip \cdot y} O_W(x, p) d^m p. \quad (2.103)$$

The Weyl symbol therefore contains the same information as the operator; it is merely a change of representation. The correspondence may also be extended to operator products, commutators, and dynamics.¹⁰

A Weyl symbol is not in general a probability or quasi-probability density. Even when $\hat{O}$ is positive, its symbol need not be pointwise positive. Moreover, the symbol of an observable need not integrate to one and may be unbounded. Its role is instead to represent the operator in phase space. If $\hat{O}$ is Hermitian, then $O_W(r)$ is real.

Several elementary operators have particularly simple Weyl symbols. The identity operator is represented by the constant function

$$(\mathbb{I})_W(r) = 1. \quad (2.104)$$

The quadrature operators have symbols

$$(\hat{x}_j)_W(r) = x_j, \quad (\hat{p}_j)_W(r) = p_j. \quad (2.105)$$

More generally, symmetrically ordered products of quadrature operators are represented by the corresponding ordinary phase-space polynomials. For example,

$$\left(\frac{\hat{x}_j \hat{p}_j + \hat{p}_j \hat{x}_j}{2} \right)_W = x_j p_j. \quad (2.106)$$

¹⁰Ordinary pointwise multiplication of Weyl symbols does not represent operator multiplication, since operators need not commute. Instead, the product of two operators is represented by the noncommutative Moyal, or star, product of their symbols. This also allows the von Neumann equation to be written entirely in phase space [148]. These extensions will not be required in this thesis.

For a single mode, the number operator $\hat{n} = a^\dagger a$ has symbol

$$n_W(x, p) = \frac{x^2 + p^2 - 1}{2}, \quad (2.107)$$

while the harmonic-oscillator Hamiltonian $\hat{H} = \omega(\hat{n} + \frac{1}{2})$ has symbol

$$H_W(x, p) = \frac{\omega}{2}(x^2 + p^2). \quad (2.108)$$

An important property of the Weyl transform is that the trace of the product of two operators becomes an integral of the product of their Weyl symbols:

$$\text{Tr}(\hat{A}\hat{B}) = \frac{1}{(2\pi)^m} \int_{\mathbb{R}^{2m}} A_W(r) B_W(r) \, d^{2m}r. \quad (2.109)$$

This identity is a consequence of the transform rather than an additional rule [87]. Substituting the definitions of A_W and B_W into the right-hand side, the integration over p produces a Dirac delta function that identifies the two relative position coordinates. The remaining change of variables then gives

$$\text{Tr}(\hat{A}\hat{B}) = \int_{\mathbb{R}^m} \int_{\mathbb{R}^m} \langle u | \hat{A} | v \rangle \langle v | \hat{B} | u \rangle \, d^m u \, d^m v, \quad (2.110)$$

which is precisely the position-basis expression for the trace of $\hat{A}\hat{B}$. Equation (2.109) may therefore be viewed as the phase-space analogue of Parseval's identity.

The Wigner function of a state

A density operator is itself an operator and consequently also has a Weyl symbol. For quantum states, however, it is convenient to use a different normalisation so that the resulting function integrates to one. The Wigner function [193] of a density operator ρ is defined by

$$W_\rho(r) = \frac{1}{(2\pi)^m} \rho_W(r), \quad (2.111)$$

or equivalently,

$$\rho_W(r) = (2\pi)^m W_\rho(r), \quad (2.112)$$

where ρ_W denotes the Weyl symbol of ρ . Thus the Wigner function is not a separate transform: it is the normalised Weyl symbol associated with a density operator.

For a single-mode density operator, this definition gives

$$W_\rho(x, p) = \frac{1}{\pi} \int_{\mathbb{R}} e^{2ipy} \langle x - y | \rho | x + y \rangle \, dy. \quad (2.113)$$

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For m modes, it becomes

$$W_\rho(x, p) = \frac{1}{\pi^m} \int_{\mathbb{R}^m} e^{2ip \cdot y} \langle x - y | \rho | x + y \rangle d^m y. \quad (2.114)$$

Like the Weyl symbol, the Wigner function is real whenever ρ is Hermitian, and it is normalised according to

$$\int_{\mathbb{R}^{2m}} W_\rho(r) d^{2m} r = \text{Tr } \rho. \quad (2.115)$$

In particular, for a normalised quantum state,

$$\int_{\mathbb{R}^{2m}} W_\rho(r) d^{2m} r = 1. \quad (2.116)$$

The Wigner function is also linear in the density operator. Thus, if $\rho = \sum_i p_i \rho_i$, then

$$W_\rho(r) = \sum_i p_i W_{\rho_i}(r). \quad (2.117)$$

This is the phase-space analogue of the linearity of the density-operator description.

For a quantum state, the Wigner function behaves in many ways like a probability density on phase space: it is normalised, its marginals give quadrature measurement distributions, and expectation values can be computed as phase-space integrals. The important difference is that the Wigner function need not be non-negative. It is therefore a quasi-probability distribution rather than an ordinary probability distribution.

The expectation-value formula now follows directly from the general trace identity. Taking $\hat{A} = \rho$ and $\hat{B} = \hat{O}$ in Eq. (2.109) gives

$$\begin{aligned} \langle \hat{O} \rangle_\rho &= \text{Tr}(\rho \hat{O}) \\ &= \frac{1}{(2\pi)^m} \int_{\mathbb{R}^{2m}} \rho_W(r) O_W(r) d^{2m} r \\ &= \int_{\mathbb{R}^{2m}} W_\rho(r) O_W(r) d^{2m} r. \end{aligned} \quad (2.118)$$

The last equality follows from Eq. (2.112). This expression resembles a classical phase-space average, but it is not an average over an underlying joint probability distribution. The Wigner function or the operator symbol may take negative values, while the full integral nevertheless reproduces the quantum-mechanical expectation value.

The same trace identity allows us to write the purity of a state as

$$\text{Tr}(\rho^2) = (2\pi)^m \int_{\mathbb{R}^{2m}} W_\rho(r)^2 d^{2m} r. \quad (2.119)$$

Likewise, if M_y is a POVM element, the probability of obtaining outcome y can be written as

$$p(y) = \text{Tr}(\rho M_y) = \int_{\mathbb{R}^{2m}} W_\rho(r)(M_y)_W(r) d^{2m}r. \quad (2.120)$$

Although the integrand need not be pointwise non-negative, its integral is a non-negative probability because it equals the Born-rule expression $\text{Tr}(\rho M_y)$.

The connection with quadrature measurement probabilities appears through the marginals. For a single mode,

$$\int_{\mathbb{R}} W_\rho(x, p) dp = \langle x | \rho | x \rangle, \quad (2.121)$$

and

$$\int_{\mathbb{R}} W_\rho(x, p) dx = \langle p | \rho | p \rangle. \quad (2.122)$$

Thus the Wigner function reproduces the probability densities for ideal measurements of the x and p quadratures, called homodyne measurements. In the special case of a pure state $\rho = |\psi\rangle\langle\psi|$, these marginals become

$$\int_{\mathbb{R}} W_{|\psi\rangle\langle\psi|}(x, p) dp = |\psi(x)|^2, \quad \int_{\mathbb{R}} W_{|\psi\rangle\langle\psi|}(x, p) dx = |\tilde{\psi}(p)|^2, \quad (2.123)$$

where $\psi(x) = \langle x | \psi \rangle$ and $\tilde{\psi}(p) = \langle p | \psi \rangle$. We will explore homodyne detection and its counterpart, heterodyne detection, in more detail in Section 2.5.

The Wigner representation also makes the partial trace especially simple. Suppose that a bipartite continuous-variable system consists of subsystems A and B , with phase-space coordinates $r_A \in \mathbb{R}^{2m_A}$ and $r_B \in \mathbb{R}^{2m_B}$. If the joint state is ρ_{AB} , then the reduced state $\rho_A = \text{Tr}_B \rho_{AB}$ has Wigner function

$$W_{\rho_A}(r_A) = \int_{\mathbb{R}^{2m_B}} W_{\rho_{AB}}(r_A, r_B) d^{2m_B}r_B. \quad (2.124)$$

That is, tracing out a subsystem corresponds to marginalising the Wigner function over the phase-space variables of that subsystem [80].

To see this, let $\hat{O}_A$ be any operator on subsystem A . Since the Weyl symbol of the identity is one, the Weyl symbol of $\hat{O}_A \otimes \mathbb{I}_B$ is

$$(\hat{O}_A \otimes \mathbb{I}_B)_W(r_A, r_B) = O_{A,W}(r_A), \quad (2.125)$$

which is independent of r_B . It follows that

$$\begin{aligned} \text{Tr}[\rho_A \hat{O}_A] &= \text{Tr}[\rho_{AB}(\hat{O}_A \otimes \mathbb{I}_B)] \\ &= \int W_{\rho_{AB}}(r_A, r_B) O_{A,W}(r_A) d^{2m_A}r_A d^{2m_B}r_B. \end{aligned} \quad (2.126)$$

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Comparing this expression with Eq. (2.118) gives Eq. (2.124).

We now define the first and second moments of a continuous-variable state. The first-moment vector is

$$d_j = \text{Tr}(\rho \hat{r}_j) = \int_{\mathbb{R}^{2m}} r_j W_\rho(r) d^{2m}r. \quad (2.127)$$

The covariance matrix is defined by

$$\Gamma_{jk} = \text{Tr}[\rho((\hat{r}_j - d_j)(\hat{r}_k - d_k) + (\hat{r}_k - d_k)(\hat{r}_j - d_j))]. \quad (2.128)$$

In the Wigner representation this becomes

$$\Gamma_{jk} = 2 \int_{\mathbb{R}^{2m}} (r_j - d_j)(r_k - d_k) W_\rho(r) d^{2m}r. \quad (2.129)$$

The factor of two comes from the covariance-matrix convention used in this thesis. With this convention, the vacuum covariance matrix of an m -mode system is

$$\Gamma_{\text{vac}} = \mathbb{I}_{2m}. \quad (2.130)$$

A state is called Gaussian if its Wigner function is a Gaussian function on phase space. An m -mode Gaussian state with first moment d and covariance matrix Γ has Wigner function

$$W_\rho(r) = \frac{1}{\pi^m \sqrt{\det \Gamma}} \exp[-(r - d)^T \Gamma^{-1} (r - d)]. \quad (2.131)$$

Thus, like a classical Gaussian distribution, a Gaussian quantum state is completely specified by its first and second moments. In the convention used here, the covariance matrix of the Gaussian function $W_\rho(r)$ is $\Gamma/2$.

The covariance matrix of any physical Gaussian state must satisfy the Robertson–Schrödinger uncertainty relation [189]

$$\Gamma + i\Omega \geq 0. \quad (2.132)$$

This is the covariance-matrix form of the canonical uncertainty principle. It ensures that the Gaussian function in Eq. (2.131) corresponds to a valid quantum state, rather than merely to a classical Gaussian distribution on phase space.

Although the Wigner function can be negative for general quantum states, Gaussian states always have non-negative Wigner functions. For Gaussian states, the Wigner representation can therefore be used much like an ordinary probability density on phase space, while still retaining the quantum constraint imposed by Eq. (2.132).

2.3.3. Gaussian states

We defined a Gaussian state in the previous section and learnt that it is completely specified by its first moment d and covariance matrix Γ . We now introduce the particular Gaussian states that occur most often in continuous-variable cryptography, using these moments.

2

Vacuum and thermal states

The vacuum is the reference Gaussian state of a single mode. It is the state with zero displacement and minimum symmetric quadrature noise.¹¹ In our covariance-matrix convention, its first moment is

$$d_{\text{vac}} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad (2.133)$$

and its covariance matrix is

$$\Gamma_{\text{vac}} = \mathbb{I}_2. \quad (2.134)$$

Thus the single-mode vacuum Wigner function is

$$W_{\text{vac}}(x, p) = \frac{1}{\pi} \exp[-x^2 - p^2]. \quad (2.135)$$

The vacuum is important because it sets the shot-noise level: in the quadrature variance convention used here, $\text{Var}(\hat{x}) = \text{Var}(\hat{p}) = 1/2$.

A thermal state is also centred and phase-space symmetric, but it has more noise than the vacuum¹². A single-mode thermal state with mean photon number $\bar{n}$ has first moment zero and covariance matrix

$$\Gamma_{\text{th}} = (2\bar{n} + 1)\mathbb{I}_2. \quad (2.136)$$

Its Wigner function is therefore a broader Gaussian:

$$W_{\text{th}}(x, p) = \frac{1}{\pi(2\bar{n} + 1)} \exp\left[-\frac{x^2 + p^2}{2\bar{n} + 1}\right]. \quad (2.137)$$

Thermal states naturally appear as environmental states in noisy Gaussian channels.

¹¹Equivalently, the vacuum is the zero-photon Fock state $|0\rangle$ and is characterised by $a|0\rangle = 0$.

¹²In the Fock basis, a thermal state is diagonal:

$$\rho_{\text{th}}(\bar{n}) = \sum_{n=0}^{\infty} \frac{\bar{n}^n}{(\bar{n} + 1)^{n+1}} |n\rangle\langle n|.$$

This representation is useful for interpreting $\bar{n}$ as the mean photon number, but in the thesis we will mostly use the covariance-matrix description.

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Coherent states

A coherent state (denoted by $|\alpha\rangle$) is a displaced vacuum state¹³. It has the same covariance matrix as the vacuum, but its Wigner function is centred at a nonzero point in phase space. The coherent state has first moment

$$d_\alpha = \begin{pmatrix} x_0 \\ p_0 \end{pmatrix}, \quad (2.138)$$

and covariance matrix

$$\Gamma_\alpha = \mathbb{I}_2. \quad (2.139)$$

Its Wigner function is

$$W_\alpha(x, p) = \frac{1}{\pi} \exp[-(x - x_0)^2 - (p - p_0)^2]. \quad (2.140)$$

Thus coherent states can be considered as phase-space points blurred by the unavoidable vacuum noise. They are central in many continuous-variable protocols and are the cheapest states to produce. An ideal, well-stabilised laser operating far above threshold is commonly modelled as producing a coherent state, although strictly speaking the laser output is more properly described as a phase-averaged mixture of coherent states [182].

Squeezed states

A squeezed state is still Gaussian, but its noise is no longer symmetric in the two quadratures. Squeezing reduces the variance of one quadrature below the vacuum level while increasing the variance of the conjugate quadrature. For a single-mode squeezed vacuum squeezed along the x axis, the first moment is zero and the covariance matrix is

$$\Gamma_{\text{sq}}(r) = \begin{pmatrix} e^{-2r} & 0 \\ 0 & e^{2r} \end{pmatrix}. \quad (2.141)$$

The corresponding Wigner function is

$$W_{\text{sq}}(x, p) = \frac{1}{\pi} \exp[-e^{2r} x^2 - e^{-2r} p^2]. \quad (2.142)$$

¹³Equivalently, a coherent state $|\alpha\rangle$ is an eigenstate of the annihilation operator, $a|\alpha\rangle = \alpha|\alpha\rangle$, and can be written as $|\alpha\rangle = D(\alpha)|0\rangle$. Its Fock-basis expansion is

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle.$$

For $r > 0$, the x quadrature is squeezed and the p quadrature is anti-squeezed. The product of the two quadrature variances remains fixed, so the uncertainty principle is respected. A displaced squeezed state is obtained by shifting the first moment while keeping the squeezed covariance matrix.¹⁴

Two-mode squeezed vacuum and EPR correlations

The two-mode squeezed vacuum is the basic entangled Gaussian state¹⁵. In phase-space language, it is a two-mode zero-mean Gaussian state with strong correlations between the quadratures of the two modes. In the ordering (x_A, p_A, x_B, p_B) , its covariance matrix is

$$\Gamma_{\text{TMSV}}(r) = \begin{pmatrix} \cosh(2r)\mathbb{I}_2 & \sinh(2r)\sigma_z \\ \sinh(2r)\sigma_z & \cosh(2r)\mathbb{I}_2 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (2.143)$$

The signs in the off-diagonal block encode the fact that the x quadratures are positively correlated while the p quadratures are anti-correlated. As $r \rightarrow \infty$, the correlations approach the ideal EPR form

$$x_A - x_B \rightarrow 0, \quad p_A + p_B \rightarrow 0, \quad (2.144)$$

in the sense that the variances of these combinations tend to zero. The ideal EPR state itself is not normalisable and would require infinite energy, but the finite-squeezing TMSV state is physical and is the entangled Gaussian resource used in many continuous-variable protocols (see Chapters 4 and 6).

The TMSV state is especially useful because measurements on one mode can prepare Gaussian states on the other mode. Heterodyning one mode conditionally prepares coherent states with Gaussian-distributed displacements, while homodyning one mode conditionally prepares squeezed states. These entanglement-based descriptions are often equivalent to prepare-and-measure descriptions and will be discussed in more detail in Section 2.5.

¹⁴The corresponding squeezing unitary is

$$S(\zeta) = \exp \left[\frac{1}{2} \left(\zeta^* a^2 - \zeta (a^\dagger)^2 \right) \right], \quad \zeta = r e^{i\phi}.$$

For $\phi = 0$, $S(\zeta)|0\rangle$ is squeezed along one quadrature. Its Fock-basis expansion contains only even photon-number states.

¹⁵In the Fock basis, the TMSV state can be written as

$$|\Psi_\lambda\rangle = \sqrt{1 - \lambda^2} \sum_{n=0}^{\infty} \lambda^n |n\rangle|n\rangle, \quad \lambda = \tanh r.$$

This expression makes the photon-number correlations explicit, but the covariance matrix is the more useful description for Gaussian calculations.

2.3.4. Continuous-variable teleportation

The two-mode squeezed vacuum is also the basic resource for continuous-variable teleportation. The purpose of teleportation is to transfer an unknown input state from one party to another using pre-shared entanglement and classical communication. In the continuous-variable setting, the entangled resource is an EPR-like two-mode squeezed state, and the Bell measurement is implemented by a beam splitter followed by two homodyne measurements [181, 189].

Let mode C contain the input state to be teleported, and let modes A and B share a two-mode squeezed state. Alice holds modes C and A , while Bob holds mode B . We use the convention in which the EPR correlations of the resource are

$$\hat{x}_A - \hat{x}_B \approx 0, \quad \hat{p}_A + \hat{p}_B \approx 0. \quad (2.145)$$

In the ideal infinite-squeezing limit, these correlations become exact.

Alice first mixes the input mode C and her half A of the entangled resource on a balanced beam splitter. She then measures the commuting quadratures

$$\hat{x}_- = \frac{\hat{x}_C - \hat{x}_A}{\sqrt{2}}, \quad \hat{p}_+ = \frac{\hat{p}_C + \hat{p}_A}{\sqrt{2}}. \quad (2.146)$$

This is the continuous-variable analogue of a Bell measurement. Let the measurement outcomes be x_m and p_m . Alice sends these two real numbers to Bob through a classical channel. Bob then applies the displacement

$$\hat{x}_B \mapsto \hat{x}_B + \sqrt{2}x_m, \quad \hat{p}_B \mapsto \hat{p}_B + \sqrt{2}p_m. \quad (2.147)$$

The effect of the protocol is easiest to see in the Heisenberg picture. Substituting the measurement outcomes into Bob's correction gives the output quadratures

$$\hat{x}_{\text{out}} = \hat{x}_C - (\hat{x}_A - \hat{x}_B), \quad \hat{p}_{\text{out}} = \hat{p}_C + (\hat{p}_A + \hat{p}_B). \quad (2.148)$$

Thus the teleported state is equal to the input state up to the residual EPR noise of the resource. In the ideal limit where $\hat{x}_A - \hat{x}_B = 0$ and $\hat{p}_A + \hat{p}_B = 0$, one obtains

$$\hat{x}_{\text{out}} = \hat{x}_C, \quad \hat{p}_{\text{out}} = \hat{p}_C, \quad (2.149)$$

and teleportation is perfect.

For a finitely squeezed two-mode squeezed state with squeezing parameter r , the EPR variances are

$$\text{Var}(\hat{x}_A - \hat{x}_B) = \text{Var}(\hat{p}_A + \hat{p}_B) = e^{-2r}. \quad (2.150)$$

Therefore finite-squeezing teleportation implements an additive-noise Gaussian channel. At the level of measured quadrature variances, each quadrature receives an additional noise contribution e^{-2r} . Equivalently, the mean and covariance matrices can be written as

$$d_{\text{out}} = d_{\text{in}}, \quad \Gamma_{\text{out}} = \Gamma_{\text{in}} + 2e^{-2r}\mathbb{I}_2. \quad (2.151)$$

The infinite-squeezing limit $r \rightarrow \infty$ removes this added noise and recovers the ideal teleportation channel.

In Chapter 4, the same mechanism appears in an attack strategy. Adversaries who share sufficiently strong EPR correlations can use teleportation-like procedures to transfer quantum information while only exchanging classical data.

2.4. Gaussian operations

In Section 2.3 (Eq. (2.92)), we introduced the quadrature vector $\hat{r}$ and the symplectic form $[\hat{r}_j, \hat{r}_k] = i\Omega_{jk}$.

The matrix Ω therefore encodes the canonical structure of phase space. If a reversible linear transformation of the quadratures is to describe a valid closed-system evolution, it must preserve these commutation relations. For a transformation $\hat{r} \mapsto S\hat{r}$, to preserve these commutation relations, we require

$$S\Omega S^T = \Omega. \quad (2.152)$$

Real matrices satisfying Eq. (2.152) form the symplectic group $\text{Sp}(2m, \mathbb{R})$. The associated quantum operations are the reversible linear transformations of phase space, and in the continuous-variable setting they are precisely the Gaussian unitaries, up to displacements.

This statement relies on the representation theory of the canonical commutation relations. For a finite number of modes, the Stone–von Neumann theorem says that every irreducible, strongly continuous representation of the Weyl form of the canonical commutation relations, with the same value of $\hbar$, is unitarily equivalent to the Schrödinger representation [172, 183, 74]. This justifies treating the usual quadrature representation as canonical. Moreover, symplectic transformations are implemented on Hilbert space by Gaussian unitaries, up to an irrelevant global phase [17, 189]. This is shown below.

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A Gaussian unitary acts affinely¹⁶ on the quadrature vector:

$$U^\dagger \hat{r} U = S \hat{r} + s, \quad S \in \text{Sp}(2m, \mathbb{R}), \quad s \in \mathbb{R}^{2m}. \quad (2.153)$$

The displacement vector s shifts the phase-space origin, while the symplectic matrix S changes the shape and orientation of the state in phase space. For a Gaussian state with first moment d and covariance matrix Γ , the transformation is

$$d \mapsto Sd + s, \quad (2.154)$$

and

$$\Gamma \mapsto S\Gamma S^T. \quad (2.155)$$

At the Wigner-function level this is an affine change of phase-space variables. Since affine transformations map Gaussian functions to Gaussian functions, Gaussian unitaries preserve Gaussianity.

A second important structural result is Williamson's theorem [195]. It states that every real, symmetric, strictly positive $2m \times 2m$ matrix Γ can be diagonalised by a symplectic transformation. That is, there exists $S_W \in \text{Sp}(2m, \mathbb{R})$ such that

$$\Gamma = S_W \left(\bigoplus_{j=1}^m \nu_j \mathbb{I}_2 \right) S_W^T. \quad (2.156)$$

The positive numbers ν_j are called the symplectic eigenvalues of Γ . For a physical covariance matrix in the convention used here, they satisfy

$$\nu_j \geq 1. \quad (2.157)$$

¹⁶A Gaussian unitary is generated by a Hamiltonian that is at most quadratic in the quadratures, $\hat{H} = \frac{1}{2} \hat{r}^T G \hat{r} + h^T \hat{r} + c$, where $G = G^T$ and $h \in \mathbb{R}^{2m}$. For $U(t) = \exp(-it\hat{H})$, the Heisenberg equation gives $\dot{\hat{r}}(t) = \Omega G \hat{r}(t) + \Omega h$. Since this is a linear inhomogeneous differential equation, its solution is affine:

$$U(t)^\dagger \hat{r} U(t) = S(t) \hat{r} + s(t), \quad S(t) = \exp(\Omega G t),$$

with $s(t) = \int_0^t \exp[\Omega G(t-\tau)] \Omega h \, d\tau$.

It remains to show that $S(t)$ is symplectic. Writing $K = \Omega G$, the symmetry of G and antisymmetry of Ω imply $K\Omega + \Omega K^T = 0$. Consequently,

$$\frac{d}{dt} (S(t) \Omega S(t)^T) = S(t) (K\Omega + \Omega K^T) S(t)^T = 0.$$

Hence $S(t) \Omega S(t)^T$ is constant in time. Since $S(0) = \mathbb{I}$, its value at $t = 0$ is Ω , and therefore $S(t) \Omega S(t)^T = \Omega$ for every t . Thus the linear part of the quadrature transformation preserves the canonical commutation relations and is symplectic. For any fixed evolution time τ , one may simply write $U = U(\tau)$, $S = S(\tau)$, and $s = s(\tau)$. Conversely, every symplectic transformation is implemented, up to a global phase, by a Gaussian unitary, and a displacement supplies the affine term.

Williamson's theorem shows that any Gaussian state is equivalent, up to a Gaussian unitary and a displacement, to a product of thermal states [195, 189]. It is therefore the covariance-matrix analogue of diagonalising a density matrix, but with the diagonalisation restricted to transformations that preserve the canonical commutation relations.

A useful tool is the Euler, or Bloch–Messiah, decomposition of symplectic transformations [103]. It states that any symplectic matrix $S \in \text{Sp}(2m, \mathbb{R})$ can be written as

$$S = K \left[\bigoplus_{j=1}^m \begin{pmatrix} e^{-r_j} & 0 \\ 0 & e^{r_j} \end{pmatrix} \right] L, \quad r_j \in \mathbb{R}, \quad (2.158)$$

where

$$K, L \in \text{Sp}(2m, \mathbb{R}) \cap \text{SO}(2m). \quad (2.159)$$

The matrices K and L are passive Gaussian transformations. They mix modes but do not squeeze them. The middle factor is a collection of independent single-mode squeezing operations. Hence every multimode Gaussian unitary can be implemented as a passive interferometer, followed by parallel single-mode squeezers, followed by another passive interferometer [17, 189]. This decomposition explains why displacements, phase rotations, squeezers, and beam splitters are the elementary operations from which general Gaussian unitaries are built. For a review of the symplectic group in quantum theory, see [17].

Displacements

A displacement shifts the mean of a state in phase space without changing its covariance matrix. For a single mode, the displacement operator is

$$D(\alpha) = \exp(\alpha a^\dagger - \alpha^* a). \quad (2.160)$$

Writing

$$\alpha = \frac{x_0 + ip_0}{\sqrt{2}}, \quad (2.161)$$

the action on the quadratures is

$$D(\alpha)^\dagger \begin{pmatrix} \hat{x} \\ \hat{p} \end{pmatrix} D(\alpha) = \begin{pmatrix} \hat{x} \\ \hat{p} \end{pmatrix} + \begin{pmatrix} x_0 \\ p_0 \end{pmatrix}. \quad (2.162)$$

Thus displacements change d but leave Γ unchanged.

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Phase rotations

A phase rotation is a passive single-mode Gaussian operation. It is generated by the number operator:

$$R(\theta) = \exp(-i\theta a^\dagger a). \quad (2.163)$$

Its action on the quadratures is

$$R(\theta)^\dagger \begin{pmatrix} \hat{x} \\ \hat{p} \end{pmatrix} R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \hat{x} \\ \hat{p} \end{pmatrix}. \quad (2.164)$$

It rotates phase space without changing the total quadrature variance.

Single-mode squeezing

Single-mode squeezing changes the relative noise in the two conjugate quadratures while preserving the canonical commutation relation. Squeezing along the x quadrature is represented by

$$S_{\text{sq}}(r) = \begin{pmatrix} e^{-r} & 0 \\ 0 & e^r \end{pmatrix}. \quad (2.165)$$

Thus

$$\hat{x} \mapsto e^{-r} \hat{x}, \quad \hat{p} \mapsto e^r \hat{p}. \quad (2.166)$$

The noise of one quadrature is reduced, while the noise of the conjugate quadrature is increased by the inverse factor. This is how the uncertainty principle is preserved.

Beam splitters

A beam splitter is a passive two-mode Gaussian operation. In the quadrature ordering (x_1, p_1, x_2, p_2) , a beam splitter with transmittance $T \in [0, 1]$ is represented by

$$S_{\text{BS}}(T) = \begin{pmatrix} \sqrt{T} \mathbb{I}_2 & \sqrt{1-T} \mathbb{I}_2 \\ -\sqrt{1-T} \mathbb{I}_2 & \sqrt{T} \mathbb{I}_2 \end{pmatrix}. \quad (2.167)$$

For $T = 1/2$, this is a balanced beam splitter. Beam splitters, together with phase rotations, generate general passive multimode interferometers. They play a central role in the physical implementation of Gaussian transformations and in measurement schemes such as heterodyne detection.

The operations above are Gaussian unitaries. More general Gaussian operations also include Gaussian channels and Gaussian measurements. These need not be symplectic on the system alone, because they may involve coupling to an environment or conditioning on a measurement outcome. They are nevertheless physical quantum operations and will be described in the following sections.

2.4.1. Gaussian channels and noise

A Gaussian channel is a CPTP map that sends Gaussian states to Gaussian states. At the level of first and second moments, an m -mode Gaussian channel is specified by real matrices X, Y and a displacement vector d_0 , acting as

$$d \mapsto Xd + d_0, \quad (2.168)$$

and

$$\Gamma \mapsto X\Gamma X^T + Y. \quad (2.169)$$

Complete positivity imposes the condition

$$Y + i\Omega - iX\Omega X^T \geq 0. \quad (2.170)$$

Gaussian channels provide the standard description of loss and noise in optical continuous-variable communication [189].

Pure loss

A pure-loss channel with transmittance $t \in [0, 1]$ can be modelled by mixing the input mode with an environmental vacuum mode on a beam splitter of transmittance t , and then tracing out the environment. Its action on moments is

$$d \mapsto \sqrt{t} d, \quad (2.171)$$

and

$$\Gamma \mapsto t\Gamma + (1 - t)\mathbb{I}_2. \quad (2.172)$$

Thus coherent-state displacements are attenuated by $\sqrt{t}$, while the output covariance of a coherent input remains the vacuum covariance.

Excess noise

If the environment is thermal rather than vacuum, the loss channel adds thermal noise. A common single-mode model is

$$\Gamma \mapsto t\Gamma + (1 - t)(2\bar{n}_{\text{env}} + 1)\mathbb{I}_2, \quad (2.173)$$

where $\bar{n}_{\text{env}}$ is the mean photon number of the environmental mode. In cryptographic analyses it is often convenient to separate attenuation from excess noise and write, for a coherent input,

$$d \mapsto \sqrt{t} d, \quad \Gamma \mapsto (1 + 2u)\mathbb{I}_2. \quad (2.174)$$

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With the covariance convention of Eq. (2.128), a homodyne measurement of a coherent state through such a channel has variance $\frac{1}{2} + u$.

The parameter u therefore represents excess quadrature noise above the vacuum shot-noise contribution. Throughout this thesis, we model imperfect quantum channels as lossy and noisy. The loss is characterised by the channel transmittance t , while the noise is quantified by the excess-noise parameter u , as defined above. Moreover, throughout this thesis, we use the term pulse to denote a single quantum state transmitted through the quantum channel. In other treatments, the same unit is referred to as a *channel* use or a *symbol*.

2.4.2. Entropy of Gaussian states

Computing the von Neumann entropy of a general continuous-variable state requires diagonalising an infinite-dimensional density operator. For Gaussian states this problem simplifies considerably. In (2.156), we saw that the covariance matrix of any m -mode Gaussian state can be written as a direct sum of single-mode covariance matrices by a symplectic transformation.

We also saw in (2.153) that a symplectic matrix is implemented by a Gaussian unitary. Since unitary transformations and displacements do not change the von Neumann entropy, the entropy of the original Gaussian state is equal to the entropy of the state with covariance matrix

$$\bigoplus_{j=1}^m \nu_j \mathbb{I}_2. \quad (2.175)$$

From Section 2.3.3, this state is a product of m thermal states. The j th thermal mode has mean photon number

$$\bar{n}_j = \frac{\nu_j - 1}{2}. \quad (2.176)$$

Therefore the entropy of a Gaussian state is the sum of the entropies of these thermal modes.

A thermal state with mean photon number $\bar{n}$ has eigenvalues

$$q_k = \frac{\bar{n}^k}{(\bar{n} + 1)^{k+1}}, \quad k = 0, 1, \dots \quad (2.177)$$

Substituting this geometric distribution into the von Neumann entropy gives

$$S(\rho_{\text{th}}) = g(\bar{n}), \quad (2.178)$$

where

$$g(n) = (n + 1) \log(n + 1) - n \log n. \quad (2.179)$$

Consequently, an m -mode Gaussian state with symplectic eigenvalues $\nu_1, \dots, \nu_m$ has entropy

$$S(\rho) = \sum_{j=1}^m g\left(\frac{\nu_j - 1}{2}\right). \quad (2.180)$$

The first moments do not appear in this expression because they can be changed by a displacement, which is unitary and therefore leaves the entropy invariant.

For a single-mode covariance matrix Γ , there is only one symplectic eigenvalue, given by

$$\nu = \sqrt{\det \Gamma}. \quad (2.181)$$

Hence

$$S(\rho) = g\left(\frac{\sqrt{\det \Gamma} - 1}{2}\right), \quad (2.182)$$

for every single-mode Gaussian state. A pure Gaussian state has $\nu = 1$ and therefore zero entropy, while $\nu > 1$ indicates mixedness.

Example: entropy after a homodyne measurement

As a simple example, consider a two-mode Gaussian state on systems A and B with covariance matrix in standard form

$$\Gamma_{AB} = \begin{pmatrix} a\mathbb{I}_2 & cZ \\ cZ & b\mathbb{I}_2 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (2.183)$$

The two symplectic eigenvalues of Γ_{AB} are

$$\nu_{1,2} = \sqrt{\frac{\Delta \pm \sqrt{\Delta^2 - 4D}}{2}}, \quad (2.184)$$

where

$$\Delta = a^2 + b^2 - 2c^2, \quad D = \det \Gamma_{AB} = (ab - c^2)^2. \quad (2.185)$$

The entropy of the joint state is therefore

$$S(AB) = g\left(\frac{\nu_1 - 1}{2}\right) + g\left(\frac{\nu_2 - 1}{2}\right). \quad (2.186)$$

Suppose now that the x quadrature of mode B is measured by ideal homodyne detection. Using the Gaussian conditioning rule introduced in Section 2.5, the conditional covariance matrix of mode A is

$$\Gamma_{A|x_B} = A - C(XBX)^{\text{MP}}C^T, \quad X = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \quad (2.187)$$

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For the covariance matrix in Eq. (2.183), this gives

$$\Gamma_{A|x_B} = \begin{pmatrix} a - \frac{c^2}{b} & 0 \\ 0 & a \end{pmatrix}. \quad (2.188)$$

Its symplectic eigenvalue is

$$\nu_3 = \sqrt{\det \Gamma_{A|x_B}} = \sqrt{a \left(a - \frac{c^2}{b} \right)}. \quad (2.189)$$

The corresponding conditional entropy is

$$S(A|x_B) = g\left(\frac{\nu_3 - 1}{2}\right). \quad (2.190)$$

For a Gaussian state and an ideal Gaussian measurement, the conditional covariance matrix does not depend on the numerical measurement outcome; only the conditional first moment changes. The entropy in Eq. (2.190) is therefore the same for every outcome.

This calculation is frequently used when evaluating the information available to an environment. Let E purify the joint state ρ_{AB} . Since the global state on ABE is pure,

$$S(E) = S(AB). \quad (2.191)$$

Moreover, an ideal homodyne measurement is rank one. Conditioned on an outcome x_B , the remaining state on AE is therefore pure, so that

$$S(E|x_B) = S(A|x_B). \quad (2.192)$$

The Holevo information between the classical homodyne outcome X_B and the purifying system E is consequently

$$\begin{aligned} \chi(X_B; E) &= S(E) - \int_{\mathbb{R}} p(x_B) S(E|x_B) dx_B \\ &= S(AB) - S(A|x_B) \\ &= g\left(\frac{\nu_1 - 1}{2}\right) + g\left(\frac{\nu_2 - 1}{2}\right) - g\left(\frac{\nu_3 - 1}{2}\right). \end{aligned} \quad (2.193)$$

Thus an apparently infinite-dimensional entropy calculation is reduced to computing three symplectic eigenvalues.

Continuous classical–quantum states

The preceding example naturally produces a state containing both classical and quantum information: the homodyne outcome is classical, while the side information held by E remains quantum. Such situations are described by classical–quantum states.

Let X be a continuous classical random variable with probability density $p_X(x)$, and let ρ_B^x be the normalised state of a quantum system B conditioned on the value $X = x$. The corresponding classical–quantum state is written formally as

$$\rho_{XB} = \int_{\mathbb{R}} p_X(x) |x\rangle\langle x|_X \otimes \rho_B^x dx. \quad (2.194)$$

Here $\{|x\rangle_X\}$ denotes the formal orthogonal basis of a classical register; it should not be confused with the quadrature eigenstates of a bosonic mode. The marginal state of the quantum system is

$$\rho_B = \int_{\mathbb{R}} p_X(x) \rho_B^x dx. \quad (2.195)$$

The Holevo information between X and B is

$$\chi(X; B) = S(\rho_B) - \int_{\mathbb{R}} p_X(x) S(\rho_B^x) dx. \quad (2.196)$$

It quantifies the information about the classical variable that is contained in the quantum system. The conditional differential entropy of X given B is then

$$h(X|B)_\rho = h(X) - \chi(X; B). \quad (2.197)$$

This is the continuous counterpart of the discrete relation $H(X|B) = H(X) - I(X; B)$.

An equivalent relative-entropy expression is obtained by introducing the subnormalised conditional operators

$$\tilde{\rho}_B^x = p_X(x) \rho_B^x. \quad (2.198)$$

Then

$$h(X|B)_\rho = - \int_{\mathbb{R}} D(\tilde{\rho}_B^x \| \rho_B) dx, \quad (2.199)$$

where the relative entropy is extended to positive subnormalised operators in the natural way. This form is particularly useful in one-shot and continuous-variable security analyses [77].

2.5. Quadrature measurements

The continuous-variable protocols studied in this thesis produce classical data by measuring field quadratures. The two most important quadrature-based measurement schemes are homodyne and heterodyne detection. In the idealised mathematical description, homodyne detection measures one quadrature of a mode, while heterodyne detection gives simultaneous, but noisier, information about both conjugate quadratures. These are the continuous-variable analogues of choosing and measuring a basis in finite-dimensional quantum information.

It is useful to keep in mind the basic optical picture, even though our analysis will use only the idealised measurement models. A photodetector directly measures intensity, not the signed field quadrature. To access a quadrature, the signal mode is interfered with a strong classical reference field, called a local oscillator. In balanced homodyne detection, the signal and local oscillator are combined on a balanced beam splitter and the difference of the two photocurrents is measured. In the limit of a strong local oscillator, this difference current is proportional to a field quadrature of the signal. The phase of the local oscillator determines which rotated quadrature is measured. Thus, changing the local-oscillator phase corresponds to changing the measurement angle in phase space [124, 133]. For a schematic representation, see Fig. 2.1.

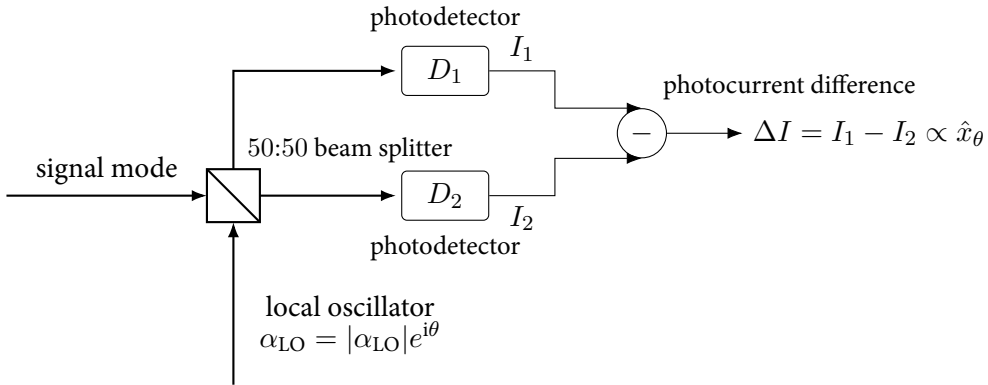


Figure 2.1: Schematic of balanced homodyne detection. The signal is mixed with a strong local oscillator of complex amplitude $\alpha_{LO} = |\alpha_{LO}|e^{i\theta}$ on a balanced beam splitter, where $|\alpha_{LO}|$ denotes the local-oscillator amplitude and θ its phase. The difference of the two photocurrents is proportional to the signal quadrature $\hat{x}_\theta$ selected by the phase θ .

Heterodyne detection [124, 189] can be understood as a double homodyne measurement. The input mode is first mixed with vacuum on a balanced beam splitter. One then homodynes one output mode in the x quadrature and the other output mode in the p quadrature. This allows both quadratures to be inferred in a single run, but the price is an additional

vacuum-noise contribution. Equivalently, heterodyne detection is a Gaussian measurement whose outcome is a joint classical probability distribution.

In what follows we use these ideal measurement models. Within the single-mode model, optical loss, finite detector efficiency, and imperfect mode matching can be represented by Gaussian loss channels acting before an ideal homodyne or heterodyne measurement. Electronic noise can similarly be incorporated as an equivalent Gaussian loss or additive-noise process [133, 189]. Therefore, for the purposes of the thesis, it is enough to describe homodyne and heterodyne detection through their action on Wigner functions and on the first and second moments of Gaussian states. We note that to avoid ambiguity, the chapters that follow state explicitly whether noise is quoted as a quadrature variance, for which the vacuum shot-noise contribution is $1/2$, or in shot-noise units (SNU), for which the same vacuum-noise level is normalised to 1. Throughout this thesis, we occasionally use *homodyne* and *heterodyne* as verbs. In this usage, to homodyne a state means to perform a homodyne measurement on it, while to heterodyne a state means to perform a heterodyne measurement on it. Similarly, the terms *homodyned* and *heterodyned* indicate that the corresponding measurement has been performed.

2.5.1. Homodyne measurement

A homodyne measurement measures one quadrature of a mode. For angle θ , the measured quadrature is

$$\hat{x}_\theta = \hat{x} \cos \theta + \hat{p} \sin \theta. \quad (2.200)$$

For $\theta = 0$ this is an x -quadrature measurement, and for $\theta = \pi/2$ it is a p -quadrature measurement.

For an m -mode state, the measurement statistics are obtained by integrating the Wigner function over all phase-space variables except x_i :

$$f_{X_i}(x_i) = \int_{\mathbb{R}^{2m-1}} W_\rho(x_1, p_1, \dots, x_m, p_m) \prod_{j=1}^m dp_j \prod_{j \neq i} dx_j. \quad (2.201)$$

For a single-mode Gaussian state with the following first moment and covariance matrix:

$$d = \begin{pmatrix} d_x \\ d_p \end{pmatrix} \quad \text{and} \quad \Gamma = \begin{pmatrix} \Gamma_{xx} & \Gamma_{xp} \\ \Gamma_{px} & \Gamma_{pp} \end{pmatrix}, \quad (2.202)$$

the corresponding marginal distributions are Gaussian:

$$f_X(x) = \frac{1}{\sqrt{\pi \Gamma_{xx}}} \exp \left[-\frac{(x - d_x)^2}{\Gamma_{xx}} \right], \quad (2.203)$$

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and

$$f_P(p) = \frac{1}{\sqrt{\pi\Gamma_{pp}}} \exp\left[-\frac{(p - d_p)^2}{\Gamma_{pp}}\right]. \quad (2.204)$$

Thus the homodyne variances are

$$\text{Var}(X) = \frac{\Gamma_{xx}}{2}, \quad \text{Var}(P) = \frac{\Gamma_{pp}}{2}. \quad (2.205)$$

For a vacuum-noise input, $\Gamma_{xx} = \Gamma_{pp} = 1$, so homodyne detection has quadrature variance $1/2$.

2.5.2. Heterodyne measurement

For the same single-mode Gaussian input state as above, the heterodyne marginal distributions are

$$f_{X_{\text{het}}}(x) = \frac{1}{\sqrt{\pi(\Gamma_{xx} + 1)}} \exp\left[-\frac{(x - d_x)^2}{\Gamma_{xx} + 1}\right], \quad (2.206)$$

and

$$f_{P_{\text{het}}}(p) = \frac{1}{\sqrt{\pi(\Gamma_{pp} + 1)}} \exp\left[-\frac{(p - d_p)^2}{\Gamma_{pp} + 1}\right]. \quad (2.207)$$

The variances are given by

$$\text{Var}(X_{\text{het}}) = \frac{\Gamma_{xx} + 1}{2}, \quad \text{Var}(P_{\text{het}}) = \frac{\Gamma_{pp} + 1}{2}. \quad (2.208)$$

Thus heterodyne detection adds one vacuum unit to the covariance matrix. For a vacuum-noise input, $\Gamma_{xx} = \Gamma_{pp} = 1$, and hence

$$\text{Var}(X_{\text{het}}) = \text{Var}(P_{\text{het}}) = 1. \quad (2.209)$$

This is twice the corresponding homodyne variance. The factor of two is the usual noise penalty for measuring both conjugate quadratures simultaneously.

2.5.3. Conditional preparation by homodyne measurement

Homodyne measurements are also useful as state-preparation tools. In particular, if one mode of a two-mode squeezed state is measured, the other mode is projected onto a conditional Gaussian state. This gives an entanglement-based description of preparing squeezed states. In the case of a homodyne measurement, one may prepare a two-mode squeezed state and homodyne one of its modes to prepare a single-mode squeezed state.

Consider a two-mode squeezed state on modes A and B with covariance matrix

$$\Gamma_{AB} = \begin{pmatrix} A & C \\ C^T & B \end{pmatrix} = \begin{pmatrix} \cosh(2s)\mathbb{I}_2 & \sinh(2s)\sigma_z \\ \sinh(2s)\sigma_z & \cosh(2s)\mathbb{I}_2 \end{pmatrix}, \quad (2.210)$$

as described in Section 2.3.3. We perform an ideal homodyne measurement of the x quadrature on mode B . The measured quadrature is represented at the covariance-matrix level by the projector

$$X = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \quad (2.211)$$

For Gaussian states, the conditional covariance matrix of mode A is obtained from the Gaussian conditioning rule [189]

$$A' = A - C(XBX)^{\text{MP}}C^T, \quad (2.212)$$

where MP denotes the Moore–Penrose inverse. Substituting the present case into Eq. (2.212) gives

$$A' = \begin{pmatrix} \cosh(2s)^{-1} & 0 \\ 0 & \cosh(2s) \end{pmatrix}. \quad (2.213)$$

Thus, if the two-mode squeezing parameter s is chosen such that $\cosh(2s) = e^{2r}$, then the conditional covariance matrix becomes

$$A' = \begin{pmatrix} e^{-2r} & 0 \\ 0 & e^{2r} \end{pmatrix}. \quad (2.214)$$

This is the covariance matrix of a single-mode squeezed state in the convention used in this chapter.

The measurement outcome affects only the displacement of the conditional state, not its covariance matrix. Let the initial first-moment vector be

$$d = \begin{pmatrix} d_A \\ d_B \end{pmatrix} = \begin{pmatrix} d_{A_x} \\ d_{A_p} \\ d_{B_x} \\ d_{B_p} \end{pmatrix}. \quad (2.215)$$

If the homodyne outcome on mode B is x_m , then the conditional displacement of mode A is

$$d'_A = d_A + C(XBX)^{\text{MP}} \left[\begin{pmatrix} x_m \\ 0 \end{pmatrix} - d_B \right]. \quad (2.216)$$

For the two-mode squeezed covariance matrix above, this reduces to

$$d'_{A_x} = d_{A_x} + \tanh(2s)(x_m - d_{B_x}), \quad d'_{A_p} = d_{A_p}. \quad (2.217)$$

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Hence the homodyne outcome determines the x displacement of the remotely prepared squeezed state. Averaging over measurement outcomes recovers the unconditioned mean of mode A , while conditioning on a particular outcome gives a definite displaced squeezed state.

2.5.4. Conditional preparation by heterodyne measurement

The previous subsection showed that homodyning one mode of a two-mode squeezed state conditionally prepares a single-mode squeezed state on the other mode. A closely related construction is obtained by heterodyne measurement. In this case, measuring one mode of a two-mode squeezed state conditionally prepares a coherent state on the other mode. This is the standard entanglement-based description of Gaussian modulation by coherent states in continuous-variable protocols [189].

We again consider a two-mode squeezed state on modes A and B , with covariance matrix

$$\Gamma_{AB} = \begin{pmatrix} A & C \\ C^T & B \end{pmatrix} = \begin{pmatrix} \cosh(2s)\mathbb{I}_2 & \sinh(2s)\sigma_z \\ \sinh(2s)\sigma_z & \cosh(2s)\mathbb{I}_2 \end{pmatrix}. \quad (2.218)$$

We now perform a heterodyne measurement on mode B . Since heterodyne detection can be viewed as a noisy simultaneous measurement of both quadratures, its conditioning rule differs from homodyne conditioning by the addition of one vacuum unit to the measured covariance block. Thus the conditional covariance matrix of mode A is

$$A' = A - C(B + \mathbb{I}_2)^{-1}C^T. \quad (2.219)$$

Substituting

$$A = B = \cosh(2s)\mathbb{I}_2, \quad C = \sinh(2s)\sigma_z, \quad (2.220)$$

we obtain

$$A' = \cosh(2s)\mathbb{I}_2 - \frac{\sinh^2(2s)}{\cosh(2s) + 1}\mathbb{I}_2. \quad (2.221)$$

Using

$$\sinh^2(2s) = \cosh^2(2s) - 1, \quad (2.222)$$

this simplifies to $A' = \mathbb{I}_2$. Therefore, after a heterodyne measurement on mode B , the conditional state of mode A has vacuum covariance. It is therefore a coherent state, with its displacement determined by the measurement outcome.

Let the initial first-moment vector be

$$d = \begin{pmatrix} d_A \\ d_B \end{pmatrix}. \quad (2.223)$$

If the heterodyne outcome on mode B is

$$m = \begin{pmatrix} x_m \\ p_m \end{pmatrix}, \quad (2.224)$$

then the conditional displacement of mode A is

$$d'_A = d_A + C(B + \mathbb{I}_2)^{-1}(m - d_B). \quad (2.225)$$

For the two-mode squeezed state this becomes

$$d'_A = d_A + \tanh(s)\sigma_z(m - d_B). \quad (2.226)$$

Equivalently, in components,

$$d'_{A_x} = d_{A_x} + \tanh(s)(x_m - d_{B_x}), \quad d'_{A_p} = d_{A_p} - \tanh(s)(p_m - d_{B_p}). \quad (2.227)$$

For a centred two-mode squeezed state, $d_A = d_B = 0$, the heterodyne outcome is Gaussian-distributed with covariance

$$\frac{B + \mathbb{I}_2}{2} = \frac{\cosh(2s) + 1}{2}\mathbb{I}_2 = \cosh^2(s)\mathbb{I}_2. \quad (2.228)$$

The resulting coherent-state displacement on mode A is therefore Gaussian-distributed with covariance

$$\tanh^2(s) \cosh^2(s)\mathbb{I}_2 = \sinh^2(s)\mathbb{I}_2. \quad (2.229)$$

Thus heterodyning one mode of a two-mode squeezed state prepares coherent states on the other mode with Gaussian-distributed displacements. In the large-squeezing limit, the two-mode squeezed state approaches the ideal EPR state, and the measurement outcome on one mode determines the displacement of the other mode with increasingly strong correlations.

3

Quantum Key Distribution

When elementary quantum systems are used to transmit digital information, the uncertainty principle gives rise to novel cryptographic phenomena unachievable with traditional transmission media.

— Charles H. Bennett and Gilles Brassard

Quantum key distribution (QKD) is a family of cryptographic protocols in which two distant parties use a quantum channel, together with authenticated classical communication, to establish a shared secret key. Its main conceptual appeal is that security is derived from the laws of quantum mechanics rather than from unproven assumptions about computational hardness. This distinguishes QKD from conventional and post-quantum key exchange protocols, whose security relies on the presumed difficulty of specific mathematical problems [21]. At the same time, QKD comes with practical challenges, including limited range, the need for specialised hardware, sensitivity to implementation imperfections, and a still-developing certification and deployment ecosystem [16, 163].

Continuous-variable QKD (CVQKD) is a particularly attractive branch of QKD because it uses coherent or squeezed optical states together with homodyne or heterodyne detection, making it compatible with standard telecom components. This contrasts with discrete-variable QKD, where single-photon detectors play a central role and can be expensive. In Gaussian-modulated CVQKD, information is encoded in the field quadratures of weak coherent states, and the receiver obtains continuous measurement outcomes. This setting is especially natural from the perspective of optical communications, but it also leads to a reconciliation problem that differs significantly from the discrete-variable case.

Error correction in continuous-variable QKD

After the quantum communication stage of a CVQKD protocol, Alice and Bob do not yet share identical keys. Instead, they possess correlated continuous random variables, and a classical information reconciliation step is needed to convert these into matching discrete data while leaking as little information as possible to an eavesdropper. This reconciliation step is one of the central bottlenecks in long-distance CVQKD.

3

The reason is closely tied to how loss manifests itself in the continuous-variable setting. In discrete-variable QKD, photon loss typically leads to no-click events, which can simply be discarded. In CVQKD, by contrast, every homodyne or heterodyne measurement produces an outcome. Channel attenuation and excess noise therefore do not primarily create erasures; instead, they degrade the signal-to-noise ratio of Bob's data. As the transmission distance increases, the relevant reconciliation problem becomes an extremely low-SNR decoding problem. To obtain useful secret key rates in this regime, one needs error correction that operates close to the Shannon limit.

A large body of work has therefore focused on highly efficient codes for CVQKD, including turbo codes, LDPC codes, polar codes, and related constructions, often combined with multidimensional reconciliation [131, 20, 127, 107, 145, 106, 91, 90, 108, 202, 200]. These methods have been very successful in pushing reconciliation efficiencies upward. However, long-distance operation places pressure on more than efficiency alone. The decoder must also be implementable at high throughput, ideally in real time, and this is challenging when decoding relies on low-rate iterative schemes that are difficult to parallelise efficiently. Moreover, many practical reconciliation systems reject blocks that decode unsuccessfully. Such frame rejection can improve performance, but from the viewpoint of security it acts as a form of post-selection and therefore has to be treated carefully in the leakage analysis.

This chapter explores a different route to reconciliation for long-distance Gaussian-modulated CVQKD, motivated by ideas from random coding. The primary motivation is the well-known fact that random coding, although impractical, can achieve capacity. The protocol studies random coding by formulating the error-correction step as a statistical identification problem. Bob embeds his observation among many candidate strings that are drawn from the appropriate distribution, and Alice uses her correlated data and the corresponding components of the candidate strings to identify the most plausible candidate through a score function closely related to likelihood-based decoding. Viewed this way, reconciliation becomes a large collection of independent matching problems, one for each block, which makes the procedure highly parallelisable.

To the best of our knowledge, this is the first detailed study of random-codebook error correction for long-distance Gaussian-modulated CVQKD. As a first investigation, we make

3. Quantum Key Distribution

several simplifying assumptions in order to isolate the asymptotic performance and practical appeal of the reconciliation method. We neglect finite-size effects, including the statistical uncertainty associated with parameter estimation, and leave their treatment for future work. We further assume that the channel statistics and the relevant calibration parameters remain stationary throughout the protocol, so that the distributions used to generate the random codebook entries and determine the decoding threshold accurately describe Bob's measurement data. Channel monitoring is therefore treated implicitly rather than modelled as part of the protocol. Pulses dedicated to parameter estimation are assumed to be publicly revealed and discarded; they are excluded from the denominator of the reported secret-key ratios. The quoted ratios should consequently be interpreted as secret bits per key-generation pulse rather than per total emitted pulse.

Within this scope, the method is attractive for two reasons. First, its blockwise structure offers a possible path towards implementations in which the most demanding computations can be distributed across many parallel processing units. Second, it leads to a comparatively clean statistical description of acceptance and decoding errors. In particular, the chapter studies a version of the protocol in which the accept/reject information is hidden from an eavesdropper, including the number of accepted blocks, so that the leakage analysis remains within a Gaussian framework. A more conventional outer error-correcting layer can then be used to remove the residual errors in the accepted blocks.

Outline of the chapter

The chapter begins in Section 3.1 with notation and background on Gaussian-modulated CVQKD with homodyne detection and reverse reconciliation. It also introduces the decoupled form of the reconciliation data, which will be used throughout the analysis. Section 3.2 then presents the reconciliation method. The starting point is an idealised version in which Bob uses genuinely random tables. This formulation is not intended as a practical protocol in itself; rather, it provides a clean setting in which the statistical structure of the method can be analysed transparently. The section also explains the role of the score function that is used by Alice for her accept/reject decisions and the decoupling property on which the later analysis relies. In Section 3.3, the statistical behaviour of the scheme is studied in more detail. There we analyse the acceptance probability and the probability of decoding errors within the accepted set of blocks. These quantities are then used in Section 3.4 to derive the asymptotic secret key ratio, including the cost of the one-time-pad encryption needed to hide the acceptance pattern. Finally, the section presents a comparison with the current state of the art. Section 3.5 turns to implementation aspects. In particular, it introduces a more practical variant in which the large random tables are replaced by pseudorandomly generated ones derived from short seeds, and it discusses how the resulting decoding task may be organised efficiently in parallel hardware. Finally,

Section 3.6 concludes the chapter with a discussion of the scope and limitations of the approach, as well as possible directions for further development.

3.1. Preliminaries

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3.1.1. Notation

For an integer q , we write $[q]$ as shorthand for the set $\{1, \dots, q\}$. Stochastic variables are written in capital letters, and their numerical values in lowercase. The normal distribution is denoted by $\varphi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$. Its cumulative distribution is written as $\Phi(x) = \int_{-\infty}^x du \varphi(u)$.

As a figure of merit for a CVQKD scheme, we use the secret key **ratio** (SKR), which equals the length of the constructed secret key divided by the number of laser pulses. This is different from the key *rate* (key bits per second). The key *ratio* allows a fair comparison between post-processing schemes for CVQKD setups with different pulse rates. Note that we do not take into account any pulses that are used solely for channel monitoring.

3.1.2. CVQKD with Gaussian modulation, homodyne detection, and reverse reconciliation

We briefly review CVQKD basics. We consider Gaussian modulation for simplicity. We consider the usual attacker model: Eve can manipulate quantum states in transit, but cannot access Alice and Bob's laboratories. Alice and Bob communicate over a classical channel that is noiseless, public, and authenticated.

Alice draws Gaussian $P, Q \in \mathbb{R}$ with zero mean and variance σ_X^2 . She sends a coherent state with quadrature values centred on (P, Q) . Bob randomly measures one of the two quadratures. This procedure is repeated n times. Bob's outcomes are $Y \in \mathbb{R}^n$. Alice's displacement components that match Bob's choice of quadratures are denoted as $X \in \mathbb{R}^n$.

Alice and Bob monitor the noise in the quantum channel. The channel can be characterised by two parameters [125]: the attenuation T (also called the transmittance or transmission parameter) and the excess noise power ξ . It holds that $y_i = \sqrt{T}x_i + N_{\text{shot}} + N_{\text{exc}}$, where N_{shot} is Gaussian shot noise with zero mean and variance $\frac{1}{2}$, and N_{exc} is Gaussian excess noise with zero mean and variance $T\xi/2$. We do not include the quantum efficiency and electronic noise of the detector in the model, as they can be absorbed in T [199] due to the noise-loss equivalence. This remodelling leads to a small value of the remaining excess noise power ξ ; in the remainder of the chapter we will set ξ to be negligible. The

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variance of Y_i is given by

$$\sigma_Y^2 = T\sigma_X^2 + \frac{1}{2} + \frac{1}{2}T\xi \quad \sigma_{Y|X}^2 = \frac{1}{2} + \frac{1}{2}T\xi. \quad (3.1)$$

In long-distance CVQKD we have $T \ll 1$. We consider the regime where $T\sigma_X^2 < 1$, $\xi \ll \sigma_X^2$, and we write the signal-to-noise ratio as

$$\varepsilon \stackrel{\text{def}}{=} \frac{T\sigma_X^2}{\sigma_{Y|X}^2} \quad (3.2)$$

with $\varepsilon \ll 1$. Bob informs Alice of his quadrature choices. Bob generates a secret key $K \in \mathcal{K}$, which is typically a hash of some U , where U is an error-correction codeword or a discretisation of Y . He computes information reconciliation data P and sends P to Alice. Typically, P contains a seed for the hash together with, for example, the syndrome of the discretised Y or a codeword XORed into Y . From X and P Alice reconstructs K by first reconstructing U and then hashing U with the provided seed.

It has been shown that the optimal attack against Gaussian-modulated CVQKD is a Gaussian attack, i.e. an attack that results in a Gaussian state. Furthermore, we restrict the present analysis¹ to Gaussian collective (i.i.d.) attacks [151]. The maximum achievable size of K , in bits, divided by n is called the secret key capacity. It has been shown [57] that the secret key capacity for Gaussian modulation and one-way reverse reconciliation ($C_{\text{secr}}^{\text{Gauss}}$) is bounded by

$$C_{\text{secr}}^{\text{Gauss}} \geq \Delta I \stackrel{\text{def}}{=} I(X_i; Y_i) - I(E_i; Y_i). \quad (3.3)$$

The mutual information between the classical X_i and Y_i follows from the signal-to-noise ratio,

$$I(X_i; Y_i) = \frac{1}{2} \log_2(1 + \varepsilon) = \frac{\varepsilon}{2 \ln 2} - \mathcal{O}(\varepsilon^2). \quad (3.4)$$

The mutual information $I(E_i; Y_i)$ follows from the symplectic eigenvalues ν_1, ν_2 of Eve's pre-measurement state ρ^{E_i} and the symplectic eigenvalue ν_3 of the (homodyne) post-measurement state $\rho_{y_i}^{A_i}$ [125].

$$I(E_i; Y_i) = g\left(\frac{\nu_1 - 1}{2}\right) + g\left(\frac{\nu_2 - 1}{2}\right) - g\left(\frac{\nu_3 - 1}{2}\right), \quad (3.5)$$

where g is the thermal entropy function $g(x) \stackrel{\text{def}}{=} (x + 1) \log_2(x + 1) - x \log_2 x$ (Section 2.4.2), and

$$\nu_{1,2}^2 = \frac{\Delta \pm \sqrt{\Delta^2 - 4D}}{2} \quad \nu_3^2 = V \left[V - \frac{T(V^2 - 1)}{2\sigma_Y^2} \right] \quad (3.6)$$

¹We discuss general attacks in Section 3.2.5

$$\Delta \stackrel{\text{def}}{=} V^2 + [2\sigma_Y^2]^2 - 2T(V^2 - 1) \quad D \stackrel{\text{def}}{=} \left[V \cdot 2\sigma_Y^2 - T(V^2 - 1) \right]^2 \quad (3.7)$$

where $V = 1 + 2\sigma_X^2$.

At low SNR, it is approximately given by

$$I(E_i; Y_i) = \frac{\varepsilon}{2 \ln 2} \cdot \sigma_X^2 \ln \frac{1 + \sigma_X^2}{\sigma_X^2} + \mathcal{O}(T\xi \log \frac{1}{T\xi}) - \mathcal{O}(\varepsilon^2). \quad (3.8)$$

In the limit of large modulation variance, it holds that

$$I(X_i; Y_i) - I(E_i; Y_i) \xrightarrow{\sigma_X^2 \rightarrow \infty} \frac{T}{2 \ln 2} + \mathcal{O}(T^2). \quad (3.9)$$

We call $\frac{T}{2 \ln 2}$ the *low- T Devetak-Winter value* [57]. It is $\frac{1}{2}$ of the PLOB bound [156], which represents the upper bound on the QKD key ratio.

Fig. 3.1 shows how large the information leakage $I(E_i; Y_i)$ is relative to the maximally available information $I(X_i; Y_i)$. The gap $I(X_i; Y_i) - I(E_i; Y_i)$ increases as a function of σ_X^2 and saturates at the Devetak-Winter value. However, in practice it is difficult to operate at large σ_X^2 for two reasons: (i) Large modulation causes phase noise [141], which manifests as excess noise; (ii) Working with large σ_X^2 demands very high reconciliation efficiency. In order to have a positive key rate, the reconciliation efficiency has to be larger than $\frac{I(E_i; Y_i)}{I(X_i; Y_i)}$. In Fig. 3.1, we see that, for example, when $\sigma_X^2 \geq 9.37$, the efficiency needs to be better than 95%. At $\sigma_X^2 < 1$, it is possible to obtain a positive key rate even with a highly inefficient error-correction code.

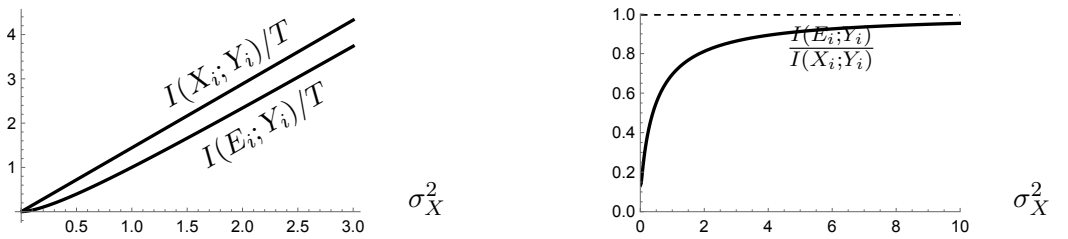


Figure 3.1: The mutual information quantities $I(X_i; Y_i)$ (3.4) and $I(E_i; Y_i)$ (3.5) as functions of the modulation variance σ_X^2 , for $T \ll 1$, $\xi \ll \sigma_X^2$.

3.1.3. ‘Decoupled’ reconciliation data

The SKR of CVQKD in practice is affected by the efficiency β of the employed error-correction code ($\beta < 1$) and by the leakage caused by the public reconciliation data P .

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A high SKR is obtained by using a highly efficient error-correcting code and leakage-free reconciliation data. Good codes for CVQKD achieve $\beta \approx 0.95$ to 0.98 . Leakage-free reconciliation is achieved by sending a one-time-pad-encrypted syndrome and by applying the method described below.

Lemma 3.1.1 (Lemma 1 in [127]). *Let U and P be independent classical random variables. Let E be a quantum system. Then $I(U; PE) \leq I(UP; E)$.*

Corollary 3.1.2 (Adapted from [127] to the case of reverse reconciliation). *Let X and Y be the measurement results of Alice and Bob, respectively. Let E be Eve's quantum system. Let U be sampled by Bob using fresh private randomness, independently of the joint state of X, Y , and E . Let P be public information reconciliation data computed from Y and U such that Alice is able to reconstruct U from P and X . Let P be independent of U . Then*

$$I(U; PE) \leq I(Y; E). \quad (3.10)$$

Proof: Lemma 3.1.1 gives $I(U; PE) \leq I(UP; E)$. Since P is computed from U and Y , it holds that $H(UP) \leq H(UY)$ and $I(UP; E) \leq I(UY; E)$. Finally, we use the fact that Bob chooses U independently of Y, X , and E , which allows us to write $I(UY; E) = I(Y; E)$. $\square$

For the SKR in the case of collective attacks, this yields

$$R_{\text{key}}^{(\text{decoupled})} \geq \beta I(X_i; Y_i) - I(E_i; Y_i). \quad (3.11)$$

In [127], the decoupling of P from U is achieved by taking 8-dimensional subspaces and using the algebraic properties of octonions.

3.1.4. Useful lemmas

Lemma 3.1.3 (Neyman-Pearson). *Let $X \in \mathcal{X}$ be a random variable distributed according to $X \sim P_\theta$, where θ is either 0 or 1. A hypothesis test function is a function $f : \mathcal{X} \rightarrow \{0, 1\}$ that tries to determine the parameter θ from the observed data x . Let $\mathbb{E}_{X \sim P_0} f(X)$ be the type I error probability of the test function, and let $\mathbb{E}_{X \sim P_1} [1 - f(X)]$ be the type II error probability. The likelihood ratio test with threshold k is given by*

$$\varphi_k(x) = \begin{cases} 1 & : \frac{P_1(x)}{P_0(x)} > k \\ 0 & : \frac{P_1(x)}{P_0(x)} < k \end{cases}. \quad (3.12)$$

At a fixed type I error probability, the likelihood ratio test (3.12) has the lowest type II error probability of all test functions.

Lemma 3.1.4. *A test function obtained by applying a monotone function to the likelihood ratio $\frac{P_1(x)}{P_0(x)}$ and to the threshold is equivalent to the likelihood ratio test.*

3.2. The proposed information reconciliation method

3.2.1. Hiding accept/reject decisions

As has often been noted (see, e.g., [125]), it is difficult to give a security proof for CVQKD schemes whose reconciliation involves ‘post-selection’, i.e. the rejection of blocks. Eve’s state, when conditioned on the fact that a block has been accepted, is no longer Gaussian; instead, it becomes, for instance, a classical mixture of Gaussian states, which is intractable.

For a scheme with block acceptance probability P_{acc} , it is tempting to postulate a rate formula $R = P_{\text{acc}}[\beta I(X_i; Y_i) - I(Y_i; E_i)]$ as a modification of (3.11). However, this is only allowed if the set of *all public data* is statistically independent of the secret key. The accept/reject decisions for blocks do *not* have this independence. For this reason, we introduce a reconciliation scheme in which Alice encrypts all the accept/reject decisions using a one-time pad. This consumes pre-shared key material and hence inflicts a heavy penalty on the achievable key ratio.

3.2.2. Protocol steps

Initialisation. Alice and Bob already share pre-shared key material. Alice and Bob determine the channel properties T and ξ . They agree on the modulation variance σ_X^2 . They agree on parameters $n, N \in \mathbb{N}$. They agree on a function $\text{Bin}: [q] \rightarrow \{0, 1\}^{\lceil \log q \rceil}$ that maps q -ary symbols to their binary representations. They work with a family of binary linear error-correcting codes. Each member of the family is parameterised by an integer N_{acc} that represents the number of accepted blocks. We denote the syndrome function by $\text{Syn}: \{0, 1\}^{N_{\text{acc}} \lceil \log q \rceil} \rightarrow \{0, 1\}^{\kappa(N_{\text{acc}})}$ and the syndrome-decoding function by $\text{SynDec}: \{0, 1\}^{\kappa(N_{\text{acc}})} \rightarrow \{0, 1\}^{N_{\text{acc}} \lceil \log q \rceil}$. For privacy amplification, they use a collection of universal hash functions, again parameterised by N_{acc} , which we denote by $\text{hash}: \mathcal{R} \times \{0, 1\}^{N_{\text{acc}} \lceil \log q \rceil} \rightarrow \{0, 1\}^{L(N_{\text{acc}})}$, where $\mathcal{R}$ is the space from which the UHF seed is drawn.

Execution. (See Fig. 3.2)

1. Alice initialises a local counter $a = 0$.
2. For $r \in \{1, \dots, N\}$, Alice and Bob perform the following procedure.
 - (a) **Alice.** Draw $q_1, \dots, q_n$ and $p_1, \dots, p_n$ from the distribution $\mathcal{N}_{0, \sigma_X^2}$. Send n coherent states, where the i th state has displacement (q_i, p_i) .
 - (b) **Bob.** For each pulse independently, choose randomly which quadrature (q or p) to measure. The measurement outcome is a vector $\mathbf{y} \in \mathbb{R}^n$. Draw a random integer $u \in \{1, \dots, q\}$. Set $w_r = u$. Draw random vectors $\mathbf{z}_1, \dots, \mathbf{z}_{q-1}$, where $\mathbf{z}_j \in \mathbb{R}^n$ and each component of each vector is independently

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drawn from $\mathcal{N}_{0, \sigma_Y^2}$.

Construct a matrix $m \in \mathbb{R}^{q \times n}$ whose rows $\mathbf{m}_1, \dots, \mathbf{m}_q$ are the vectors $\mathbf{z}_1, \dots, \mathbf{z}_{q-1}$, with $\mathbf{y}$ inserted in position u .² Send m and the quadrature choices.

- (c) **Alice.** Collect the correct quadrature component from each pair (q_i, p_i) ; the result is a vector $\mathbf{x} \in \mathbb{R}^n$. Compute the threshold $\theta(\mathbf{x}^2)$. For each $\ell \in \{1, \dots, q\}$, compute the score $s_\ell = J(\mathbf{x}, \mathbf{m}_\ell)$, where

$$J(\mathbf{x}, \mathbf{m}) \stackrel{\text{def}}{=} \frac{\frac{\mathbf{m}}{\sigma_Y} \cdot \frac{\mathbf{x}}{\sigma_X \sqrt{n}} + \frac{1}{2} \sqrt{\frac{n\varepsilon}{1+\varepsilon}} \left(1 - \frac{\mathbf{m}^2}{n\sigma_Y^2}\right)}{\sqrt{\frac{\mathbf{x}^2}{n\sigma_X^2} + \frac{1}{2} \frac{\varepsilon}{1+\varepsilon}}}. \quad (3.13)$$

If there is exactly one index ℓ such that $s_\ell > \theta(\mathbf{x}^2)$, then {set $\alpha_r = 1$; increase the counter a ; set $\hat{v}_a = \ell$ }, otherwise {set $\alpha_r = 0$ }.

3. **Alice.** Set $N_{\text{acc}} = a$. Create a binary string v_A by concatenating the strings $\text{Bin}(\hat{v}_j)$, with j running from 1 to N_{acc} . Prepare a short encoding³ $\mathcal{A}$ of the accept/reject indicator string $\alpha \in \{0, 1\}^N$. Compute ciphertext $c_0 = \text{Encrypt}(N_{\text{acc}} | \mathcal{A})$, a one-time-pad encryption of the concatenation $N_{\text{acc}} | \mathcal{A}$. Set $c = c_0 | \text{padding}$, where the padding is random and serves to create a fixed-length string c . Send c .
4. **Bob.** Recover N_{acc} and $\mathcal{A}$ by decrypting c_0 . Recover α from $\mathcal{A}$. Create a binary string v_B by concatenating $\{\text{Bin}(w_r) | \alpha_r = 1\}$. Compute $\sigma = \text{Syn}(v_B)$. Draw a random seed g . Compute the key $k_B = \text{hash}(g, v_B)$. Compute ciphertext $\tilde{\sigma}$ as a one-time-pad encryption of σ . Set $S = \tilde{\sigma} | \text{padding}$, where the padding is random and serves to create a fixed-length string. Send g and S . Here, the seed g is also randomly padded to a fixed length.
5. **Alice.** Recover σ by decrypting $\tilde{\sigma}$. Compute the error pattern $e = \text{SynDec}(\sigma \oplus \text{Syn } v_A)$. If SynDec returns an error, abort. Otherwise, compute $\hat{v} = v_A \oplus e$ and the key $k_A = \text{hash}(g, \hat{v})$. Send an encrypted success/fail bit to Bob.

Note that the lengths of $\mathcal{A}$, σ , and g depend on N_{acc} . The padding in Steps 3 and 4 serves to hide N_{acc} . When Bob receives c , he recovers N_{acc} from the first $\log N$ bits of the ciphertext; once he knows N_{acc} , he knows the size of $\mathcal{A}$ and ignores the padding. When Alice receives S , she already knows N_{acc} and hence the length of σ . Similarly, she knows the length of g beforehand.

3.2.3. Motivation for the score function

Our choice of score function is motivated by the fact that $J(\mathbf{x}, \mathbf{m}_\ell)$ is equivalent to the Neyman-Pearson score for the hypothesis test with the null hypothesis “ $u = \ell$ ” and the competing hypothesis “ $u \neq \ell$ ”. We briefly discuss it below.

²That is, $\mathbf{m}_j = \mathbf{z}_j$ for $j < u$; $\mathbf{m}_u = \mathbf{y}$; and $\mathbf{m}_j = \mathbf{z}_{j-1}$ for $j > u$.

³The size of $\mathcal{A}$ is $\log \binom{N}{N_{\text{acc}}}$ bits.

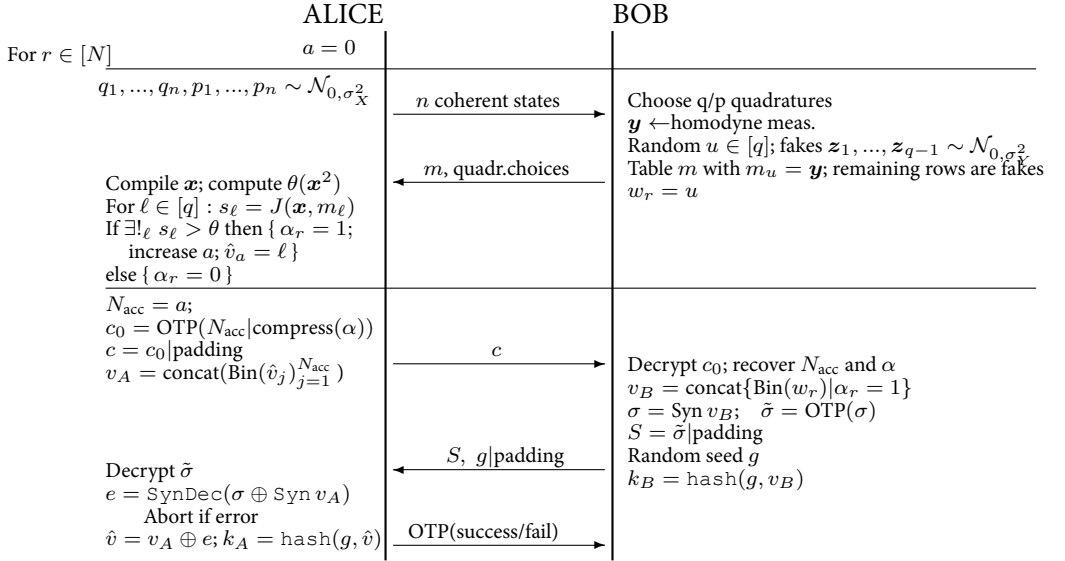


Figure 3.2: Protocol steps.

For hypothesis tests involving two competing hypotheses, the *likelihood-ratio test* is known to be uniformly most powerful. Alice knows $\mathbf{x}$ and the matrix m . Based on this information, she wants to test the null hypothesis $H_0: u = \ell$ against the competing hypothesis $H_1: u \neq \ell$, for some fixed $\ell \in [q]$.

The likelihood ratio is $\text{LR} = \frac{\Pr[H_0 | \mathbf{x}, m]}{\Pr[H_1 | \mathbf{x}, m]} = \frac{\Pr[H_0 | \mathbf{x}, m]}{1 - \Pr[H_0 | \mathbf{x}, m]}$. Since $\frac{x}{1-x}$ is an increasing function, thresholding $\Pr[H_0 | \mathbf{x}, m]$ is equivalent to thresholding R . It holds that $\Pr[H_0 | \mathbf{x}, m] \propto f(m | \mathbf{x}, H_0)$, where the proportionality factor does not depend on ℓ .

We have $f(m | \mathbf{x}, H_0) = f_{Y|X}(\mathbf{m}_\ell | \mathbf{x}) \prod_{j \neq \ell} f_Y(\mathbf{m}_j) = \frac{f_{Y|X}(\mathbf{m}_\ell | \mathbf{x})}{f_Y(\mathbf{m}_\ell)} \prod_j f_Y(\mathbf{m}_j) \propto \frac{f_{Y|X}(\mathbf{m}_\ell | \mathbf{x})}{f_Y(\mathbf{m}_\ell)} \propto \exp\left(\frac{\mathbf{m}_\ell^2}{2\sigma_Y^2} - \frac{(\mathbf{m}_\ell - \mathbf{x}\sqrt{T})^2}{2\sigma_{Y|X}^2}\right) \propto \exp\left[-\frac{T\sigma_X^2}{2\sigma_Y^2\sigma_{Y|X}^2} (\mathbf{m}_\ell - \mathbf{x}\sqrt{T} \frac{\sigma_Y}{T\sigma_X})^2\right] \propto \exp[J(\mathbf{x}, \mathbf{m}_\ell) \sqrt{n\varepsilon}]$

again with proportionality factors that do not depend on ℓ . The function J is the score function defined in (3.13). We conclude that thresholding the score $J(\mathbf{x}, \mathbf{m}_\ell)$ is equivalent to thresholding the likelihood ratio LR.

When choosing between more than two hypotheses, there is no known optimal algorithm. Therefore, we have settled on a possibly suboptimal solution by splitting Alice's q -fold hypothesis test into q binary tests that we know how to handle.

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3.2.4. The ‘decoupling’ property

The classical messages visible to Eve satisfy the ‘decoupling’ property, i.e., all the messages are independent of u , which then allows us to invoke Corollary 3.1.2. Eve sees the quadrature choices, all the tables m , the ciphertext c , the ciphertext S , the seed g , and the encrypted success/fail bit. The index u is obviously independent of the quadrature choices. Furthermore, all the ciphertexts are produced using a one-time pad, which makes them independent variables. The seed g is chosen independently. Finally, the tables satisfy $I(M; U) = H(M) - H(M|U)$, with $H(M|U) = H(M)$. The latter holds because the fake rows follow the same distribution as the row that contains Y .

3.2.5. Extension to coherent attacks

The security analysis in this chapter is carried out against Gaussian collective attacks, in which Eve applies the same interaction independently to every transmitted quantum state and may postpone her measurement until the end of the protocol. Gaussian optimality [151] then allows the secret-key rate to be bounded using the observed covariance matrix.

In the asymptotic regime, this analysis can be extended to coherent attacks using the infinite-dimensional de Finetti reduction of Ref. [162]. To apply this reduction, Alice and Bob first uniformly permute all signal records before dividing them into reconciliation blocks. The random permutation ensures that the complete collection of signals is treated symmetrically, while the subsequent reconciliation procedure processes every block using the same codebook construction, score function, threshold, and acceptance rule. Thus, no block is distinguished in advance by the protocol.

Permutation symmetry alone is not sufficient in the continuous-variable setting, because each optical mode is described by an infinite-dimensional Hilbert space. An additional energy test is therefore required. Alice and Bob may select a uniformly random subset of the transmitted signals, reveal the corresponding data, and verify that the measured quadrature values satisfy an appropriate energy or tail bound. This test restricts the remaining state, except with vanishing probability, to an effectively finite-dimensional low-energy subspace. The usual permutation-based de Finetti argument can then be applied to show that the remaining state is well approximated by a mixture of states that are independent and identically distributed on all but a vanishing fraction of the signals.

The energy test may be incorporated into the parameter-estimation procedure, since Bob already measures randomly chosen quadratures. The tested signals must, however, constitute a uniformly random sample of the same transmission used for key generation, and the protocol must abort when the prescribed bound is violated. In the asymptotic regime, the number of signals used for this test can grow while remaining a vanishing fraction of

the total number of signals. Neither the test nor the de Finetti reduction therefore changes the limiting secret-key rate.

3.3. Statistical properties of the scores; error probabilities

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We briefly analyse the statistics of the scores s_ℓ (3.13). We distinguish between the case where $\mathbf{m}_\ell$ is a random vector independent of $\mathbf{x}$ and the case where $\mathbf{m}_\ell = \mathbf{y}$. In the former case we write $\mathbf{m}_\ell = \mathbf{Z}$. In the latter case we write $\mathbf{y} = \mathbf{x}\sqrt{T} + \mathbf{N}$, where the noise components N_i are Gaussian with zero mean and variance $\sigma_{Y|X}^2$. In terms of the ε parameter this can be written as

$$\frac{\mathbf{y}}{\sigma_Y} = \frac{\frac{\mathbf{x}}{\sigma_X}\sqrt{\varepsilon} + \mathbf{N}/\sigma_{Y|X}}{\sqrt{1+\varepsilon}}. \quad (3.14)$$

The score function J has been constructed so that, for a given $\mathbf{x}$, the scores of the fake table entries have zero mean and unit variance,

$$\mathbb{E}_Z J(\mathbf{x}, \mathbf{Z}) = 0 \quad \text{Var}_Z J(\mathbf{x}, \mathbf{Z}) = 1. \quad (3.15)$$

For a given $\mathbf{x}$, the mean and variance of the correct-symbol score $J(\mathbf{x}, \mathbf{Y})$ are given by

$$\mathbb{E}_{Y|\mathbf{x}} J(\mathbf{x}, \mathbf{Y}) = \frac{\sqrt{n\varepsilon}}{(1+\varepsilon)^{3/2}} \frac{\frac{\mathbf{x}^2}{n\sigma_X^2}(1+\frac{\varepsilon}{2}) + \frac{\varepsilon}{2}}{\sqrt{\frac{\mathbf{x}^2}{n\sigma_X^2} + \frac{1}{2}\frac{\varepsilon}{1+\varepsilon}}} = \sqrt{n\varepsilon} \left[\sqrt{\frac{\mathbf{x}^2}{n\sigma_X^2}} + \mathcal{O}(\varepsilon) \right] \quad (3.16)$$

$$\text{Var}_{Y|\mathbf{x}} J(\mathbf{x}, \mathbf{Y}) = \frac{1}{(1+\varepsilon)^3} \frac{\frac{\mathbf{x}^2}{n\sigma_X^2} + \frac{\varepsilon}{2}}{\frac{\mathbf{x}^2}{n\sigma_X^2} + \frac{1}{2}\frac{\varepsilon}{1+\varepsilon}} = 1 + \mathcal{O}(\varepsilon). \quad (3.17)$$

We further examine the distribution of the fake-entry score $J(\mathbf{x}, \mathbf{Z})$. We introduce the notation $\frac{\mathbf{Z}}{\sigma_Y} = \zeta_{\parallel} \frac{\mathbf{x}}{|\mathbf{x}|} + \zeta_{\perp}$, where $\zeta_{\parallel}$ represents the part of $\frac{\mathbf{Z}}{\sigma_Y}$ parallel to $\mathbf{x}$ and $\zeta_{\perp}$ the part perpendicular to $\mathbf{x}$. This gives $\frac{\mathbf{Z}}{\sigma_Y} \cdot \frac{\mathbf{x}}{\sigma_X\sqrt{n}} = \zeta_{\parallel} \frac{|\mathbf{x}|}{\sigma_X\sqrt{n}}$. (One can rotate the basis of the n -dimensional vector space and let the first basis vector point in the direction of $\mathbf{x}$. Then $\zeta_{\parallel}$ is the component in the first basis direction.) Note that $\zeta_{\parallel}$ is normally distributed. Note that $\zeta_{\perp}^2$ is a scalar variable with a χ_{n-1}^2 distribution. With this notation, the fake-entry score is expressed as

$$J(\mathbf{x}, \mathbf{Z}) = \frac{\zeta_{\parallel} \frac{|\mathbf{x}|}{\sigma_X\sqrt{n}} + \frac{1}{2}\sqrt{\frac{n\varepsilon}{1+\varepsilon}}(1 - \frac{\zeta_{\parallel}^2 + \zeta_{\perp}^2}{n})}{\sqrt{\frac{\mathbf{x}^2}{n\sigma_X^2} + \frac{1}{2}\frac{\varepsilon}{1+\varepsilon}}} = \zeta_{\parallel} + \mathcal{O}(\sqrt{\varepsilon}) + \mathcal{O}(\frac{1}{\sqrt{n}}). \quad (3.18)$$

3. Quantum Key Distribution

Here, we have used that $\frac{x^2}{n\sigma_X^2} = 1 + \mathcal{O}(1/\sqrt{n})$, $\zeta_{\parallel}^2 = 1 + \mathcal{O}(1)$, and $\zeta_{\perp}^2 = n - 1 + \mathcal{O}(\sqrt{n})$ with overwhelming probability. From (3.18) we observe that the fake-entry score $J(\mathbf{x}, \mathbf{Z})$ consists almost entirely of the normally distributed variable $\zeta_{\parallel}$.

The amount of mutual information between Alice and Bob contained in n pulses is given by $I(\mathbf{X}; \mathbf{Y}) = nI(X_i; Y_i) = n \cdot \frac{1}{2} \log(1 + \varepsilon)$. Ideally, $\log q$ is set equal to $I(\mathbf{X}; \mathbf{Y})$, so that capacity is achieved. We will introduce a parameter γ that encodes how far away from capacity we operate,

$$\log q = (1 + \gamma)n \cdot \frac{1}{2} \log(1 + \varepsilon). \quad (3.19)$$

$\gamma < 0$ means operating below capacity. We will express n in terms of ε , q , and γ , i.e., $n = \frac{2 \ln q}{(1+\gamma) \ln(1+\varepsilon)}$. With this notation, the mean (3.16) is expressed as

$$\mathbb{E}_{Y|X} J(\mathbf{x}, \mathbf{Y}) = \frac{\sqrt{2 \ln q}}{\sqrt{1+\gamma}} + \mathcal{O}(\sqrt{\varepsilon}). \quad (3.20)$$

We also introduce the notation $\frac{N}{\sigma_{Y|X}} = \nu_{\parallel} \frac{x}{|x|} + \nu_{\perp}$, where $\nu_{\parallel}$ is normally distributed.

$$\begin{aligned} J(\mathbf{x}, \mathbf{Y}) &= \frac{1}{\sqrt{\frac{x^2}{n\sigma_X^2}(1+\varepsilon) + \varepsilon/2}} \\ &\times \left[\frac{x^2}{n\sigma_X^2} \sqrt{n\varepsilon} \frac{1+\varepsilon/2}{1+\varepsilon} + \nu_{\parallel} \sqrt{\frac{x^2}{n\sigma_X^2} \frac{1+\varepsilon/2}{1+\varepsilon}} + \frac{\sqrt{n\varepsilon}}{2} \left(1 - \frac{\nu_{\parallel}^2 + \nu_{\perp}^2}{n(1+\varepsilon)}\right) \right] \\ &= \frac{\sqrt{2 \ln q}}{\sqrt{1+\gamma}} + \nu_{\parallel} + \mathcal{O}(\sqrt{\varepsilon}). \end{aligned} \quad (3.21)$$

Motivated by (3.18) and (3.21), we model all the scores as Gaussian random variables, neglecting terms of order $\sqrt{\varepsilon}$. (See Fig. 3.3 for a visualisation.)

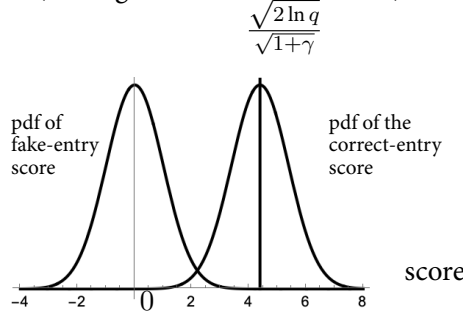


Figure 3.3: Probability density functions of the scores for a fake table entry and for the correct table entry at position u . Contributions of order $\mathcal{O}(\sqrt{\varepsilon})$ are neglected. The pdfs are almost Gaussian and have variance 1. The plot is for $q = 2^{10}$, $\gamma = -0.28$.

Error probabilities.

We place the threshold θ near the average true-entry score, $\mathbb{E}_{Y|x} J(x, Y)$. Although the protocol allows the threshold to depend on x^2 , in the following performance analysis we restrict ourselves to a single threshold that is applied to every block. This is a valid special case of $\theta(x^2)$. We parametrise this threshold relative to the leading-order mean of the true-entry score as

$$\theta(x^2) \equiv \theta = \frac{\sqrt{2 \ln q}}{\sqrt{1 + \gamma}} + \delta, \quad (3.22)$$

where the offset δ is optimised together with σ_X^2 and γ . The resulting threshold is therefore optimal within the class of constant thresholds but is not necessarily optimal among all possible functions $\theta(x^2)$. With the Gaussian approximation, we compute the True Accept (TA) and False Accept (FA) probabilities as follows:

$$P_{\text{TA}} = [\Phi(\theta)]^{q-1} [1 - \Phi(\delta)] \quad ; \quad P_{\text{FA}} = (q - 1) [\Phi(\theta)]^{q-2} [1 - \Phi(\theta)] \Phi(\delta), \quad (3.23)$$

where Φ is the cumulative distribution function of the normal distribution. A True Accept occurs when all $q - 1$ fake-entry scores lie below θ and the true-entry score lies above θ . A False Accept occurs when $q - 2$ fake-entry scores lie below θ , one fake-entry score lies above θ , and the true-entry score lies below θ . The factor $q - 1$ occurs because there are $q - 1$ choices for which fake entry has a high score.

For the total acceptance probability, we write $P_{\text{acc}} = P_{\text{TA}} + P_{\text{FA}}$. Each acceptance results in a q -ary symbol being retained as part of the raw key. If Bob's reconstructed entry does not equal Alice's u , we say that a *symbol error* has occurred. The symbol error probability σ is given by

$$\sigma = \frac{P_{\text{FA}}}{P_{\text{FA}} + P_{\text{TA}}} = \left[1 + \frac{1}{q - 1} \frac{\Phi(\theta)}{1 - \Phi(\theta)} \frac{1 - \Phi(\delta)}{\Phi(\delta)} \right]^{-1}. \quad (3.24)$$

A symbol error leads to bit errors in approximately 50% of the bits in the binarisation of the symbol.

Lemma 3.3.1. *Let q be a power of 2. The bit error rate (BER) is related to the symbol error rate by*

$$\text{BER} = \frac{\sigma}{2} \cdot \frac{q}{q - 1}. \quad (3.25)$$

Proof: Let u be Alice's q -ary symbol, and let $\hat{u}$ be Bob's reconstruction of u . The binarisation is $\text{Bin}(u) \in \{0, 1\}^{\log q}$. For any bit position j , it holds that $\Pr[\text{Bin}(\hat{u})_j \neq \text{Bin}(u)_j] = \Pr[\hat{u} \neq u] \frac{|\{\hat{u}: \text{Bin}(\hat{u})_j \neq \text{Bin}(u)_j\}|}{|\{\hat{u}: \hat{u} \neq u\}|} = \sigma \frac{q/2}{q-1}$. $\square$

3.4. Asymptotic Secret Key Ratio

Since our protocol and the subsequent key-combination procedure hide N_{acc} and the individual accept/reject decisions from the adversary, we can use Eve's *Gaussian* states ρ^E and ρ_y^E without further conditioning. The classical messages accessible to Eve satisfy the 'decoupling' property (Section 3.2.4), which allows us to invoke Corollary 3.1.2. Combining Corollary 3.1.2 with the standard formula for the leakage gives the following expression:

$$I(v_B; E, \text{transcript}) \leq N_{\text{acc}} n I(E_i; Y_i). \quad (3.26)$$

The size of the raw key (before privacy amplification) shared by Alice and Bob is $N_{\text{acc}} \log q$. However, because of the non-zero symbol error rate, a second layer of error correction is needed, e.g. in the form of a binary linear code or a non-binary burst-correction code. Since we are considering the asymptotic regime, we assume that this code operates close to the capacity $1 - h(\text{BER})$. Taking this into account, the size of the syndrome σ is $h(\text{BER}) N_{\text{acc}} \log q$.

In calculating the key ratio, we have to subtract the amount of key material that is consumed in the encryption of the integer N_{acc} , the compressed acceptance vector α , the syndrome σ , and the final success/fail bit. The value N_{acc} is represented by at most $\log N$ bits. The size of the compressed α is $\log \binom{N}{N_{\text{acc}}} \leq N h(P_{\text{acc}})$ bits; the inequality is tight. Combining these ingredients, we obtain the following provably safe asymptotic key ratio (SKR_{∞}) for our scheme:

$$\begin{aligned} \text{SKR}_{\infty} &= \frac{1}{Nn} \left[N_{\text{acc}} (\log q - n I(E_i; Y_i) - h(\text{BER}) \log q) - \log N - N h(P_{\text{acc}}) - 1 \right] \\ &= P_{\text{acc}} \left[(1 + \gamma) \frac{1}{2} \log(1 + \varepsilon) \left\{ 1 - h\left(\frac{\sigma}{2} \cdot \frac{q}{q-1}\right) - \frac{h(P_{\text{acc}})/P_{\text{acc}}}{\log q} \right\} - I(E_i; Y_i) \right] \\ &\quad - \frac{\log 2N}{Nn}. \end{aligned} \quad (3.27)$$

Here, we used (3.19) to write n in terms of q , ε , and γ , and we substituted (3.25). Asymptotically, the term $\frac{\log 2N}{Nn}$ is negligible. It is worth noting that privacy amplification introduces an additional penalty in the final key length. In particular, the quantum Leftover Hash Lemma implies that, when a two-universal hash function is used with a privacy-amplification secrecy parameter ε_{PA} , the extractable key length is reduced by $2 \log(1/\varepsilon_{\text{PA}})$ bits [175, Section 7.3.2]. Since privacy amplification is applied once to the entire reconciled string obtained from Nn transmitted pulses, the corresponding penalty in the secret key ratio is

$$\Delta_{\text{PA}} = \frac{2}{Nn} \log \frac{1}{\varepsilon_{\text{PA}}}. \quad (3.28)$$

For a fixed ε_{PA} , it vanishes as $Nn \rightarrow \infty$ and is therefore not included in SKR_{∞} .

In Table 3.1, we show optimised key ratios as a function of q for $T = 10^{-6}$ and $\xi = 10^{-5}$. When T , ξ , and q are fixed, optimisation consists of finding $\sigma_X^2, \gamma, \delta$ such that the SKR is maximised. The table lists the SKR normalised with respect to the Devetak-Winter value; this has the advantage that we do not have to present a table for different values of T . The results at $T = 10^{-6}, \xi = 10^{-5}$ are representative of the regime $T \ll 1, \xi \ll 1$. Although we omit the details, the numbers in Table 3.1 hardly change when T is varied.

Fig. 3.4 shows an example of the key rate in bits per second at $q = 2^{15}$, illustrating the orders of magnitude at a value of q compatible with real-time implementation.

Fig. 3.5 zooms in on the SKR ‘landscape’ as a function of γ and δ at optimal σ_X^2 . We observe the following trends. (i) The SKR is several percent of the DW value, which is a good result given the low T . The SKR slowly increases with growing q . From the listed $\text{SKR}/\Delta I$ values, we see that this is partly due to the low value of σ_X^2 , which lowers the achievable rate, and partly due to the fact that the scheme performs at 13% to 29% of ΔI . (ii) The average photon number is below 1, even at very large q . (iii) The parameters γ and δ are negative, getting closer to zero as q increases. Negative γ moves the threshold θ to the right, but negative δ moves it to the left. The net effect is that the symbol error rate is kept low, around 1% or 2%, while P_{acc} ranges from 59% to 74%. We note that our assumption about the asymptotic size of the syndrome is well justified at low BER.

We briefly comment on a scheme variant without thresholding. That variant always accepts and is therefore very simple to analyse. It has a symbol error rate $\omega = 1 - \mathbb{E}_{\lambda \sim \mathcal{N}_{01}}[\Phi(\frac{\sqrt{2 \ln q}}{\sqrt{1+\gamma}} + \lambda)]^{q-1}$ and an asymptotic key ratio $(1+\gamma)\frac{1}{2} \log(1+\varepsilon)[1 - h(\frac{\omega}{2} \frac{q}{q-1})] - I(E_i; Y_i)$. At large q , the ω is small enough to justify the asymptotic expression $1 - h(\cdot)$, but the key ratio is worse than for the thresholded scheme. At small q , the asymptotic key ratio is slightly better than for the thresholded scheme, but the ω grows, making it difficult to justify the $1 - h(\cdot)$ expression.

Table 3.1: Values of σ_X^2, γ , and δ that optimise SKR_∞ , listed for several values of q . The channel parameters are $T = 10^{-6}$ and $\xi = 10^{-5}$. The fraction $\text{SKR}_\infty/\text{DW}$ is the secret key ratio (3.27) divided by the Devetak-Winter value $T/(2 \ln 2)$. In (3.27), we use (??), (3.24), and (3.5). The acceptance probability and symbol error rate σ at the optimum are also listed. “ ΔI ” stands for $I(X_i; Y_i) - I(E_i; Y_i)$ at the listed σ_X^2 .

q	σ_X^2	γ	δ	$\text{SKR}_\infty/\text{DW}$	P_{acc}	σ	$\text{SKR}_\infty/\Delta I$
2^5	0.095	-0.45	-0.78	0.019	0.74	0.024	0.13
2^{10}	0.21	-0.28	-0.50	0.053	0.67	0.022	0.20
2^{15}	0.31	-0.21	-0.40	0.082	0.64	0.020	0.24
2^{20}	0.40	-0.17	-0.30	0.11	0.61	0.014	0.26
2^{30}	0.56	-0.13	-0.24	0.14	0.59	0.0097	0.29

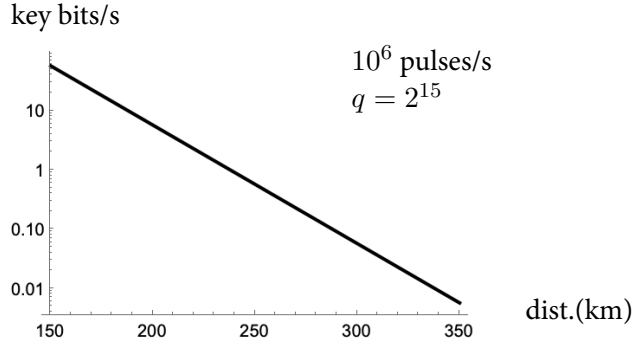


Figure 3.4: Secret key rate in bits per second, according to Table 3.1, at $q = 2^{15}$, given a pulse rate of 10^6 pulses per second. The loss is taken to be 0.2 dB/km. Excess noise is neglected.

3.4.1. Comparison with the state of the art

Table 3.2 places the present results alongside a number of reported long-distance CVQKD reconciliation systems. For each entry, the table lists the transmission distance, the corresponding attenuation parameter T , the total loss in dB, the reported key throughput, the pulse rate, and the achieved fraction of the Devetak–Winter benchmark. Here ‘DW’ denotes the Devetak–Winter value of the key ratio introduced in Section 3.1, and %DW denotes the reported ratio as a percentage of that value.

A first point to note is that reconciliation efficiency, usually denoted by β , is only one aspect of performance. It measures how closely the employed code rate approaches the channel capacity. In the CVQKD literature, turbo codes have typically reached efficiencies of around 70–80% [131, 20], while low-rate LDPC, polar, and Raptor-based schemes often report much higher values, commonly in the range of roughly 94–98%. However, Table 3.2 shows that the achieved fraction of the Devetak–Winter value, which is the maximum theoretical channel capacity, can still be much smaller than the nominal reconciliation efficiency.

In the very low-SNR regime relevant for long-distance CVQKD, block rejections (or frame errors) become increasingly significant. As noted in Section 3, discarding unsuccessfully decoded blocks is a form of post-selection [185]. As such, it conditions the joint state shared by Alice, Bob, and Eve, and may therefore alter the relevant leakage terms in the secret-key analysis. For this reason, the %DW values reported in the literature should be interpreted with some care: they indicate reconciliation performance, but they do not automatically translate into provable secret key ratios unless the effect of frame rejection has been incorporated consistently into the security proof. To the best of our knowledge, this point is generally not treated explicitly in the works listed in Table 3.2, with the notable exception of systems with a zero frame error rate, such as that in Ref. [93].

Table 3.2: CVQKD at various distances. $T \in (0, 1)$ is the attenuation parameter of the quantum channel. The entries marked with ‘*’ or ‘**’ use ultra-low-loss optical fibre, with rated losses of 0.16 dB/km and 0.14 dB/km, respectively (at 1550 nm). The remaining entries use more standard optical fibre with a loss of 0.2 dB/km. The present work is denoted by RC (random codebook) at $q = 2^{15}$. The “R-time?” column indicates whether the protocol was or could be run in real time. For the present work, we predict real-time operation with commercially available hardware. In Section 3.5.1, we show a sample calculation.

Paper	Code	Dist. (km)	T	Loss (dB)	Key bits/s	Pulses/s	%DW	R-time?
[203]	Polar	27	0.37	4.4*	2.8×10^5	5.0×10^6	17	No
[203]	Polar	49	0.15	8.3*	6.2×10^4	5.0×10^6	11	No
[75]	LDPC	13	0.56	2.6	5.8×10^6	7.5×10^7	13	No
[109]	LDPC	25	0.32	5.0	10^4	1.0×10^6	3.7	Yes
[75]	LDPC	26	0.31	5.1	5×10^5	7.6×10^7	2.5	No
[186]	LDPC	50	0.10	10	5.2×10^4	2.5×10^7	2.7	No
[203]	LDPC	70	0.068	12*	4.3×10^4	5.0×10^6	17	No
[109]	LDPC	81	0.025	16	5×10^2	1.0×10^6	2.8	Yes
[93]	LDPC	100	0.029	15**	10^5	1.0×10^8	4.7	No
[203]	LDPC	141	0.0045	23*	3.2×10^2	5.0×10^6	2.0	No
[203]	Raptor	203	0.00057	32*	6.2	5.0×10^6	0.30	No
Our work	RC	300	10^{-6}	60	0.059	1.0×10^6	8.2	Yes (pred.)

3. Quantum Key Distribution

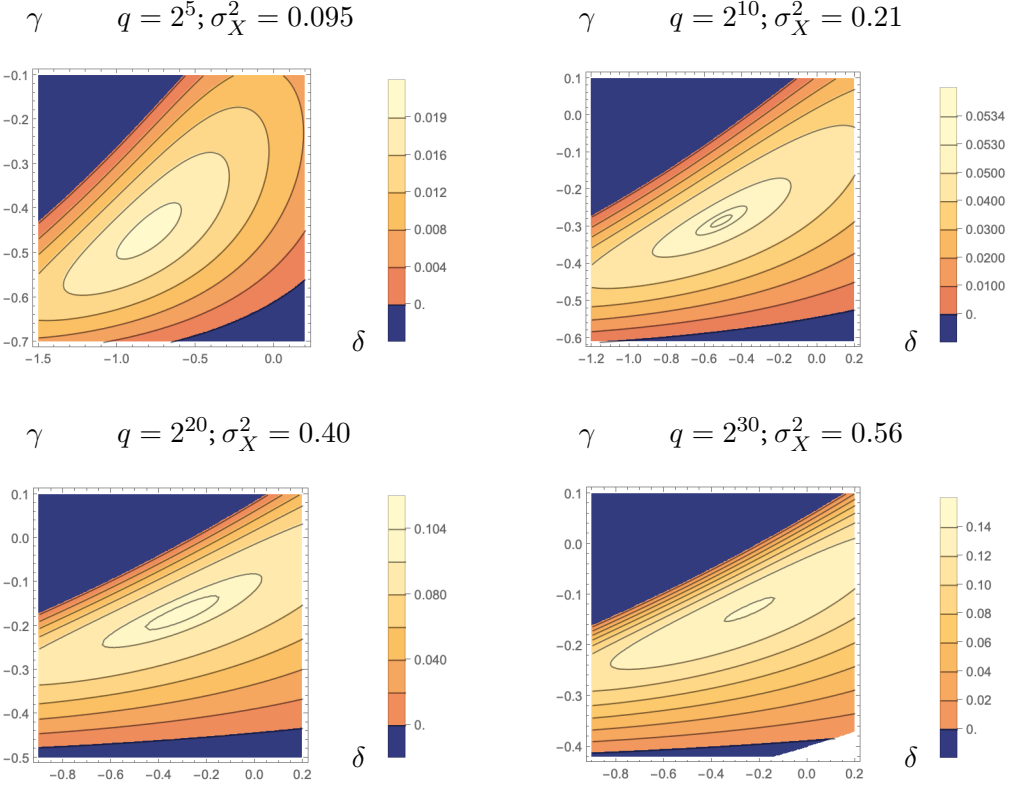


Figure 3.5: SKR_∞/DW as a function of γ and δ for various combinations of q and σ_X^2 (see Table 3.1). $T = 10^{-6}$, $\xi = 10^{-5}$.

A second point concerns implementation. In QKD, successful reconciliation is not enough by itself; the decoding must also be fast enough to keep pace with the optical system. This is especially demanding in long-distance CVQKD, where the low-SNR regime typically requires very low-rate codes and correspondingly expensive decoding. For LDPC-based schemes, for example, the decoder often relies on many iterative message-passing rounds, which limits the degree of parallelism and makes real-time operation difficult at large distances. As Table 3.2 illustrates, reported long-distance demonstrations either do not aim at real-time decoding or only achieve it at considerably shorter distances than the most loss-tolerant non-real-time systems.

It is in this context that the present construction should be viewed. The random-codebook approach explored in this chapter is not introduced as a replacement for the highly optimised structured codes used in the literature. Rather, it is meant as an alternative reconciliation paradigm tailored to the long-distance regime, where both extremely low SNR

and implementation constraints become central. Its main attraction is that the decoding task decomposes into a large number of simple score evaluations that can be carried out in parallel. This makes the scheme naturally compatible with real-time implementation, while the analysis keeps explicit track of acceptance, residual errors, and the resulting secret key ratio.

Within the parameter regime considered here, this leads to an operating point at 300 km, corresponding to approximately 60 dB loss, with an attained key ratio of about 8.2% of the Devetak–Winter value and a decoding architecture that is compatible with real-time processing. In that sense, the comparison in Table 3.2 highlights a trade-off that is central to this chapter: highly optimised structured codes may achieve excellent nominal efficiencies, but a random-coding approach can offer a different balance between loss tolerance, analytical transparency, and parallelisability.

3.5. Implementation aspects

3.5.1. Decoding

In CVQKD, the main computational bottleneck is the error-correction decoding on Alice’s side. Our scheme does not alter that fact, but it allows for massive parallelisation: all the scores $(s_\ell)_{\ell=1}^q$ can be computed independently, and even within $s_\ell = J(\mathbf{x}, \mathbf{m}_\ell)$ all the terms in the inner products $\mathbf{x} \cdot \mathbf{m}_\ell = \sum_{i=1}^n x_i m_{\ell i}$ and $\mathbf{m}_\ell^2 = \sum_{i=1}^n (m_{\ell i})^2$ can be computed in parallel. (Note that $\mathbf{x}^2$ can be precomputed.)

In a real-world CVQKD system, Alice has to perform the decoding in real time. This means that the time spent on computations pertaining to Nn pulses should not exceed the time required to send those Nn pulses. (Latency is not important. A time lag can be tolerated.) Hence, in the time interval between two pulses, $P_{\text{acc}}q$ basic multiplications $x_i m_{\ell i}$ and $(m_{\ell i})^2$ must be performed, plus additions.⁴ For example, if Alice fires 10^6 pulses per second and implements our scheme with $q = 2^{15}$, she will need to perform $2^{15} \approx 10^5$ multiplications in 1 microsecond. On a 5 GHz processor, that means 5000 clock cycles, which implies around 4 multiplications per clock cycle. With a handful of cores, that task should be achievable.

Several options can make the implementation faster.

- In the computation of $J(\mathbf{x}, \mathbf{m})$, initially ignore the contribution from $1 - \frac{\mathbf{m}^2}{n\sigma_Y^2}$, since it is often negligible. Compute this contribution only if the $\mathbf{x} \cdot \mathbf{m}$ part of the score is close to the threshold θ .

⁴This does not include the outer error correction, which we consider to be ‘easy’ given the low BER.

3. Quantum Key Distribution

- If two scores have already been found that exceed the threshold, there is no need to compute the remaining scores.
- As the index i runs from 1 to n , the scores ‘grow’ over time. It can be advantageous to postpone the full computation of scores that, at an early stage (e.g. $i = n/4$ or $n/2$), start out negative, since they are unlikely to exceed θ by the end. It may even be possible to discard them.
- The scheme as described in Section 3.2.2 works with real numbers, but of course we have to discretise. A good way to discretise continuous variables while retaining a large amount of information is to use quantiles so that the binary representation has a uniform distribution. For instance, the component z_i/σ_Y of a fake table entry is normally distributed, and its discretisation as a b -bit binary string is specified by an integer in $\{0, \dots, 2^b - 1\}$ determined as $\lfloor 2^b \Phi(z_i/\sigma_Y) \rfloor$. The same is done with x_i/σ_X and y_i/σ_Y . Note that the discretisation does not affect the security analysis. Eq. (3.27) remains valid, but the symbol error rate σ may become worse due to the loss of information. After the discretisation, it will not be necessary to actually *compute* multiplications $\frac{x_i}{\sigma_X} \frac{m_{\ell i}}{\sigma_Y}$. Instead, we can precompute a $2^b \times 2^b$ lookup table (LUT) for all combinations of the discretised inputs $\frac{x_i}{\sigma_X}$ and $\frac{m_i}{\sigma_Y}$, and look up values from the LUT. Optionally, for small b , computing a score s_ℓ could then be implemented as (i) aggregating tallies $(t_{jk})_{j,k \in \{0, \dots, 2^b - 1\}}$ that count how often input combination (j, k) occurs; (ii) computing $\sum_{jk} t_{jk} \text{LUT}_{jk}$.

Ideally, one would have a large number of mini-cores, each of which receives inputs in a pipelined way and produces either LUT lookups or tallies as defined above. At a downstream layer, the results are aggregated and decisions are made about which table entries to discontinue.

3.5.2. Communication

Our scheme requires far more communication than other reconciliation methods. For each pulse that contributes to the final key, Bob needs to send q numbers to Alice. At large q , this becomes expensive or even infeasible. We propose a protocol modification that replaces the truly random fake entries with pseudorandom sequences.

Let the ‘tilde’ notation denote b -bit discretisation as above. Let $\psi(\cdot, \cdot, \cdot)$ be a function that takes as input a seed and two integers and outputs a pseudorandom sequence. Steps 2b and 2c of the protocol are modified as follows.

- Bob draws a random seed τ . For $j \in \{1, \dots, k\}$, he computes $a_j = \psi(\tau, u, j)$ and $a = a_1 | \dots | a_k$. Here, k is an integer chosen so that a has the same length as $\tilde{y}$. Bob computes $\mu = \tilde{y} \oplus a$, where $\oplus$ stands for bitwise XOR. He sends τ, μ .
- For $\ell \in [q]$, Alice computes scores $J(x, \mathbf{m}_\ell)$, where the vector $\mathbf{m}_\ell$ is obtained as $\tilde{\mathbf{m}}_\ell =$

$\mu \oplus (g_{\ell 1} | \cdots | g_{\ell k}), \text{ with } g_{\ell j} = \psi(\tau, \ell, j).$

At $\ell = u$, Alice retrieves $\tilde{\mathbf{m}}_u = \tilde{y}$. At $\ell \neq u$, she obtains $\tilde{y}$ XORed with two combined pseudorandom sequences.

This construction resembles schemes that XOR $\tilde{y}$, or some other representation of y , with a randomly chosen codeword from a linear error-correcting code. In our case, however, the code is entirely unstructured and allows for parallel decoding. In particular, the pieces $\psi(\tau, \ell, j)$ can be computed in parallel for all j , which implies that the corresponding contributions to the score s_ℓ can be computed in parallel. Note that spinal codes [155, 22, 190] (another type of pseudorandom code) do not have this property; they need to generate a codeword sequentially.

Note that $I(U; \tau\mu) = 0$.⁵ This implies that we are still allowed to use Corollary 3.1.2, and that the bound on the leakage, $I(U; \tau\mu E) \leq I(E; Y)$, remains valid. One must still be careful with the choice of pseudorandom function ψ , since bad pseudorandomness introduces patterns into the sequences $\psi(\tau, \ell, j)$, which means that the vectors $\mathbf{m}_\ell \in \mathbb{R}^n$ are not evenly spread out, leading to inefficiencies in the decoding. A good choice might be to use unbiased sets [9, 10, 150, 11], pairwise independent hash functions, or a fast stream cipher, e.g. ChaCha20 [28]. However, cryptographic-strength pseudorandomness is probably not required for coding efficiency. PRNG algorithms exist that produce multiple bytes of pseudorandom output per clock cycle on a CPU [31, 60, 32, 176].

The secret key ratio measures net secret bits per transmitted quantum pulse and therefore does not include the bandwidth required by the authenticated classical channel. We briefly comment on the required classical bandwidth. Let s_τ denote the length of the public seed τ used for each n -pulse block, let s_g denote the padded transmitted length of the privacy-amplification seed g , and let C_{tag} denote the total length of the authentication tags. With b -bit discretisation, the pseudorandom construction transmits the nb -bit string μ , the n quadrature choices, and the seed τ for every block. Excluding channel-monitoring communication, the total classical communication is therefore

$$C_{\text{cl}} = N[n(b+1) + s_\tau] + |c| + |S| + s_g + 1 + C_{\text{tag}}, \quad (3.29)$$

where c denotes the fixed-length message sent by Alice, consisting of the one-time-pad encryption of N_{acc} and the compressed acceptance vector α , while S denotes the fixed-length message sent by Bob, consisting of the one-time-pad-encrypted reconciliation syndrome σ ; both messages include random padding. By comparison, transmitting the complete random table would require approximately $qbn + n$ bits per block.

⁵ t is independent of u . Furthermore, $\tilde{y}$ is uniform and serves as a one-time pad, completely hiding the $\psi(\tau, u, j)$.

3.6. Discussion

The main result of this chapter is the key ratio (3.27) obtained using the random codebook with MAP score thresholding and encryption of the accept/reject decisions. We emphasise again that many works on reconciliation in CVQKD do not fully take into account that, at a non-zero block rejection rate, the relevant part of Eve's state is conditioned on the 'accept' outcome, which negatively affects the key-ratio analysis. Hence, it is difficult to gauge the accuracy of the claimed key rates and %DW values listed in Table 3.2, except when the frame rejection rate is close to zero, as in Ref. [93].

Against this backdrop, the pseudorandom scheme variant presented in Section 3.5.2 is the easiest to implement and should be seen as the default version of the construction studied in this chapter. It strongly resembles existing schemes in which a randomly chosen codeword from an error-correcting code is XORed into Bob's measurement Y . The main operational difference lies in the choice of code: in the present setting, the decoding is massively parallelisable and works even at very low SNR. The decoding complexity scales favourably as the SNR decreases: the sequence length n is proportional to $1/\varepsilon$, while the decoding workload (at fixed q) is linear in n . As the example in Section 3.5.1 shows, the $q = 2^{15}$ scheme is expected to be implementable by sheer 'brute force' on a small number of multicore CPUs, without any optimisations such as pruning. It is left for future work to determine how much can be gained from aggressive pruning and from an optimised parallel implementation on an FPGA.

A separate practical issue is privacy amplification. It is applied not to each n -pulse reconciliation block separately, but to the concatenated corrected string of length $L = N_{\text{acc}} \log q$, which can be very large. A standard implementation uses universal hashing with a Toeplitz matrix, whose matrix-vector product can be evaluated using the fast Fourier transform, reducing the computational complexity from $\mathcal{O}(L^2)$ to $\mathcal{O}(L \log L)$. For example, a GPU implementation developed specifically for CVQKD reported an average throughput of approximately $1.35 \text{ Gbit, s}^{-1}$ for its benchmark parameters and was tested for input lengths up to 10^{10} bits [187]. This suggests that privacy amplification need not be the dominant bottleneck at the rates considered here. Nevertheless, its memory requirements and end-to-end integration should be included in a complete implementation study.

The obtained SKR values are competitive (see Table 3.2), given that the operating regime corresponds to extremely low T . At $q = 2^{15}$, the scheme attains about 8% of the Devetak-Winter value, and this fraction slowly increases with larger q . The analysis shows why it is difficult to get much closer to the DW value. The distance between the two Gaussian curves is $\sqrt{2 \ln q}$ if we set $\gamma = 0$, i.e., if we try to operate at Shannon capacity $I(X_i; Y_i)$. At 'reasonable' values of q , this distance is not sufficient to get a good separation between the $q - 1$ fake scores and the true-entry score. Therefore, we take negative γ . That lowers

the efficiency of the code. The low efficiency then forces us to work with small modulation variance σ_X^2 , which further lowers the SKR. The only way out of this spiral is to increase q , but that has practical limits because of the real-time constraint. We also emphasise that encrypting all the accept/reject decisions is a drastic measure that consumes a lot of key material, which hurts the SKR. This is quite a conservative approach. Thus, it is reasonable to expect that better methods can be developed.

3

However, making the fake entries *pseudorandom*⁶ should not strongly affect the acceptance probability P_{acc} and symbol error rate σ , which are the two statistical quantities that enter the SKR computation.

⁶The use of pseudorandom fake entries does not introduce a computational assumption into the security proof. Since $\tilde{Y}$ is uniform, XOR with $\psi(\tau, U, \cdot)$ leaves μ uniform for each fixed τ ; consequently, the public pair (τ, μ) is independent of U , irrespective of the statistical quality of ψ . Poor pseudorandomness may nevertheless degrade reconciliation performance and hence reduce the achievable key rate, but, provided that the uniform discretisation and the decoupling construction above are retained, it does not weaken the information-theoretic leakage bound.

4

Quantum Position Verification

*No one has observed a place except at a particular time, or
has observed a time except at a particular place.*

— Hermann Minkowski

Position-based cryptography uses the geographic location of a party as a cryptographic credential. In such protocols, one aims to authenticate not who a party is, but where that party is. This makes position-based cryptography an appealing primitive in settings where location itself is the relevant trust assumption, for instance when one wants to verify that a message was generated at a claimed site, or that a device or server is physically located where it is supposed to be. A central task in this area is *position verification*, where an untrusted prover tries to convince a set of verifiers that he is present at a designated position P .

The primitive was first introduced by Chandran, Goyal, Moriarty, and Ostrovsky [48]. In a typical position-verification protocol, the verifiers exploit both relativistic constraints and the laws of information processing in order to test whether a response could only have originated from the claimed location. Usually one studies the simplest setting of one spatial dimension, with two verifiers V_0 and V_1 located on either side of the claimed position P , since this already captures the main ideas while keeping the analysis manageable.

It is known that classical position verification is impossible in full generality, due to a universal attack based on copying and redistributing the challenge information. Quantum physics initially appeared to offer a way around this impossibility, since the no-cloning theorem [198] rules out the perfect copying of unknown quantum states. This observation led to the development of *quantum position verification* (QPV), also studied in the early literature under the name *quantum tagging* [112, 137, 138, 118]. However, a universal quantum attack was later discovered [43, 24]. That attack relies on pre-shared entan-

glement and instantaneous nonlocal quantum computation, and in general requires an amount of entanglement that grows exponentially with the size of the challenge. For this reason, the attack may not be feasible, and the question of security in restricted adversarial models remains highly relevant.

This has led to a substantial body of work on the amount of entanglement required to attack QPV protocols and on the design of protocols that remain secure under realistic restrictions on the adversaries' resources [44, 170, 64, 33, 54, 5, 15, 19, 18]. Many different protocols and security models have been proposed [47, 6, 85, 128, 8, 4, 179, 79, 63, 7], and recent work has increasingly focused on questions of practical implementability, such as robustness to channel loss and noise [7, 72, 4]. The present chapter fits into this line of work, but shifts the focus from the usual finite-dimensional setting to continuous-variable quantum systems.

Towards continuous-variable quantum position verification

Most QPV protocols studied so far are based on finite-dimensional quantum systems, whereas only a single work considers the continuous-variable setting [159]. In CV-QPV, the protocol is formulated in terms of optical quadratures, with challenge states and responses described using coherent or squeezed states and quadrature measurements [41, 45, 14]. This gives rise to a version of position verification in which the relevant information is no longer encoded in a finite alphabet, but in continuous measurement outcomes.

This difference is not merely formal; it changes the verification task itself. In many finite-dimensional QPV protocols, the honest prover is expected to return the *correct* answer, and cheating is detected even if a single answer is incorrect. In the continuous-variable setting, however, even an honest prover produces outcomes that are intrinsically random. The verifiers therefore do not simply check correctness round by round, but must determine whether the observed data are statistically compatible with an honest prover or with spatially separated attackers.

At the same time, the CV setting brings with it characteristic limitations. Channel attenuation and excess noise directly alter the honest response distribution and therefore enter the security analysis at a fundamental level. Understanding CV-QPV therefore requires not only adapting the ideas of finite-dimensional position verification, but also reworking them in a setting where both honest behaviour and adversarial simulations must be compared through continuous statistics. The overarching aim of this chapter is therefore to develop a standard methodology to analyse CV-QPV protocols.

The early proposal in [159] pointed out that for this type of protocol there is a generic attack whenever the transmission satisfies $t \leq 1/2$, but the security of a concrete CV-QPV protocol was not analysed in detail there. The first part of this chapter addresses precisely this

4. Quantum Position Verification

gap. It studies a CV-QPV protocol, called QPV_{coh} , that is closely related to that proposal, and provides a formal security analysis against attackers who do not possess pre-shared entanglement before the protocol begins, while allowing them to distribute entanglement during the attack. The analysis explicitly incorporates attenuation and excess noise in the quantum channel. This model is practically meaningful because the generation and distribution of high-quality entanglement over long distances remains technologically challenging. It is also a natural starting point, and indeed this non-entangled-attacker analysis serves as the basis for stronger results obtained later. However, the generic attack for $t < 1/2$ still applies.

The second part of the chapter then turns to a more powerful protocol family. In the discrete-variable literature, a particularly interesting idea is to split the classical input information between the verifiers, so that the prover must act according to a function value $f(x, y)$ that can only be determined once both pieces of information are effectively combined. This idea underlies the protocol $\text{QPV}_{\text{BB84}}^f$ [33, 72]. Its appeal is that the quantum resources required for a successful attack increase with the size of the classical input, which makes the protocol more resistant to entanglement-based cheating strategies. The chapter studies a continuous-variable analogue of this construction, the protocol $\text{QPV}_{\text{coh}}^f$. The main message is that the favourable bounded-entanglement behaviour of the discrete-variable protocol survives in the continuous-variable setting as well: for random protocol functions, the attackers' task remains hard even when one allows pre-shared quantum resources whose dimension grows linearly with the input size. The analysis again includes attenuation and excess noise, and shows how entropy-based arguments can be used to separate the honest and dishonest response distributions.

Outline of this chapter

The remainder of this chapter is organised as follows. We first introduce the basic coherent-state protocol QPV_{coh} . Section 4.1.1 gives the prepare-and-measure description of the protocol, including the timing constraints and the statistical score used by the verifiers. Section 4.1.2 then reformulates the same protocol in an entanglement-based picture, which is the form used in the security analysis. The honest prover is analysed in Section 4.1.3: first through the probability of passing the verification test, then through the posterior uncertainty about the verifier's Gaussian displacement, and finally through the conditional entropy calculation in Section 4.1.3.

We then study the security of QPV_{coh} . The chapter first considers several concrete attacks without pre-shared entanglement, namely heterodyne measurement, state splitting, and homodyne measurement under a guessed angle. These attacks serve as useful examples, but the main security result is given in Section 4.3.1, where we prove security against general attackers who do not pre-share entanglement before the protocol begins. This

is followed by a comparison between the honest prover's uncertainty and the attackers' uncertainty, which identifies the parameter regime in which the two response distributions can be separated. Section 4.3.3 then shows that this basic protocol is broken if the attackers share an ideal continuous-variable EPR pair.

The second part of the chapter studies the stronger protocol family $\text{QPV}_{\text{coh}}^f$, introduced in Definition 4.4.1. Here the classical information determining the measurement basis is split between the two verifiers through a function $f(x, y)$, in analogy with the $\text{QPV}_{\text{BB84}}^f$ -type protocols from the finite-dimensional setting. The chapter proves that, for random functions, this structure gives security against attackers with bounded pre-shared entanglement, and then quantifies the resulting bounds for concrete channel parameters in Section 4.6. Finally, Section 4.7 discusses the implications of these results, the relation between the two CV-QPV protocol families, and the limitations that remain in the continuous-variable setting.

4

4.1. The QPV_{coh} protocol

4.1.1. Prepare-and-measure

Consider two spatially separated verifiers V_1 and V_2 , and a prover P somewhere in between them. As in the discrete case, we now define a QPV protocol QPV_{coh} utilising coherent states.

Definition 4.1.1. *Let $\mathcal{A}$ be a publicly known set of angles in $[0, 2\pi)$ such that $\alpha \in \mathcal{A} \implies \alpha + \pi/2 \in \mathcal{A}$. Let σ be a publicly known parameter, $\sigma \gg 1$. A single round of the QPV_{coh} protocol consists of the following steps (for a diagrammatic picture, see Fig. 4.1):*

1. *The verifiers draw random $\theta \in \mathcal{A}$ and two random variables $(r, r^\perp)$ from the Gaussian distribution $\mathcal{N}_{0, \sigma^2}$. Verifier V_1 prepares a coherent state $|\psi\rangle$ with quadratures $(x_0, p_0) = (r \cos \theta + r^\perp \sin \theta, r \sin \theta - r^\perp \cos \theta)$. Then V_1 sends $|\psi\rangle$ to the prover, and V_2 sends θ to the prover.*
2. *The prover receives θ and $|\psi\rangle$ and performs a homodyne measurement on $|\psi\rangle$ in the θ direction, resulting in a value $r' \in \mathbb{R}$. The prover sends r' to both verifiers.*

After n rounds, the verifiers have received a sample of responses, which we denote by $(r'_i)_{i=1}^n$. The verifiers check whether all prover responses arrived at the correct (and at the same) time, and whether the reported values $(r'_i)_{i=1}^n$ satisfy

$$\frac{1}{n} \sum_{i=1}^n \frac{(r'_i - r_i \sqrt{t})^2}{\frac{1}{2} + u} < \gamma \quad \text{with } \gamma \stackrel{\text{def}}{=} 1 + \frac{2}{\sqrt{n}} \sqrt{\ln \frac{1}{\varepsilon_{\text{hon}}}} + \frac{2}{n} \ln \frac{1}{\varepsilon_{\text{hon}}}. \quad (4.1)$$

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Here ε_{hon} is an upper bound on the honest prover's failure probability. The verifiers additionally compare the responses received from the two sides and reject unless they coincide in every round.

ε_{hon} is a protocol parameter and can be set to a desired value. The verifiers reject if not all these checks are satisfied. We refer to the sum in (4.1) as the *score*.

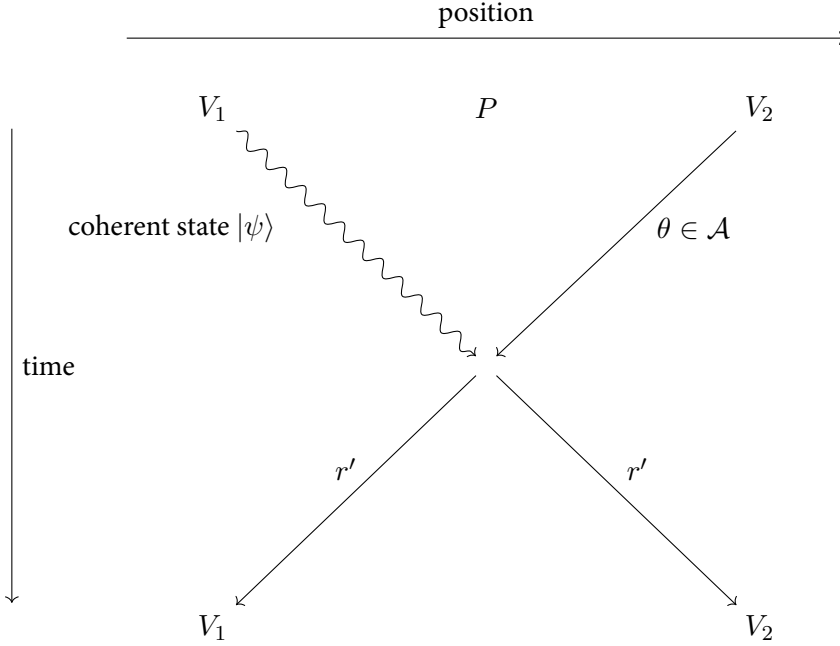


Figure 4.1: Schematic representation of the protocol described in Section 4.1.1. Undulated lines represent quantum information, whereas straight lines represent classical information.

4.1.2. Entanglement-based version of the protocol

In security proofs for qubit-based schemes, it is customary to reformulate a protocol into an EPR-based form. The act of one party (V) preparing and sending a qubit state in a particular basis $\mathcal{B}$ is equivalent to V preparing a maximally entangled two-qubit state (EPR pair) and then measuring one of the qubits in the $\mathcal{B}$ basis while the other qubit is sent. The act of measuring can be postponed. This has the advantage that in the security analysis the basis choice can be delayed, and it is then possible to base security on the properties of entangled states.

We will do an analogous reformulation for CV states. In fact, we work with exactly the same states as in Gaussian-modulated CV-QKD [119]. We tune the squeezing parameter ζ

such that $\sinh \zeta = \sigma$. Preparing a coherent state with Gaussian-distributed displacements $x_0, p_0 \sim \mathcal{N}_{0, \sigma^2}$ is equivalent to preparing a two-mode squeezed state with squeezing parameter ζ and then performing a heterodyne $(\hat{x}, \hat{p})$ measurement on one mode, with measurement outcome $\frac{(x_0, -p_0)}{\sqrt{2} \tanh \zeta}$.

In our particular case, the verifier V_1 prepares the two-mode squeezed state ρ_{VP} and performs the heterodyne measurement with quadratures that are rotated by the angle θ on the V subsystem. The measurement outcomes are $r/(\sqrt{2} \tanh \zeta)$ and $-r^\perp/(\sqrt{2} \tanh \zeta)$, resulting in displacement $(r, r^\perp)$ in the state sent to the prover (i.e. subsystem P). The prover then performs a homodyne measurement at angle θ to recover r , similar to the prepare-and-measure scheme.

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In the security analysis in Section 4.3.1, we will explicitly write V_1 's heterodyne measurement as a double-homodyne measurement. First V_1 mixes its own mode with the vacuum using a beamsplitter, resulting in a two-mode state. On one of these modes, V_1 then performs a homodyne measurement in the θ -direction and, on the other mode, in the $\theta + \frac{\pi}{2}$ direction.

4.1.3. Honest prover

Success probability

We show that the honest prover has a failure probability smaller than ε_{hon} .

Lemma 4.1.2. (Eq. (4.3) in [120]) *Let X be a χ_n^2 -distributed random variable. It holds that*

$$\mathbb{P}[X - n \geq 2\sqrt{na} + 2a] \leq e^{-a}. \quad (4.2)$$

In round i , the honest prover performs a homodyne measurement at angle θ_i . We assume that each signal reaches the prover through an independent Gaussian channel with transmission t and excess-noise variance u . The displacement in the measured direction is therefore attenuated from r_i to $\sqrt{t} r_i$, while the quadrature variance is $\frac{1}{2} + u$. Hence,

$$R'_i \sim \mathcal{N}_{\sqrt{t} r_i, \frac{1}{2} + u}. \quad (4.3)$$

It follows that the random variables $\frac{R'_i - \sqrt{t} r_i}{\sqrt{\frac{1}{2} + u}}$ are independent standard normal random variables. Therefore,

$$Z = \sum_{i=1}^n \frac{(R'_i - \sqrt{t} r_i)^2}{\frac{1}{2} + u} \sim \chi_n^2. \quad (4.4)$$

The probability that the honest prover fails to pass verification is given by

$$\mathbb{P}[Z \geq n\gamma] = \mathbb{P}\left[Z \geq n + 2\sqrt{n \ln \frac{1}{\varepsilon_{\text{hon}}}} + 2 \ln \frac{1}{\varepsilon_{\text{hon}}}\right]. \quad (4.5)$$

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By Lemma 4.1.2, this is upper bounded by ε_{hon} .

A posteriori distribution and entropy of R conditioned on measurement

We determine how much uncertainty the honest prover has about the displacements r_i , given the measurement outcomes r'_i . For notational brevity we omit the round index i . We write the probability density for R as f_R . Since r' is the result of a measurement at angle θ , conditioning on θ is implicit and will be omitted from the notation.

The prover's posterior distribution of R , given r' , is

$$f_{R|R'}(r|r') = \frac{f_{RR'}(r, r')}{f_{R'}(r')} = \frac{f_R(r)f_{R'|R}(r'|r)}{f_{R'}(r')}. \quad (4.6)$$

Using $f_R = \mathcal{N}_{0, \sigma^2}$, $f_{R'|R}(r'|r) = \mathcal{N}_{r\sqrt{t}, \frac{1}{2}+u}(r')$ and $f_{R'} = \mathcal{N}_{0, t\sigma^2 + \frac{1}{2}+u}$ we get,

$$\begin{aligned} f_{R|R'}(r|r') &= \frac{1}{f_{R'}(r')} \frac{1}{2\pi\sigma\sqrt{\frac{1}{2}+u}} \exp\left(-\frac{r^2}{2\sigma^2} - \frac{(r' - \sqrt{t}r)^2}{2(\frac{1}{2}+u)}\right) \\ &= \frac{1}{f_{R'}(r')} \frac{1}{2\pi\sigma\sqrt{\frac{1}{2}+u}} \exp\left[-\frac{1}{2}\left(\left(\frac{1}{\sigma^2} + \frac{t}{\frac{1}{2}+u}\right)r^2 - 2\frac{\sqrt{t}r'}{\frac{1}{2}+u}r + \frac{r'^2}{\frac{1}{2}+u}\right)\right]. \end{aligned} \quad (4.7)$$

Now define

$$\Sigma^2 \stackrel{\text{def}}{=} \left(\frac{1}{\sigma^2} + \frac{t}{\frac{1}{2}+u}\right)^{-1}, \quad M \stackrel{\text{def}}{=} \Sigma^2 \frac{\sqrt{t}r'}{\frac{1}{2}+u}. \quad (4.8)$$

Then

$$\left(\frac{1}{\sigma^2} + \frac{t}{\frac{1}{2}+u}\right)r^2 - 2\frac{\sqrt{t}r'}{\frac{1}{2}+u}r = \frac{(r - M)^2}{\Sigma^2} - \frac{M^2}{\Sigma^2}, \quad (4.9)$$

so (4.7) becomes

$$f_{R|R'}(r|r') \propto \exp\left(-\frac{(r - M)^2}{2\Sigma^2}\right). \quad (4.10)$$

Since this is a normalised Gaussian density in r , we conclude

$$f_{R|R'}(r|r') = \mathcal{N}_{M, \Sigma^2}(r). \quad (4.11)$$

For $t\sigma^2 \gg 1$ this tends to a normal distribution centred on $r'/\sqrt{t}$, with variance $(\frac{1}{2}+u)/t$. From the Gaussian probability density function (4.10) we directly obtain the differential entropy of R given R' ,

$$h(R|R') = \frac{1}{2} \log 2\pi e \Sigma^2. \quad (4.12)$$

Entropy of R conditioned on the prover's quantum state

Let ρ_{VP} be the entangled state that is prepared by V_1 as described in Section 4.1.2. Here P denotes the prover's quantum system. The heterodyne measurement on the V system yields $(r, r^\perp)$. The measurement maps ρ_{VP} to $\rho_{RR^\perp P}$. We write the post-measurement state as

$$\rho_{RR^\perp P} = \int_{\mathbb{R}^2} f_{RR^\perp}(r, r^\perp) |r\rangle\langle r|_R \otimes |r^\perp\rangle\langle r^\perp|_{R^\perp} \otimes \rho_P^{rr^\perp} dr dr^\perp. \quad (4.13)$$

The (differential) entropy of R , conditioned on the prover's quantum state, can be expanded as

$$h(R|P) = h(R) + S(P|R) - S(P). \quad (4.14)$$

From the definition of conditional entropy, $S(P|R) = \mathbb{E}_r S(\mathbb{E}_{r^\perp} \rho_P^{rr^\perp})$ and $S(P) = S(\mathbb{E}_{r, r^\perp} \rho_P^{rr^\perp})$.

As discussed in Section 3.2, in the ideal case the prover would receive a coherent state. In the noisy channel, however, the output state becomes a displaced Gaussian state with covariance matrix $(1 + 2u) \mathbb{1}_2$. For fixed phase-space coordinates (x_0, p_0) , its Wigner function is therefore

$$W_{x_0, p_0}(x, p) = \frac{1}{\pi(1 + 2u)} \exp\left(-\frac{(x - \sqrt{t}x_0)^2 + (p - \sqrt{t}p_0)^2}{1 + 2u}\right). \quad (4.15)$$

Here x_0 and p_0 are distributed according to the heterodyne outcome on V , i.e.

$\frac{1}{2\pi\sigma^2} \exp\left(-\frac{x_0^2 + p_0^2}{2\sigma^2}\right)$. To obtain the state conditioned on $R = r$, we fix $x_0 = r$ and average only over the orthogonal outcome $p_0 = r^\perp$. Thus

$$W_r(x, p) = \int_{\mathbb{R}} \frac{dr^\perp}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(r^\perp)^2}{2\sigma^2}\right) W_{r, r^\perp}(x, p) \quad (4.16)$$

$$\begin{aligned} &= \frac{1}{\pi(1 + 2u)\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - r\sqrt{t})^2}{1 + 2u}\right) \\ &\quad \times \int_{\mathbb{R}} \exp\left[-\frac{(r^\perp)^2}{2\sigma^2} - \frac{(p - \sqrt{t}r^\perp)^2}{1 + 2u}\right] dr^\perp \end{aligned} \quad (4.17)$$

$$\begin{aligned} &= \frac{1}{\pi(1 + 2u)\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - r\sqrt{t})^2}{1 + 2u}\right) \\ &\quad \times \int_{\mathbb{R}} \exp\left[-\left(\frac{1}{2\sigma^2} + \frac{t}{1 + 2u}\right)(r^\perp)^2 + \frac{2\sqrt{t}p}{1 + 2u}r^\perp - \frac{p^2}{1 + 2u}\right] dr^\perp. \end{aligned} \quad (4.18)$$

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Using the Gaussian integral $\int_{\mathbb{R}} e^{-az^2+bz} dz = \sqrt{\frac{\pi}{a}} e^{b^2/(4a)}$, $a > 0$, we obtain

$$W_r(x, p) = \frac{1}{\pi \sqrt{(1+2u)(2t\sigma^2 + 2u + 1)}} \exp\left(-\frac{p^2}{2t\sigma^2 + 2u + 1} - \frac{(x - r\sqrt{t})^2}{2u + 1}\right). \quad (4.19)$$

Hence the covariance matrix of the conditional state is

$$V_{P|R=r} = \begin{pmatrix} 2u + 1 & 0 \\ 0 & 2t\sigma^2 + 2u + 1 \end{pmatrix}, \quad (4.20)$$

so its symplectic eigenvalue is

$$\nu_{P|R} = \sqrt{(2u + 1)(2t\sigma^2 + 2u + 1)}. \quad (4.21)$$

Similarly, for the unconditional state we average over both heterodyne outcomes:

$$W(x, p) = \int_{\mathbb{R}^2} \frac{dr dr^\perp}{2\pi\sigma^2} \exp\left(-\frac{r^2 + (r^\perp)^2}{2\sigma^2}\right) W_{r, r^\perp}(x, p) \quad (4.22)$$

$$\begin{aligned} &= \frac{1}{\pi(1+2u)} \int_{\mathbb{R}^2} \frac{dr dr^\perp}{2\pi\sigma^2} \exp\left[-\frac{r^2 + (r^\perp)^2}{2\sigma^2} - \frac{(x - \sqrt{t}r)^2 + (p - \sqrt{t}r^\perp)^2}{1+2u}\right] \\ &= \frac{1}{\pi(2t\sigma^2 + 2u + 1)} \exp\left(-\frac{x^2 + p^2}{2t\sigma^2 + 2u + 1}\right). \end{aligned} \quad (4.23)$$

Therefore the unconditional covariance matrix is

$$V_P = (2t\sigma^2 + 2u + 1) \mathbb{1}_2, \quad (4.24)$$

with symplectic eigenvalue

$$\nu_P = 2t\sigma^2 + 2u + 1. \quad (4.25)$$

Using the g function (cf. Lemma 2.5), the corresponding entropies are

$$S(P|R) = g\left(\frac{\sqrt{(2u+1)(2t\sigma^2 + 2u + 1)} - 1}{2}\right), \quad (4.26)$$

and

$$S(P) = g(t\sigma^2 + u). \quad (4.27)$$

Finally, by definition R is Gaussian and $h(R) = \frac{1}{2} \log 2\pi e \sigma^2$. Altogether, this yields

$$h(R|P) = \frac{1}{2} \log(2\pi e \sigma^2) + g\left(\frac{\sqrt{(2u+1)(2t\sigma^2 + 2u + 1)} - 1}{2}\right) - g(t\sigma^2 + u). \quad (4.28)$$

Now set

$$A_\sigma := \frac{\sqrt{(2u+1)(2t\sigma^2+2u+1)} - 1}{2}, \quad B_\sigma := t\sigma^2 + u. \quad (4.29)$$

The large-argument behaviour of the function g is given by $g(x) \sim \log(ex) + \mathcal{O}(1/x)$. Using this, we have

$$g(A_\sigma) - g(B_\sigma) = \log\left(\frac{A_\sigma}{B_\sigma}\right) + O\left(\frac{1}{\sigma}\right), \quad (4.30)$$

since $A_\sigma = O(\sigma)$ and $B_\sigma = O(\sigma^2)$. Moreover,

$$A_\sigma = \sqrt{\frac{(2u+1)t}{2}}\sigma + O(1), \quad B_\sigma = t\sigma^2 + O(1), \quad (4.31)$$

and therefore

$$\log\left(\frac{A_\sigma}{B_\sigma}\right) = -\log\sigma + \frac{1}{2}\log\left(\frac{2u+1}{2t}\right) + O\left(\frac{1}{\sigma}\right). \quad (4.32)$$

Hence

$$\begin{aligned} h(R|P) &= \frac{1}{2}\log(2\pi e\sigma^2) - \log\sigma + \frac{1}{2}\log\left(\frac{2u+1}{2t}\right) + O\left(\frac{1}{\sigma}\right) \\ &= \frac{1}{2}\log\left(\frac{\pi e(2u+1)}{t}\right) + O\left(\frac{1}{\sigma}\right). \end{aligned} \quad (4.33)$$

For large σ , this is essentially the same as $h(R|R')$ in (4.12).

4.2. Security of QPV_{coh} against specific attacks

Before showing security against a general attack, we highlight security against some specific attacks that one might naturally think of. We look into three specific attacks in which the adversaries do not have access to entanglement: performing a heterodyne measurement, state splitting and performing a homodyne measurement at a guessed angle. These examples provide some insight into the security but do not constitute a general security proof. A rigorous security proof for the case of adversaries who do not pre-share entanglement is given in Section 4.3.1.

The most general attack on a 1-dimensional QPV protocol consists of placing two attackers Alice A and Bob B between V_1 and P , and V_2 and P , respectively. For attackers that do not pre-share entanglement, an attack proceeds as follows. Alice intercepts the quantum state sent to the prover P . Alice applies a local operation to her quantum system and sends

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some classical and/or quantum information to the second attacker Bob. The most general action Bob can take (assuming he has no quantum register) is to intercept the message θ and broadcast it, since any quantum operation can be embedded in Alice's actions. After one round of simultaneous communication, Alice and Bob use their respective quantum and classical information to produce a classical output and respond to their closest verifier such that the answer arrives on time. For the following analysis, we describe the attacks per round of the protocol.

4.2.1. Heterodyne attack

In a *heterodyne attack*, Alice performs a heterodyne measurement on the coherent state she intercepts and sends the result (x', p') to Bob. At the end, A and B report the best guess for r that they can produce based on x', p', θ . Let us denote this estimator by $\tilde{r}$. It holds that $\tilde{r} = \tilde{x} \cos \theta + \tilde{p} \sin \theta$, where $\tilde{x}$ is an estimator for x_0 , and similarly $\tilde{p}$. The posterior distribution of x_0 given x' is

$$\begin{aligned} f_{X_0|X'}(x_0|x') &= \frac{f_{X_0}(x_0)f_{X'|X_0}(x'|x_0)}{f_{X'}(x')} = \frac{\mathcal{N}_{0,\sigma^2}(x_0)\mathcal{N}_{\frac{x_0}{\sqrt{2}},\frac{1}{2}}(x')}{\mathcal{N}_{0,\frac{\sigma^2}{2}+\frac{1}{2}}(x')} \\ &= \mathcal{N}_{x'\sqrt{2}\frac{\sigma^2}{1+\sigma^2},\frac{\sigma^2}{1+\sigma^2}}(x_0). \end{aligned} \quad (4.34)$$

Hence $\tilde{x} = x'\sqrt{2}\frac{\sigma^2}{1+\sigma^2}$ and $\tilde{p} = -p'\sqrt{2}\frac{\sigma^2}{1+\sigma^2}$. Given x_0, p_0, θ , the random variable $\tilde{R}$ is Gaussian with mean $\frac{\sigma^2}{1+\sigma^2}r$ and variance $\frac{1}{2}(\sqrt{2}\frac{\sigma^2}{1+\sigma^2}\cos\theta)^2 + \frac{1}{2}(\sqrt{2}\frac{\sigma^2}{1+\sigma^2}\sin\theta)^2 = (\frac{\sigma^2}{1+\sigma^2})^2$. This gives

$$\mathbb{E}(\tilde{R} - r)^2 = \left(\frac{\sigma^2}{1+\sigma^2}\right)^2 + r^2\left(\frac{1}{1+\sigma^2}\right)^2 \approx 1 \quad \text{for } \sigma \gg 1, \quad (4.35)$$

which is easily distinguishable from the honest prover's value $\frac{1}{2}$.¹ From (4.34) we obtain the variance of R from the attackers' point of view as $\frac{\sigma^2}{1+\sigma^2}(\cos\theta)^2 + \frac{\sigma^2}{1+\sigma^2}(\sin\theta)^2 = \frac{\sigma^2}{1+\sigma^2}$. The attackers' ignorance about R is thus quantified as

$$h(R|X'P'\Theta) = \frac{1}{2} \log\left(2\pi e \frac{\sigma^2}{1+\sigma^2}\right), \quad (4.36)$$

with conditioning on Θ being made explicit.

¹Note that the unbiased estimator $x'\sqrt{2}\cos\theta - p'\sqrt{2}\sin\theta$ would yield $\mathbb{E}(\tilde{R} - r)^2 = 1$, which is larger than (4.35).

4.2.2. Splitting as a benchmark attack

We next consider a splitting strategy. Alice mixes the intercepted coherent state with a vacuum ancilla on a balanced beam splitter, $T_{\text{BS}} = 1/2$. She retains one output mode and sends the other (losslessly) to Bob, while Bob forwards the angle θ to Alice. Once θ is known, both attackers homodyne their respective modes in the corresponding quadrature direction.²

Let the intercepted coherent state has displacement (x_0, p_0) , the two output modes have displacements $(x_0, p_0)/\sqrt{2}$ and $-(x_0, p_0)/\sqrt{2}$. After Bob compensates for the minus sign, the homodyne outcome of either attacker $j \in \{A, B\}$ is therefore

$$U_j = \frac{R}{\sqrt{2}} + Z_j, \quad Z_j \sim \mathcal{N}_{0,1/2}, \quad R = x_0 \cos \theta + p_0 \sin \theta. \quad (4.37)$$

Consequently, the distribution of their response would be given by

$$f_{U_j|R\Theta}(u|r, \theta) = \mathcal{N}_{r/\sqrt{2}, 1/2}(u), \quad \text{and} \quad f_{U_j|\Theta}(u|\theta) = \mathcal{N}_{0, (1+\sigma^2)/2}(u). \quad (4.38)$$

Using Bayes' rule, the posterior distribution is

$$\begin{aligned} f_{R|U_j\Theta}(r|u, \theta) &= \frac{\mathcal{N}_{0,\sigma^2}(r) \mathcal{N}_{r/\sqrt{2}, 1/2}(u)}{\mathcal{N}_{0, (1+\sigma^2)/2}(u)} \\ &= \mathcal{N}_{\frac{\sqrt{2}\sigma^2}{1+\sigma^2}u, \frac{\sigma^2}{1+\sigma^2}}(r). \end{aligned} \quad (4.39)$$

This is exactly the posterior distribution obtained in the heterodyne attack. Thus each attacker has the same marginal estimator, estimation error and conditional entropy as in the preceding subsection.

The equivalence is only at the level of the attackers' marginal information. In the heterodyne attack, both attackers construct their answer from the same communicated measurement result. In the splitting strategy, U_A and U_B contain independent shot noise and therefore typically would not produce identical answers. We consequently use this strategy as a benchmark for the information available locally to each attacker, rather than as a complete attack.

²Within the more general strategy in which the signal is mixed with an independent single-mode Gaussian ancilla, optimising the marginal posterior errors gives effective noise variances $\nu_A = \frac{1}{2} + \frac{1-T_{\text{BS}}}{T_{\text{BS}}} v_\theta$ and $\nu_B = \frac{1}{2} + \frac{T_{\text{BS}}}{1-T_{\text{BS}}} v_\theta$. The larger of these is minimised at $T_{\text{BS}} = 1/2$. The quadrature uncertainty relation then implies that the angle-averaged error is minimised by $v_\theta = v_{\theta+\pi/2} = 1/2$, corresponding to a vacuum ancilla.

4.2.3. Attackers perform a homodyne measurement at a guessed angle

In this attack, Alice picks a random angle φ and does a homodyne measurement at this angle. She forwards the result m to Bob. The distribution of m is given by $f_{M|X_0P_0\Phi}(m|x_0p_0\varphi) = \mathcal{N}_{x_0 \cos \varphi + p_0 \sin \varphi, \frac{1}{2}}(m) = \mathcal{N}_{r \cos(\varphi - \theta) + r^\perp \sin(\varphi - \theta), \frac{1}{2}}(m)$. The attackers' posterior distribution for R is

$$f_{R|M\Phi\Theta}(r|m\varphi\theta) = \frac{f_\Theta(\theta)f_\Phi(\varphi)f_R(r)f_{M|R\Theta\Phi}(m|r\theta\varphi)}{f_\Theta(\theta)f_\Phi(\varphi)f_{M|\Theta\Phi}(m|\theta\varphi)} \quad (4.40)$$

$$\propto f_R(r)f_{M|R\Theta\Phi}(m|r\theta\varphi) \quad (4.41)$$

$$= f_R(r)\mathbb{E}_{r^\perp} f_{M|RR^\perp\Theta\Phi}(m|rr^\perp\theta\varphi) \quad (4.42)$$

$$= f_R(r)\mathbb{E}_{r^\perp} \mathcal{N}_{r \cos(\varphi - \theta) + r^\perp \sin(\varphi - \theta), \frac{1}{2}}(m) \quad (4.43)$$

$$= \mathcal{N}_{0, \sigma^2}(r) \mathcal{N}_{r \cos(\varphi - \theta), \frac{1}{2} + \sigma^2 \sin^2(\varphi - \theta)}(m). \quad (4.44)$$

After some algebra this can be rewritten as

$$f_{R|M\Phi\Theta}(r|m\varphi\theta) = \mathcal{N}_{\mu, S^2}(r) \quad \text{with} \\ \mu = m \cos(\varphi - \theta) \frac{\sigma^2}{\frac{1}{2} + \sigma^2}, \quad \text{and} \quad S^2 = \sigma^2 \frac{\frac{1}{2} + \sigma^2 \sin^2(\varphi - \theta)}{\frac{1}{2} + \sigma^2}. \quad (4.45)$$

The attackers send μ to the verifiers. For the expected score we get

$$\mathbb{E}(R - \mu)^2 = \mathbb{E}R^2 + \mathbb{E}\mu^2 - 2\mathbb{E}\mu R \quad (4.46)$$

$$= \sigma^2 + \left(\frac{\sigma^2}{\frac{1}{2} + \sigma^2} \right)^2 \mathbb{E}m^2 \cos^2(\varphi - \theta) - 2 \frac{\sigma^2}{\frac{1}{2} + \sigma^2} \mathbb{E}mr \cos(\varphi - \theta). \quad (4.47)$$

We introduce the notation $\delta = \varphi - \theta$. We use the distribution of m conditioned on $rr^\perp\varphi\theta$ to write

$$\mathbb{E} \cos^2 \delta m^2 = \mathbb{E} \cos^2 \delta \left[\frac{1}{2} + (r \cos \delta + r^\perp \sin \delta)^2 \right] \quad (4.48)$$

$$= \frac{1}{2} \mathbb{E} \cos^2 \delta + (\mathbb{E}r^2) \mathbb{E} \cos^4 \delta + (\mathbb{E}[r^\perp]^2) \mathbb{E} \cos^2 \delta \sin^2 \delta \quad (4.49)$$

$$= \frac{1}{2} \mathbb{E} \cos^2 \delta + \sigma^2 \mathbb{E} \cos^2 \delta \quad (4.50)$$

$$= \frac{1}{2} \left(\frac{1}{2} + \sigma^2 \right). \quad (4.51)$$

4.3. Security of QPV_{coh} against general (un-entangled) attackers

Here we have used that $\mathbb{E} \cos^2 \delta = \frac{1}{2}$ because φ is uniform. Furthermore, we have

$$\mathbb{E} m r \cos \delta = \mathbb{E}(r^2 \cos^2 \delta + r r^\perp \sin \delta \cos \delta) = \sigma^2/2 + 0. \quad (4.52)$$

Substitution of (4.51,4.52) into (4.47) yields

$$\mathbb{E}(R - \mu)^2 = \frac{\sigma^2}{2} \cdot \frac{\sigma^2 + 1}{\sigma^2 + \frac{1}{2}}. \quad (4.53)$$

This is much larger than the honest prover's value $1/2 + u$ for sufficiently large σ .

These examples are intended to build intuition and illustrate representative attack strategies, but they are not exhaustive; the complete and rigorous security guarantee is provided in Section 4.3.1.

4

4.3. Security of QPV_{coh} against general (un-entangled) attackers

4.3.1. Provable security against no pre-shared entanglement-based attacks

In this section, we show that we not only have security against the above-described attacks, but that the result generalises to all attackers that do not pre-share entanglement by establishing a lower bound on their uncertainty that is higher than the prover's. The basis of our security proofs is the quantum-mechanical uncertainty principle. We use the following form for the differential entropy in a tripartite setting of a guessing game, as is often useful in the context of quantum cryptography.

Lemma 4.3.1. [77] *Let ρ_{ABC} be a tripartite density matrix on systems A , B and C . Let Q and P denote the random variables of position and momentum, respectively, resulting from a homodyne measurement on the A system, and let the following hold: $h(Q|B)_\rho, h(P|C)_\rho > -\infty$ and $H(Q_\alpha|B)_\rho, H(P_\alpha|C)_\rho < \infty$ for any $\alpha > 0$. Then*

$$h(Q|B)_\rho + h(P|C)_\rho \geq \log(2\pi). \quad (4.54)$$

Furthermore, we will make use of the following estimation inequality.

Theorem 4.3.2. [53] *Let X be a random variable and $\hat{X}(Y)$ an estimator of X given side information Y . Then*

$$\mathbb{E} \left[\left(X - \hat{X}(Y) \right)^2 \right] \geq \frac{1}{2\pi e} e^{2h_{\text{nats}}(X|Y)}, \quad (4.55)$$

where $h_{\text{nats}}(X|Y)$ is the conditional entropy in natural units. Moreover, if X is Gaussian and $\hat{X}(Y)$ is its mean, then equality holds.

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Finally, the full security proof is captured by the following theorem.

Theorem 4.3.3. *For any attackers who do not pre-share entanglement prior to the execution of the protocol, there is at least one attacker E in a general attack for whom the differential entropy of R , given the side information held by E , satisfies the inequality*

$$h(R|E) \geq \frac{1}{2} \log \frac{4\pi}{1 + \sigma^{-2}}, \quad (4.56)$$

where σ is the same as that defined in Section 4.1.2. Furthermore, this attacker's response r' satisfies the inequality

$$\mathbb{E}(R - r')^2 \geq \frac{2}{e} \cdot \frac{1}{1 + \sigma^{-2}}. \quad (4.57)$$

Proof. In the entanglement-based protocol, the verifiers perform a heterodyne measurement. This is achieved by mixing one half of the TMS state with vacuum (denoted by O) and then performing a homodyne measurement per mode, in orthogonal directions θ and $\theta + \frac{\pi}{2}$, so

$$\rho_{VP} \xrightarrow[\text{with } O]{\text{Mixing}} \rho_{\bar{V}\bar{O}P}, \quad (4.58)$$

where the bar represents the modes after mixing. Here P is the subsystem sent to the prover and $\bar{V}$ is the subsystem on which the θ measurement will be applied.

The attackers (Alice and Bob) perform a quantum operation on the mode P and any ancilla mode. We call the subsystem that Alice holds A , and the one sent to Bob B . The resulting state is $\rho_{\bar{V}\bar{O}AB}$. We are interested in the tripartite state $\rho_{\bar{V}AB}$. We write the result of a homodyne measurement on $\bar{V}$ at angle θ as $U_\theta \in \mathbb{R}$, and we write $\bar{\theta} = \theta + \frac{\pi}{2}$. Lemma 4.3.1 gives

$$\forall \theta \in \mathcal{A} \quad h(U_\theta|A) + h(U_{\bar{\theta}}|B) \geq \log 2\pi. \quad (4.59)$$

Averaging over θ , and using the fact that averaging over $\bar{\theta}$ is the same as averaging over θ , gives

$$\mathbb{E}_{\theta \in \mathcal{A}} h(U_\theta|A) + \mathbb{E}_{\theta \in \mathcal{A}} h(U_{\bar{\theta}}|B) \geq \log 2\pi \quad (4.60)$$

$$\implies \mathbb{E}_{\theta \in \mathcal{A}} h(U_\theta|A) + \mathbb{E}_{\theta \in \mathcal{A}} h(U_\theta|B) \geq \log 2\pi. \quad (4.61)$$

The last expression can be written as

$$h(U|A\Theta) + h(U|B\Theta) \geq \log 2\pi, \quad (4.62)$$

where the angle Θ is now represented as a random variable. It follows that

$$\max \left\{ h(U|A\Theta), h(U|B\Theta) \right\} \geq \frac{1}{2} \log 2\pi. \quad (4.63)$$

Finally, we note that $R = U\sqrt{2} \tanh \zeta$ (with $\sinh \zeta = \sigma$) to conclude

$$h(R|E) \geq \frac{1}{2} \log 2\pi + \frac{1}{2} \log \frac{2}{1 + \sigma^{-2}} = \frac{1}{2} \log \frac{4\pi}{1 + \sigma^{-2}}. \quad (4.64)$$

Here, we have set $\max \{h(R|A\Theta), h(R|B\Theta)\} = h(R|E)$. The result for $\mathbb{E}(R - r')^2$ follows directly from Fano's inequality (Theorem 4.3.2). $\square$

4.3.2. Comparison between attacker and honest prover

We will now work in the $\sigma \gg 1$ limit. The protocol works only if the attackers have more ignorance about the value R than the honest prover. Note that we assume that the attackers are powerful and have access to an ideal channel ($t = 1, u = 0$). For $\sigma \rightarrow \infty$, the difference between their entropies (4.56), (4.12) satisfies

$$h(R|E) - h(R|R') \geq \frac{1}{2} \log \left(\frac{4}{e} \cdot \frac{t}{1 + 2u} \right). \quad (4.65)$$

The argument of the logarithm needs to be larger than 1. This is the case when

$$t > \frac{e}{4} \approx 0.680 \quad \wedge \quad u \leq \frac{t \cdot 4/e - 1}{2}. \quad (4.66)$$

Note that Fano's inequality applied to the honest prover's entropy (4.12) would yield the expression $\mathbb{E}(\sqrt{t}R - r'/\sqrt{t})^2 \approx \Sigma^2$ (as $\sqrt{t}R|R'$ is Gaussian with mean r' in the large- σ limit), with Σ^2 as defined in (4.8), evaluating to $\Sigma^2 \approx (1/2 + u)/t$. On the other hand, the expected error of the attacker is lower-bounded by $2/e \approx 0.74$, which is strictly greater than $(1/2 + u)/t$ for certain parameter ranges, as depicted in Figure 4.2. This proves security of the protocol against a general attack in these parameter ranges.

As long as Eq. (4.65) is positive, i.e.

$$\frac{t}{1 + 2u} > \frac{e}{4}, \quad (4.67)$$

there is a finite gap between the attacker's and the honest prover's entropies about R .

Theorem 4.3.4 (Soundness under independent repetition). *Consider n rounds of the QPV_{coh} protocol in the limit $\sigma \rightarrow \infty$. Suppose that the attackers do not pre-share entanglement, have access to ideal channels, and repeat the same attack independently in every round. If the single-round score has finite variance $\tilde{\sigma}^2$ and $\Delta := \frac{2t}{e(1/2+u)} - \gamma > 0$, then their success probability satisfies*

$$p_{\text{att}}^{(n)} \leq \frac{\tilde{\sigma}^2}{n\Delta^2}. \quad (4.68)$$

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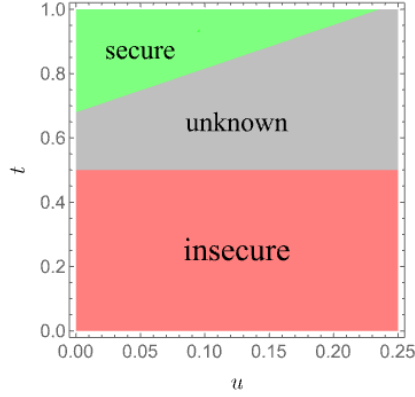


Figure 4.2: Security of the QPV_{coh} protocol. For $t \leq 1/2$ it is insecure, as shown in [159]. For values in the upper triangular region, we prove security. Currently, no conclusions can be drawn about the leftover region.

Proof. By Theorem 4.57, there is at least one attacker E whose estimator of R has mean-square error at least $2/e$. Since $r'/\sqrt{t}$ is an estimator of R , the response r' satisfies

$$\mathbb{E}\left[(\sqrt{t}R - r')^2\right] = t \mathbb{E}\left[\left(R - \frac{r'}{\sqrt{t}}\right)^2\right] \geq \frac{2t}{e}. \quad (4.69)$$

For each round, let

$$X_i := \frac{(\sqrt{t}R_i - R_i)^2}{1/2 + u}, \quad \tilde{\sigma}^2 = \text{Var}(X_i) < \infty. \quad (4.70)$$

It follows that

$$\mathbb{E}[X_i] \geq \frac{2t}{e(1/2 + u)} = \gamma + \Delta. \quad (4.71)$$

If the attackers pass the verification test, then attacker E 's average score is at most γ . Since the X_i are independent and identically distributed, Chebyshev's inequality gives

$$\begin{aligned} p_{\text{att}}^{(n)} &\leq \Pr\left[\frac{1}{n} \sum_{i=1}^n X_i \leq \gamma\right] \\ &\leq \Pr\left[\left|\frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}[X_i]\right| \geq \Delta\right] \leq \frac{\tilde{\sigma}^2}{n\Delta^2}. \end{aligned} \quad (4.72)$$

□

Because $\gamma \rightarrow 1$ as $n \rightarrow \infty$, the condition $\Delta > 0^3$ holds for sufficiently large n whenever $\frac{t}{1+2u} > \frac{e}{4}$.

Corollary 4.3.5. *Under the assumptions of Theorem 4.3.4, for any $\varepsilon_{\text{attack}} > 0$, if $n \geq \frac{\bar{\sigma}^2}{\varepsilon_{\text{attack}} \Delta^2}$, then the QPV_{coh} protocol is $\varepsilon_{\text{attack}}$ -sound, i.e.,*

$$p_{\text{att}}^{(n)} \leq \varepsilon_{\text{attack}}. \quad (4.73)$$

Proof. This follows immediately from Eq. (4.68). $\square$

4.3.3. Perfect attack with a single EPR pair

It turns out that the QPV_{coh} protocol can be attacked if Alice and Bob pre-share one CV EPR pair (see Section 2.3.3 for formal descriptions of CV entanglement and Section 2.3.4 for a description of CV teleportation). The entanglement attack proceeds as follows (see Fig. 4.3 for a schematic representation):

1. Alice and Bob pre-share an ideal EPR pair.
2. Alice teleports $|\psi\rangle$ to Bob. She forwards the measured displacement (d_x, d_p) to Bob.
3. Bob intercepts θ and immediately performs a homodyne measurement at angle θ on his own half of the EPR pair, obtaining outcome $\mu \in \mathbb{R}$. He forwards θ, μ to Alice.
4. Alice receives θ, μ . She computes $r' = \mu - d_x \cos \theta - d_p \sin \theta$ and sends r' to V_1 .
5. Bob receives d_x, d_p . He computes $r' = \mu - d_x \cos \theta - d_p \sin \theta$ and sends r' to V_2 .

The state $|\psi\rangle$ is a coherent state with displacement (x_0, p_0) . The effect of the teleportation is that Bob's half of the EPR pair becomes a coherent state with displacement $(x_0 + d_x, p_0 + d_p)$. Bob's homodyne measurement commutes with the teleport-induced displacement: the undoing of the displacement can be done *after* Bob's measurement. The noise in r' with respect to r is just shot noise, exactly as for the honest prover. Other noise originating from loss or excess noise can just be simulated by the attacker.

Hence, in the case of an ideal pre-shared EPR pair, the responses from the attackers are statistically indistinguishable from honest prover responses.

4.4. The $\text{QPV}_{\text{coh}}^f$ protocol

Based on the ideas in [44, 179, 33], we introduce a variation of the quantum position verification protocol studied in the previous sections. Instead of only a single verifier

³For $t = 1$, this condition becomes $u < 2/e - 1/2 \approx 0.24$, as can also be observed in Fig. 6.4.

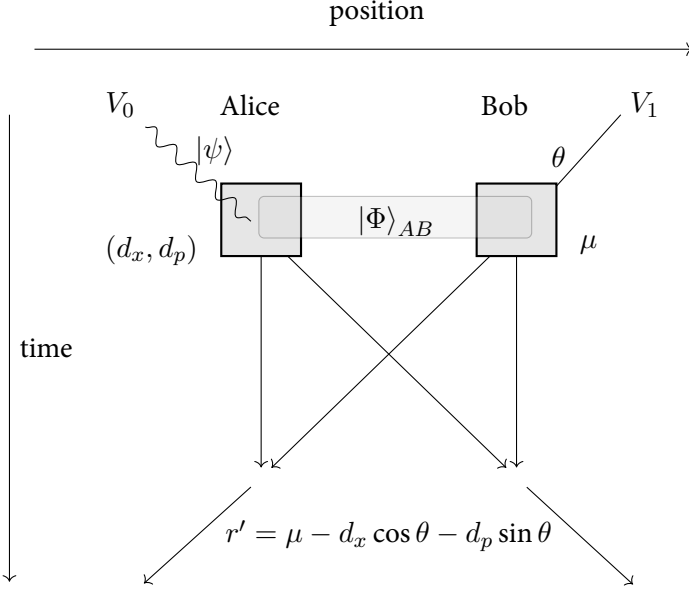


Figure 4.3: Schematic representation of a perfect attack using an EPR pair $|\Phi\rangle_{AB}$, where Alice and Bob hold registers A and B , respectively, prior to the execution of the attack.

sending classical information, both verifiers send classical information that, combined, determines the action of the prover. When sent through a channel, a continuous-variable state gets attenuated and acquires excess noise. We will denote by $t \in [0, 1]$ the attenuation parameter, and by $u \geq 0$ the excess noise power of the quantum channel connecting V_0 and P . Then, the protocol is described as follows:

Definition 4.4.1. Let $n \in \mathbb{N}$ and consider a $2n$ -bit Boolean function $f : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}$. A round of the $\text{QPV}_{\text{coh}}^f$ protocol is described as follows.

1. The verifiers V_0 and V_1 randomly choose bit strings $x, y \in \{0, 1\}^n$, respectively. They draw two random variables $(r, r^\perp)$ from the Gaussian distribution $\mathcal{N}_{0, \sigma^2}$, for $\sigma \gg 1$, and compute $f(x, y)$. Verifier V_0 prepares a coherent state $|\psi\rangle$ with quadratures $(x_0, p_0) = (r \cos \theta + r^\perp \sin \theta, r \sin \theta - r^\perp \cos \theta)$, where $\theta = 0$ if $f(x, y) = 0$ and $\theta = \frac{\pi}{2}$ if $f(x, y) = 1$.
2. The verifier V_0 sends $|\psi\rangle$ and x to P , and the verifier V_1 sends y to P such that all information arrives at P simultaneously. The classical information is required to travel at the speed of light whereas the quantum information can be sent arbitrarily slowly.
3. Immediately, P computes $f(x, y)$ and performs a homodyne measurement on $|\psi\rangle$ in the direction $\theta = 0$ if $f(x, y) = 0$ or $\theta = \frac{\pi}{2}$ if $f(x, y) = 1$, resulting in a value

$r' \in \mathbb{R}$. The prover broadcasts r' to both verifiers at the speed of light.

After N rounds, the verifiers have received a sample of responses, which we denote by $(r'_i)_{i=1}^N$. The verifiers check whether all prover responses arrived at the correct time, and whether the reported values $(r'_i)_{i=1}^N$ satisfy

$$\frac{1}{N} \sum_{i=1}^N \frac{(r'_i - r_i \sqrt{t})^2}{\frac{1}{2} + u} < \gamma, \quad \text{with} \quad \gamma \stackrel{\text{def}}{=} 1 + \frac{2}{\sqrt{N}} \sqrt{\ln \frac{1}{\varepsilon_{\text{hon}}}} + \frac{2}{N} \ln \frac{1}{\varepsilon_{\text{hon}}}. \quad (4.74)$$

Here ε_{hon} is an upper bound on the honest prover's failure probability (as a parameter of the protocol). See Fig. 4.4 for a schematic representation of the $\text{QPV}_{\text{coh}}^f$ protocol.

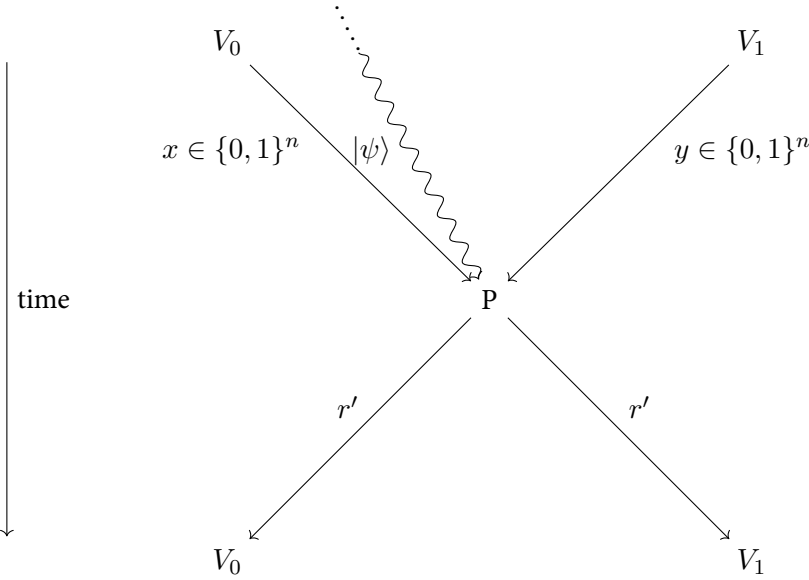


Figure 4.4: One round of the $\text{QPV}_{\text{coh}}^f$ protocol. The coherent state $|\psi\rangle$ originates from V_0 in the past.

The excess noise can be modelled as $u = u_0 \sigma^2$, with, for instance, a reasonably small value, due to the prevalence of phase noise [119]. The protocol can be proven secure only when the excess noise is under a certain threshold, and thus the u_0 places a practical upper limit on the modulation variance σ .

The protocol in [8] is as in Definition 4.4.1 but instead of sending x and y , V_1 directly sends θ . For the honest party, the only difference is that he has to compute $f(x, y)$ to determine θ . We will use the following notation: for $\theta \in \{0, \frac{\pi}{2}\}$ we will denote by $\bar{\theta}$ the remaining value, i.e. if $\theta = 0$, then $\bar{\theta} = \frac{\pi}{2}$ and if $\theta = \frac{\pi}{2}$, then $\bar{\theta} = 0$.

The honest prover's uncertainty about the displacements r , drawn by the random variable denoted by R in each round, conditioned on his measurement outcomes r' , drawn from

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the random variable denoted by R' in each round, is the same as in the protocol where θ is sent directly from V_1 , since the only change for P is to compute $f(x, y)$ to determine θ . This uncertainty was calculated in Section 4.1.3 to be

$$h(R|R')_\psi = \frac{1}{2} \log 2\pi e \Sigma^2, \quad (4.75)$$

where

$$\Sigma^2 = \left(\frac{1}{\sigma^2} + \frac{t}{1/2 + u} \right)^{-1}. \quad (4.76)$$

It is well known that the preparation of a coherent state with Gaussian-distributed displacements $x_0, p_0 \sim \mathcal{N}_{0, \sigma^2}$ is equivalent to preparing a two-mode squeezed state with squeezing parameter $\sinh^{-1} \sigma$ and then performing a heterodyne $(\hat{x}, \hat{p})$ measurement on one mode, with measurement outcome $\frac{(x_0, -p_0)}{\sqrt{2} \tanh \sinh^{-1} \sigma}$. For notational brevity, we set $\lambda = \tanh \sinh^{-1} \sigma$.

In the purified version of QPV_{coh}^f, the verifier V_0 prepares the two-mode squeezed state $|\Psi\rangle_{V_0 P}$ with the above-mentioned λ -dependent squeezing, which is given by

$$|\Psi\rangle = \sqrt{1 - \lambda^2} \sum_{m=0}^{\infty} \lambda^m |mm\rangle, \quad (4.77)$$

in the Fock space. Notice that $\lambda < 1$. The verifier V_0 performs a heterodyne measurement with quadratures rotated by an angle θ on its register. The measurement outcomes are $r/(\sqrt{2}\lambda)$ and $-r^\perp/(\sqrt{2}\lambda)$, resulting in displacement $(r, r^\perp)$ in the state sent to the prover P . P then performs a homodyne measurement at angle θ to recover r , as in the original protocol.

V_0 's heterodyne measurement can be described as a double-homodyne measurement. First V_0 mixes its own mode with the vacuum using a beamsplitter, resulting in a two-mode state. On one of these modes, V_0 then performs a homodyne measurement in the θ -direction and, on the other mode, in the $\theta + \frac{\pi}{2}$ direction.

From (4.33) we saw that the honest prover's uncertainty about R evaluates to

$$h(R|P)_\Psi = \frac{1}{2} \log \frac{\pi e(1 + 2u)}{t} + O\left(\frac{1}{\sigma}\right). \quad (4.78)$$

Let $U = R/(\lambda\sqrt{2})$ be the displacement in the θ direction as measured by V_0 .

Then

$$\begin{aligned} h(U|P)_\Psi &= h\left(\frac{R}{\sqrt{2}\lambda} \middle| P\right)_\Psi = h(R|P)_\Psi - \log(\sqrt{2}\lambda) \\ &= \frac{1}{2} \log \frac{\pi e(1 + 2u)}{t} + O\left(\frac{1}{\sigma}\right) - \log(\sqrt{2}\lambda). \end{aligned} \quad (4.79)$$

In the regime $\sigma \gg 1$ (i.e. $\lambda \rightarrow 1$),

$$h(U|P)_\Psi \rightarrow h(U|P)_\psi = \frac{1}{2} \log \frac{\pi e(1+2u)}{2t}. \quad (4.80)$$

Unless stated otherwise, we will work in the regime $\sigma \gg 1$. Recall that σ is under the control of the verifiers.

4.5. Security of $\text{QPV}_{\text{coh}}^f$ against bounded entanglement

4

In this section, we prove the security of the $\text{QPV}_{\text{coh}}^f$ protocol, showing that with high probability, attackers who possess a bounded amount of entanglement will not be able to attack the protocol. If counted in entangled qubits, the bound is linear in n .

To do so, we consider an ‘imaginary world’ where the $\text{QPV}_{\text{coh}}^f$ protocol, instead of using the state $|\Psi\rangle$, is executed with a cutoff at photon number 2^{m_0} using the state $|\Psi_{m_0}\rangle$, given by

$$|\Psi_{m_0}\rangle = \sqrt{\frac{1-\lambda^2}{1-(\lambda^2)^{2^{m_0}}}} \sum_{m=0}^{2^{m_0}-1} \lambda^m |mm\rangle. \quad (4.81)$$

We will denote this variation of the protocol by $\text{QPV}_{\text{coh}_{m_0}}^f$. The state $|\Psi_{m_0}\rangle$ is an approximation of the state $|\Psi\rangle$ and can be made arbitrarily close to it by increasing m_0 . Note that

$$\mathcal{P}(|\Psi\rangle, |\Psi_{m_0}\rangle) = \lambda^{2^{m_0}}, \quad (4.82)$$

i.e. $|\Psi_{m_0}\rangle$ is doubly exponentially close⁴ (in m_0) to $|\Psi\rangle$ (recall that $\lambda < 1$). Here $\mathcal{P}$ is the purified distance (Section 2.2). If one replaces the state $|\Psi\rangle$ by $|\Psi_{m_0}\rangle$, the probability that the verifiers accept the action of an honest party will change by at most $O(\lambda^{2^{m_0}})$. The cutoff reduces the dimension of the Hilbert space from infinite to 2^{m_0} , which is the dimension of an m_0 -qubit state space.

The energy of the V_0 subsystem is given by

$$\langle \Psi_{m_0} | H_{V_0} | \Psi_{m_0} \rangle_{V_0 P} = \frac{\lambda^2 + ((2^{m_0} - 1)\lambda^2 - 2^{m_0})\lambda^{2^{m_0}+1}}{(\lambda^2 - 1)(\lambda^{2^{m_0}+1} - 1)} = \frac{2^{m_0} \left(\frac{\sigma^2}{\sigma^2 + 1} \right)^{2^{m_0}}}{\left(\frac{\sigma^2}{\sigma^2 + 1} \right)^{2^{m_0}} - 1} + \sigma^2, \quad (4.83)$$

which tends to σ^2 (the expected energy of the challenge state chosen by the verifiers) as m_0 tends to infinity.

⁴Note that we chose $\sigma > 1$, and since $\lambda \approx 1 - \frac{1}{2\sigma^2}$ which implies $2^{m_0} > \sigma^2$.

4. Quantum Position Verification

The most general attack on $\text{QPV}_{\text{coh}_{m_0}}^f$ for adversaries with a photon-number cutoff such that their Hilbert space is isomorphic to a multi-qubit Hilbert space consists of an adversary Alice between V_0 and P , and an adversary Bob between V_1 and P . They proceed as follows:

1. *Preparing*: The attackers prepare a joint (possibly entangled) CV state with a cutoff at the photon number of q qubits each.
2. *Intercepting*: Alice intercepts the quantum information sent from V_0 . At this stage, V_0 , Alice and Bob share a state $|\chi\rangle_{VPAA_cBB_c}$ of dimension 2^{2q+2m_0} . Here V is the register kept by V_0 , and P is the challenge register that V_0 sends. Alice controls the registers P , A and A_c , and Bob possesses the registers B and B_c . Moreover, Alice and Bob intercept x and y and perform arbitrary quantum channels depending on the intercepted classical information: $U_{PAA_c}^x$ and $V_{BB_c}^y$, respectively, ending up with the state $|\phi\rangle_{VPAA_cBB_c}$.
3. *Communicating*: Alice and Bob send a copy of x and y to the other attacker, respectively. Alice keeps registers P and A and sends register A_c to Bob, and Bob keeps register B and sends B_c to Alice.
4. *Measuring*: Upon receiving the information sent by the other party, Alice and Bob locally apply arbitrary POVMs $\{A_{PAB_c}^{xy}\}$ and $\{B_{A_cB}^{xy}\}$ to obtain classical answers, which will be sent to their closest verifier, respectively.

Due to the Stinespring dilation, we can consider the quantum channels to be unitaries.

Definition 4.5.1. *The tuple $\{|\chi\rangle_{VPAA_cBB_c}, U_{PAA_c}^x, V_{BB_c}^y, A_{PAB_c}^{xy}, B_{A_cB}^{xy}\}_{xy}$ is a q -qubit strategy for $\text{QPV}_{\text{coh}_{m_0}}^f$. Moreover, we say that a q -qubit strategy for $\text{QPV}_{\text{coh}_{m_0}}^f$ is (ε, l) -perfect if for l pairs of strings (x, y) , for $\theta \in \{0, \frac{\pi}{2}\}$,*

$$h(U_\theta|PAB_c)_\phi \leq h(U|P)_\psi + \varepsilon \quad \text{and} \quad h(U_\theta|A_cB)_\phi \leq h(U|P)_\psi + \varepsilon. \quad (4.84)$$

Notice that there is no θ -dependence on the right-hand side of the inequality since $h(U|P)_\psi$ does not depend on θ . The parameter ε (see the analysis below) will quantify the difference between the uncertainty of the honest prover and that of the attackers. It will have to be picked depending on the number of rounds for which the protocol is run sequentially. Next, we define ‘good’ states for attacking the protocol for either $\theta = 0$ or $\theta = \pi/2$. We will see that a good state for the $\theta = 0$ case cannot be too close (in trace distance) to a good state for the $\theta = \pi/2$ case. This hinders the attackers.

Remark 4.5.2. *Notice that we consider strategies starting with the state $|\chi\rangle_{VPAA_cBB_c}$ instead of $|\Psi_{m_0}\rangle_{VP} \otimes |\chi\rangle_{AA_cBB_c}$. This will give more power to the attackers, and it will account for the fact that the quantum information sent from V_0 can travel arbitrarily slowly, and the attackers are allowed to modify $|\Psi_{m_0}\rangle_{VP} \otimes |\chi\rangle_{AA_cBB_c}$ to end up with any arbitrary state $|\chi\rangle_{VPAA_cBB_c}$.*

4.5. Security of $\text{QPV}_{\text{coh}}^f$ against bounded entanglement

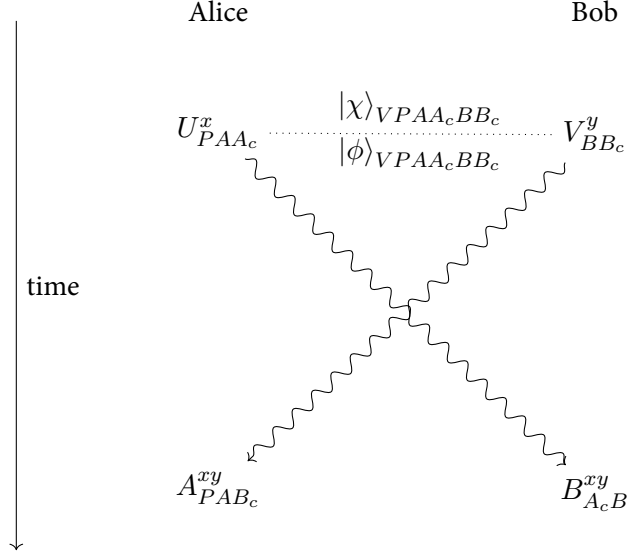


Figure 4.5: Schematic depiction of a generic attack on $\text{QPV}_{\text{coh}_{m_0}}^f$.

Definition 4.5.3. Let $\varepsilon \geq 0$, and let q be the number of qubits that Alice and Bob each hold at the preparing stage of a multi-qubit attack on $\text{QPV}_{\text{coh}_{m_0}}^f$. We define $\mathcal{S}_\theta^\varepsilon$ as

$$\mathcal{S}_\theta^\varepsilon := \{|\phi\rangle_{VPAA_cBB_c} \in \mathbb{C}^{2q+2m_0} \mid \exists \text{ POVMs } \{A_{PAB_c}^{xy}\}, \{B_{A_cB}^{xy}\} \text{ s.t. (4.84) holds}\}. \quad (4.85)$$

Proposition 4.5.4. Let $\varepsilon > 0$, $|\phi_0\rangle_{VPAA_cBB_c} \in \mathcal{S}_0^\varepsilon$, and $|\phi_{\frac{\pi}{2}}\rangle_{VPAA_cBB_c} \in \mathcal{S}_{\frac{\pi}{2}}^\varepsilon$, with bounded energies $\text{Tr}[\rho^0 H], \text{Tr}[\rho^1 H] \leq E$, where ρ^0 and ρ^1 are the respective density matrices of $|\phi_0\rangle$ and $|\phi_{\frac{\pi}{2}}\rangle$. The corresponding Hamiltonian is the harmonic oscillator on system V and identity on the other systems. Let $\frac{1}{2} \geq \alpha \geq 0$ and $\tilde{\varepsilon} > 0$ be such that

$$\varepsilon < \frac{1}{2} \log \frac{4t}{e(1+2u)} - \left(\frac{\tilde{\alpha}}{2} + \alpha \right) \left[2\tilde{\varepsilon} \left(\log(E+1) + \log \frac{e}{\alpha(1-\tilde{\varepsilon})} \right) + 6\tilde{h}(\tilde{\alpha}\tilde{\varepsilon}) \right], \quad (4.86)$$

where $\tilde{\alpha} = \frac{1+\alpha}{1-\alpha}$. Then,

$$\frac{1}{2} \| |\phi_0\rangle - |\phi_{\frac{\pi}{2}}\rangle \|_1 > \tilde{\varepsilon}. \quad (4.87)$$

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Notice that the energy bound E is the energy given by the Hamiltonian corresponding to one harmonic oscillator in the V system (4.83), which approaches σ^2 from below. Moreover, from (4.86), we see that ε must be chosen to be at most $\frac{1}{2} \log \frac{4t}{e(1+2u)}$. In order to have non-negative $\tilde{\varepsilon}$, we need

$$4t > e(1 + 2u), \quad (4.88)$$

see Fig. 4.6. The maximum value of ε will be upper bounded by

$$\varepsilon < \frac{1}{2} \log \frac{4t}{e(1+2u)} \leq \frac{1}{2} \log \frac{4}{e} \simeq 0.278652, \quad (4.89)$$

since the maximum value is reached at $t = 1$ and $u = 0$. We will also use the following lemma for the proof of the above proposition.

Lemma 4.5.5. (Lemma 18, [197]) Let $\alpha \in [0, \frac{1}{2}]$. Consider a Hamiltonian $H = H_A \otimes \mathbb{I}_B$, with system A composed of one harmonic oscillator and an arbitrary system B . Let there be states ρ and σ on the bipartite system $\mathcal{H}_A \otimes \mathcal{H}_B$ with $\text{Tr}[\rho H], \text{Tr}[\sigma H] \leq E$. If $\frac{1}{2} \|\rho - \sigma\|_1 \leq \tilde{\varepsilon}$, then

$$|h(A|B)_\rho - h(A|B)_\sigma| \leq (\tilde{\alpha} + 2\alpha) \left[2\tilde{\varepsilon} \left(\log(E + 1) + \log \frac{e}{\alpha(1 - \tilde{\varepsilon})} \right) + 6\tilde{h}(\tilde{\alpha}\tilde{\varepsilon}) \right], \quad (4.90)$$

where

$$\tilde{h}(x) := \begin{cases} -x \log x - (1 - x) \log(1 - x) & \text{if } x \leq \frac{1}{2} \\ 1 & \text{if } x \geq \frac{1}{2} \end{cases} \quad (4.91)$$

and $\tilde{\alpha} = \frac{1+\alpha}{1-\alpha}$.

Proof. Let ρ^θ and $\rho^{\bar{\theta}}$ be the density matrices of $|\phi_\theta\rangle_{VPAAcBB_c}$ and $|\phi_{\bar{\theta}}\rangle_{VPAAcBB_c}$, respectively. By hypothesis,

$$h(U_\theta|PAB_c)_{\rho^\theta} \leq h(U|P)_\psi + \varepsilon, \text{ and } h(U_{\bar{\theta}}|PAB_c)_{\rho^{\bar{\theta}}} \leq h(U|P)_\psi + \varepsilon. \quad (4.92)$$

By Lemma 4.3.1,

$$h(U_\theta|PAB_c)_{\rho^\theta} + h(U_{\bar{\theta}}|A_cB)_{\rho^\theta} \geq \log 2\pi. \quad (4.93)$$

Then,

$$h(U_{\bar{\theta}}|A_cB)_{\rho^\theta} \geq \log 2\pi - h(U_\theta|PAB_c)_{\rho^\theta} \geq \log 2\pi - h(U|P)_\psi - \varepsilon, \quad (4.94)$$

where in the last inequality we used (4.92). Therefore,

$$h(U_{\bar{\theta}}|A_cB)_{\rho^\theta} - h(U_{\bar{\theta}}|PAB_c)_{\rho^{\bar{\theta}}} \geq \log 2\pi - 2h(U|P)_\psi - 2\varepsilon. \quad (4.95)$$

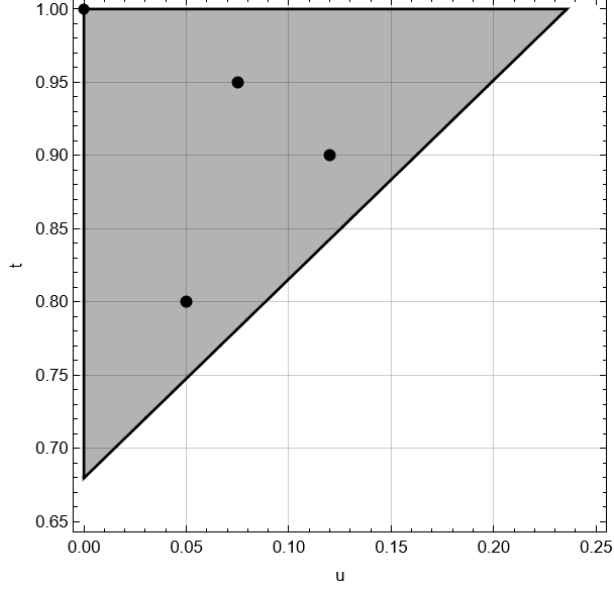


Figure 4.6: Necessary condition for u and t so that (4.86) is fulfilled (grey region). The blue dots are specific (u, t) for which we do numerical analysis in Section 4.6.

In the regime $\sigma \gg 1$, $h(U|P)_\psi \rightarrow \frac{1}{2} \log \frac{\pi e(1+2u)}{2t}$, and thus,

$$h(U_{\bar{\theta}}|A_c B)_{\rho^\theta} - h(U_{\bar{\theta}}|PAB_c)_{\rho^{\bar{\theta}}} \geq \log \frac{4t}{e(1+2u)} - 2\varepsilon > 0, \quad (4.96)$$

where the last inequality comes from the fact that, by hypothesis, $\frac{1}{2} \log \frac{4t}{e(1+2u)} > \varepsilon$. This leads to

$$|h(U_{\bar{\theta}}|A_c B)_{\rho^\theta} - h(U_{\bar{\theta}}|PAB_c)_{\rho^{\bar{\theta}}}| \geq \log \frac{4t}{e(1+2u)} - 2\varepsilon. \quad (4.97)$$

By hypothesis,

$\log \frac{4t}{e(1+2u)} - 2\varepsilon > \left(\frac{1+\alpha}{1-\alpha} + 2\alpha \right) \left[2\tilde{\varepsilon} \left(\log(E+1) + \log \frac{e}{\alpha(1-\tilde{\varepsilon})} \right) + 6\tilde{h} \left(\frac{1+\alpha}{1-\alpha} \tilde{\varepsilon} \right) \right]$. Thus, by the contrapositive of Lemma 4.5.5,

$$\frac{1}{2} \|\rho^\theta - \rho^{\bar{\theta}}\|_1 > \tilde{\varepsilon}. \quad (4.98)$$

□

Definition 4.5.6. [33] Let $q, k, n \in \mathbb{N}$, $\varepsilon > 0$. Then, $g : \{0, 1\}^{3k} \rightarrow \{0, 1\}$ is an (ε, l) -classical rounding of size k if for all $f : \{0, 1\}^{2n} \rightarrow \{0, 1\}$, for all states $|\psi\rangle$ on $2q + 2m_0$ qubits, for all $l \in \{1, \dots, 2^{2n}\}$ and for all (ε, l) -perfect q -qubit strategies for $\text{QPV}_{\text{coh}_{m_0}}^f$,

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there are functions $f_A : \{0, 1\}^n \rightarrow \{0, 1\}^k$, $f_B : \{0, 1\}^n \rightarrow \{0, 1\}^k$ and $\gamma \in \{0, 1\}^k$ such that $g(f_A(x), f_B(y), \gamma) = f(x, y)$ on at least l pairs (x, y) .

Lemma 4.5.7. [121] Let $\|\cdot\|$ be any norm on $\mathbb{R}^{n_0}$, for $n_0 \in \mathbb{N}$. There is a δ -net S of the unit sphere of $(\mathbb{R}^{n_0}, \|\cdot\|)$ of cardinality at most $(1 + 2/\delta)^{n_0}$.

Lemma 4.5.8. [33] Let $|x\rangle, |y\rangle \in \mathbb{C}^d$, for $d \in \mathbb{N}$, be two unit vectors. Then, $\mathcal{P}(|x\rangle, |y\rangle) \leq \| |x\rangle - |y\rangle \|_2$.

Note the following two results. The proofs can be found in the full version [71].

Proposition 4.5.9. Let $\varepsilon, \tilde{\varepsilon}$ be such that $|\varphi_\theta\rangle \in \mathcal{S}_\theta^\varepsilon$ and $|\varphi_{\bar{\theta}}\rangle \in \mathcal{S}_{\bar{\theta}}^{\tilde{\varepsilon}}$ imply $\mathcal{P}(|\varphi_\theta\rangle, |\varphi_{\bar{\theta}}\rangle) > \tilde{\varepsilon}$. Then there is an (ε, q) -classical rounding of size $k = 2^{2q+2m_0} \log \left(1 + \frac{4}{\sqrt[3]{4(2+\tilde{\varepsilon})}-2}\right)$.

Lemma 4.5.10. Let $\varepsilon \in [0, 1]$, $E, t, u > 0$ be such that there exist $\tilde{\varepsilon} > 0$ and α such that (4.86) holds. Let $k, q, n \in \mathbb{N}$. Moreover, fix an (ε, q) -classical rounding g of size k with

$$k = 2^{2q+2m_0} \log \left(1 + \frac{4}{\sqrt[3]{4(2+\tilde{\varepsilon})}-2}\right).$$

Assume that

$$q + m_0 \leq \frac{n}{2} - c_{\tilde{\varepsilon}}, \quad (4.99)$$

where $c_{\tilde{\varepsilon}} > 0$ is a sufficiently large constant depending only on $\tilde{\varepsilon}$. Then, with probability $1 - O(\lambda^{2^{m_0}})$, the following holds: A uniformly random $f : \{0, 1\}^{2n} \rightarrow \{0, 1\}$ fulfils the following with probability at least $1 - O(2^{-2^n})$: for any $f_A : \{0, 1\}^n \rightarrow \{0, 1\}^k$, $f_B : \{0, 1\}^n \rightarrow \{0, 1\}^k$, and $\gamma \in \{0, 1\}^k$, the equality $g(f_A(x), f_B(y), \gamma) = f(x, y)$ holds on less than $3/4$ of all pairs (x, y) .

To obtain an explicit bound on q in Eq. (4.99), we have to fix the value of $\tilde{\varepsilon}$, since this determines the size of the classical rounding and hence the constant $c_{\tilde{\varepsilon}}$. To obtain better bounds, we are interested in picking $\tilde{\varepsilon}$ as large as possible. Given parameters E, t, u and ε , known by the verifiers, we will be interested in picking values of α such that (4.86) holds for $\tilde{\varepsilon}$ as large as possible. This needs to be done numerically, since (4.86) leads to a transcendental equation; see Section 4.6 for this analysis. This also applies to the theorem below, which, in short, states that if the number of qubits the attackers pre-share at the beginning of the protocol is bounded as specified, then, with high probability, at least one of them will have a finite gap from the uncertainty of the honest prover regarding the value of the random variable U . That is, at least one attacker will have strictly larger uncertainty than the prover.

Theorem 4.5.11. Let $\varepsilon \in [0, 1]$, $E, t, u > 0$ (under the control of the verifiers) be such that there exist $\tilde{\varepsilon} > 0$ and α such that (4.86) holds. Let $n, m_0 \in \mathbb{N}$. Let the number of qubits that Alice and Bob each control at the beginning of the protocol be $q \in \mathbb{N}$, and assume that

$$q + m_0 \leq \frac{n}{2} - c_{\tilde{\varepsilon}}, \quad (4.100)$$

where c_{ε} is as in Lemma 4.5.10. Then, with probability $1 - O(\lambda^{2^{m_0}})$, the following holds. A random function f fulfils the following with probability at least $1 - O(2^{-2^n})$: the uncertainties for Alice and Bob when attacking the protocol $\text{QPV}_{\text{coh}}^f$ are such that

$$\max\{h(U_{\theta}|PAB_c)_{\phi}, h(U_{\theta}|A_cB)_{\phi}\} \geq h(U|P)_{\psi} + \frac{\varepsilon}{4}, \quad (4.101)$$

for every state $|\phi\rangle \in \mathbb{C}^{2^{2q+2m_0}}$ and every $\theta \in \{0, \frac{\pi}{2}\}$.

Proof. By Lemma 4.5.10, with probability at least $1 - O(2^{-2^n})$, the function f is such that there are no $(\varepsilon, \frac{3}{4}2^{2n})$ -perfect q -qubit strategies for $\text{QPV}_{\text{coh}_{m_0}}^f$. That means that for every strategy, on at least a fraction $\frac{1}{4}$ of (x, y) , either $h(U_{\theta}|PAB_c)_{\phi} \geq h(U|P)_{\psi} + \frac{\varepsilon}{4}$ or $h(U_{\theta}|A_cB)_{\phi} \geq h(U|P)_{\psi} + \frac{\varepsilon}{4}$. $\square$

4.5.1. Bounding attack success probability after repeated i.i.d. rounds

For the following, remember that $\sigma \gg 1$, equivalently $\lambda \rightarrow 1$. To estimate the number of (independent) rounds N we have to run for the attack success probability to become vanishingly small, we cannot assume a specific attack distribution, and we have to assume the attackers have access to an ideal channel. By Eq. (4.101) we know that

$$h(U_{\theta}|E)_{\phi} := \max\{h(U_{\theta}|PAB_c)_{\phi}, h(U_{\theta}|A_cB)_{\phi}\} \geq h(U|P)_{\psi} + \frac{\varepsilon}{4} \quad (4.102)$$

$$= \frac{1}{2} \log \left(\pi e \frac{1/2 + u}{t} \right) + \frac{\varepsilon}{4}. \quad (4.103)$$

Now re-substituting $R = \sqrt{2}\lambda U$ yields

$$h(R|E)_{\phi} \geq h(R|P)_{\psi} + \frac{\varepsilon}{4} \quad (4.104)$$

$$= \frac{1}{2} \log \left(2\pi e \frac{1/2 + u}{t} \right) + \frac{\varepsilon}{4} \quad (4.105)$$

$$\geq \frac{1}{2} \log(\pi e) + \frac{\varepsilon}{4}, \quad (4.106)$$

where the equality follows from Eq. (4.80), and the lower bound in the final line is minimised for the ideal channel with $t = 1$ and $u = 0$, which we assume attackers can use. Via the continuous-variable version of Fano's inequality, Theorem 4.3.2, applied to the quantity $\sqrt{t}R$ appearing in the score, we obtain

$$h(\sqrt{t}R|E)_{\phi} = h(R|E)_{\phi} + \frac{1}{2} \log t, \quad (4.107)$$

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and hence

$$\begin{aligned}\mathbb{E}\left[\left(\sqrt{t}R - r'\right)^2\right] &\geq \frac{1}{2\pi e}e^{2h_{\text{nats}}(\sqrt{t}R|E)_\phi} \\ &= \frac{t}{2\pi e}e^{2h_{\text{nats}}(R|E)_\phi} \geq \frac{t}{2}e^{\varepsilon/2}.\end{aligned}\quad (4.108)$$

The probability that the attackers' score falls below the threshold γ is therefore at most the probability that the score differs from its expectation by more than

$$\Delta := \frac{t}{2(1/2 + u)}e^{\varepsilon/2} - \gamma = \frac{t}{1 + 2u}e^{\varepsilon/2} - \gamma. \quad (4.109)$$

Let $(r_i^{\text{att}})_{i=1}^N$ be the attackers' sample after N i.i.d. rounds. Then we can use Chebyshev's inequality for the random variable of the score to get

$$\mathbb{P}\left[\left|\frac{1}{N}\sum_{i=1}^N \frac{(\sqrt{t}r_i - r_i^{\text{att}})^2}{1/2 + u} - \frac{\mathbb{E}(\sqrt{t}R - r')^2}{1/2 + u}\right| \geq \Delta\right] \leq \frac{\tilde{\sigma}^2}{N\Delta^2} = O\left(\frac{1}{N\Delta^2}\right), \quad (4.110)$$

where $\tilde{\sigma}^2 = \mathbb{V}\left[\frac{(\sqrt{t}R - r')^2}{1/2 + u}\right]$ is the variance. We set the tolerance for the attack success probability to $\varepsilon_{\text{att}} = \varepsilon_{\text{hon}}$ for simplicity. If we then set $N\Delta^2 = \Omega\left(\frac{1}{\varepsilon_{\text{hon}}}\right)$, we get

$$\mathbb{P}\left[\frac{1}{N}\sum_{i=1}^N \frac{(\sqrt{t}R_i - r_i')^2}{1/2 + u} \leq \gamma\right] \leq O(\varepsilon_{\text{hon}}). \quad (4.111)$$

The required number of rounds N can be obtained by first setting the tolerated ε_{hon} and then solving $N\Delta^2 = \Omega\left(\frac{1}{\varepsilon_{\text{hon}}}\right)$ for N . This means we accept the honest prover with probability at least $1 - \varepsilon_{\text{hon}}$, while accepting attackers with probability at most ε_{hon} after N i.i.d. rounds.

4.6. Concrete linear bounds for given experimental parameters

In the above section, we proved that, if V_0 prepares a two-mode squeezed state with squeezing parameter ζ and $\lambda = \tanh \zeta$, then, with probability $1 - O(\lambda^{2^{m_0}})$, attackers who each pre-share q qubits satisfying $q + m_0 \leq \frac{n}{2} - c_\varepsilon$ will not be able to closely mimic an honest prover. For that, we need ε , E , t , and u to be such that there exist $\tilde{\varepsilon} > 0$ and α for which (4.86) holds. In order to have parameters fulfilling (4.86) for $\varepsilon > 0$, we need the attenuation parameter t and the excess noise power u of the quantum channel connecting V_0 and P to fulfil the relation (4.88). In this section, we analyse perfect and imperfect channels.

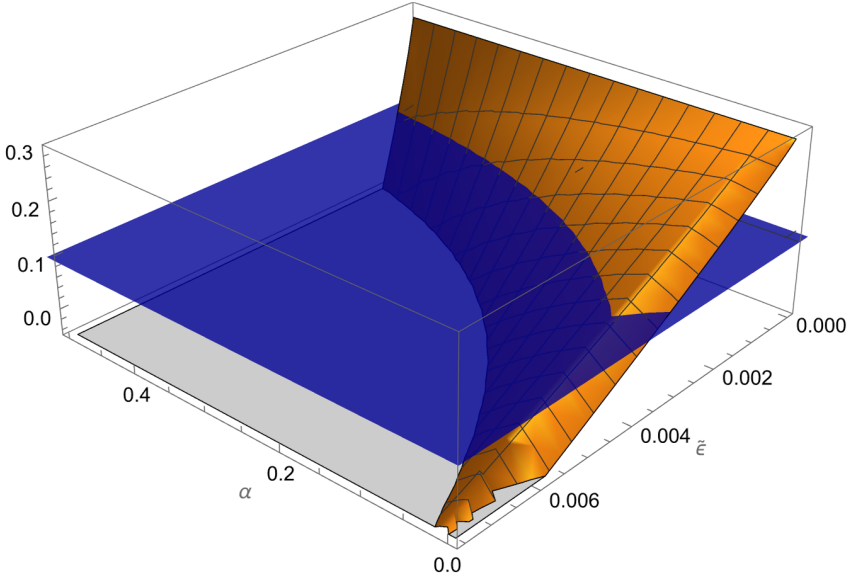


Figure 4.7: Representation of the left-hand side (horizontal plane) and right-hand side (textured surface) of the inequality (4.86) for $\varepsilon = 0.1$, $E = 10^3$, $t = 1$ and $u = 0$. The region where the textured surface is above the plane gives the $(\alpha, \tilde{\varepsilon})$ such that the inequality (4.86) holds.

4.6.1. Perfect channel

We start by considering a perfect channel connecting V_0 and P , given by $t = 1$ and $u = 0$. We fix $\varepsilon = 0.1$ and assume the protocol is played for enough rounds to statistically distinguish the honest party's uncertainty

$$h(U|P)_\psi \rightarrow \frac{1}{2} \log \frac{\pi e}{2} \simeq 1.0471, \quad (4.112)$$

from that of at least one of the attackers, which is lower-bounded by

$$\frac{1}{2} \log \frac{\pi e}{2} + \frac{\varepsilon}{4} \simeq 1.0721. \quad (4.113)$$

Fix an energy bound $E = 10^3$ in units such that $\hbar\omega = 1$. Then, the largest $\tilde{\varepsilon}$ that fulfils (4.86) is

$$\tilde{\varepsilon} \simeq 0.0037, \quad (4.114)$$

for $\alpha \simeq 0.036$. See Fig. 4.7 for a representation of the inequality (4.86) where the fixed value of ε is represented by the blue plane and the orange surface represents the right-hand side of the inequality.

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Notice that the energy term in (4.86) scales as $\tilde{\varepsilon} \log(E + 1)$, i.e. logarithmically in E with a small factor $\tilde{\varepsilon}$ in front. Therefore, the inequality remains very stable with respect to E . For instance, if one picks $E = 10, 10^2, 10^4$, the values of the maximum $\tilde{\varepsilon}$ remain almost unchanged.

For the values of $\varepsilon = 0.1$ and $\tilde{\varepsilon} = 0.004$, the size of the (ε, q) -classical rounding in Proposition 4.5.9 is $k = 12 \cdot 2^{2q+2m_0}$. Then, for the above energies, Theorem 4.5.11 can be restated as follows.

Corollary 4.6.1. *Let $\varepsilon = 0.1$, $t = 1$, $u = 0$. Let $n > 2(m_0 + 5)$. Let the number of qubits that Alice and Bob each control at the beginning of the protocol be*

$$q \leq \frac{n}{2} - m_0 - 5. \quad (4.115)$$

Then, with probability $1 - O(\lambda^{2m_0})$ the following holds. A random function f fulfils the following with probability at least $1 - 2^{-2^n}$: the uncertainties for Alice and Bob when attacking the protocol $\text{QPV}_{\text{coh}}^f$ are such that

$$\max\{h(U_\theta|PAB_c)_\phi, h(U_\theta|A_cB)_\phi\} \geq h(U|P)_\psi + \frac{\varepsilon}{4}, \quad (4.116)$$

for every state $|\phi\rangle \in \mathbb{C}^{2q+2m_0}$, for $\theta \in \{0, \frac{\pi}{2}\}$.

4.6.2. Imperfect channel

We do the analysis for an imperfect channel for some (u, t) such that the condition (4.88) is fulfilled. We pick the parameters plotted in Fig. 4.6. For the values of ε , t and u in Table 4.1, we find the maximum $\tilde{\varepsilon}$ for $E = 10^3$, and we have the same linear bounds as in Corollary 4.6.1.

ε	t	u	α	$\tilde{\varepsilon}$
0.03	0.8	0.05	0.013	0.00031
0.03	0.9	0.12	0.013	0.00029
0.07	0.95	0.075	0.025	0.00131

Table 4.1: Maximum value of $\tilde{\varepsilon}$ fulfilling (4.86) given ε , t , and u , with the corresponding value of α that attains it.

4.7. Discussion

The aim of this chapter was to investigate whether quantum position verification can be carried out in a meaningful and provably secure way using continuous-variable quantum states. The overall strategy was to begin with the simplest natural CV construction

and understand what can already be proved there, and then to move to a more structured protocol in order to obtain security against attackers with pre-shared entanglement. The first protocol (QPV_{coh}) establishes the basic feasibility of CV-QPV, while the second ($\text{QPV}_{\text{coh}}^f$) shows how additional protocol structure can be used to recover the kind of bounded-entanglement security that has made discrete-variable QPV particularly interesting.

A central lesson of the chapter is that the security analysis of CV-QPV differs in an essential way from the discrete-variable case. In the discrete setting, the honest prover is typically expected to return the correct answer, and security amounts to showing that attackers necessarily introduce too many errors. In the continuous-variable setting, however, the honest prover already responds with a sample from a probability distribution. Here the relevant question becomes whether they can reproduce the honest distribution closely enough to pass the verifiers' statistical test. For the QPV_{coh} protocol, this meant proving that attackers without pre-shared entanglement necessarily generate a distribution that differs from the honest one, and that the verifiers can distinguish the two. We showed that this distinguishing can indeed be done by combining an entropic uncertainty relation for differential entropy with a continuum version of Fano's inequality.

An important aspect of this first result is that it was obtained in a physically non-ideal setting. The analysis explicitly included attenuation and excess noise in the honest channel, and showed that the protocol remains secure for a small but realistic range of such parameters. In that sense, the proof technique was strong enough to address the more relevant question of whether security survives under imperfect channel conditions. At the same time, the first protocol also makes clear where the limitations lie: we showed that it can be broken if the attackers pre-share a single strongly entangled CV-EPR pair. Thus, the initial construction succeeds in establishing the possibility of CV-QPV in the weakest adversarial model considered here, but it also sharply reveals that this is not yet enough to withstand shared entanglement. This naturally motivates the second part of the chapter.

If the most direct CV protocol becomes insecure once the attackers are given even a modest amount of pre-shared entanglement, then stronger security must come from more structure in the protocol itself. This is precisely what the split-input construction achieves. By adapting the $\text{QPV}_{\text{BB84}}^f$ idea to the continuous-variable setting, the protocol forces the prover's action to depend on classical information distributed across the verifiers, thereby making the attackers' task more (quantum) resource-intensive. This is a very favourable trade-off. With present technology, it is much easier for the verifiers to sample a longer bit string (classical information) than for the adversaries to share entanglement. This means that the favourable resource-scaling phenomenon known from the discrete-variable setting persists in the CV case as well. Here too, the proof had to be phrased in genuinely

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continuous-variable terms: the point is to show that the attackers retain enough uncertainty about the requested quadrature value that their response distribution necessarily differs from that of the honest prover. Moreover, this result was again obtained in the presence of channel imperfections. Thus, both protocols studied in the chapter are secure within suitable regimes under imperfect channel conditions.

5

Unitary two-designs

Mathematics is the art of giving the same name to different things.

— Jules Henri Poincaré

Random unitaries drawn from the Haar measure on the unitary group play a central role in quantum information theory, because they provide a mathematically clean notion of generic dynamics and enable uniform averages that underlie a wide range of tasks, including benchmarking, tomography, decoupling, and various average-case statements in quantum information processing [144, 56]. In practice, however, Haar-random unitaries are rarely available as a physical primitive, and even sampling them exactly becomes computationally expensive in high dimensions [144]. A standard remedy is therefore to replace Haar randomness by *unitary designs*: structured ensembles of unitaries that reproduce Haar averages up to some fixed integer degree. In many applications, matching only the first few moments is already sufficient, so low-order designs provide a useful bridge between idealised Haar arguments and more structured, potentially implementable randomisation schemes [56].

One can define unitary designs through the t -fold twirling channel. Let μ be a probability measure on $U(d)$, the group of unitary operators on a d -dimensional Hilbert space. The associated t -fold twirl is the channel

$$\mathcal{G}_\mu^{(t)}(X) = \int_{U(d)} U^{\otimes t} X (U^\dagger)^{\otimes t} d\mu(U), \quad X \in \mathcal{L}((\mathbb{C}^d)^{\otimes t}). \quad (5.1)$$

Here $\mathcal{L}$ denotes the space of linear operators. The measure μ is called an exact *unitary t -design* if $\mathcal{G}_\mu^{(t)}$ coincides with the Haar twirl $\mathcal{G}_{\text{Haar}}^{(t)}$. Equivalently, all balanced polynomials¹

¹A balanced polynomial is a polynomial that contains the same number of matrix entries of U and

in the matrix entries of U of degree at most t have the same average under μ as under Haar [56, 50]. Among these notions, the case $t = 2$ is especially important. A unitary two-design captures second-moment Haar statistics and is therefore the lowest-order notion of pseudorandomness that already supports a substantial range of average-case constructions and estimation tasks [56, 35, 51].

In finite dimensions, unitary designs are by now a mature subject. Exact constructions are known in several settings, and approximate constructions are especially useful because they often admit efficient implementations while retaining the properties needed in applications [56, 50, 95]. A particularly simple and influential example is the alternating construction of Nakata *et al.* [149], where one alternates random unitaries diagonal in the Pauli- Z basis with random unitaries diagonal in the Pauli- X basis. Repetition of this procedure yields an efficient approximate unitary two-design, with the approximation error decaying polynomially in the dimension.

These developments naturally raise the question whether analogous notions and constructions exist for continuous-variable quantum systems.

Towards continuous-variable unitary two-designs

Continuous-variable (CV) quantum systems differ from their finite-dimensional counterparts in a fundamental way: they are modelled on separable infinite-dimensional Hilbert spaces, such as $L^2(\mathbb{R})$ for a single bosonic mode. Because of this, several tools that are straightforward in finite dimensions no longer carry over directly. In particular, there is no direct analogue of the finite-dimensional Haar measure that gives a finite, well-behaved notion of uniform averaging over all unitaries on the full CV Hilbert space. Thus, before even asking for a CV unitary two-design, one must first ask what the right formulation of such a notion should be.

A natural first guess is to look at Gaussian states and Gaussian unitaries, since these occupy a role in CV somewhat analogous to that of stabiliser states and Clifford unitaries in finite dimensions. This hope turns out to be too optimistic. Blume-Kohout and Turner [35] showed that Gaussian states cannot form a CV two-design on $L^2(\mathbb{R})$, and more broadly that the relevant invariant averages fail to produce a normalisable object on the full infinite-dimensional space. Thus, the most accessible and experimentally familiar family of CV operations already fails to reproduce the required second-moment Haar behaviour.

A related perspective was developed by Zhuang *et al.* in their study of scrambling and pseudorandomness in CV phase space [204]. Their analysis makes explicit that the phase

complex-conjugated matrix entries of U . For example, a degree- t balanced polynomial is built from monomials containing t factors of the form U_{ij} and t factors of the form $\bar{U}_{kl}$.

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space has infinite volume, so even ensembles that look like natural CV analogues of uniform finite-dimensional ones, such as displacements, are not normalisable without some form of regularisation. They showed that strongly squeezed random Gaussian circuits can exhibit signatures reminiscent of higher-design behaviour, but still do not give an exact CV unitary two-design. This points to a recurring theme: in the continuous-variable setting, any meaningful notion of design-like randomness must involve some restriction, cutoff, or regularised notion of approximation.

This viewpoint has been sharpened considerably by the more recent no-go results of Iosue *et al.* [105]. They proved that, under natural infinite-dimensional extensions of the standard finite-dimensional definitions, no exact CV state t -designs exist for any $t \geq 2$, and likewise no exact CV unitary t -designs exist for any $t \geq 2$ on separable infinite-dimensional Hilbert spaces. At the same time, their work also suggests the correct direction forward: rather than insisting on exact equality on the full infinite-dimensional space, one should look for regularised or approximate notions that respect the physical fact that any realistic CV implementation necessarily involves finite energy and finite resolution.

This is precisely the perspective adopted in this chapter. Rather than searching for an exact CV analogue of the Haar measure on the full Hilbert space, we ask whether one can meaningfully reproduce two-design behaviour on a controlled finite-dimensional shadow of a CV system. More specifically, the goal is to identify a mathematically controlled regime in which finite-dimensional design ideas can be translated into the CV language and which tends to the full CV Hilbert space under certain limits.

Outline of this chapter

As discussed, the main objective of this chapter is to develop an explicit approximate unitary two-design construction that is naturally adapted to continuous-variable systems. The basic strategy is to impose a physically motivated discretisation of quadrature space, thereby obtaining a finite-dimensional effective Hilbert space on which one can formulate and analyse design properties. To that end, we introduce a discretisation of the CV one-mode Hilbert space, by discretising the q -quadrature space into d ‘tiles’ of size Δ . This effectively replaces a state $\rho(q, q')$ by a piecewise constant approximation. Furthermore, we impose a cutoff $q_{\max} \propto 1/\Delta$. Within this discretised setting, the construction is built from unitaries diagonal in the discretised position and momentum bases, in clear analogy with the alternating complementary-basis constructions known from the finite-dimensional case [149].

The remainder of the chapter is organised as follows. In Section 5.1 we collect the necessary preliminaries on norms, twirling channels, and approximate designs. In Section 5.2 we introduce the discretised CV Hilbert space, and in Section 5.3 we construct the approx-

imate unitary two-design. In Section 5.4 we discuss how the required unitaries relate to physically meaningful CV operations and what the main implementation obstacles are. We conclude in Section 5.5 with a short discussion.

5.1. Preliminaries

5.1.1. Hilbert spaces; operators; norms

This chapter uses a variety of linear algebraic quantities not used elsewhere. Therefore, we attempt to provide a comprehensive list of tools and notation here. Like the rest of the thesis, we use Dirac notation for quantum states, and $\mathcal{H}_d$ stands for a Hilbert space (of dimension d). The $k \times k$ identity matrix is written as I_k . The set of linear operators on a vector space $\mathcal{H}$ is denoted by $\mathcal{L}(\mathcal{H})$; the set of bounded operators is $\mathcal{B}(\mathcal{H})$. The set of density matrices on a Hilbert space $\mathcal{H}$ is written as $\mathcal{D}(\mathcal{H})$. A state $\rho \in \mathcal{D}(\mathcal{H})$ satisfies $\rho \geq 0$, $\rho^\dagger = \rho$, and $\text{Tr } \rho = 1$. The fully mixed state of a register A is denoted by τ^A . The Hilbert-Schmidt inner product on operators is $\langle X, Y \rangle = \text{Tr } X^\dagger Y$. The Schatten p -norm ($p \geq 1$) of an operator M is $\|M\|_p = (\text{Tr } |M|^p)^{1/p}$, with $|M| = \sqrt{M^\dagger M}$. The trace norm is $\|M\|_{\text{tr}} = \frac{1}{2}\|M\|_1$. A superoperator maps operators to operators, potentially on a different space. The most important superoperators for quantum physics are the completely positive trace-preserving (CPTP) maps. A CPTP map $\Phi : \mathcal{B}(\mathcal{H}_A) \rightarrow \mathcal{B}(\mathcal{H}_B)$ is a linear map that satisfies $\text{Tr } \Phi(\rho) = \text{Tr } \rho$ and $\forall_{k \in \mathbb{N}} \forall_{\rho \geq 0} (\Phi \otimes \text{id}_k)(\rho) \geq 0$, where id_k is the identity map on a k -dimensional Hilbert space. All quantum channels are CPTP maps. Several norms exist for superoperators. The ' $q \rightarrow p$ ' norm is defined as $\|\Phi\|_{q \rightarrow p} = \sup_{X \neq 0} \frac{\|\Phi(X)\|_p}{\|X\|_q}$. The diamond norm is defined as $\|\Phi\|_{\diamond} = \sup_k \|\Phi \otimes \text{id}_k\|_{1 \rightarrow 1}$.

5.1.2. Structure of approximate unitary 2-designs

Definition 5.1.1 (Approximate unitary 2-design). *Let μ be a measure on $U(d)$, the space of unitary operators acting on $\mathcal{H}_d$. Let $\mathcal{G}_\mu^{(2)}$ be the two-fold twirl map as defined by (5.1), with $t=2$, and let $\mathcal{G}_{\text{Haar}}^{(2)}$ be the corresponding map using the Haar measure. The measure μ is called an ε -approximate unitary 2-design if*

$$\left\| \mathcal{G}_\mu^{(2)} - \mathcal{G}_{\text{Haar}}^{(2)} \right\|_{2 \rightarrow 2} \leq \varepsilon. \quad (5.2)$$

A useful finite-dimensional construction of approximate unitary 2-designs was given by Nakata *et al.* [149]. Their construction uses two families of random unitaries: $U_Z(\varphi)$, which are diagonal in the computational basis, and $U_X(\chi)$, which are diagonal in the Hadamard basis. The parameters φ and χ are sampled uniformly from a suitable discrete

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subset of $[0, 2\pi)^d$, where $d = 2^n$. The corresponding two-fold twirl maps are denoted by $\mathcal{G}_Z$ and $\mathcal{G}_X$, and one considers the compound channel

$$\mathcal{R} = \mathcal{G}_Z \circ \mathcal{G}_X \circ \mathcal{G}_Z. \quad (5.3)$$

This is itself a two-fold twirl, corresponding to random unitaries of the form

$$U_Z(\varphi') U_X(\chi) U_Z(\varphi), \quad (5.4)$$

where φ', χ, φ are sampled independently. Repeating this procedure ℓ times yields the channel $\mathcal{R}^\ell$, which corresponds to random products of the form

$$U_Z(\varphi'_\ell) U_X(\chi_\ell) U_Z(\varphi_\ell) \cdots U_Z(\varphi'_1) U_X(\chi_1) U_Z(\varphi_1). \quad (5.5)$$

Nakata *et al.* showed that this construction is an ε -approximate unitary 2-design, with ε decaying as $d^{-\ell}$ [149].

The mechanism behind this construction becomes much clearer once one decomposes the operator space into the part that is already Haar-invariant and the complementary part that must be suppressed. Heuristically, one may think of $\mathcal{G}_Z$ and $\mathcal{G}_X$ as non-commuting projections: each preserves certain components of an operator and removes others, and alternating them progressively contracts everything except the Haar-invariant sector.

To make this precise, recall that the study of unitary t -designs is closely tied to the representation theory of the symmetric group S_t [50]. In the present case $t = 2$, so the relevant permutation group is S_2 . Let $\{|a\rangle\}$ be an orthonormal basis of $\mathcal{H}_d$, and write $|ab\rangle = |a\rangle \otimes |b\rangle$. On $\mathcal{H}_d \otimes \mathcal{H}_d$ we define the operators

$$I = \sum_{a,b} |ab\rangle \langle ab|, \quad F = \sum_{a,b} |ab\rangle \langle ba|. \quad (5.6)$$

Here I is the identity operator on $\mathcal{H}_d \otimes \mathcal{H}_d$, hence a $d^2 \times d^2$ identity matrix, and F is the swap operator. One immediately checks that $F^2 = I$.

Definition 5.1.2. On the space $\mathcal{L}(\mathcal{H}_d \otimes \mathcal{H}_d)$ we use the Hilbert–Schmidt inner product $\langle X, Y \rangle = \text{Tr}(X^\dagger Y)$. We then define

$$\mathcal{A} = \text{span}\{I, F\}, \quad \mathcal{K} = \mathcal{A}^\perp. \quad (5.7)$$

Accordingly, every operator $X \in \mathcal{L}(\mathcal{H}_d \otimes \mathcal{H}_d)$ has a unique orthogonal decomposition

$$X = X_{\mathcal{A}} + X_{\mathcal{K}}, \quad X_{\mathcal{A}} \in \mathcal{A}, \quad X_{\mathcal{K}} \in \mathcal{K}. \quad (5.8)$$

The significance of this decomposition is that the action of any two-fold twirl is trivial on $\mathcal{A}$, while all nontrivial contraction takes place on $\mathcal{K}$.

Lemma 5.1.3. *Let μ be any measure on $\mathbf{U}(d)$. Then the following statements hold.*

1. *The operators I and F are fixed points of the two-fold twirl: $\mathcal{G}_\mu^{(2)}(I) = I$, $\mathcal{G}_\mu^{(2)}(F) = F$.*
2. *The subspace $\mathcal{A}$ is invariant under $\mathcal{G}_\mu^{(2)}$.*
3. *The subspace $\mathcal{K}$ is invariant under $\mathcal{G}_\mu^{(2)}$.*
4. *For every $X = X_{\mathcal{A}} + X_{\mathcal{K}}$ with respect to the decomposition of Definition 5.1.2,*

$$\mathcal{G}_\mu^{(2)}(X) = X_{\mathcal{A}} + \mathcal{G}_\mu^{(2)}(X_{\mathcal{K}}), \quad \mathcal{G}_\mu^{(2)}(X_{\mathcal{K}}) \in \mathcal{K}. \quad (5.9)$$

Proof. We prove the statements one by one.

Proof of (1). By definition of the two-fold twirl

$$\mathcal{G}_\mu^{(2)}(F) = \int_{\mathbf{U}(d)} U^{\otimes 2} F (U^\dagger)^{\otimes 2} d\mu(U). \quad (5.10)$$

Since F swaps the two tensor factors, it commutes with $U \otimes U$ for every $U \in \mathbf{U}(d)$, and therefore $U^{\otimes 2} F (U^\dagger)^{\otimes 2} = F$. Hence $\mathcal{G}_\mu^{(2)}(F) = F$. The case for I is trivial.

Proof of (2). Every $X_{\mathcal{A}} \in \mathcal{A}$ can be written as

$$X_{\mathcal{A}} = \alpha I + \beta F \quad (5.11)$$

for some $\alpha, \beta \in \mathbb{C}$. By linearity of $\mathcal{G}_\mu^{(2)}$ and part (1),

$$\mathcal{G}_\mu^{(2)}(X_{\mathcal{A}}) = \alpha \mathcal{G}_\mu^{(2)}(I) + \beta \mathcal{G}_\mu^{(2)}(F) = \alpha I + \beta F = X_{\mathcal{A}}. \quad (5.12)$$

In particular, $\mathcal{G}_\mu^{(2)}(X_{\mathcal{A}}) \in \mathcal{A}$, so $\mathcal{A}$ is invariant.

Proof of (3). Let $Y \in \mathcal{K}$. By definition of $\mathcal{K} = \mathcal{A}^\perp$, we have

$$\langle I, Y \rangle = 0, \quad \langle F, Y \rangle = 0. \quad (5.13)$$

We show that the same holds for $\mathcal{G}_\mu^{(2)}(Y)$. First,

$$\langle I, \mathcal{G}_\mu^{(2)}(Y) \rangle = \text{Tr}(\mathcal{G}_\mu^{(2)}(Y)) = \int_{\mathbf{U}(d)} \text{Tr}(U^{\otimes 2} Y (U^\dagger)^{\otimes 2}) d\mu(U) = \text{Tr}(Y) = \langle I, Y \rangle = 0. \quad (5.14)$$

Next,

$$\langle F, \mathcal{G}_\mu^{(2)}(Y) \rangle = \int_{\mathbf{U}(d)} \text{Tr}(F U^{\otimes 2} Y (U^\dagger)^{\otimes 2}) d\mu(U). \quad (5.15)$$

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Using the commutation relation $F(U \otimes U) = (U \otimes U)F$ and cyclicity of the trace,

$$\mathrm{Tr}(F U^{\otimes 2} Y (U^\dagger)^{\otimes 2}) = \mathrm{Tr}(U^{\otimes 2} F Y (U^\dagger)^{\otimes 2}) = \mathrm{Tr}(F Y). \quad (5.16)$$

Therefore,

$$\langle F, \mathcal{G}_\mu^{(2)}(Y) \rangle = \mathrm{Tr}(F Y) = \langle F, Y \rangle = 0. \quad (5.17)$$

So $\mathcal{G}_\mu^{(2)}(Y)$ is orthogonal to both I and F , and hence belongs to $\mathcal{K}$.

Proof of (4). Let $X = X_{\mathcal{A}} + X_{\mathcal{K}}$ be the orthogonal decomposition from Definition 5.1.2. By linearity,

$$\mathcal{G}_\mu^{(2)}(X) = \mathcal{G}_\mu^{(2)}(X_{\mathcal{A}}) + \mathcal{G}_\mu^{(2)}(X_{\mathcal{K}}). \quad (5.18)$$

Part (2) gives $\mathcal{G}_\mu^{(2)}(X_{\mathcal{A}}) = X_{\mathcal{A}}$, while part (3) gives $\mathcal{G}_\mu^{(2)}(X_{\mathcal{K}}) \in \mathcal{K}$. Hence

$$\mathcal{G}_\mu^{(2)}(X) = X_{\mathcal{A}} + \mathcal{G}_\mu^{(2)}(X_{\mathcal{K}}), \quad \mathcal{G}_\mu^{(2)}(X_{\mathcal{K}}) \in \mathcal{K}. \quad (5.19)$$

This proves the claim. $\square$

The previous lemma shows that every two-fold twirl acts trivially on $\mathcal{A}$. To compare a given ensemble with the Haar measure, we now need one further fact: for $t = 2$, the Haar twirl keeps exactly the $\mathcal{A}$ -component and removes the $\mathcal{K}$ -component.

Lemma 5.1.4. *For every $X \in \mathcal{L}(\mathcal{H}_d \otimes \mathcal{H}_d)$, $\mathcal{G}_{\mathrm{Haar}}^{(2)}(X) = X_{\mathcal{A}}$.*

Proof. By Schur–Weyl duality, the image of the Haar twirl on $\mathcal{H}_d^{\otimes 2}$ is the commutant of the representation $U \mapsto U^{\otimes 2}$, which is precisely $\mathrm{span}\{I, F\} = \mathcal{A}$ [50]. Hence $\mathcal{G}_{\mathrm{Haar}}^{(2)}(X) \in \mathcal{A}$ for every X . Now write $X = X_{\mathcal{A}} + X_{\mathcal{K}}$. By Lemma 5.1.3, the Haar twirl fixes $\mathcal{A}$ pointwise and preserves $\mathcal{K}$. Therefore,

$$\mathcal{G}_{\mathrm{Haar}}^{(2)}(X) = \mathcal{G}_{\mathrm{Haar}}^{(2)}(X_{\mathcal{A}}) + \mathcal{G}_{\mathrm{Haar}}^{(2)}(X_{\mathcal{K}}) = X_{\mathcal{A}} + \mathcal{G}_{\mathrm{Haar}}^{(2)}(X_{\mathcal{K}}), \quad (5.20)$$

with $\mathcal{G}_{\mathrm{Haar}}^{(2)}(X_{\mathcal{K}}) \in \mathcal{K}$. But we also know that $\mathcal{G}_{\mathrm{Haar}}^{(2)}(X_{\mathcal{K}}) \in \mathcal{A}$, since the image of the Haar twirl lies in $\mathcal{A}$. Thus

$$\mathcal{G}_{\mathrm{Haar}}^{(2)}(X_{\mathcal{K}}) \in \mathcal{A} \cap \mathcal{K} = \{0\}, \quad (5.21)$$

which implies $\mathcal{G}_{\mathrm{Haar}}^{(2)}(X) = X_{\mathcal{A}}$. $\square$

Thus, in order to show that an ensemble forms an approximate unitary 2-design, it is enough to understand how strongly its two-fold twirl contracts the complementary subspace $\mathcal{K}$. This is precisely the content of the following corollary.

Corollary 5.1.5. *Let $\mathcal{G}_{\mu, \mathcal{K}}^{(2)} : \mathcal{K} \rightarrow \mathcal{K}$ denote the mapping $\mathcal{G}_\mu^{(2)}$ restricted to the space $\mathcal{K}$. If $\|\mathcal{G}_{\mu, \mathcal{K}}^{(2)}\|_{2 \rightarrow 2} \leq \varepsilon$, then μ is an ε -approximate unitary two-design.*

5.2. Discrete phase space: Boxed operator formalism

In order to formulate a unitary design construction for a continuous-variable system, we first need a finite-dimensional approximation of the one-mode Hilbert space. The aim is not to replace the CV system by an unrelated discrete model, but to extract from it a d -dimensional subspace that still retains the basic quadrature structure. To do this, we partition the q axis into intervals of width Δ , associate to each interval a normalised state obtained by averaging the quadrature eigenstates over that interval, and then use the discrete Fourier transform to define the complementary basis. This gives a finite-dimensional space on which discrete analogues of the q - and p -quadrature operators can be defined.

We begin with a one-mode CV density operator written in the q basis as

$$\hat{\rho} = \int_{\mathbb{R}} dx \int_{\mathbb{R}} dx' \rho(x, x') |x\rangle\langle x'|. \quad (5.22)$$

We will be interested in states whose kernel is sufficiently regular on the scale Δ and whose weight is essentially concentrated in a finite region of phase space. More precisely, we say that a density operator $\hat{\rho}$ is *compatible with the discretisation scale* Δ if its kernel $\rho(x, x')$ satisfies the following two properties:

1. the derivatives

$$\frac{\partial}{\partial x} \rho(x, x'), \quad \frac{\partial}{\partial x'} \rho(x, x') \quad (5.23)$$

are bounded in absolute value;

2. there exists a cutoff $q_{\max} > 0$ such that

$$\int_{|x|+|x'|>q_{\max}} dx dx' |\rho(x, x')| \ll 1. \quad (5.24)$$

The first condition ensures that the kernel can be approximated as nearly constant on sufficiently small intervals of width Δ . The second condition ensures that only finitely many such intervals are needed. We therefore restrict attention to a finite set of bins covering the relevant region of the q axis.

Throughout this chapter we take d to be even (without loss of generality), and define $\mathcal{I}_d = \{-\frac{d}{2}, \dots, \frac{d}{2} - 1\}$. We then choose $\Delta = \sqrt{\frac{2\pi}{d}}$, so that the total covered range is of order

$$q_{\max} \approx \frac{d}{2} \Delta = \sqrt{\frac{\pi d}{2}}. \quad (5.25)$$

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Definition 5.2.1 (Box kets). For each $i \in \mathcal{I}_d$, we define the q -boxed ket

$$|i\rangle \stackrel{\text{def}}{=} \frac{1}{\sqrt{\Delta}} \int_0^\Delta dx |q = i\Delta + x\rangle. \quad (5.26)$$

These states represent the intervals $[i\Delta, (i+1)\Delta)$.

The terminology “boxed ket” reflects the fact that each state is obtained by averaging the quadrature eigenstates over a small box of width Δ . The normalisation factor is chosen so that these states become orthonormal.

Proposition 5.2.2. The family $\{|i\rangle\}_{i \in \mathcal{I}_d}$ is orthonormal: $\langle i|j\rangle = \delta_{ij}$.

Proof. Using the definition of the boxed kets and the relation $\langle q|q'\rangle = \delta(q - q')$, we obtain

$$\begin{aligned} \langle i|j\rangle &= \frac{1}{\Delta} \int_0^\Delta dx \int_0^\Delta dy \langle q = i\Delta + x|q = j\Delta + y\rangle \\ &= \frac{1}{\Delta} \int_0^\Delta dx \int_0^\Delta dy \delta((i-j)\Delta + x - y). \end{aligned} \quad (5.27)$$

If $i \neq j$, then $(i-j)\Delta + x - y$ cannot vanish for $x, y \in [0, \Delta)$, except possibly on a boundary set of measure zero, so the integral is zero. If $i = j$, then

$$\langle i|i\rangle = \frac{1}{\Delta} \int_0^\Delta dx \int_0^\Delta dy \delta(x - y) = \frac{1}{\Delta} \int_0^\Delta dx = 1. \quad (5.28)$$

Hence $\langle i|j\rangle = \delta_{ij}$. □

The boxed kets therefore span a d -dimensional Hilbert space.

Definition 5.2.3 (Discretised Hilbert space). We define

$$\mathcal{H}_d \stackrel{\text{def}}{=} \text{span}\{|i\rangle : i \in \mathcal{I}_d\}. \quad (5.29)$$

The orthogonal projector onto this subspace is

$$\Pi \stackrel{\text{def}}{=} \sum_{i \in \mathcal{I}_d} |i\rangle\langle i|. \quad (5.30)$$

With this discretisation, the projected state $\Pi\hat{\rho}\Pi$ can be written as

$$\Pi\hat{\rho}\Pi = \sum_{i, i' \in \mathcal{I}_d} \rho_i^{i'} |i\rangle\langle i'|, \quad (5.31)$$

where the matrix elements are given by bin averages of the continuous kernel,

$$\rho_i^{i'} = \langle i | \hat{\rho} | i' \rangle = \frac{1}{\Delta} \int_0^\Delta dx \int_0^\Delta dx' \rho(i\Delta + x, i'\Delta + x'). \quad (5.32)$$

Thus the discretised state keeps only the coarse-grained information of the original kernel on the chosen grid.

We now define the complementary basis that will play the role of the discretised p basis. As in finite dimension, the natural choice is the discrete Fourier transform of the q basis.

Definition 5.2.4 (Fourier-conjugate basis). *Let $\omega = \exp(i\frac{2\pi}{d})$. For each $k \in \mathcal{I}_d$, define*

$$|\widetilde{k}\rangle \stackrel{\text{def}}{=} \frac{1}{\sqrt{d}} \sum_{j \in \mathcal{I}_d} \omega^{kj} |j\rangle. \quad (5.33)$$

These states form another orthonormal basis of $\mathcal{H}_d$.

Proposition 5.2.5. *The family $\{|\widetilde{k}\rangle\}_{k \in \mathcal{I}_d}$ is orthonormal: $\langle \widetilde{k'} | \widetilde{k} \rangle = \delta_{k'k}$. Moreover,*

$$\Pi = \sum_{i \in \mathcal{I}_d} |i\rangle \langle i| = \sum_{k \in \mathcal{I}_d} |\widetilde{k}\rangle \langle \widetilde{k}|. \quad (5.34)$$

Proof. Using Definition 5.2.4 and Proposition 5.2.2,

$$\begin{aligned} \langle \widetilde{k'} | \widetilde{k} \rangle &= \frac{1}{d} \sum_{j, j' \in \mathcal{I}_d} \omega^{-k'j'} \omega^{kj} \langle j' | j \rangle \\ &= \frac{1}{d} \sum_{j \in \mathcal{I}_d} \omega^{(k-k')j}. \end{aligned} \quad (5.35)$$

This is the standard finite Fourier sum, which equals 1 if $k = k'$ and 0 otherwise. Hence

$$\langle \widetilde{k'} | \widetilde{k} \rangle = \delta_{k'k}. \quad (5.36)$$

Since there are d orthonormal vectors in the d -dimensional space $\mathcal{H}_d$, they form a basis, and the second identity in (5.34) follows. $\square$

We now introduce the discrete quadrature operators. These are not the original CV quadrature operators themselves, but their coarse-grained counterparts acting on the discretised Hilbert space.

Definition 5.2.6 (Discrete quadrature operators). *On $\mathcal{H}_d$ we define*

$$\hat{Q} \stackrel{\text{def}}{=} \sum_{i \in \mathcal{I}_d} i |i\rangle \langle i|, \quad \hat{P} \stackrel{\text{def}}{=} \sum_{k \in \mathcal{I}_d} k |\widetilde{k}\rangle \langle \widetilde{k}|. \quad (5.37)$$

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Thus $\hat{Q}$ and $\hat{P}$ return the integer labels of the q - and Fourier-conjugate bins, respectively. The relation between $\hat{Q}$ and the original CV position operator is obtained by projection.

Proposition 5.2.7. *Let $\hat{q} = \int_{-\infty}^{\infty} dq \, q |q\rangle\langle q|$ be the continuous position quadrature operator. Then*

$$\Pi \hat{q} \Pi = \Delta \sum_{j \in \mathcal{I}_d} \left(j + \frac{1}{2} \right) |j\rangle\langle j| = \Delta \hat{Q} + \frac{\Delta}{2} \Pi. \quad (5.38)$$

Proof. Starting from (5.30) and (5.26),

$$\Pi \hat{q} \Pi = \sum_{j \in \mathcal{I}_d} \int_0^\Delta dx \, (j\Delta + x) \Pi |q = j\Delta + x\rangle\langle q = j\Delta + x| \Pi. \quad (5.39)$$

Now

$$\Pi |q = j\Delta + x\rangle = \sum_{m \in \mathcal{I}_d} |m\rangle\langle m|q = j\Delta + x\rangle. \quad (5.40)$$

Since the state $|q = j\Delta + x\rangle$ lies in the j th interval, only the term $m = j$ contributes, and from (5.26) one gets

$$\Pi |q = j\Delta + x\rangle = \frac{1}{\sqrt{\Delta}} |j\rangle. \quad (5.41)$$

Hence

$$\Pi |q = j\Delta + x\rangle\langle q = j\Delta + x| \Pi = \frac{1}{\Delta} |j\rangle\langle j|. \quad (5.42)$$

Substituting this back gives

$$\begin{aligned} \Pi \hat{q} \Pi &= \sum_{j \in \mathcal{I}_d} \int_0^\Delta dx \, (j\Delta + x) \frac{1}{\Delta} |j\rangle\langle j| = \sum_{j \in \mathcal{I}_d} \left(\frac{1}{\Delta} \int_0^\Delta dx \, (j\Delta + x) \right) |j\rangle\langle j| \\ &= \sum_{j \in \mathcal{I}_d} \left(j\Delta + \frac{\Delta}{2} \right) |j\rangle\langle j| = \Delta \sum_{j \in \mathcal{I}_d} j |j\rangle\langle j| + \frac{\Delta}{2} \sum_{j \in \mathcal{I}_d} |j\rangle\langle j|. \end{aligned} \quad (5.43)$$

Using (5.30) and (5.37), this becomes

$$\Pi \hat{q} \Pi = \Delta \hat{Q} + \frac{\Delta}{2} \Pi. \quad (5.44)$$

□

Equation (5.38) shows that $\hat{Q}$ is the projected position operator measured in units of the bin size, up to the constant shift $\Delta/2$ coming from the fact that the interval associated with $|j\rangle$ is $[j\Delta, (j+1)\Delta)$ rather than being centred at $j\Delta$. The operator $\hat{P}$ plays the analogous role in the Fourier-conjugate basis and will be the natural discrete momentum operator used in the design construction below.

We now show that the error made in the transition from a state ρ to its discretised version is related to the expected energy of the state divided by the number of discretisation intervals. To prove this, we require the Gentle Measurement Lemma.

Lemma 5.2.8 (Gentle Measurement Lemma [196, 1]). *Let Λ be a measurement operator. Consider a state ρ such that $\text{Tr } \Lambda \rho \geq 1 - \varepsilon$. Then the post-measurement state $\rho' \equiv \frac{\sqrt{\Lambda} \rho \sqrt{\Lambda}}{\text{Tr}\{\Lambda \rho\}}$ is $\sqrt{\varepsilon}$ -close to the original state ρ in trace distance: $\|\rho - \rho'\|_{\text{tr}} \leq \sqrt{\varepsilon}$.*

In other words, the Gentle Measurement Lemma states that if a quantum measurement yields a specific, expected outcome with high probability, the quantum state is not significantly disturbed (collapsed) by that measurement. Now we are ready to prove our result.

Lemma 5.2.9. *Let ρ be a density matrix with matrix elements $\rho(q, q') = \langle q | \rho | q' \rangle$. Let $\rho(q, q')$ be negligible outside of the interval $|q| < \sqrt{\pi d/2} \wedge |q'| < \sqrt{\pi d/2}$.*

Let $\rho_{\text{discr}} = \frac{\Pi \rho \Pi^\dagger}{\text{Tr}(\Pi \rho \Pi^\dagger)}$ be the discretised version of ρ . The trace distance between the discretised and the original state is upper bounded as

$$\|\rho_{\text{discr}} - \rho\|_{\text{tr}} \leq \sqrt{\frac{2}{\pi} \cdot \frac{\text{Tr } \hat{p}^2 \rho}{d}}, \quad (5.45)$$

where $\hat{p}$ is the CV p -quadrature operator.

Proof. We first consider a pure state $|\psi\rangle = \int dq |q\rangle \psi(q)$. Let $x \in [0, \Delta)$. We note that $\langle i | q = i\Delta + x \rangle = \frac{1}{\sqrt{\Delta}}$ and $\Pi |q = i\Delta + x\rangle = \frac{1}{\sqrt{\Delta}} |i\rangle$. Projection of $|\psi\rangle$ gives $\Pi |\psi\rangle = \sum_j \int_0^\Delta dx \frac{|j\rangle}{\sqrt{\Delta}} \psi(j\Delta + x) = \sum_j |j\rangle \bar{\psi}_j \sqrt{\Delta}$, where we have defined $\bar{\psi}_j = \frac{1}{\Delta^2} \int_0^\Delta dx \psi(j\Delta + x)$. This yields

$$\langle \psi | \Pi^\dagger \Pi | \psi \rangle = \sum_j |\bar{\psi}_j|^2 \Delta. \quad (5.46)$$

Clearly, $|\bar{\psi}_j|^2 \Delta$ is the probability mass in the j th box. We write

$$1 - \text{Tr } \Pi |\psi\rangle \langle \psi | \Pi^\dagger = 1 - \langle \psi | \Pi^\dagger \Pi | \psi \rangle \quad (5.47)$$

$$\stackrel{(5.46)}{=} \sum_j \int_0^\Delta dx |\psi(j\Delta + x)|^2 - \sum_j \int_0^\Delta dx |\bar{\psi}_j|^2 \quad (5.48)$$

$$= \sum_j \int_0^\Delta dx |\psi(j\Delta + x) - \bar{\psi}_j|^2 \quad (5.49)$$

$$\stackrel{(a)}{\leq} \sum_j \frac{\Delta^2}{\pi^2} \int_0^\Delta dx |\psi'(j\Delta + x)|^2 = \frac{\Delta^2}{\pi^2} \int dq \left| \frac{d\psi}{dq} \right|^2 \quad (5.50)$$

$$\stackrel{(b)}{=} \frac{\Delta^2}{\pi^2} \langle \psi | \hat{p}^2 | \psi \rangle. \quad (5.51)$$

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In (a) we used the Poincaré–Wirtinger inequality [94, 27] for a one-dimensional interval of size Δ . In (b) we used $\hat{p} = \frac{1}{i} \frac{d}{dq}$ and integration by parts, with $\psi(q)\psi'(q) = 0$ at the integration boundaries². The result (5.51) generalises straightforwardly from pure states $|\psi\rangle\langle\psi|$ to mixed states $\sum_r \lambda_r |\psi_r\rangle\langle\psi_r|$, yielding $1 - \text{Tr } \Pi \rho \Pi^\dagger \leq \frac{\Delta^2}{\pi^2} \sum_r \lambda_r \langle\psi_r|\hat{p}^2|\psi_r\rangle$. Finally, we use the Gentle Measurement Lemma (Lemma 5.2.8) and $\Delta^2 = 2\pi/d$. $\square$

Since $\frac{1}{2}\hat{p}^2 + \frac{1}{2}\hat{q}^2 = \hat{n} + \frac{1}{2}$, with $\hat{n}$ the photon number operator, we have $\text{Tr } \rho \hat{p}^2 \leq 2\bar{n} + 1$, where $\bar{n}$ is the expected photon number. This establishes a relation between the error caused by the discretisation and the energy of the state, leading to a discretisation error less than $\sqrt{\frac{4}{\pi}(\bar{n} + \frac{1}{2})/d}$. Hence, d should be chosen large enough to accommodate $\bar{n}$. Conversely, given a fixed d , this relation indicates which states are going to be approximated well.

5.3. Approximate unitary 2-design using boxed unitaries

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We now construct the family of random unitaries that will generate our approximate unitary 2-design on $\mathcal{H}_d$. The guiding idea is the same as in the finite-dimensional construction of Nakata *et al.* [149]: one alternates random unitaries that are diagonal in two complementary bases. In the present setting, however, we want both bases to have a direct quadrature interpretation. For this reason we work with the discrete quadrature operators $\hat{Q}$ and $\hat{P}$ introduced in Section 5.2, rather than with an abstract computational basis and Hadamard basis.

The goal of this section is the following. We want to define a family of random unitaries on $\mathcal{H}_d$ which

1. are naturally generated by the discretised quadrature operators $\hat{Q}$ and $\hat{P}$,
2. act as phase operators in the q -boxed basis and in the Fourier-conjugate basis, respectively,
3. and, when alternated, produce a two-fold twirl that approaches the Haar twirl as d increases.

Although the construction was motivated by the discretised CV space, once the space $\mathcal{H}_d$ and the operators $\hat{Q}, \hat{P}$ have been defined, the discussion may equally well be viewed as taking place entirely inside the d -dimensional Hilbert space $\mathcal{H}_d$.

We begin by defining the unitaries.

²Here we neglect the part of ρ outside the boundary.

Definition 5.3.1 (Quadrature phase unitaries). Let $\omega = \exp(i\frac{2\pi}{d})$, where i is the imaginary unit and not an index. For parameters $\alpha, \beta \in [0, d)$, define

$$V_{\alpha\beta} \stackrel{\text{def}}{=} \omega^{\alpha\hat{Q}+\beta\hat{Q}^2} = \exp\left(\frac{2\pi i}{d}(\alpha\hat{Q} + \beta\hat{Q}^2)\right), \quad (5.52)$$

and

$$\tilde{V}_{\alpha\beta} \stackrel{\text{def}}{=} \omega^{\alpha\hat{P}+\beta\hat{P}^2} = \exp\left(\frac{2\pi i}{d}(\alpha\hat{P} + \beta\hat{P}^2)\right). \quad (5.53)$$

The interval $[0, d)$ is chosen for convenience; any interval of length d would lead to the same averaged maps.

The next proposition records the most basic properties of these operators.

Proposition 5.3.2. Let $\alpha, \beta, \alpha', \beta' \in \mathbb{R}$. The operators $V_{\alpha\beta}$ and $\tilde{V}_{\alpha\beta}$ have the following properties.

1. They are unitary and act diagonally in the corresponding quadrature bases:

$$V_{\alpha\beta}|i\rangle = \omega^{\alpha i + \beta i^2}|i\rangle, \quad i \in \mathcal{I}_d, \quad (5.54)$$

$$\tilde{V}_{\alpha\beta}|\widetilde{k}\rangle = \omega^{\alpha k + \beta k^2}|\widetilde{k}\rangle, \quad k \in \mathcal{I}_d. \quad (5.55)$$

2. They satisfy the composition laws

$$V_{\alpha\beta}V_{\alpha'\beta'} = V_{\alpha+\alpha', \beta+\beta'}, \quad (5.56)$$

$$\tilde{V}_{\alpha\beta}\tilde{V}_{\alpha'\beta'} = \tilde{V}_{\alpha+\alpha', \beta+\beta'}, \quad (5.57)$$

and are hence also Abelian. Consequently, $V_{00} = I$, $V_{\alpha\beta}^\dagger = V_{-\alpha, -\beta}$, $\tilde{V}_{00} = I$, $\tilde{V}_{\alpha\beta}^\dagger = \tilde{V}_{-\alpha, -\beta}$.

3. Their parametrisations are periodic with period d in each variable. Namely, for every $m, n \in \mathbb{Z}$,

$$V_{\alpha+md, \beta+nd} = V_{\alpha\beta}, \quad (5.58)$$

$$\tilde{V}_{\alpha+md, \beta+nd} = \tilde{V}_{\alpha\beta}. \quad (5.59)$$

Proof. By definition,

$$V_{\alpha\beta} = \omega^{\alpha\hat{Q}+\beta\hat{Q}^2}, \quad \tilde{V}_{\alpha\beta} = \omega^{\alpha\hat{P}+\beta\hat{P}^2}.$$

Since $\hat{Q}$ and $\hat{P}$ are self-adjoint, so are $\alpha\hat{Q} + \beta\hat{Q}^2$ and $\alpha\hat{P} + \beta\hat{P}^2$. As $\omega = e^{2\pi i/d}$, exponentiating these self-adjoint operators with the purely imaginary factor $2\pi i/d$ gives unitary operators.

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Using the spectral decompositions

$$\hat{Q} = \sum_{i \in \mathcal{I}_d} i |i\rangle \langle i|, \quad \hat{P} = \sum_{k \in \mathcal{I}_d} k |\widetilde{k}\rangle \langle \widetilde{k}|,$$

we obtain $(\alpha \hat{Q} + \beta \hat{Q}^2)|i\rangle = (\alpha i + \beta i^2)|i\rangle$ and $(\alpha \hat{P} + \beta \hat{P}^2)|\widetilde{k}\rangle = (\alpha k + \beta k^2)|\widetilde{k}\rangle$. Exponentiating these relations gives (5.54) and (5.55).

The operators $\alpha \hat{Q} + \beta \hat{Q}^2$ and $\alpha' \hat{Q} + \beta' \hat{Q}^2$ commute, since they are both polynomials in $\hat{Q}$. Therefore,

$$V_{\alpha\beta} V_{\alpha'\beta'} = \omega^{(\alpha+\alpha')\hat{Q} + (\beta+\beta')\hat{Q}^2} = V_{\alpha+\alpha', \beta+\beta'}.$$

The same argument with $\hat{P}$ proves (5.57). Setting all parameters to zero gives the identities, while composing $V_{\alpha\beta}$ with $V_{-\alpha, -\beta}$ and similarly for $\widetilde{V}_{\alpha\beta}$ gives the inverse and adjoint relations.

Finally, since every $i \in \mathcal{I}_d$ is an integer,

$$\omega^{(\alpha+md)i + (\beta+nd)i^2} = \omega^{\alpha i + \beta i^2} \omega^{d(mi + ni^2)} = \omega^{\alpha i + \beta i^2}.$$

Thus $V_{\alpha+md, \beta+nd}$ and $V_{\alpha\beta}$ agree on every basis vector $|i\rangle$ and are therefore equal. Replacing i by $k \in \mathcal{I}_d$ in the Fourier-conjugate basis proves the corresponding periodicity of $\widetilde{V}_{\alpha\beta}$. $\square$

Proposition 5.3.2 shows that the parameters are naturally defined modulo d . We therefore introduce the two-dimensional torus

$$\mathbb{T}_d^2 \stackrel{\text{def}}{=} (\mathbb{R}/d\mathbb{Z})^2, \quad (5.60)$$

with group operation given by componentwise addition modulo d .

Corollary 5.3.3. *The maps*

$$\Phi_Q : \mathbb{T}_d^2 \longrightarrow \mathcal{U}(\mathcal{H}_d), \quad (\alpha, \beta) \longmapsto V_{\alpha\beta}, \quad (5.61)$$

$$\Phi_P : \mathbb{T}_d^2 \longrightarrow \mathcal{U}(\mathcal{H}_d), \quad (\alpha, \beta) \longmapsto \widetilde{V}_{\alpha\beta}, \quad (5.62)$$

are continuous group homomorphisms. Consequently, their images

$$\mathcal{V}_Q \stackrel{\text{def}}{=} \{V_{\alpha\beta} : (\alpha, \beta) \in \mathbb{T}_d^2\}, \quad \mathcal{V}_P \stackrel{\text{def}}{=} \{\widetilde{V}_{\alpha\beta} : (\alpha, \beta) \in \mathbb{T}_d^2\}, \quad (5.63)$$

are compact abelian subgroups of the unitary group $\mathcal{U}(\mathcal{H}_d)$.

Proof. By Proposition 5.3.2, every $V_{\alpha\beta}$ and $\tilde{V}_{\alpha\beta}$ is unitary and hence belongs to $\mathcal{U}(\mathcal{H}_d)$.

A point of $\mathbb{T}_d^2$ is represented by a pair $(\alpha, \beta) \in \mathbb{R}^2$, where two pairs are identified whenever they differ by integer multiples of d :

$$(\alpha, \beta) \sim (\alpha + md, \beta + nd), \quad m, n \in \mathbb{Z}. \quad (5.64)$$

Proposition 5.3.2 gives $V_{\alpha+md, \beta+nd} = V_{\alpha\beta}$ and $\tilde{V}_{\alpha+md, \beta+nd} = \tilde{V}_{\alpha\beta}$. Thus, the value assigned by Φ_Q or Φ_P does not depend on which representative of a point in $\mathbb{T}_d^2$ is chosen. The maps are therefore well-defined.

The composition laws in Proposition 5.3.2 imply

$$\Phi_Q((\alpha, \beta) + (\alpha', \beta')) = \Phi_Q(\alpha, \beta)\Phi_Q(\alpha', \beta'), \quad (5.65)$$

and analogously for Φ_P . Hence both maps are group homomorphisms.

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We now verify directly that their images are subgroups of $\mathcal{U}(\mathcal{H}_d)$. First, $V_{00} = I$, so $I \in \mathcal{V}_Q$. Second, the composition law gives $V_{\alpha\beta}V_{\alpha'\beta'} = V_{\alpha+\alpha', \beta+\beta'} \in \mathcal{V}_Q$, so $\mathcal{V}_Q$ is closed under multiplication. Finally, $V_{\alpha\beta}^{-1} = V_{-\alpha, -\beta} \in \mathcal{V}_Q$, so it is closed under inverses. Therefore $\mathcal{V}_Q$ is a subgroup of $\mathcal{U}(\mathcal{H}_d)$. The same argument proves that $\mathcal{V}_P$ is a subgroup.

The composition laws also show that $V_{\alpha\beta}V_{\alpha'\beta'} = V_{\alpha'\beta'}V_{\alpha\beta}$, and similarly for the operators $\tilde{V}_{\alpha\beta}$. Thus, as already established in Proposition 5.3.2, both subgroups are abelian.

The maps Φ_Q and Φ_P are continuous because their matrix elements are finite sums of continuous functions of α and β . Since $\mathbb{T}_d^2$ is compact, their images $\mathcal{V}_Q = \Phi_Q(\mathbb{T}_d^2)$ and $\mathcal{V}_P = \Phi_P(\mathbb{T}_d^2)$ are compact. Hence $\mathcal{V}_Q$ and $\mathcal{V}_P$ are compact abelian subgroups of $\mathcal{U}(\mathcal{H}_d)$. $\square$

Since $\mathcal{V}_Q$ and $\mathcal{V}_P$ are compact groups, each admits a unique normalised Haar measure. These measures can be obtained directly from the normalised Haar measure on the parameter group $\mathbb{T}_d^2$. On the representative region $[0, d)^2$, this measure is the uniform measure

$$d\mu_{\mathbb{T}_d^2}(\alpha, \beta) = \frac{1}{d^2} d\alpha d\beta. \quad (5.66)$$

The normalised Haar measures on $\mathcal{V}_Q$ and $\mathcal{V}_P$ are the pushforwards of this measure under Φ_Q and Φ_P , respectively. In particular, for any integrable function $f : \mathcal{V}_Q \rightarrow \mathbb{C}$,

$$\int_{\mathcal{V}_Q} f(V) d\mu_{\mathcal{V}_Q}(V) = \frac{1}{d^2} \int_0^d \int_0^d f(V_{\alpha\beta}) d\alpha d\beta, \quad (5.67)$$

and analogously,

$$\int_{\mathcal{V}_P} f(V) d\mu_{\mathcal{V}_P}(V) = \frac{1}{d^2} \int_0^d \int_0^d f(\tilde{V}_{\alpha\beta}) d\alpha d\beta. \quad (5.68)$$

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Thus, uniformly sampling the parameters $(\alpha, \beta) \in [0, d]^2$ implements the Haar average over the corresponding compact subgroup of $\mathcal{U}(\mathcal{H}_d)$.

With a well-defined notion of uniform averaging, we are now in a position to define two-fold twirling maps.

Definition 5.3.4 (Quadrature twirling maps). *Let $X \in \mathcal{L}(\mathcal{H}_d \otimes \mathcal{H}_d)$. We define*

$$G(X) \stackrel{\text{def}}{=} \mathbb{E}_{\alpha, \beta} \left[V_{\alpha\beta}^{\otimes 2} X (V_{\alpha\beta}^\dagger)^{\otimes 2} \right], \quad (5.69)$$

and

$$\tilde{G}(X) \stackrel{\text{def}}{=} \mathbb{E}_{\alpha, \beta} \left[\tilde{V}_{\alpha\beta}^{\otimes 2} X (\tilde{V}_{\alpha\beta}^\dagger)^{\otimes 2} \right], \quad (5.70)$$

where the expectation is taken with respect to the uniform distribution of α, β on $[0, d]$.

Since the Haar measure is left- and right-invariant, one can also immediately see that the maps $G, \tilde{G}$ are projections. Since the unitary $V_{\alpha\beta}$ is diagonal in the $|i\rangle$ basis, the map G is easiest to understand on the operator basis $|ij\rangle\langle i'j'|$, where $|ij\rangle \stackrel{\text{def}}{=} |i\rangle \otimes |j\rangle$. Similarly, $\tilde{G}$ is simplest in the Fourier-conjugate basis $|\widetilde{k\ell}\rangle \stackrel{\text{def}}{=} |\widetilde{k}\rangle \otimes |\widetilde{\ell}\rangle$.

Proposition 5.3.5. *For all $i, j, i', j' \in \mathcal{I}_d$,*

$$G(|ij\rangle\langle i'j'|) = \delta_{i+j, i'+j'} \delta_{i^2+j^2, (i')^2+(j')^2} |ij\rangle\langle i'j'|. \quad (5.71)$$

Similarly, for all $k, \ell, k', \ell' \in \mathcal{I}_d$,

$$\tilde{G}(|\widetilde{k\ell}\rangle\langle\widetilde{k'\ell'}|) = \delta_{k+\ell, k'+\ell'} \delta_{k^2+\ell^2, (k')^2+(\ell')^2} |\widetilde{k\ell}\rangle\langle\widetilde{k'\ell'}|. \quad (5.72)$$

Proof. Using Proposition 5.3.2, we have

$$V_{\alpha\beta}^{\otimes 2} |ij\rangle = \omega^{\alpha(i+j)+\beta(i^2+j^2)} |ij\rangle, \quad (5.73)$$

$$\langle i'j'| (V_{\alpha\beta}^\dagger)^{\otimes 2} = \omega^{-\alpha(i'+j')-\beta((i')^2+(j')^2)} \langle i'j'|. \quad (5.74)$$

Therefore,

$$\begin{aligned} & V_{\alpha\beta}^{\otimes 2} |ij\rangle\langle i'j'| (V_{\alpha\beta}^\dagger)^{\otimes 2} \\ &= \omega^{\alpha(i+j-i'-j')+\beta(i^2+j^2-(i')^2-(j')^2)} |ij\rangle\langle i'j'|. \end{aligned} \quad (5.75)$$

Averaging over $\alpha, \beta \in [0, d]$ yields

$$G(|ij\rangle\langle i'j'|) = \left(\frac{1}{d} \int_0^d d\alpha \omega^{\alpha(i+j-i'-j')} \right) \left(\frac{1}{d} \int_0^d d\beta \omega^{\beta(i^2+j^2-(i')^2-(j')^2)} \right) |ij\rangle\langle i'j'|. \quad (5.76)$$

For any integer m ,

$$\frac{1}{d} \int_0^d dx \omega^{mx} = \frac{1}{d} \int_0^d dx e^{2\pi i mx/d} = \delta_{m,0}. \quad (5.77)$$

Applying (5.77) to both integrals gives (5.71). The proof of (5.72) is identical in the Fourier-conjugate basis. $\square$

Proposition 5.3.5 shows that G and $\tilde{G}$ are highly structured dephasing maps. They preserve only those matrix elements for which both the linear and quadratic discrete moments match. This is the first indication that the quadratic phases in (5.52) and (5.53) contain just enough randomness to eliminate most operator components, while still remaining closely tied to the quadrature structure.

5

The approximate design itself is obtained by alternating these two twirls.

Definition 5.3.6 (Alternating quadrature twirl). *We define the superoperator*

$$\mathcal{R} \stackrel{\text{def}}{=} G \circ \tilde{G} \circ G. \quad (5.78)$$

Equivalently,

$$\mathcal{R}(X) = \mathbb{E}_{\alpha, \beta, \alpha', \beta', \alpha'', \beta''} \left[(V_{\alpha''\beta''} \tilde{V}_{\alpha'\beta'} V_{\alpha\beta})^{\otimes 2} X (V_{\alpha\beta}^\dagger \tilde{V}_{\alpha'\beta'}^\dagger V_{\alpha''\beta''}^\dagger)^{\otimes 2} \right]. \quad (5.79)$$

Thus $\mathcal{R}$ is the two-fold twirl associated with random unitaries of the form

$$V_{\alpha''\beta''} \tilde{V}_{\alpha'\beta'} V_{\alpha\beta}, \quad (5.80)$$

where all parameters are sampled independently and uniformly from $[0, d)$.

Proposition 5.3.7. *For every $\ell \in \mathbb{N}$, the map $\mathcal{R}^\ell$ is again a two-fold twirl. More precisely,*

$$\mathcal{R}^\ell(X) = \mathbb{E}_{\alpha\beta} \left[W_\ell^{\otimes 2} X (W_\ell^\dagger)^{\otimes 2} \right], \quad (5.81)$$

where

$$W_\ell = \prod_{r=\ell}^1 \left(V_{\alpha_r''\beta_r''} \tilde{V}_{\alpha_r'\beta_r'} V_{\alpha_r\beta_r} \right), \quad (5.82)$$

and all parameters appearing in the product are independent and uniformly distributed on $[0, d)$.

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Proof. The case $\ell = 1$ is exactly Definition 5.3.6. Suppose the statement holds for some ℓ . Then

$$\begin{aligned}\mathcal{R}^{\ell+1}(X) &= \mathcal{R}(\mathcal{R}^\ell(X)) \\ &= \mathbb{E}_W \left[W^{\otimes 2} \mathbb{E}_{W_\ell} \left[W_\ell^{\otimes 2} X (W_\ell^\dagger)^{\otimes 2} \right] (W^\dagger)^{\otimes 2} \right] \\ &= \mathbb{E}_{W, W_\ell} \left[(WW_\ell)^{\otimes 2} X ((WW_\ell)^\dagger)^{\otimes 2} \right],\end{aligned}\tag{5.83}$$

which is again a two-fold twirl, now with respect to the product ensemble. This proves the claim by induction. $\square$

It is useful to compare this construction with that of Nakata *et al.* [149]. In their case, the diagonal phases in each basis are completely arbitrary: one samples an independent random phase for every basis vector. In contrast, our construction uses only two real parameters in each layer, through the quadratic phase profiles $\alpha x + \beta x^2$ applied to the spectra of $\hat{Q}$ and $\hat{P}$. This makes the ensemble much more structured. The advantage is that both families of unitaries have a direct quadrature interpretation and therefore admit a natural connection with the underlying CV picture. Although the randomness is much more constrained, approximate two-design behaviour can still be proved.

We will now prove that $\mathcal{R}$ acts almost like a two-design, and that the resemblance becomes better with increasing d . Furthermore, applying $\mathcal{R}$ again gives an improvement by a factor $\frac{1}{d}$.

Lemma 5.3.8. *Let $i \in \mathcal{I}_d$, $u \in \mathcal{I}_d \setminus \{0\}$ and $E = \sum_i |ii\rangle\langle ii|$. For $\{i', j'\} \neq \{i, j\}$ it holds that $\mathcal{R}(|ij\rangle\langle i'j'|) = 0$. Furthermore,*

$$\mathcal{R}(|i, i+u\rangle\langle i, i+u|) = \frac{1}{d^2} \sum_a |a, a+u\rangle\langle a, a+u| + \frac{I}{d^2} - \frac{I+F-E}{d^3}\tag{5.84}$$

$$\mathcal{R}(|i, i+u\rangle\langle i+u, i|) = \frac{1}{d^2} \sum_a |a, a+u\rangle\langle a+u, a| + \frac{F}{d^2} - \frac{I+F-E}{d^3}\tag{5.85}$$

$$\mathcal{R}(|ii\rangle\langle ii|) = \frac{I+F}{d^2} - \frac{I+F-E}{d^3},\tag{5.86}$$

where the additions $i+u$ and $a+u$ are understood to be modulo d , resulting in values in $\mathcal{I}_d$.

Proof. We begin by recalling the action of the first twirl G on a matrix unit. For convenience, define

$$C_{ij}^{i'j'} \stackrel{\text{def}}{=} \delta_{i+j, i'+j'} \delta_{i^2+j^2, (i')^2+(j')^2}.\tag{5.87}$$

Then, from Proposition 5.3.5,

$$G(|ij\rangle\langle i'j'|) = C_{ij}^{i'j'} |ij\rangle\langle i'j'|. \quad (5.88)$$

The meaning of the coefficient $C_{ij}^{i'j'}$ is important. If $C_{ij}^{i'j'} = 1$, then

$$i + j = i' + j', \quad i^2 + j^2 = (i')^2 + (j')^2. \quad (5.89)$$

From these two equalities we also obtain equality of the products, since

$$2ij = (i + j)^2 - (i^2 + j^2) = (i' + j')^2 - ((i')^2 + (j')^2) = 2i'j'. \quad (5.90)$$

Hence the pairs (i, j) and (i', j') have the same sum and the same product, so they are the two roots of the same quadratic polynomial. Therefore $\{i', j'\} = \{i, j\}$. This proves the first statement of the lemma: if $\{i', j'\} \neq \{i, j\}$, then $C_{ij}^{i'j'} = 0$, and hence $\mathcal{R}(|ij\rangle\langle i'j'|) = 0$.

Thus only three types of inputs can survive:

- (I) $j \neq i$ and $(i', j') = (i, j)$;
- (II) $j \neq i$ and $(i', j') = (j, i)$;
- (III) $j = i$ and $(i', j') = (i, i)$.

These are precisely the three cases appearing in (5.84)–(5.86).

We now continue with the action of $\tilde{G}$. Using the Fourier basis $|\widetilde{p}\rangle$, we write

$$|ij\rangle\langle i'j'| = \frac{1}{d^2} \sum_{p,k,p',k'} |\widetilde{pk}\rangle\langle\widetilde{p'k'}| \omega^{-pi-kj+p'i'+k'j'}. \quad (5.91)$$

Similarly to (5.88), but now in the $|\widetilde{p}\rangle$ basis,

$$\tilde{G}(|\widetilde{pk}\rangle\langle\widetilde{p'k'}|) = C_{pk}^{p'k'} |\widetilde{pk}\rangle\langle\widetilde{p'k'}|. \quad (5.92)$$

Therefore

$$(\tilde{G} \circ G)(|ij\rangle\langle i'j'|) = C_{ij}^{i'j'} \frac{1}{d^2} \sum_{p,k,p',k'} |\widetilde{pk}\rangle\langle\widetilde{p'k'}| C_{pk}^{p'k'} \omega^{-pi-kj+p'i'+k'j'}. \quad (5.93)$$

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Transforming back to the q basis gives

$$(\tilde{G} \circ G)(|ij\rangle\langle i'j'|) = C_{ij}^{i'j'} \frac{1}{d^4} \sum_{a,b,a',b'} |ab\rangle\langle a'b'| \times \sum_{p,k,p',k'} C_{pk}^{p'k'} \omega^{p(a-i)+k(b-j)+p'(i'-a')+k'(j'-b')}. \quad (5.94)$$

Applying G one more time inserts the factor $C_{ab}^{a'b'}$, so

$$\mathcal{R}(|ij\rangle\langle i'j'|) = C_{ij}^{i'j'} \frac{1}{d^4} \sum_{a,b,a',b'} C_{ab}^{a'b'} |ab\rangle\langle a'b'| \times \sum_{p,k,p',k'} C_{pk}^{p'k'} \omega^{p(a-i)+k(b-j)+p'(i'-a')+k'(j'-b')}. \quad (5.95)$$

The remaining task is to evaluate the free summation over p, k, p', k' . The coefficient $C_{pk}^{p'k'}$ is nonzero only when

- (a) $(p', k') = (p, k)$; or
- (b) $(p', k') = (k, p)$.

These two possibilities are not disjoint: they coincide when $p = k$, in which case $p = k = p' = k'$. Consequently, adding the contributions from (a) and (b) counts these diagonal terms twice, and their contribution must therefore be subtracted once. For the index combinations for which the outer coefficients $C_{ij}^{i'j'} C_{ab}^{a'b'}$ are nonzero, the phase factor of every diagonal term is equal to one. Since there are d such terms and the summation carries the prefactor $1/d^2$, the diagonal contribution to be subtracted is $d/d^2 = 1/d$.

Using the finite Fourier identity $\frac{1}{d} \sum_{r \in \mathcal{I}_d} \omega^{rm} = \delta_{m,0}^{(\text{mod } d)}$, one obtains

$$\begin{aligned} & \frac{1}{d^2} \sum_{p,k,p',k'} C_{pk}^{p'k'} \omega^{p(a-i)+k(b-j)+p'(i'-a')+k'(j'-b')} \\ &= \delta_{a'-a, i'-i}^{(\text{mod } d)} \delta_{b'-b, j'-j}^{(\text{mod } d)} + \delta_{a-b', i-j'}^{(\text{mod } d)} \delta_{b-a', j-i'}^{(\text{mod } d)} - \frac{1}{d}. \end{aligned} \quad (5.96)$$

Substituting (5.96) into (5.95) yields

$$\begin{aligned} \mathcal{R}(|ij\rangle\langle i'j'|) &= C_{ij}^{i'j'} \frac{1}{d^2} \sum_{a,b,a',b'} C_{ab}^{a'b'} |ab\rangle\langle a'b'| \left(\delta_{a'-a, i'-i}^{(\text{mod } d)} \delta_{b'-b, j'-j}^{(\text{mod } d)} \right. \\ &\quad \left. + \delta_{a-b', i-j'}^{(\text{mod } d)} \delta_{b-a', j-i'}^{(\text{mod } d)} - \frac{1}{d} \right). \end{aligned} \quad (5.97)$$

It is convenient to isolate the operator $\sum_{a,b,a',b'} C_{ab}^{a'b'} |ab\rangle\langle a'b'|$. Since $C_{ab}^{a'b'} = 1$ exactly when $\{a', b'\} = \{a, b\}$, this sum contains the diagonal terms $|ab\rangle\langle ab|$ and the swapped terms $|ab\rangle\langle ba|$. The diagonal sector $a = b$ is counted twice, so

$$\sum_{a,b,a',b'} C_{ab}^{a'b'} |ab\rangle\langle a'b'| = I + F - E. \quad (5.98)$$

We now evaluate the three surviving input cases separately.

Case (I): $|i, i + u\rangle\langle i, i + u|$. Here $u \neq 0$, $i' = i$, and $j' = i + u$. The first pair of modular deltas in (5.97) enforces

$$a' = a, \quad b' = b, \quad (5.99)$$

and since $C_{ab}^{ab} = 1$, this contributes $\sum_{a,b} |ab\rangle\langle ab| = I$. The second pair of modular deltas gives

$$b' = a + u, \quad a' = b - u \pmod{d}. \quad (5.100)$$

Together with $C_{ab}^{a'b'}$, and using $u \neq 0$, this forces

$$b = a + u, \quad a' = a, \quad b' = a + u, \quad (5.101)$$

so the second contribution is $\sum_a |a, a + u\rangle\langle a, a + u|$. Finally, the last term in (5.97) gives $-\frac{1}{d^3}(I + F - E)$. Combining these proves

$$\mathcal{R}(|i, i + u\rangle\langle i, i + u|) = \frac{1}{d^2} \sum_a |a, a + u\rangle\langle a, a + u| + \frac{I}{d^2} - \frac{I + F - E}{d^3}. \quad (5.102)$$

Case (II): $|i, i + u\rangle\langle i + u, i|$. Now $i' = i + u$ and $j' = i$. The first pair of modular deltas in (5.97) gives

$$a' = a + u, \quad b' = b - u. \quad (5.103)$$

Imposing $C_{ab}^{a'b'}$ then forces

$$b = a + u, \quad a' = a + u, \quad b' = a, \quad (5.104)$$

hence the contribution $\sum_a |a, a + u\rangle\langle a + u, a|$. The second pair of modular deltas now gives

$$a' = b, \quad b' = a. \quad (5.105)$$

This automatically satisfies $C_{ab}^{a'b'}$, and therefore contributes $\sum_{a,b} |ab\rangle\langle ba| = F$. Again the last term contributes $-\frac{1}{d^3}(I + F - E)$. Hence

$$\mathcal{R}(|i, i + u\rangle\langle i + u, i|) = \frac{1}{d^2} \sum_a |a, a + u\rangle\langle a + u, a| + \frac{F}{d^2} - \frac{I + F - E}{d^3}. \quad (5.106)$$

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Case (III): $|ii\rangle\langle ii|$. Here $i' = i$ and $j' = i$. The first pair of modular deltas in (5.97) gives $a' = a$, $b' = b$, which contributes I . The second pair gives $a' = b$, $b' = a$, which contributes F . The last term again yields $-\frac{1}{d^3}(I + F - E)$. Therefore

$$\mathcal{R}(|ii\rangle\langle ii|) = \frac{I + F}{d^2} - \frac{I + F - E}{d^3}. \quad (5.107)$$

This completes the proof. $\square$

In (5.84)–(5.86) we note that the right-hand sides do not depend on i . This makes it particularly easy to compute the action of $\mathcal{R}$ on some general matrix X .

Corollary 5.3.9. *Let $X \in \mathcal{L}(\mathcal{H}_d^{\otimes 2})$ and $E = \sum_i |ii\rangle\langle ii|$. It holds that*

$$\begin{aligned} \mathcal{R}(X) &= \frac{1}{d^2} \sum_{u \neq 0} \left(\sum_i X_{i,i+u}^{i,i+u} \right) \sum_a |a, a+u\rangle\langle a, a+u| \\ &\quad + \frac{1}{d^2} \sum_{u \neq 0} \left(\sum_i X_{i,i+u}^{i+u,i} \right) \sum_a |a, a+u\rangle\langle a+u, a| \\ &\quad + \left(\sum_i X_{ii}^{ii} \right) \frac{I + F - E}{d^3} + \left(\frac{I}{d^2} - \frac{I + F - E}{d^3} \right) \text{Tr } X \\ &\quad + \left(\frac{F}{d^2} - \frac{I + F - E}{d^3} \right) \text{Tr } F X. \end{aligned} \quad (5.108)$$

For $X \in \mathcal{K}$ we can write

$$\begin{aligned} \mathcal{R}(X) &= \sum_{u \neq 0} \left(\frac{1}{d^2} \sum_i X_{i,i+u}^{i,i+u} + \frac{\sum_i X_{ii}^{ii}}{d^3} \right) \sum_a |a, a+u\rangle\langle a, a+u| \\ &\quad + \sum_{u \neq 0} \left(\frac{1}{d^2} \sum_i X_{i,i+u}^{i+u,i} + \frac{\sum_i X_{ii}^{ii}}{d^3} \right) \sum_a |a, a+u\rangle\langle a+u, a| + \frac{\sum_i X_{ii}^{ii}}{d^3} E. \end{aligned} \quad (5.109)$$

Here $X = \sum_{i,j,i',j'} X_{ij}^{i'j'} |ij\rangle\langle i'j'|$.

Proof. Write X in the standard basis. By linearity of $\mathcal{R}$,

$$\mathcal{R}(X) = \sum_{i,j,i',j'} X_{ij}^{i'j'} \mathcal{R}(|ij\rangle\langle i'j'|). \quad (5.110)$$

Now Lemma 5.3.8 tells us that $\mathcal{R}(|ij\rangle\langle i'j'|) = 0$ unless $\{i', j'\} = \{i, j\}$. Hence only the following three types of matrix elements contribute:

1. $|i, i + u\rangle\langle i, i + u|$ with $u \neq 0$;
2. $|i, i + u\rangle\langle i + u, i|$ with $u \neq 0$;
3. $|ii\rangle\langle ii|$.

Therefore (5.110) reduces to

$$\begin{aligned} \mathcal{R}(X) &= \sum_{u \neq 0} \sum_i X_{i,i+u}^{i,i+u} \mathcal{R}(|i, i + u\rangle\langle i, i + u|) \\ &\quad + \sum_{u \neq 0} \sum_i X_{i,i+u}^{i+u,i} \mathcal{R}(|i, i + u\rangle\langle i + u, i|) + \sum_i X_{ii}^{ii} \mathcal{R}(|ii\rangle\langle ii|). \end{aligned} \quad (5.111)$$

We now substitute the three formulas from Lemma 5.3.8. This gives

$$\begin{aligned} \mathcal{R}(X) &= \frac{1}{d^2} \sum_{u \neq 0} \left(\sum_i X_{i,i+u}^{i,i+u} \right) \sum_a |a, a + u\rangle\langle a, a + u| \\ &\quad + \frac{1}{d^2} \sum_{u \neq 0} \left(\sum_i X_{i,i+u}^{i+u,i} \right) \sum_a |a, a + u\rangle\langle a + u, a| \\ &\quad + \frac{I}{d^2} \sum_{u \neq 0} \sum_i X_{i,i+u}^{i,i+u} + \frac{F}{d^2} \sum_{u \neq 0} \sum_i X_{i,i+u}^{i+u,i} + \frac{I + F}{d^2} \sum_i X_{ii}^{ii} \\ &\quad - \frac{I + F - E}{d^3} \left(\sum_{u \neq 0} \sum_i X_{i,i+u}^{i,i+u} + \sum_{u \neq 0} \sum_i X_{i,i+u}^{i+u,i} + \sum_i X_{ii}^{ii} \right). \end{aligned} \quad (5.112)$$

At this point it is convenient to rewrite the scalar coefficients in terms of traces. First,

$$\text{Tr } X = \sum_{i,j} X_{ij}^{ji} = \sum_{u \neq 0} \sum_i X_{i,i+u}^{i,i+u} + \sum_i X_{ii}^{ii}, \quad (5.113)$$

since every ordered pair (i, j) is either of the form (i, i) or $(i, i + u)$ with $u \neq 0$. Similarly,

$$\text{Tr}(FX) = \sum_{i,j} X_{ij}^{ji} = \sum_{u \neq 0} \sum_i X_{i,i+u}^{i+u,i} + \sum_i X_{ii}^{ii}. \quad (5.114)$$

Using (5.113) and (5.114), we can rewrite the third line of (5.112) as $\frac{I}{d^2} \text{Tr } X + \frac{F}{d^2} \text{Tr}(FX)$, while the coefficient of $-(I + F - E)/d^3$ becomes

$$\sum_{u \neq 0} \sum_i X_{i,i+u}^{i,i+u} + \sum_{u \neq 0} \sum_i X_{i,i+u}^{i+u,i} + \sum_i X_{ii}^{ii} = \text{Tr } X + \text{Tr}(FX) - \sum_i X_{ii}^{ii}. \quad (5.115)$$

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Substituting this into (5.112) yields

$$\begin{aligned}
 \mathcal{R}(X) = & \frac{1}{d^2} \sum_{u \neq 0} \left(\sum_i X_{i,i+u}^{i,i+u} \right) \sum_a |a, a+u\rangle \langle a, a+u| \\
 & + \frac{1}{d^2} \sum_{u \neq 0} \left(\sum_i X_{i,i+u}^{i+u,i} \right) \sum_a |a, a+u\rangle \langle a+u, a| \\
 & + \left(\sum_i X_{ii}^{ii} \right) \frac{I + F - E}{d^3} + \left(\frac{I}{d^2} - \frac{I + F - E}{d^3} \right) \text{Tr } X \\
 & + \left(\frac{F}{d^2} - \frac{I + F - E}{d^3} \right) \text{Tr } FX,
 \end{aligned} \tag{5.116}$$

which is the first claimed formula.

Now suppose that $X \in \mathcal{K}$. By definition of $\mathcal{K} = \mathcal{A}^\perp$, this means $\text{Tr } X = 0$, $\text{Tr}(FX) = 0$. Hence the previous formula simplifies to

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$$\begin{aligned}
 \mathcal{R}(X) = & \frac{1}{d^2} \sum_{u \neq 0} \left(\sum_i X_{i,i+u}^{i,i+u} \right) \sum_a |a, a+u\rangle \langle a, a+u| \\
 & + \frac{1}{d^2} \sum_{u \neq 0} \left(\sum_i X_{i,i+u}^{i+u,i} \right) \sum_a |a, a+u\rangle \langle a+u, a| + \left(\sum_i X_{ii}^{ii} \right) \frac{I + F - E}{d^3}.
 \end{aligned} \tag{5.117}$$

Finally, we decompose the operator $I + F - E$ into the same three families of terms:

$$I + F - E = \sum_{u \neq 0} \sum_a |a, a+u\rangle \langle a, a+u| + \sum_{u \neq 0} \sum_a |a, a+u\rangle \langle a+u, a| + E. \tag{5.118}$$

Substituting (5.118) into (5.117) and grouping like terms gives

$$\begin{aligned}
 \mathcal{R}(X) = & \sum_{u \neq 0} \left(\frac{1}{d^2} \sum_i X_{i,i+u}^{i,i+u} + \frac{\sum_i X_{ii}^{ii}}{d^3} \right) \sum_a |a, a+u\rangle \langle a, a+u| \\
 & + \sum_{u \neq 0} \left(\frac{1}{d^2} \sum_i X_{i,i+u}^{i+u,i} + \frac{\sum_i X_{ii}^{ii}}{d^3} \right) \sum_a |a, a+u\rangle \langle a+u, a| + \frac{\sum_i X_{ii}^{ii}}{d^3} E,
 \end{aligned} \tag{5.119}$$

which is exactly (5.109). $\square$

The form (5.109) allows us to directly compute the norm $\|\mathcal{R}(X)\|_2$.

Theorem 5.3.10. *The set of unitaries $\{\omega^{\alpha''\hat{Q}+\beta''\hat{Q}^2}\omega^{\alpha'\hat{P}+\beta'\hat{P}^2}\omega^{\alpha\hat{Q}+\beta\hat{Q}^2}\}$, with $\alpha, \beta, \alpha', \beta', \alpha'', \beta''$ chosen uniformly in $[0, d)$, is an ε -approximate unitary two-design with $\varepsilon = \frac{1}{d}$.*

Proof. Consider $X = \sum_{i,j,i',j'} X_{ij}^{i'j'} |ij\rangle\langle i'j'| \in \mathcal{K}$. We introduce the notation $Y = \mathcal{R}(X)$ and $c = \sum_i X_{ii}^{ii}$. From $\text{Tr } Y^\dagger Y = \sum_{ij,i'j'} |Y_{ij}^{i'j'}|^2$ and (5.109) we obtain

$$\text{Tr } Y^\dagger Y = \frac{1}{d^4} \sum_a \sum_{u \neq 0} \left| \sum_i X_{i,i+u}^{i,i+u} + \frac{c}{d} \right|^2 + \frac{1}{d^4} \sum_a \sum_{u \neq 0} \left| \sum_i X_{i,i+u}^{i+u,i} + \frac{c}{d} \right|^2 + d \frac{|c|^2}{d^6} \quad (5.120)$$

$$= \frac{1}{d^3} \sum_{u \neq 0} \left| \sum_i X_{i,i+u}^{i,i+u} \right|^2 + \frac{1}{d^3} \sum_{u \neq 0} \left| \sum_i X_{i,i+u}^{i+u,i} \right|^2 + 2d(d-1) \frac{|c|^2}{d^6} \\ + \left(\frac{c^*}{d^4} \sum_{u \neq 0} \sum_i X_{i,i+u}^{i,i+u} + cc \right) + \left(\frac{c^*}{d^4} \sum_{u \neq 0} \sum_i X_{i,i+u}^{i+u,i} + cc \right) + d \frac{|c|^2}{d^6} \quad (5.121)$$

$$\stackrel{(a)}{=} -\frac{|c|^2}{d^5} - 2\frac{|c|^2}{d^4} + \frac{1}{d^3} \sum_{u \neq 0} \left| \sum_s X_{s,s+u}^{s,s+u} \right|^2 + \frac{1}{d^3} \sum_{u \neq 0} \left| \sum_s X_{s,s+u}^{s+u,s} \right|^2 \quad (5.122)$$

$$\stackrel{(b)}{\leq} \frac{1}{d^3} \sum_{u \neq 0} \left(\sum_s |X_{s,s+u}^{s,s+u}| \right)^2 + \frac{1}{d^3} \sum_{u \neq 0} \left(\sum_s |X_{s,s+u}^{s+u,s}| \right)^2 \quad (5.123)$$

$$\stackrel{(c)}{\leq} \frac{1}{d^2} \sum_{u \neq 0} \sum_s |X_{s,s+u}^{s,s+u}|^2 + \frac{1}{d^2} \sum_{u \neq 0} \sum_s |X_{s,s+u}^{s+u,s}|^2 \quad (5.124)$$

$$\stackrel{(d)}{\leq} \frac{1}{d^2} \text{Tr } X^\dagger X. \quad (5.125)$$

The cc stands for the complex conjugate. In (a) we used

$$\sum_{u \neq 0} \sum_i X_{i,i+u}^{i,i+u} = \text{Tr } X - c = -c \text{ and} \quad (5.126)$$

$$\sum_{u \neq 0} \sum_i X_{i,i+u}^{i+u,i} = \text{Tr } F X - c = -c. \quad (5.127)$$

Step (b) is obtained by using the triangle inequality and neglecting the negative terms. In (c) we have used $\|x\|_1^2 \leq d\|x\|_2^2$. Finally, (d) follows from

$$\text{Tr } X^\dagger X = \sum_{aba'b'} |X_{ab}^{a'b'}|^2. \quad (5.128)$$

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We therefore conclude that

$$\frac{\text{Tr } \mathcal{R}^\dagger(X) \mathcal{R}(X)}{\text{Tr } X^\dagger X} \leq \frac{1}{d^2} \quad (5.129)$$

and hence

$$\sup_{X \in \mathcal{K}, X \neq 0} \frac{\|\mathcal{R}(X)\|_2}{\|X\|_2} \leq \frac{1}{d}. \quad (5.130)$$

Finally, we use Corollary 5.1.5. $\square$

Lemma 5.3.11. *Let $X = \sum_{i,j,i',j'} X_{ij}^{i'j'} |ij\rangle\langle i'j'| \in \mathcal{K}$. For $\ell \geq 1$, applying $\mathcal{R}$ ℓ times results in*

$$\begin{aligned} \mathcal{R}^\ell(X) = & \sum_{u \neq 0} \xi_u^{(\ell)} \sum_a |a, a+u\rangle\langle a, a+u| + \sum_{u \neq 0} \lambda_u^{(\ell)} \sum_a |a, a+u\rangle\langle a+u, a| \\ & + \eta^{(\ell)} \sum_a |aa\rangle\langle aa| \end{aligned} \quad (5.131)$$

$$\xi_u^{(\ell)} = \frac{1}{d^{\ell+1}} \sum_i X_{i,i+u}^{i,i+u} + \frac{\sum_i X_{ii}^{ii}}{d^{\ell+2}} \sum_{t=0}^{\ell-1} \left(\frac{1}{d}\right)^t = \frac{\sum_i X_{i,i+u}^{i,i+u}}{d^{\ell+1}} + \frac{\sum_i X_{ii}^{ii}}{d^{\ell+1}} \cdot \frac{1 - (1/d)^\ell}{d - 1} \quad (5.132)$$

$$\lambda_u^{(\ell)} = \frac{1}{d^{\ell+1}} \sum_i X_{i,i+u}^{i+u,i} + \frac{\sum_i X_{ii}^{ii}}{d^{\ell+2}} \sum_{t=0}^{\ell-1} \left(\frac{1}{d}\right)^t = \frac{\sum_i X_{i,i+u}^{i+u,i}}{d^{\ell+1}} + \frac{\sum_i X_{ii}^{ii}}{d^{\ell+1}} \cdot \frac{1 - (1/d)^\ell}{d - 1} \quad (5.133)$$

$$\eta^{(\ell)} = \frac{\sum_i X_{ii}^{ii}}{d^{2\ell+1}}. \quad (5.134)$$

Proof. From (5.109), the case $\ell = 1$ already has the required form:

$$\begin{aligned} \mathcal{R}(X) = & \sum_{u \neq 0} \xi_u^{(1)} \sum_a |a, a+u\rangle\langle a, a+u| + \sum_{u \neq 0} \lambda_u^{(1)} \sum_a |a, a+u\rangle\langle a+u, a| \\ & + \eta^{(1)} \sum_a |aa\rangle\langle aa|, \end{aligned} \quad (5.135)$$

with

$$\xi_u^{(1)} = \frac{1}{d^2} \sum_i X_{i,i+u}^{i,i+u} + \frac{1}{d^3} \sum_i X_{ii}^{ii}, \quad (5.136)$$

$$\lambda_u^{(1)} = \frac{1}{d^2} \sum_i X_{i,i+u}^{i+u,i} + \frac{1}{d^3} \sum_i X_{ii}^{ii}, \quad (5.137)$$

and

$$\eta^{(1)} = \frac{1}{d^3} \sum_i X_{ii}^{ii}. \quad (5.138)$$

Since $\mathcal{R}(X) \in \mathcal{K}$, we may apply (5.109) again to $\mathcal{R}(X)$ itself. Comparing coefficients, we obtain the recursion relations

$$\eta^{(s+1)} = \frac{1}{d^2} \eta^{(s)}, \quad (5.139)$$

and, for every $u \neq 0$,

$$\xi_u^{(s+1)} = \frac{1}{d} \xi_u^{(s)} + \frac{1}{d^2} \eta^{(s)}, \quad \lambda_u^{(s+1)} = \frac{1}{d} \lambda_u^{(s)} + \frac{1}{d^2} \eta^{(s)}. \quad (5.140)$$

We now solve these recursions.

1: the coefficient $\eta^{(\ell)}$. Equation (5.139) is a geometric recursion. Using (5.138),

$$\eta^{(\ell)} = \left(\frac{1}{d^2} \right)^{\ell-1} \eta^{(1)} = \left(\frac{1}{d^2} \right)^{\ell-1} \frac{1}{d^3} \sum_i X_{ii}^{ii} = \frac{\sum_i X_{ii}^{ii}}{d^{2\ell+1}}, \quad (5.141)$$

which proves (5.134).

2: the coefficient $\xi_u^{(\ell)}$. Substituting the formula for $\eta^{(s)}$ into (5.140) gives

$$\xi_u^{(s+1)} = \frac{1}{d} \xi_u^{(s)} + \frac{1}{d^2} \cdot \frac{\sum_i X_{ii}^{ii}}{d^{2s+1}} = \frac{1}{d} \xi_u^{(s)} + \frac{\sum_i X_{ii}^{ii}}{d^{2s+3}}. \quad (5.142)$$

To simplify this recursion, multiply both sides by d^{s+2} . Then

$$d^{s+2} \xi_u^{(s+1)} = d^{s+1} \xi_u^{(s)} + \frac{\sum_i X_{ii}^{ii}}{d^{s+1}}. \quad (5.143)$$

Thus the quantity $d^{s+1} \xi_u^{(s)}$ satisfies a telescoping recursion:

$$d^{s+2} \xi_u^{(s+1)} - d^{s+1} \xi_u^{(s)} = \frac{\sum_i X_{ii}^{ii}}{d^{s+1}}. \quad (5.144)$$

Summing this from $s = 1$ to $s = \ell - 1$, we obtain

$$d^{\ell+1} \xi_u^{(\ell)} - d^2 \xi_u^{(1)} = \sum_{s=1}^{\ell-1} \frac{\sum_i X_{ii}^{ii}}{d^{s+1}}. \quad (5.145)$$

Now substitute the initial value (5.136):

$$d^2 \xi_u^{(1)} = \sum_i X_{i,i+u}^{i,i+u} + \frac{1}{d} \sum_i X_{ii}^{ii}. \quad (5.146)$$

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Hence

$$d^{\ell+1}\xi_u^{(\ell)} = \sum_i X_{i,i+u}^{i,i+u} + \frac{1}{d} \sum_i X_{ii}^{ii} + \sum_{s=1}^{\ell-1} \frac{\sum_i X_{ii}^{ii}}{d^{s+1}}. \quad (5.147)$$

Combining the last two terms into a single sum gives

$$d^{\ell+1}\xi_u^{(\ell)} = \sum_i X_{i,i+u}^{i,i+u} + \sum_i X_{ii}^{ii} \sum_{t=0}^{\ell-1} \frac{1}{d^{t+1}}. \quad (5.148)$$

Dividing by $d^{\ell+1}$ yields

$$\xi_u^{(\ell)} = \frac{1}{d^{\ell+1}} \sum_i X_{i,i+u}^{i,i+u} + \frac{\sum_i X_{ii}^{ii}}{d^{\ell+2}} \sum_{t=0}^{\ell-1} \left(\frac{1}{d}\right)^t, \quad (5.149)$$

which is the first expression in (5.132).

To obtain the closed form, use the geometric-series identity

$$\sum_{t=0}^{\ell-1} \left(\frac{1}{d}\right)^t = \frac{1 - (1/d)^\ell}{1 - 1/d} = \frac{d}{d-1} \left(1 - (1/d)^\ell\right). \quad (5.150)$$

Substituting this gives

$$\xi_u^{(\ell)} = \frac{1}{d^{\ell+1}} \sum_i X_{i,i+u}^{i,i+u} + \frac{\sum_i X_{ii}^{ii}}{d^{\ell+1}} \cdot \frac{1 - (1/d)^\ell}{d-1}, \quad (5.151)$$

which is exactly (5.132).

3: the coefficient $\lambda_u^{(\ell)}$. The recursion for $\lambda_u^{(s)}$ is identical to that for $\xi_u^{(s)}$, except that the initial value (5.137) contains $\sum_i X_{i,i+u}^{i+u,i}$ instead of $\sum_i X_{i,i+u}^{i,i+u}$. Repeating the same steps therefore gives

$$\lambda_u^{(\ell)} = \frac{1}{d^{\ell+1}} \sum_i X_{i,i+u}^{i+u,i} + \frac{\sum_i X_{ii}^{ii}}{d^{\ell+2}} \sum_{t=0}^{\ell-1} \left(\frac{1}{d}\right)^t \quad (5.152)$$

and hence

$$\lambda_u^{(\ell)} = \frac{1}{d^{\ell+1}} \sum_i X_{i,i+u}^{i+u,i} + \frac{\sum_i X_{ii}^{ii}}{d^{\ell+1}} \cdot \frac{1 - (1/d)^\ell}{d-1}, \quad (5.153)$$

which proves (5.133).

This completes the proof. $\square$

Corollary 5.3.12. Applying $\mathcal{R}^\ell$ yields an ε -approximate unitary two-design with $\varepsilon = (\frac{1}{d})^\ell$.

Proof. Let $\mathcal{K}' = \mathcal{K} \setminus \{0\}$ and let $\mathcal{R}_{\mathcal{K}}$ denote the restriction of $\mathcal{R}$ to the subspace $\mathcal{K}$. From the previous result we know that

$$\|\mathcal{R}_{\mathcal{K}}\|_{2 \rightarrow 2} \leq \frac{1}{d}. \quad (5.154)$$

We now apply this bound recursively. By definition,

$$\begin{aligned} \|\mathcal{R}_{\mathcal{K}}^{\ell}\|_{2 \rightarrow 2} &\stackrel{\text{def}}{=} \sup_{X \in \mathcal{K}'} \frac{\|\mathcal{R}^{\ell}(X)\|_2}{\|X\|_2} \\ &= \sup_{X \in \mathcal{K}'} \frac{\|\mathcal{R}(\mathcal{R}^{\ell-1}(X))\|_2}{\|\mathcal{R}^{\ell-1}(X)\|_2} \cdot \frac{\|\mathcal{R}^{\ell-1}(X)\|_2}{\|X\|_2} \\ &\leq \left(\sup_{Y \in \mathcal{K}'} \frac{\|\mathcal{R}(Y)\|_2}{\|Y\|_2} \right) \left(\sup_{X \in \mathcal{K}'} \frac{\|\mathcal{R}^{\ell-1}(X)\|_2}{\|X\|_2} \right) \\ &= \|\mathcal{R}_{\mathcal{K}}\|_{2 \rightarrow 2} \cdot \|\mathcal{R}_{\mathcal{K}}^{\ell-1}\|_{2 \rightarrow 2} \\ &\leq \frac{1}{d} \|\mathcal{R}_{\mathcal{K}}^{\ell-1}\|_{2 \rightarrow 2}. \end{aligned} \quad (5.155)$$

Iterating this estimate gives

$$\|\mathcal{R}_{\mathcal{K}}^{\ell}\|_{2 \rightarrow 2} \leq \frac{1}{d^{\ell}}. \quad (5.156)$$

Finally, Corollary 5.1.5 implies that applying $\mathcal{R}^{\ell}$ yields an ε -approximate unitary 2-design with $\varepsilon = \left(\frac{1}{d}\right)^{\ell}$. $\square$

5.3.1. A basis of the $\mathcal{K}$ subspace

For a more comprehensive understanding of the operator $\mathcal{R}$, it may be useful to work with an explicit basis of the subspace $\mathcal{K}$. We thus build an example basis using orthogonal subspaces. Throughout this subsection we work entirely in the q -boxed basis.

We begin by identifying those matrix units that already lie in $\mathcal{K}$.

Lemma 5.3.13. *A matrix unit $|ij\rangle\langle i'j'|$ lies in $\mathcal{K}$ provided it is neither of the following two forms:*

$$(diagonal-diagonal) \quad |ab\rangle\langle ab|, \quad a, b \in \mathcal{I}_d, \quad (5.157)$$

$$(swap \text{ pattern}) \quad |ij\rangle\langle ji|, \quad i, j \in \mathcal{I}_d. \quad (5.158)$$

Proof. The Hilbert–Schmidt inner products with I and F are

$$\langle |ij\rangle\langle i'j'|, I \rangle = \delta_{ii'}\delta_{jj'}, \quad \langle |ij\rangle\langle i'j'|, F \rangle = \delta_{ij'}\delta_{ji'}. \quad (5.159)$$

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Therefore,

$$\langle |ij\rangle\langle i'j'|, I \rangle \neq 0 \iff i = i' \text{ and } j = j', \quad (5.160)$$

which is exactly the diagonal–diagonal pattern (5.157). Likewise,

$$\langle |ij\rangle\langle i'j'|, F \rangle \neq 0 \iff i = j' \text{ and } j = i', \quad (5.161)$$

which is exactly the swap pattern (5.158). Hence any matrix unit that is of neither form is orthogonal to both I and F , and therefore lies in $\mathcal{K}$. $\square$

We next single out the part of $\mathcal{K}$ that is spanned by individual rank-one matrix units. We denote this subspace by $\mathcal{K}_1$. A convenient generating set is obtained by separating the possible placements of the diagonal index structure.

1. *Right diagonal, left off-diagonal:* for all $i, j, a \in \mathcal{I}_d$ with $i \neq j$,

$$B_{aij}^{\text{R-diag}} := |ia\rangle\langle ja|. \quad (5.162)$$

2. *Left diagonal, right off-diagonal:* for all $a, i, j \in \mathcal{I}_d$ with $i \neq j$,

$$B_{aij}^{\text{L-diag}} := |ai\rangle\langle aj|. \quad (5.163)$$

3. *Both sides off-diagonal:* for all $i, j, i', j' \in \mathcal{I}_d$ with $i \neq i', j \neq j'$, and $(i, j) \neq (j', i')$,

$$B_{ij i' j'}^{\text{off-diag}} := |ij\rangle\langle i'j'|. \quad (5.164)$$

Definition 5.3.14. Define the set

$$B_{\mathcal{K}_1} := \{B_{aij}^{\text{R-diag}}\}_{i \neq j, a \in \mathcal{I}_d} \cup \{B_{aij}^{\text{L-diag}}\}_{i \neq j, a \in \mathcal{I}_d} \cup \{B_{ij i' j'}^{\text{off-diag}}\}_{i \neq i', j \neq j', (i, j) \neq (j', i')}. \quad (5.165)$$

We define $\mathcal{K}_1$ to be the span of these vectors:

$$\mathcal{K}_1 := \text{span}(B_{\mathcal{K}_1}). \quad (5.166)$$

Lemma 5.3.15. $\mathcal{K}_1 \subset \mathcal{K}$, and the elements of $B_{\mathcal{K}_1}$ are pairwise orthogonal and therefore linearly independent.

Proof. We first show that $\mathcal{K}_1 \subset \mathcal{K}$. Since $\mathcal{K}_1 = \text{span}(B_{\mathcal{K}_1})$, it is enough to check that each generator belongs to $\mathcal{K}$. By Lemma 5.3.13, a matrix unit lies in $\mathcal{K}$ whenever it is neither of the forbidden forms (5.157) and (5.158). This is immediate for each of the three families in (5.165).

For orthogonality, observe that distinct parameter choices in (5.165) produce distinct matrix units, and no matrix unit appears in more than one family. Hence any two distinct elements of $B_{\mathcal{K}_1}$ differ in at least one index. But we know that distinct matrix units are orthogonal. Therefore the elements of $B_{\mathcal{K}_1}$ are pairwise orthogonal. Since they are also nonzero, they are linearly independent. $\square$

We now introduce a second subspace $\mathcal{K}_2$, generated by certain higher-rank combinations of the exceptional matrix units excluded by Lemma 5.3.13. We fix the reference indices $(0, 1)$ and define the following vectors.

1. *Diagonal–diagonal off-diagonal differences:* for all $a, b \in \mathcal{I}_d$ with $a \neq b$ and $(a, b) \neq (0, 1)$,

$$B_{ab}^{\text{diag}} := |ab\rangle\langle ab| - |01\rangle\langle 01|. \quad (5.167)$$

2. *Swap-pattern off-diagonal differences:* for all $i, j \in \mathcal{I}_d$ with $i \neq j$ and $(i, j) \neq (0, 1)$,

$$B_{ij}^{\text{swap}} := |ij\rangle\langle ji| - |01\rangle\langle 10|. \quad (5.168)$$

3. *Diagonal differences:* for $r \in \mathcal{I}_d \setminus \{0\}$,

$$B_r := |00\rangle\langle 00| - |rr\rangle\langle rr|. \quad (5.169)$$

4. *Coupling vector:*

$$B_0 := \sum_{\substack{a, b \in \mathcal{I}_d \\ a \neq b}} |ab\rangle\langle ab| + \sum_{\substack{i, j \in \mathcal{I}_d \\ i \neq j}} |ij\rangle\langle ji| - (d-1) \sum_{r \in \mathcal{I}_d} |rr\rangle\langle rr|. \quad (5.170)$$

Definition 5.3.16. Define the set

$$B_{\mathcal{K}_2} := \{B_{ab}^{\text{diag}}\}_{a \neq b, (a,b) \neq (0,1)} \cup \{B_{ij}^{\text{swap}}\}_{i \neq j, (i,j) \neq (0,1)} \cup \{B_r\}_{r \in \mathcal{I}_d \setminus \{0\}} \cup \{B_0\}. \quad (5.171)$$

We define $\mathcal{K}_2$ to be the span of these vectors:

$$\mathcal{K}_2 := \text{span}(B_{\mathcal{K}_2}). \quad (5.172)$$

Lemma 5.3.17. $\mathcal{K}_2 \subset \mathcal{K}$, and the vectors in $B_{\mathcal{K}_2}$ are linearly independent. Moreover, the families

$$\{B_{ab}^{\text{diag}}\}_{a \neq b, (a,b) \neq (0,1)}, \quad \{B_{ij}^{\text{swap}}\}_{i \neq j, (i,j) \neq (0,1)}, \quad \{B_r\}_{r \in \mathcal{I}_d \setminus \{0\}}, \quad \{B_0\}$$

are mutually orthogonal.

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Proof. We first show that $\mathcal{K}_2 \subset \mathcal{K}$. Since $\mathcal{K}_2 = \text{span}(B_{\mathcal{K}_2})$, it suffices to check that each generator lies in $\mathcal{K}$.

Recall from Lemma 5.3.13 that the only matrix units not belonging to $\mathcal{K}$ are those of the diagonal–diagonal form (5.157) and the swap form (5.158). For B_{ab}^{diag} and B_{ij}^{swap} , the two summands have exactly the same nonzero overlap with I or F , respectively, so their difference lies in $\mathcal{K}$. The same argument applies to B_r , whose two summands have identical overlap with I and zero overlap with F .

For the coupling vector B_0 , using (5.159) we compute

$$\langle B_0, I \rangle = \sum_{\substack{a, b \in \mathcal{I}_d \\ a \neq b}} 1 + 0 - (d-1) \sum_{r \in \mathcal{I}_d} 1 = d(d-1) - (d-1)d = 0, \quad (5.173)$$

$$\langle B_0, F \rangle = 0 + \sum_{\substack{i, j \in \mathcal{I}_d \\ i \neq j}} 1 - (d-1) \sum_{r \in \mathcal{I}_d} 1 = d(d-1) - (d-1)d = 0. \quad (5.174)$$

Thus $B_0 \in \mathcal{K}$ as well. Hence every generator lies in $\mathcal{K}$, and therefore $\mathcal{K}_2 \subset \mathcal{K}$.

To prove linear independence, suppose that

$$\sum_{a \neq b, (a,b) \neq (0,1)} \alpha_{ab} B_{ab}^{\text{diag}} + \sum_{i \neq j, (i,j) \neq (0,1)} \beta_{ij} B_{ij}^{\text{swap}} + \sum_{r \in \mathcal{I}_d \setminus \{0\}} \gamma_r B_r + \delta B_0 = 0. \quad (5.175)$$

We first isolate the coefficient of $|ab\rangle\langle ab|$ with $a \neq b$ and $(a,b) \neq (0,1)$. This matrix unit appears only in B_{ab}^{diag} and in B_0 , so its coefficient in (5.175) is $\alpha_{ab} + \delta$. Hence

$$\alpha_{ab} = -\delta \quad \text{for all } a \neq b, (a,b) \neq (0,1). \quad (5.176)$$

Now consider the coefficient of the reference unit $|01\rangle\langle 01|$. It appears with coefficient -1 in every B_{ab}^{diag} , with coefficient $+1$ in B_0 , and nowhere else. Therefore,

$$- \sum_{a \neq b, (a,b) \neq (0,1)} \alpha_{ab} + \delta = 0. \quad (5.177)$$

There are $d(d-1) - 1$ terms in this sum, and each $\alpha_{ab} = -\delta$ by (5.176). Hence

$$-(d(d-1) - 1)(-\delta) + \delta = d(d-1)\delta = 0,$$

so $\delta = 0$, and therefore $\alpha_{ab} = 0$ for all $a \neq b, (a,b) \neq (0,1)$.

Next consider the coefficient of $|ij\rangle\langle ji|$ with $i \neq j$ and $(i,j) \neq (0,1)$. This matrix unit appears only in B_{ij}^{swap} and in B_0 , but now $\delta = 0$, so its coefficient in (5.175) is simply β_{ij} . Therefore

$$\beta_{ij} = 0 \quad \text{for all } i \neq j, (i,j) \neq (0,1). \quad (5.178)$$

Finally, for each $r \in \mathcal{I}_d \setminus \{0\}$, the matrix unit $|rr\rangle\langle rr|$ appears only in B_r with coefficient -1 (and in B_0 with coefficient $-(d-1)$, but again $\delta = 0$), so (5.175) implies $\gamma_r = 0$ for all r .

Thus all coefficients vanish, proving linear independence.

It remains to show the orthogonality statement. The first three families are supported on disjoint sets of matrix units:

$$|ab\rangle\langle ab| \quad (a \neq b), \quad |ij\rangle\langle ji| \quad (i \neq j), \quad |rr\rangle\langle rr|.$$

Distinct matrix units are orthogonal, so these three families are mutually orthogonal. It therefore only remains to check orthogonality with B_0 .

Fix $a \neq b$ with $(a, b) \neq (0, 1)$. Using (5.170) and (5.167), only the terms $|ab\rangle\langle ab|$ and $|01\rangle\langle 01|$ contribute, and hence

$$\langle B_0, B_{ab}^{\text{diag}} \rangle = \langle B_0, |ab\rangle\langle ab| \rangle - \langle B_0, |01\rangle\langle 01| \rangle = 1 - 1 = 0.$$

Similarly, for $i \neq j$ with $(i, j) \neq (0, 1)$,

$$\langle B_0, B_{ij}^{\text{swap}} \rangle = \langle B_0, |ij\rangle\langle ji| \rangle - \langle B_0, |01\rangle\langle 10| \rangle = 1 - 1 = 0.$$

Finally, for $r \in \mathcal{I}_d \setminus \{0\}$,

$$\langle B_0, B_r \rangle = \langle B_0, |00\rangle\langle 00| \rangle - \langle B_0, |rr\rangle\langle rr| \rangle = -(d-1) - (-(d-1)) = 0.$$

Thus B_0 is orthogonal to all other generators in $B_{\mathcal{K}_2}$, completing the proof. $\square$

Theorem 5.3.18 (Decomposition of $\mathcal{K}$). *The subspace $\mathcal{K}$ decomposes as the direct sum*

$$\mathcal{K} = \mathcal{K}_1 \oplus \mathcal{K}_2, \quad \text{and in particular} \quad \mathcal{K} = \text{span}(B_{\mathcal{K}_1} \cup B_{\mathcal{K}_2}). \quad (5.179)$$

Proof. By Lemma 5.3.15 and Lemma 5.3.17, we have $\mathcal{K}_1 \subset \mathcal{K}$ and $\mathcal{K}_2 \subset \mathcal{K}$, hence

$$\mathcal{K}_1 + \mathcal{K}_2 \subset \mathcal{K}. \quad (5.180)$$

Next we show that $\mathcal{K}_1 \perp \mathcal{K}_2$, and therefore $\mathcal{K}_1 \cap \mathcal{K}_2 = \{0\}$. Every generator in $B_{\mathcal{K}_1}$ is a single matrix unit $|ij\rangle\langle i'j'|$ which is neither of the forbidden patterns (5.157)–(5.158). By contrast, every generator in $B_{\mathcal{K}_2}$ is a linear combination of matrix units of diagonal–diagonal type $|ab\rangle\langle ab|$ and/or swap type $|ij\rangle\langle ji|$; see (5.167)–(5.170). Since distinct matrix units are orthogonal, it follows that

$$\langle b_1, b_2 \rangle = 0 \quad \text{for all } b_1 \in B_{\mathcal{K}_1}, b_2 \in B_{\mathcal{K}_2}. \quad (5.181)$$

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Hence $\mathcal{K}_1 \perp \mathcal{K}_2$.

We now count generators. From (5.165) we obtain

$$|B_{\mathcal{K}_1}| = \underbrace{d \cdot d(d-1)}_{\text{family (5.162)}} + \underbrace{d \cdot d(d-1)}_{\text{family (5.163)}} + \underbrace{(d(d-1) \cdot d(d-1) - d(d-1))}_{\text{family (5.164)}} = d^4 - 2d^2 + d. \quad (5.182)$$

Likewise, from (5.171),

$$|B_{\mathcal{K}_2}| = \underbrace{(d(d-1) - 1)}_{\text{family (5.167)}} + \underbrace{(d(d-1) - 1)}_{\text{family (5.168)}} + \underbrace{(d-1)}_{\text{family (5.169)}} + \underbrace{1}_{\text{vector (5.170)}} = 2d^2 - d - 2. \quad (5.183)$$

Adding (5.182) and (5.183), we find

$$|B_{\mathcal{K}_1}| + |B_{\mathcal{K}_2}| = d^4 - 2. \quad (5.184)$$

Finally, since $\mathcal{K}$ is the Hilbert–Schmidt orthogonal complement of $\text{span}\{I, F\}$ inside the d^4 -dimensional operator space $\mathcal{L}(\mathcal{H}_d^{\otimes 2})$, we have

$$\dim(\mathcal{K}) = d^4 - 2. \quad (5.185)$$

Combining (5.180), the trivial intersection $\mathcal{K}_1 \cap \mathcal{K}_2 = \{0\}$, and (5.184)–(5.185), we conclude that

$$\dim(\mathcal{K}_1 + \mathcal{K}_2) = \dim(\mathcal{K}_1) + \dim(\mathcal{K}_2) = d^4 - 2 = \dim(\mathcal{K}).$$

Since $\mathcal{K}_1 + \mathcal{K}_2$ is a subspace of $\mathcal{K}$ with the same dimension, it must equal $\mathcal{K}$. Together with orthogonality, this proves (5.179). $\square$

Corollary 5.3.19. *The set $B_{\mathcal{K}_1} \cup B_{\mathcal{K}_2}$ is a basis of $\mathcal{K}$.*

Proof. By Lemma 5.3.15, the set $B_{\mathcal{K}_1}$ is linearly independent. By Lemma 5.3.17, the same holds for $B_{\mathcal{K}_2}$. Moreover, Theorem 5.3.18 shows that their spans are disjoint and together span all of $\mathcal{K}$. Hence $B_{\mathcal{K}_1} \cup B_{\mathcal{K}_2}$ is a linearly independent spanning set of $\mathcal{K}$, and therefore a basis. $\square$

5.3.2. Alternative construction with discrete α, β

The construction of Section 5.3 uses continuous parameters $\alpha, \beta \in [0, d)$. It is natural to ask whether these parameters can instead be chosen from a finite set. Doing so would replace the continuous ensemble of unitaries by a finite ensemble, which is conceptually cleaner and closer in spirit to the usual finite-dimensional notion of a unitary design.

This modification is indeed possible, but it comes with a restriction. In order for the averaging identities used in the proof of Section 5.3 to remain valid, one must take d to be an odd prime. Thus the main advantage of the present variant is that the set of unitaries becomes finite, while the main drawback is that the Hilbert-space dimension can no longer be chosen arbitrarily. In particular, this odd-prime restriction does not fit naturally with the unclonable encryption construction of [30], which will become important in Chapter 6.

In this variant, we therefore assume throughout that d is an odd prime. The index set is redefined as $\mathcal{I}_d^{\text{odd}} = \{-\frac{d-1}{2}, \dots, \frac{d-1}{2}\}$. The parameters α, β are now sampled uniformly from $\mathcal{I}_d^{\text{odd}}$ rather than from a continuous interval. To match this symmetric choice of labels, the q -intervals in the underlying CV space are also redefined as

$$\left(i\Delta - \frac{\Delta}{2}, i\Delta + \frac{\Delta}{2}\right), \quad i \in \mathcal{I}_d^{\text{odd}}. \quad (5.186)$$

5

With these changes, the unitaries $V_{\alpha\beta}$ and $\tilde{V}_{\alpha\beta}$ are defined exactly as before, except that the random parameters now range over the finite set $\mathcal{I}_d^{\text{odd}}$. The resulting superoperators $G, \tilde{G}$, and $\mathcal{R}$ are again defined by uniform averaging, now with finite sums instead of integrals. Apart from this change, the construction is unchanged.

In fact, all results of Section 5.3 continue to hold, with the same statements and the same proofs. The only point that requires attention is the basic averaging identity underlying the action of the first twirl. In the continuous-parameter case, one uses

$$\mathbb{E}_{\alpha,\beta} \omega^{\alpha(i+j-i'-j')} \omega^{\beta(i^2+j^2-(i')^2-(j')^2)} = \delta_{i'i} \delta_{j'j} + \delta_{i'j} \delta_{ji'} - \delta_{ij} \delta_{i'i} \delta_{j'j}. \quad (5.187)$$

For discrete α, β , the corresponding average becomes

$$\begin{aligned} \frac{1}{d^2} \sum_{\alpha,\beta \in \mathcal{I}_d^{\text{odd}}} \omega^{\alpha(i+j-i'-j')} \omega^{\beta(i^2+j^2-(i')^2-(j')^2)} \\ = \delta_{i'+j', i+j}^{(\text{mod } d)} \delta_{(i')^2+(j')^2, i^2+j^2}^{(\text{mod } d)}. \end{aligned} \quad (5.188)$$

Thus the relevant constraints are now imposed only modulo d .

This is precisely where the primality of d becomes important. If d is prime, then the modular conditions in (5.188) imply the same pair structure as before, namely that

$$\{i', j'\} = \{i, j\} \pmod{d}, \quad (5.189)$$

so no new solutions appear. By contrast, if d were composite, say $d = pq$, then the modular constraints could admit additional solutions. For example, when

$$j - i = p + q \cdot \text{integer}, \quad (5.190)$$

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one could also have a solution of the form

$$(i', j') = (i + p, j - p) \pmod{d}, \quad (5.191)$$

which is not simply a permutation of (i, j) . Such extra solutions would alter the structure of the twirling map and would invalidate the previous proofs.

For this reason, the discrete-parameter version of the construction is restricted to odd prime d . Under this assumption, the same analysis as in Section 5.3 goes through unchanged, and one obtains the same approximate unitary 2-design behaviour, now generated by a finite ensemble of unitaries.

5.4. Physical implementation and constraints

The first question we must ask is whether the boxed unitaries (or a close enough approximation) can be physically implemented by Gaussian operators only. This is an important point for the viability of this construction, since Gaussian operations are easy to set up and cheap to implement³. From the literature, we already know that true Gaussians cannot form a 2-design. We thus need to investigate whether they can be *close* to true Gaussians. Recall that $\omega := e^{2\pi i/d}$. The boxed Gaussian gates on the boxed subspace are $V_{\alpha\beta} := \omega^{\alpha\hat{Q} + \beta\hat{Q}^2}$, $\hat{Q}|i\rangle = i|i\rangle$. Fix a true CV Gaussian phase gate $U_{ab} := \exp(i(a\hat{q} + b\hat{q}^2))$, $a, b \in \mathbb{R}$. Unfortunately, we find that for large β (hence large b), the action of U_{ab} on at least one boxed ket cannot be close to *any* “constant-phase” action on that bin, and therefore cannot be close to $V_{\alpha\beta}$.

Consider the bin $i = 0$, so $q = x \in [0, \Delta)$ in the boxed parametrisation. For simplicity we also set $a = 0$, since adding a linear phase cannot remove the quadratic oscillation. Then

$$U_{0b}|0\rangle = \frac{1}{\sqrt{\Delta}} \int_0^\Delta dx e^{ibx^2} |q = x\rangle. \quad (5.192)$$

For an arbitrary phase $\theta \in \mathbb{R}$, orthonormality $\langle q|q'\rangle = \delta(q - q')$ gives the exact identity

$$\|(U_{0b} - e^{i\theta})|0\rangle\|^2 = \frac{1}{\Delta} \int_0^\Delta dx |e^{ibx^2} - e^{i\theta}|^2 = 2 - 2 \operatorname{Re}(e^{-i\theta} J(b)), \quad (5.193)$$

where

$$J(b) := \frac{1}{\Delta} \int_0^\Delta dx e^{ibx^2}. \quad (5.194)$$

³Any Gaussian operation can be implemented as a passive interferometer, followed by parallel single-mode squeezers, followed by another passive interferometer. See Section 2.4

Optimising over θ yields

$$\inf_{\theta \in \mathbb{R}} \|(U_{0b} - e^{i\theta})|0\rangle\|^2 = 2 - 2|J(b)|.$$

In particular, since $V_{\alpha\beta}|0\rangle = |0\rangle$ for all α, β , we obtain

$$\|(U_{0b} - V_{\alpha\beta})\Pi\|_{\infty} \geq \|(U_{0b} - V_{\alpha\beta})|0\rangle\| = \|(U_{0b} - \mathbf{1})|0\rangle\| \geq \sqrt{2 - 2|J(b)|}, \quad (5.195)$$

where Π is as defined in Definition 5.2.3. Making the change of variables $x = \Delta t$ in (5.194) shows that $J(b)$ depends only on the dimensionless combination $\kappa := b\Delta^2$:

$$J(b) = \int_0^1 dt e^{i\kappa t^2}, \quad \kappa := b\Delta^2. \quad (5.196)$$

5

It remains to identify the relevant regime of b for our boxed family. On the boxed side, $V_{\alpha\beta}$ applies the phase $\exp(\frac{2\pi i}{d}(\alpha i + \beta i^2))$ to $|i\rangle$, so the quadratic coefficient in i^2 equals $\frac{2\pi}{d}\beta$. Matching the quadratic midpoint phase of the CV gate $\exp(ibq^2)$ at $q_i = i\Delta + \Delta/2$ forces

$$b\Delta^2 = \frac{2\pi}{d}\beta \iff \kappa = \frac{2\pi}{d}\beta. \quad (5.197)$$

Since $\beta \in [0, d]$, the rescaled parameter $\kappa = \frac{2\pi}{d}\beta$ ranges over $[0, 2\pi)$. The numerical plot of the lower bound (Fig. 5.1) shows that it is strictly positive for every $\kappa \neq 0$ (and even > 1 for some κ). Therefore, for any fixed d there exist boxed unitaries (namely, those with $\beta \neq 0$) whose corresponding $\kappa = \frac{2\pi}{d}\beta$ yields a nonzero lower bound in (5.195), meaning that these boxed unitaries remain a constant distance away (in operator norm on the boxed subspace) from any true CV Gaussian approximation. Consequently, as $d \rightarrow \infty$, no true Gaussian approximation scheme exists for the boxed unitaries⁴.

This obstruction is complementary to the state-side no-go theorem of [35], which rules out Gaussian *state* ensembles approximating a projective 2-design via rigidity of second moments in the Fock basis. Here the mismatch appears already at the level of Gaussian *unitaries*: the boxed gates impose phases that are constant over each bin, while any true

⁴It is important here that the parameter β is not held fixed as d increases: at dimension d , it ranges over $[0, d]$. Writing $c = \beta/d$, we have

$$\kappa = b\Delta^2 = 2\pi c. \quad (5.198)$$

Hence, for any fixed $c \in (0, 1)$, the sequence $\beta_d = cd$ gives a value of κ that is independent of d . For example, $\beta_d = d/4$ gives $\kappa = \pi/2$ and therefore

$$\|(U_{0b} - V_{\alpha\beta_d})\Pi\|_{\infty} \geq \sqrt{2 - 2|J(\pi/2)|} > 0 \quad (5.199)$$

for every d .

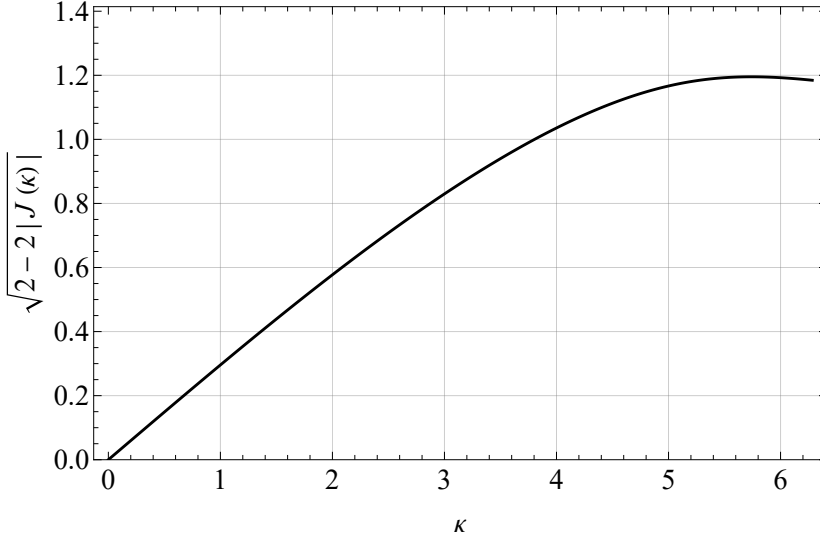


Figure 5.1: Numerical evaluation of the one-bin operator-norm lower bound $\sqrt{2 - 2|J(\kappa)|}$ as a function of $\kappa = b\Delta^2$, where $J(\kappa) := \int_0^1 e^{i\kappa t^2} dt$. For $\kappa \neq 0$ the bound is strictly positive, illustrating that a quadratic CV phase gate induces nontrivial intra-bin phase winding and therefore cannot be uniformly close (even up to a global phase) to a bin-constant boxed phase action.

quadratic CV phase gate that matches the boxed quadratic coefficient necessarily produces a nontrivial intra-bin phase profile.

This, however, is not very surprising. In the photonic CV setting, the boxed unitaries $V_{\alpha\beta}$, $\tilde{V}_{\alpha\beta}$ should be understood as *phase profiles that are constant on each quadrature bin*. This perspective is useful because it identifies $V_{\alpha\beta}$ and $\tilde{V}_{\alpha\beta}$ as instances of (non-Gaussian) *quadrature-phase gates* $e^{if(\hat{q})}$ and $e^{if(\hat{p})}$, for explicitly defined staircase functions f . We can write

$$V_{\alpha\beta} = e^{if_{\alpha\beta}(\hat{q})}, \quad f_{\alpha\beta}(q) = 2\pi \left(\frac{\alpha}{d} \lfloor q/\Delta \rfloor + \frac{\beta}{d} \lfloor q/\Delta \rfloor^2 \right) \text{ for } q \in [-\sqrt{\pi d/2}, \sqrt{\pi d/2}). \quad (5.200)$$

It can be seen that $\exp i2\pi \frac{\beta}{d} \lfloor \frac{q}{\Delta} \rfloor^2$ is typically *not* close to the Gaussian expression $\exp i2\pi \frac{\beta}{d} (\frac{q}{\Delta})^2$; the latter expression can have many oscillations in a q -interval of width Δ , whereas the former stays constant. Hence our scheme *cannot* be implemented with Gaussian operators in $\hat{p}$ and $\hat{q}$. We therefore turn to the literature on implementing non-Gaussian operators.

In travelling-wave photonic CV, implementing $e^{if(\hat{q})}$ for a general real function f is a “nonlinear phase gate” task. The prevailing implementation is *measurement-induced* (or teleportation-based): one combines linear optics with ancilla states, quadrature measure-

ments, and classical feedforward so that the desired non-Gaussian transformation is applied deterministically (or near-deterministically) to the target mode [143]. This approach is rooted in continuous-variable teleportation [40], which has been experimentally well established [78], and measurement-based CV processing [41].

Fortunately, *one does not need to implement a discontinuous staircase gate perfectly*; it suffices to implement an approximation on the finite discretisation window. There are two complementary theoretical guarantees that justify this.

(i) *Universality from polynomial Hamiltonians.* Lloyd and Braunstein [129] have shown that Gaussian operations together with an appropriate nonlinearity (e.g. a cubic phase interaction, Kerr-type nonlinearity, etc.) yield universality for transformations generated by polynomial Hamiltonians in $\hat{q}, \hat{p}$; in particular one can approximate single-mode unitaries of the form $e^{i\text{poly}(\hat{q})}$ to high accuracy [129, 41], where even high-order polynomials can be implemented in a limited number of steps. To apply this universality result, one may choose a polynomial $z_{\alpha\beta}(q)$ that approximates a smoothed version of the staircase profile $f_{\alpha\beta}(q)$ on the finite window, and then target $e^{iz_{\alpha\beta}(\hat{q})}$ as an implementable proxy⁵ for $e^{if_{\alpha\beta}(\hat{q})}$. This does not establish uniform approximation of the original staircase gate; the remaining error must instead be evaluated on the relevant class of input states.

(ii) *Direct measurement-induced synthesis of nonlinear quadrature phase gates.* Beyond universality as an existence statement, there are explicit measurement-induced protocols to realise nonlinear quadrature phase gates. Filip *et al.* [73] have developed a measurement-induced toolbox for CV interactions using linear optics, homodyne detection, and offline resources. Marek and Filip [139] have proposed a deterministic method for implementing weak cubic nonlinearity using ancilla states and feedforward. Miyata *et al.* propose an implementation of a cubic gate by adaptive non-Gaussian measurement [147]. Most directly aligned with our needs, Marek *et al.* [140] presented a *general methodology for deterministic realisation of arbitrary-order nonlinear quadrature phase gates*, enabling direct implementations of single-mode transformations that nonlinearly modify one quadrature variable, using ancillary modes, quadrature measurements, and adaptive control.

While strong deterministic optical nonlinearities at the single-photon level are challenging, measurement-induced strategies have demonstrated effective nonlinear interactions on weak optical states. A representative example is the experimental emulation of a strong Kerr nonlinearity via measurement-induced operations (combining photon addition/subtraction with homodyne-based characterisation) [52]. Such demonstrations support the practical feasibility of measurement-induced nonlinear phase processing of travelling light.

⁵And similarly for $\hat{p}$. What can be achieved for one quadrature can also be achieved for the other.

5.5. Discussion

The main message of this chapter is that, although exact continuous-variable unitary 2-designs do not exist on the full infinite-dimensional Hilbert space in the usual sense, a meaningful and explicit approximate substitute can nevertheless be constructed once one works with a physically motivated regularisation. The regularisation adopted here is a discretisation of quadrature space. This led to a finite-dimensional Hilbert space $\mathcal{H}_d$ that retains a clear interpretation in terms of the underlying q - and p -quadratures, while being sufficiently structured to support a design analysis of the usual finite-dimensional kind.

For a fixed energy cutoff, the discretisation error can be made arbitrarily small by increasing the parameter d , so that the discretised description converges to the underlying CV state in a controlled way. At the same time, increasing d also improves the quality of the approximate design itself, since the design error scales as $\varepsilon = d^{-\ell}$. In this sense, larger d simultaneously sharpens the CV approximation and strengthens the design property. On the implementation side, d determines the bin size and the finite quadrature window on which the boxed phase profiles must be realised. The experimentally relevant choice of d should depend on the available state preparation, the energy range of the states being processed, and the accuracy with which the required staircase-like nonlinear phase gates can be approximated. Thus the achievable value of d in practice will be set by the experimental platform rather than by the theory alone.

Our construction was initially motivated by continuous-variable cryptography (see Section 6.5, Chapter 6), but it is also of independent interest as a finite-dimensional scheme. Even when viewed purely in finite dimensions, without reference to CV implementations, it provides a structured family of approximate unitary 2-designs that uses fewer parameters than the construction of [149]. Conceptually, the construction also suggests a broader viewpoint. The no-go results for exact CV designs indicate that one should not expect a direct infinite-dimensional analogue of Haar-random unitary behaviour. The present chapter shows, however, that this does not prevent one from recovering a useful second-moment notion of pseudorandomness in a regularised setting. What survives is not an exact CV unitary 2-design on the full Hilbert space, but rather a family of approximate designs on finite-dimensional shadows that remain closely connected to continuous-variable quadrature physics. This gives a concrete answer to the question posed at the beginning of the chapter: although exact CV unitary 2-designs are ruled out, an explicit and quadrature-compatible approximate construction is still possible.

6

Unclonable Encryption

There is no perfect quantum copier.

— William K. Wootters

Quantum cryptography exploits features of quantum mechanics that have no classical counterpart. Among the most important of these is the no-cloning principle [58, 153, 198], which forbids the perfect copying of an unknown quantum state. This principle makes it possible to formulate cryptographic tasks in which security is not based on computational hardness, but on physical limitations imposed by quantum theory. Unclonable encryption is one such task.

In unclonable encryption, a classical plaintext is encrypted into a quantum ciphertext in such a way that an adversary cannot split the ciphertext into two parts that can both later be decrypted successfully once the key is revealed. The central idea is therefore stronger than ordinary secrecy alone: the ciphertext should not only hide the message, but should also resist duplication. This makes unclonable encryption a natural primitive for one-time uses of encrypted data, where it is important that a ciphertext cannot be copied and replayed.

The term *unclonable encryption* first appeared in work by Gottesman [86], who introduced a notion combining encryption with tamper detection. Later, Leermakers and Škorić constructed an unclonable-encryption scheme with key recycling [122]. A more refined framework was introduced by Broadbent and Lord [42], who formalised the stronger notions of *unclonable security* and *unclonable indistinguishability*. In their formulation, an adversary receives a challenge ciphertext, splits it into two quantum registers, and hands them to two non-communicating parties. After the decryption key is revealed, both parties attempt to recover the plaintext. Unclonable security requires that the probability that both parties recover the message correctly is small, while unclonable indistinguishability strengthens this by asking that both parties cannot simultaneously distinguish between

two candidate plaintexts with non-negligible advantage. Several further constructions and limitations have since been established [12, 123, 117, 13, 136].

Towards unclonable encryption in continuous variables

In line with the thesis of this dissertation, we turn our attention to continuous variables. So far, unclonable encryption has been studied almost exclusively in discrete-variable quantum systems. This leaves open the question of how the same cryptographic primitive can be realised in continuous-variable systems, where the information carriers and measurement procedures are very different. What we quickly realise is that doing so is not straightforward. On the level of the construction, one must choose parameters so that the protocol remains decryptable while still preventing successful cloning attacks. On the level of the proof, the continuous-variable setting introduces effects that do not arise in finite-dimensional schemes, most notably imperfect completeness caused by the tails of continuous measurement distributions and the need to work with finite squeezing and finite acceptance windows. As a consequence, the move from discrete to continuous variables is not a routine translation, but requires new protocol choices as well as new proof techniques.

6

This chapter develops two complementary routes to continuous-variable unclonable encryption. The first route is based on *monogamy of entanglement*. Here the goal is to construct a continuous-variable unclonable-encryption scheme that satisfies the notion of unclonable security. The proof proceeds by relating the relevant cloning experiment to a continuous-variable monogamy-of-entanglement game, for which bounds on the winning probability are available [55]. To make this connection precise, one passes through a sequence of hybrid arguments that translate the operational attack on the ciphertext into a form that can be analysed using the monogamy game. This approach yields a first continuous-variable unclonable-encryption scheme and establishes unclonable security in the Broadbent–Lord framework.

The second route is motivated by the stronger notion of *unclonable indistinguishability*. In the discrete-variable setting, this notion is closely connected to decoupling-based constructions. Majenz *et al.* [136] identified an optimal unclonable-encryption format for uniformly distributed plaintexts, in which the message is embedded into orthogonal subspaces and then hidden by a Haar-random unitary or, equivalently for second-moment purposes, by a unitary 2-design. More recently, Bhattacharyya and Culf [30] showed how *decoupling* can be used to obtain information-theoretic unclonable indistinguishability.

Decoupling is a randomisation mechanism that destroys correlations between a target system and a reference system. One-shot decoupling theorems [65] give sufficient (and essentially tight) entropic criteria guaranteeing bounds on the decoupling. Decoupling is

closely tied to state merging and the operational meaning of conditional entropy developed by Horodecki, Oppenheim, and Winter [101, 102]. While many decoupling statements are first proved for Haar-random unitaries, only second-moment information is used, and hence unitary 2-designs suffice. Moreover, approximate 2-designs yield approximate decoupling guarantees. This connection was made explicit by Szehr *et al.* [174]. The same perspective suggests a natural path in the continuous-variable setting. If one has a physically meaningful regularised notion of a continuous-variable approximate unitary 2-design (like the one developed in Chapter 5), then those unitaries can serve as encryption operations, and decoupling can be used to prove unclonable indistinguishability. This provides the conceptual basis for the second scheme discussed in this chapter.

Outline of the chapter

The aim of this chapter is to develop continuous-variable unclonable-encryption schemes that satisfy standard unclonability notions in the sense of the modern UE framework. The chapter therefore has two objectives. The first is to show that unclonable encryption can indeed be realised with continuous-variable states by giving a concrete scheme, proving its unclonable security and establishing its practicality. The second is to go beyond this first step and show how the stronger notion of unclonable indistinguishability can be obtained in the continuous-variable setting by combining decoupling arguments with continuous-variable approximate unitary 2-designs.

The chapter is organised accordingly. In Section 6.1 we collect the necessary definitions of unclonable encryption. In Section 6.2 we introduce a practically motivated CV unclonable encryption protocol. In Section 6.3 we prove unclonability of the protocol using a series of hybrids starting from a CV monogamy game. In Section 6.4, we investigate the practicality of implementing such a protocol and test its robustness against noisy conditions. In Section 6.5, we present our second protocol, which satisfies the property of unclonable indistinguishability, and demonstrate a short proof based on previous results. Finally, we conclude the chapter with Section 6.6, discussing limitations and open questions while comparing present results with the literature.

6.1. Unclonable Security

In Section 3.2 we will introduce a scheme that encrypts a classical message into a quantum ciphertext (cipherstate), using a classical key. For the formal description of such a scheme we follow the definition given in [42], with a small modification: we allow for a small probability of failure in the decryption.

Definition 6.1.1. Let $\lambda \in \mathbb{N}^+$ be a security parameter. Let $\mathcal{M}(\lambda)$ be the (classical) plaintext space. Let $\mathcal{K}(\lambda)$ be the (classical) key space. A **quantum encryption of classical messages (QECM)** scheme is a triplet of efficient quantum circuits $(Key_\lambda, Enc_\lambda, Dec_\lambda)$ implementing CPTP maps of the form

$$Key_\lambda : \mathcal{D}(\mathcal{C}) \rightarrow \mathcal{D}(\mathcal{H}_{\mathcal{K}(\lambda)}) \quad (6.1)$$

$$Enc_\lambda : \mathcal{D}(\mathcal{H}_{\mathcal{K}(\lambda)} \otimes \mathcal{H}_{\mathcal{M}(\lambda)}) \rightarrow \mathcal{D}(\mathcal{H}_{T,\lambda}) \quad (6.2)$$

$$Dec_\lambda : \mathcal{D}(\mathcal{H}_{\mathcal{K}(\lambda)} \otimes \mathcal{H}_{T,\lambda}) \rightarrow \mathcal{D}(\mathcal{H}_{\mathcal{M}(\lambda)}) \quad (6.3)$$

where $\mathcal{H}_{\mathcal{K}(\lambda)}$ is the Hilbert space of the key, $\mathcal{H}_{\mathcal{M}(\lambda)}$ is the Hilbert space of the plaintext, and $\mathcal{H}_{T,\lambda}$ is the cipherstate space.

Let E_k denote the CPTP map $\rho \mapsto Enc_\lambda(|k\rangle\langle k| \otimes \rho)$, and let D_k stand for the CPTP map $\rho \mapsto Dec_\lambda(|k\rangle\langle k| \otimes \rho)$. For all $k \in \mathcal{K}(\lambda)$, $m \in \mathcal{M}(\lambda)$, it must hold that

$$\text{Tr} [|k\rangle\langle k| Key_\lambda(1)] > 0 \quad \implies \quad \text{Tr} [|m\rangle\langle m| D_k \circ E_k(|m\rangle\langle m|)] \geq 1 - \varepsilon_{\text{DF}} \quad (6.4)$$

where ε_{DF} is the tolerated probability of decryption failure.

Unclonability of a QECM scheme is defined via a *cloning game*. A challenger prepares a cipherstate and gives it to Alice. Alice splits the cipherstate into two pieces; one piece goes to Bob, one to Charlie. Then Bob and Charlie receive the decryption key and must *both* produce the correct plaintext, without being allowed to communicate. A QECM scheme is considered to be secure against cloning if the three players ABC, acting together, have an exponentially small probability of winning the game. We follow the definitions of [42] for the cloning game and the security against cloning.

Definition 6.1.2 (Cloning attack). Let S be a QECM scheme, with Hilbert spaces $\mathcal{H}_{\mathcal{K}(\lambda)}$, $\mathcal{H}_{\mathcal{M}(\lambda)}$, $\mathcal{H}_{T,\lambda}$ as given in Def. 6.1.1. Let $\mathcal{H}_B$ and $\mathcal{H}_C$ be Hilbert spaces of arbitrary dimension. A cloning attack against S is a triplet of efficient quantum circuits $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ implementing CPTP maps of the form

$$\mathcal{A} : \mathcal{D}(\mathcal{H}_{T,\lambda}) \rightarrow \mathcal{D}(\mathcal{H}_B \otimes \mathcal{H}_C), \quad (6.5)$$

$$\mathcal{B} : \mathcal{D}(\mathcal{H}_{\mathcal{K}(\lambda)} \otimes \mathcal{H}_B) \rightarrow \mathcal{D}(\mathcal{H}_{\mathcal{M}(\lambda)}), \text{ and} \quad (6.6)$$

$$\mathcal{C} : \mathcal{D}(\mathcal{H}_{\mathcal{K}(\lambda)} \otimes \mathcal{H}_C) \rightarrow \mathcal{D}(\mathcal{H}_{\mathcal{M}(\lambda)}). \quad (6.7)$$

Definition 6.1.3. Consider the cloning attack according to Def. 6.1.2. Let B_k stand for the CPTP map $\rho \mapsto \mathcal{B}(|k\rangle\langle k| \otimes \rho)$, and analogously C_k . Let $\mathcal{M}(\lambda) = \{0, 1\}^n$. A QECM scheme S is $\tau(\lambda)$ -**unclonable secure** if for all cloning attacks $(\mathcal{A}, \mathcal{B}, \mathcal{C})$ against S , there exists a negligible function η such that

$$\mathbb{E}_{m,k} \text{Tr} \left((|m\rangle\langle m| \otimes |m\rangle\langle m|) (B_k \otimes C_k) \circ \mathcal{A} \circ E_k(|m\rangle\langle m|) \right) \leq 2^{-n+\tau(\lambda)} + \eta(\lambda). \quad (6.8)$$

Here the expectation $\mathbb{E}$ is over uniform m , and k distributed according to $\Pr[K = k] = \text{Tr}[|k\rangle\langle k| \text{Key}_\lambda(1)]$. If S is 0-unclonable secure, we simply say that it is unclonable secure.

The word *splits* should be understood in the most general operational sense: A may apply any efficient quantum channel to the cipherstate and distribute an arbitrary, possibly entangled, state across $\mathcal{H}_B \otimes \mathcal{H}_C$, rather than attempting to create two literal copies. The fact that A does not receive k as input captures the requirement that the split occurs before the key is revealed, while the tensor-product form $B_k \otimes C_k$ ensures that Bob and Charlie act locally and cannot communicate afterwards. The trace expression in the definition is their joint winning probability, meaning that both outputs must equal the original plaintext. For a uniformly random n -bit plaintext, 2^{-n} is the success probability of a coordinated blind guess, while $\tau(\lambda)$ relaxes this benchmark by a factor of $2^{\tau(\lambda)}$ and $\eta(\lambda)$ allows for only a negligible additional term.

Our security proof will make use of recent results on *entanglement monogamy games* for CV systems [55]. Such a game is played between Alice on one side and Bob and Charlie on the other. Bob and Charlie prepare a tripartite state ρ_{ABC} of their choice. Then Alice does an unpredictable measurement, and Bob and Charlie have to show that they are both sufficiently entangled with Alice to guess Alice's outcome to some degree of accuracy. Below we present the *coset monogamy game on $\mathbb{R}^N$* as introduced in Section 4 of [55]. In the original version, the measurement bases are continuous-variable coset states of $\mathbb{R}^N$,

given by $|P_{q,p}\rangle = \bigotimes_{i=1}^N \begin{cases} |q = q_i\rangle, & i \notin I, \\ |p = p_i\rangle, & i \in I. \end{cases}$ Here, I is a subset of $[N]$. We note that

this is equivalent to performing a homodyne measurement in the q direction if $i \notin I$, and otherwise in the p direction. We refer to this game as the ‘partial’ CV monogamy of entanglement game, since it requires Bob and Charlie to produce guesses for only one quadrature direction instead of two. In Section 6.3.6 we introduce a ‘full’ version of the game.

Game 6.1.4: CV Partial Monogamy of Entanglement Game

1. **Initial state preparation.**

Bob and Charlie prepare an M -mode state ρ_{ABC} across three registers: one for Alice, one for Bob, and one for Charlie. After preparation, they are no longer allowed to communicate.

2. **Alice's measurement choice and outcomes.**

Alice chooses quadrature directions $(\theta_i)_{i=1}^M$, $\theta_i \in \{0, \frac{\pi}{2}\}$, such that $|\{i : \theta_i = 0\}| = M/2$, i.e., each direction is chosen exactly $M/2$ times. She then does homodyne detection of mode i in direction θ_i , getting outcomes that we denote by

$$q_i \quad \text{if } \theta_i = 0, \quad p_i \quad \text{if } \theta_i = \frac{\pi}{2}.$$

3. **Announcement and responses.**

Alice announces $(\theta_i)_{i=1}^M$. Bob outputs $q_B \in \mathbb{R}^{M/2}$ containing his guesses for Alice's q -values. Charlie outputs $p_C \in \mathbb{R}^{M/2}$ containing his guesses for Alice's p -values.

4. **Winning condition.**

Bob and Charlie win the game if

$$\|q - q_B\|_\infty < \delta \quad \text{and} \quad \|p - p_C\|_\infty < \varepsilon. \quad (6.9)$$

The ∞ -norm in the winning condition means that all guesses $(q_B)_i$ and $(p_C)_i$ must be close to Alice's values.

As shown in [55], this game can be viewed as an *abelian coset measure* monogamy game, where Bob and Charlie attempt to guess the measurement outcomes in each quadrature within error neighbourhoods $(-\delta, \delta)^N$ and $(-\varepsilon, \varepsilon)^N$. The following upper bound was obtained on the winning probability.

Lemma 6.1.5. (Theorem 4.1 in [55]). *In the CV Partial Monogamy of Entanglement Game (6.1.4), the winning probability w is upper bounded as*

$$w \leq \frac{1}{\binom{M}{M/2}} \sum_{k=0}^{M/2} \binom{M/2}{k}^2 (2\sqrt{\delta\varepsilon})^k \leq \sqrt{e} \left(\frac{1}{2} + \sqrt{\delta\varepsilon}\right)^{\frac{M}{2}}. \quad (6.10)$$

6.2. Protocol Description

We propose a QECM scheme that makes use of squeezed coherent states. This is a more general treatment, since coherent states are just a special case of squeezed states. The scheme encrypts a message $m \in \{0, 1\}^n$ into an N -mode cipherstate. First, we apply a

6. Unclonable Encryption

symmetric classical encryption scheme; this step ensures message confidentiality independent of the unclonability. We do not specify which symmetric cipher is used. We denote the encryption and decryption operations as $\text{Enc}_{\text{base}}(\cdot, \cdot)$ and $\text{Dec}_{\text{base}}(\cdot, \cdot)$.

Then the classical ciphertext gets encoded into an N -bit codeword c . The error-correcting code is chosen such that it can correct t bit errors. The encoding and decoding algorithms are denoted by Encode , Decode .

We encode a bit value c_i into a CV mode by displacing a squeezed vacuum state by a distance $(-1)^{c_i} \alpha$, where $\alpha > 0$ is a constant. The displacement is in the ‘narrow’ direction of the state. The underlying idea is that the squeezing direction is part of the encryption key; the attacker $\mathcal{A}$, who does not know this direction, has trouble determining c and hence cannot create good clones. This encoding is similar to conjugate coding for qubits [192].

For proof-technical reasons, the squeezing directions are not chosen uniformly at random. We impose the constraint that exactly half the modes are squeezed in the q -direction, and the other half in the p -direction. Hence the corresponding key space is not $\{0, 1\}^N$ but rather the set $\mathcal{L} = \{1, \dots, \binom{N}{N/2}\}$ of labels which uniquely enumerate the strings with Hamming weight $N/2$.

The QECM algorithms (Key, Enc, Dec) are given below.

Algorithm 1 Key Generation (Key)

Input: $z(\lambda)$, $N(\lambda)$, $r(\lambda)$, $\alpha(\lambda)$.

Output: a symmetric key $s \in \{0, 1\}^z$, a binary string $\phi \in \{0, 1\}^N$ and a real vector $k \in \mathbb{R}^N$.

- 1: Sample s uniformly from $\{0, 1\}^z$.
 - 2: Sample label ℓ uniformly from $\{1, \dots, \log_2 \binom{N}{N/2}\}$. Convert ℓ to a string $\phi \in \{0, 1\}^N$ with Hamming weight $N/2$.
 - 3: **for** $i = 1$ to N **do**
 - 4: Sample k_i from the normal distribution $\mathcal{N}(0, \frac{1}{2} \cosh r \tanh^2 r)$ truncated to the interval $(-\alpha \tanh r, \alpha \tanh r)$.
 - 5: **end for**
 - 6: $k = (k_1, \dots, k_N)$.
 - 7: Output (s, ϕ, k) .
-

Note that the pdf of k_i is proportional to $\exp(-\frac{k_i^2}{\cosh r \tanh^2 r})$, i.e., is of Gaussian form, but the support $k_i \in (-\alpha \tanh r, \alpha \tanh r)$ is quite narrow compared to the width of the Gaussian.

Algorithm 2 Encryption (Enc)

Input: Key $(s, \phi, k) \in \{0, 1\}^z \times \{0, 1\}^N \times \mathbb{R}^N$; message $m \in \{0, 1\}^n$; parameters $\alpha, r \in \mathbb{R}^+$.

Output: Cipherstate $\rho \in \mathcal{D}(\mathcal{H}_1^{\otimes N})$.

- 1: Compute ciphertext $m' = \text{Enc}_{\text{base}}(s, m)$.
- 2: Compute codeword $c = \text{Encode}(m')$.
- 3: **for** $i = 1$ to N **do**
- 4: Prepare a single-mode squeezed state ρ_i with displacement d_i and covariance matrix Γ_i , where $d_i = [\alpha(-1)^{c_i} + k_i] \begin{pmatrix} \overline{\phi_i} \\ \phi_i \end{pmatrix}$, and $\Gamma_i = \begin{pmatrix} (\cosh r)^{2\phi_i-1} & 0 \\ 0 & (\cosh r)^{1-2\phi_i} \end{pmatrix}$.
- 5: **end for**
- 6: $\rho = \rho_1 \otimes \cdots \otimes \rho_N$.
- 7: Output ρ .

As a reminder, the notation $\mathcal{D}(\mathcal{H})$ stands for the set of density operators on $\mathcal{H}$. The Hilbert space of a single CV mode is denoted by $\mathcal{H}_1$.

6

Algorithm 3 Decryption (Dec)

Input: Key $(s, \phi, k) \in \{0, 1\}^z \times \{0, 1\}^N \times \mathbb{R}^N$; cipherstate $\rho \in \mathcal{D}(\mathcal{H}_1^{\otimes N})$.

Output: Message $\hat{m} \in \{0, 1\}^n$.

- 1: **for** $i = 1$ to N **do**
- 2: Perform homodyne measurement on ρ_i in the $\phi_i \cdot \frac{\pi}{2}$ direction, resulting in outcome $y_i \in \mathbb{R}$.
- 3: $\hat{c}_i = \frac{1}{2} - \frac{1}{2} \text{sign}(y_i - k_i)$.
- 4: **end for**
- 5: $\hat{c} = (\hat{c}_1, \dots, \hat{c}_N)$.
- 6: $\mu = \text{Decode}(\hat{c})$.
- 7: $\hat{m} = \text{Dec}_{\text{base}}(s, \mu)$.
- 8: Output $\hat{m}$.

Define the single-bit error probability

$$\beta := \frac{1}{2} \text{Erfc}(\alpha \sqrt{\cosh r}) = \frac{1}{2} - \frac{1}{2} \text{Erf}(\alpha \sqrt{\cosh r}). \quad (6.11)$$

Theorem 6.2.1. Assume that $(t+1)/N > \beta$. Our construction is a QECM scheme with decryption failure parameter

$$\varepsilon_{\text{DF}} = \exp \left[-N D_{\text{KL}} \left(\frac{t+1}{N} \parallel \beta \right) \right], \quad (6.12)$$

6. Unclonable Encryption

where D_{KL} denotes the binary Kullback–Leibler divergence, $D_{\text{KL}}(a\|b) = a \ln(a/b) + (1 - a) \ln((1 - a)/(1 - b))$, and Erfc is the complementary error function.

Proof. It is trivial to see that the triplet (Key, Enc, Dec) fits the format of the CPTP maps in Def. 6.1.1. All that is left to show is that the protocol satisfies the correctness condition. A bit error in the codeword bit c_i occurs when $y_i - k_i$ has the wrong sign. Consider, without loss of generality, $c_i = 0$. Then the bit-error probability is $\Pr[Y_i - k_i < 0] = \beta$. When k_i is known, the random variable $Y_i - k_i$ is Gaussian-distributed with mean α and variance $1/(2 \cosh r)$, which gives the value of β in Eq. (6.11).

A decoding error can occur only when there are more than t bit flips; this occurs with probability

$$P = \sum_{j=t+1}^N \binom{N}{j} \beta^j (1 - \beta)^{N-j} = \sum_{\ell=0}^{N-t-1} \binom{N}{\ell} \beta^{N-\ell} (1 - \beta)^\ell. \quad (6.13)$$

Since $(t + 1)/N > \beta$, applying the Chernoff bound to this upper binomial tail gives

$$P \leq \exp \left[-N D_{\text{KL}} \left(\frac{t+1}{N} \parallel \beta \right) \right] = \varepsilon_{\text{DF}}. \quad (6.14)$$

□

Note that asymptotically N gets close to $\frac{n}{1-h(\beta)}$, where h is the binary entropy function.

6.3. Proving Unclonability

6.3.1. Game hopping

We prove that our scheme satisfies the security definition of Def. 6.1.3. We do this as follows. (i) We rewrite the state preparation from the original prepare-and-send form to entanglement-based (EB) form. (ii) We show, in a series of ‘hops’ (hybrids), that the cloning game that the security definition is based on is equivalent to a CV entanglement monogamy game. (iii) Finally, we use Lemma 6.1.5, which upper bounds the winning probability in the CV entanglement monogamy game.

The sequence of games is as follows:

- The cloning game for the actual QECM scheme.
- The cloning game for the entanglement-based form of the QECM scheme.

- A variant of the above game, where now the keys s and k are *not* revealed to Bob and Charlie. Only the squeezing directions ϕ are revealed.
- The Full CV entanglement monogamy game.
- The Partial CV entanglement monogamy game (Game 6.1.4).

6.3.2. Entanglement-based version

The first hop is to replace (in the encryption algorithm) the drawing of k_i and the preparation of the single-mode state ρ_i by the following procedure. (1) Prepare a two-mode entangled state $\rho_{ChA}^{(i)}$. (2) Do a homodyne measurement on the ‘Ch’ subsystem in the ϕ_i direction, yielding outcome u_i . Compute $k_i = [u_i - \alpha(-1)^{c_i}] \tanh r$.

The state $\rho_{ChA}^{(i)}$ must be such that it yields the correct distribution for k_i (truncated Gaussian) when the Ch system is measured in the ϕ_i direction, *and* the state of the ‘A’ subsystem gets projected to the correct squeezed coherent state. Without the truncation, $\rho_{ChA}^{(i)}$ would simply be given by the displaced EPR state mentioned in Section 2.3. In order to reproduce the truncated k_i -interval, however, the EPR state needs to be modified. The details are presented below.

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6.3.3. The Entangled State

In the prepare-and-send scheme (algorithm 2), we restrict k_i to the interval $(-\alpha \tanh r, \alpha \tanh r)$. To ensure compatibility with the EB version, the corresponding outcome u_i of the homodyne measurement on the Challenger’s mode needs to be similarly restricted, $u_i - \alpha(-1)^{c_i} \in (-\alpha, \alpha)$. Consider the two-mode squeezed state without displacement

$$|\text{EPR}_{r,\varphi}(0)\rangle = \frac{1}{\cosh \frac{r}{2}} \sum_{n=0}^{\infty} \left(-e^{i\varphi} \tanh \frac{r}{2} \right)^n |n, n\rangle. \quad (6.15)$$

In the position eigenbasis of both modes, the amplitude of this state can be represented using Hermite polynomials H_n . For $\phi = 0$ we have

$$\langle x_1, x_2 | \text{EPR}_{r,0}(0) \rangle = \frac{e^{-\frac{1}{2}(x_1^2 + x_2^2)}}{\sqrt{\pi} \cosh \frac{r}{2}} \sum_{n=0}^{\infty} \frac{1}{n!} H_n(x_1) H_n(x_2) \left(\frac{1}{2} \tanh \frac{r}{2} \right)^n. \quad (6.16)$$

Using the identity $\sum_{n=0}^{\infty} \frac{1}{n!} H_n(x) H_n(y) \left(\frac{z}{2} \right)^n = \frac{1}{\sqrt{1-z^2}} \exp \left[\frac{2z}{1+z} xy - \frac{z^2}{1-z^2} (x-y)^2 \right]$, we can simplify the expression to

$$\langle x_1, x_2 | \text{EPR}_{r,0}(0) \rangle = \frac{1}{\sqrt{\pi}} e^{-\frac{1}{4} e^r (x_1 - x_2)^2} e^{-\frac{1}{4} e^{-r} (x_1 + x_2)^2}. \quad (6.17)$$

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Now we switch to the displaced EPR state, where both modes are translated by $\pm\alpha$ in the $\phi = 0$ direction,

$$\langle x_1, x_2 | \text{EPR}_{r,0}(\pm\alpha) \rangle = \frac{1}{\sqrt{\pi}} e^{-\frac{1}{4}e^r(x_1-x_2)^2} e^{-\frac{1}{4}e^{-r}(x_1+x_2 \mp 2\alpha)^2}. \quad (6.18)$$

Finally, we impose the restriction on the homodyne outcome by restricting the x_1 -range,

$$|\text{EPR}_{r,0}(\pm\alpha)\rangle_{\text{restr}} \propto \int_{\pm\alpha-\alpha}^{\pm\alpha+\alpha} dx_1 |x_1\rangle \otimes \int_{-\infty}^{\infty} dx_2 |x_2\rangle \langle x_1, x_2 | \text{EPR}_{r,0}(\pm\alpha) \rangle \quad (6.19)$$

$$\propto \int_{\pm\alpha-\alpha}^{\pm\alpha+\alpha} dx_1 |x_1\rangle e^{-\frac{(x_1-\pm\alpha)^2}{2\cosh r}} \otimes \int_{-\infty}^{\infty} dx_2 |x_2\rangle e^{-\frac{\cosh r}{2}[x_2-\pm\alpha-(x_1-\pm\alpha)\tanh r]^2} \quad (6.20)$$

$$= \int_{-\alpha}^{\alpha} dx |x \pm \alpha\rangle e^{-\frac{x^2}{2\cosh r}} \otimes \int_{-\infty}^{\infty} dx' |x' \pm \alpha\rangle e^{-\frac{\cosh r}{2}[x'-x\tanh r]^2}. \quad (6.21)$$

It is not necessary to have a Gaussian-like distribution for the quadrature in the Challenger's mode. Here we specified an entangled state that resembles the standard TMS state, because it has a special property: The effect of $|\text{EPR}_{r,\phi}(\pm\alpha)\rangle_{\text{restr}}$ can be obtained (although inefficiently) by taking the standard TMS state, homodyne measuring the Ch mode, and discarding if u_i does not lie in the right interval.

6.3.4. Unclonable Encryption game

The cloning attack (Def. 6.1.2) can be represented as the game below, between the Challenger and the three players $\mathcal{ABC}$.

Game A: Unclonable Encryption

1. **Initial state preparation.**

The challenger Ch picks a random message m .

Prepare-and-send: Ch runs Key (algorithm 1) and obtains (s, ϕ, k) . He runs Enc (algorithm 2) on m . He sends the resulting cipherstate to Alice.

Entanglement-based: Ch runs part of algorithm 1 and obtains (s, ϕ) . He prepares an entangled state $\rho_{ChA} = \bigotimes_{i=1}^N \rho_{ChA}^{(i)}$, as explained in Section 6.3.2. He sends the 'A' part to Alice.

2. **Distributing quantum information to co-players.**

Alice ($\mathcal{A}$) “splits” her quantum state and sends the two parts to her co-players, Bob ($\mathcal{B}$) and Charlie ($\mathcal{C}$). After that $\mathcal{A}, \mathcal{B}$ and $\mathcal{C}$ are no longer allowed to communicate.

3. **Key opening and response.**

Prepare-and-send: Ch announces the key (s, ϕ, k) . $\mathcal{B}$ and $\mathcal{C}$ respond with m_B and m_C , respectively.

Entanglement-based: Ch does homodyne measurements on his modes and computes k from the outcomes. Ch announces the key (s, ϕ, k) . $\mathcal{B}$ and $\mathcal{C}$ respond with m_B and m_C , respectively.

4. **Winning condition.**

The triplet of players $\mathcal{ABC}$ wins the game if $m_B = m_C = m$.

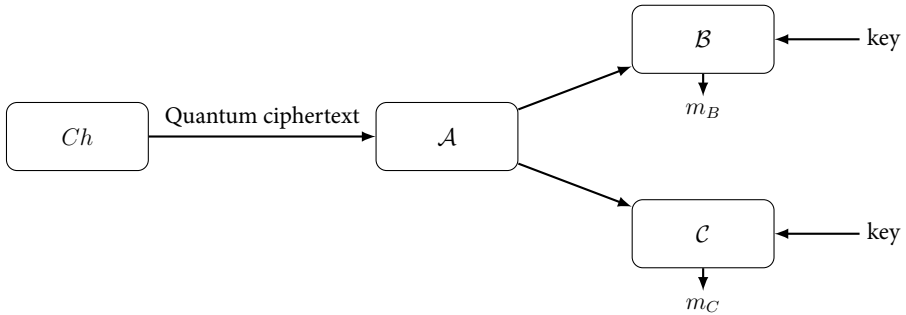


Figure 6.1: Schematic representation of the unclonable encryption game.

Note that in this game the measurement by Ch has been postponed to step 3, as opposed to step 1 in the direct EB description of the preparation of $\mathcal{A}$'s state. We will now describe the entanglement-based version (Section 6.3.2) of the above game.

Game B: EB Intermediate Unclonable Encryption

1. **Initial state preparation.**
 Ch prepares the entangled state ρ_{ChA} . He sends one half of the state to Alice.
2. **Distributing quantum information to co-players.**
 Alice ($\mathcal{A}$) “splits” her quantum state and sends the two parts to her co-players, Bob ($\mathcal{B}$) and Charlie ($\mathcal{C}$). After that $\mathcal{A}, \mathcal{B}$ and $\mathcal{C}$ are no longer allowed to communicate.
3. **Key opening (only ϕ) and response.**
 Ch measures u and announces ϕ . $\mathcal{B}$ and $\mathcal{C}$ respond with u_B and u_C , respectively. (These are their guesses for u .)
4. **Winning condition.**
 For $i \in \{1, \dots, N\}$, Ch calculates $c_{Bi} = \frac{1}{2} - \frac{1}{2} \text{sign}(u_{Bi} - \frac{k_i}{\tanh r})$ and $c_{Ci} = \frac{1}{2} - \frac{1}{2} \text{sign}(u_{Ci} - \frac{k_i}{\tanh r})$. Then $m_B = \text{Dec}_{\text{base}}(s, \text{Decode}(c_B))$ and $m_C = \text{Dec}_{\text{base}}(s, \text{Decode}(c_C))$. The players $\mathcal{ABC}$ win if $m_B = m_C = m$.

These two games are equivalent, as proved by the lemma below.

Lemma 6.3.1. *Winning the Entanglement-based Unclonable Encryption game (Game A) is equivalent to winning the EB Intermediate Unclonable Encryption game (Game B).*

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Proof. Consider the EB version of Game A. Given k_i , Bob and Charlie know that there are only two possible values for u_i , namely $u_i = \frac{k_i}{\tanh r} \pm \alpha$. Hence, getting a bit c_i correct is equivalent to making the correct binary choice for u_i . From the fact that, by construction, $u_i \in (-2\alpha, 2\alpha)$, it follows that this in turn is equivalent to determining u_i with a resolution of α or better.

In Game B it is precisely such resolution on the u -axis that is required to guess c_i . □

6.3.5. Splitting – an attack strategy

The game description allows us to analyse a simple strategy that the adversary $\mathcal{A}$ could use in an attempt to win the prepare-and-send UE game. The idea is to split each incoming mode on a balanced beam splitter and send one output mode to $\mathcal{B}$ and the other to $\mathcal{C}$.

The i th mode has displacement vector and covariance matrix

$$d_i = [\alpha(-1)^{c_i} + k_i] \begin{pmatrix} \overline{\phi_i} \\ \phi_i \end{pmatrix}, \quad \Gamma_i = \begin{pmatrix} (\cosh r)^{2\phi_i-1} & 0 \\ 0 & (\cosh r)^{1-2\phi_i} \end{pmatrix}. \quad (6.22)$$

To keep the discussion simple, we set $\phi_i = 0$ and drop the mode index. Then

$$d = \begin{pmatrix} \alpha(-1)^c + k \\ 0 \end{pmatrix}, \quad \Gamma = \begin{pmatrix} (\cosh r)^{-1} & 0 \\ 0 & \cosh r \end{pmatrix}. \quad (6.23)$$

Writing $m := \alpha(-1)^c + k$, we mix the signal with vacuum on a balanced beam splitter with symplectic matrix

$$S_{1/2} = \frac{1}{\sqrt{2}} \begin{pmatrix} I_2 & I_2 \\ -I_2 & I_2 \end{pmatrix}, \quad (6.24)$$

so that the input mean vector and covariance matrix are

$$d_{\text{in}} = \begin{pmatrix} m \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \Gamma_{\text{in}} = \Gamma \oplus I_2. \quad (6.25)$$

After the beam splitter, the mean vector becomes

$$d' = S_{1/2} d_{\text{in}} = \frac{1}{\sqrt{2}} \begin{pmatrix} m \\ 0 \\ -m \\ 0 \end{pmatrix}, \quad d_B = \begin{pmatrix} m/\sqrt{2} \\ 0 \end{pmatrix}, \quad d_C = \begin{pmatrix} -m/\sqrt{2} \\ 0 \end{pmatrix}. \quad (6.26)$$

The covariance matrix transforms as

$$\Gamma' = S_{1/2} \Gamma_{\text{in}} S_{1/2}^T = \begin{pmatrix} \Gamma_B & \gamma \\ \gamma & \Gamma_C \end{pmatrix}, \quad (6.27)$$

with

$$\Gamma_B = \Gamma_C = \frac{1}{2}(\Gamma + I_2) = \begin{pmatrix} \frac{1+(\cosh r)^{-1}}{2} & 0 \\ 0 & \frac{1+\cosh r}{2} \end{pmatrix}, \quad (6.28)$$

$$\gamma = \frac{1}{2}(I_2 - \Gamma) = \begin{pmatrix} \frac{1-(\cosh r)^{-1}}{2} & 0 \\ 0 & \frac{1-\cosh r}{2} \end{pmatrix}. \quad (6.29)$$

We now consider homodyne measurements of the x -quadrature on the two output modes. Under our Gaussian-state convention, the variance of the homodyne outcome is half of the corresponding diagonal entry of the covariance matrix. Hence

$$X_B \sim \mathcal{N}\left(\frac{m}{\sqrt{2}}, \frac{1 + (\cosh r)^{-1}}{4}\right), \quad X_C \sim \mathcal{N}\left(-\frac{m}{\sqrt{2}}, \frac{1 + (\cosh r)^{-1}}{4}\right). \quad (6.30)$$

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Once the key value k is revealed, $\mathcal{B}$ and $\mathcal{C}$ can classically post-process their data by defining

$$\tilde{X}_B := \sqrt{2} X_B - k, \quad \tilde{X}_C := -\sqrt{2} X_C - k, \quad (6.31)$$

so that both random variables have the same distribution

$$\tilde{X}_B, \tilde{X}_C \sim \mathcal{N}\left(\alpha(-1)^c, \frac{1 + (\cosh r)^{-1}}{2}\right). \quad (6.32)$$

This already shows that if $\alpha = \mathcal{O}(1)$ or greater, then Bob and Charlie can easily distinguish each bit of the ciphertext with very high probability, as the spread can be at most 1.

6.3.6. Monogamy of entanglement game

We introduce a version of the entanglement monogamy game that is closer to the encryption scheme than Game 6.1.4.

Game C: CV Full Monogamy of Entanglement game

1. Initial state preparation.

Bob and Charlie prepare an M -mode state ρ_{ABC} across three registers: one for Alice, one for Bob, and one for Charlie. After preparation, they are no longer allowed to communicate.

2. Alice's measurement choice and outcomes.

Alice chooses quadrature directions $(\theta_i)_{i=1}^M$, $\theta_i \in \{0, \frac{\pi}{2}\}$, such that $|\{i : \theta_i = 0\}| = M/2$, i.e., each direction is chosen exactly $M/2$ times. She then does homodyne detection of mode i in direction θ_i , getting outcomes that we denote by

$$q_i \quad \text{if } \theta_i = 0, \quad p_i \quad \text{if } \theta_i = \frac{\pi}{2}.$$

3. Announcement and responses.

Alice announces $(\theta_i)_{i=1}^M$. Bob outputs a list $q_B \in \mathbb{R}^{M/2}$ containing his guesses for Alice's q -values and a list $p_B \in \mathbb{R}^{M/2}$ containing his guesses for Alice's p -values. Similarly, Charlie outputs q_C and p_C .

4. Winning condition.

Bob and Charlie win the game if

$$\|q - q_B\|_\infty < \delta, \quad \|p - p_B\|_\infty < \delta, \quad \|q - q_C\|_\infty < \varepsilon, \quad \text{and} \quad \|p - p_C\|_\infty < \varepsilon. \quad (6.33)$$

Lemma 6.3.2. *Let $w^{\text{UE}}(N, t)$ denote the winning probability of Game B. Let $w^{\text{full}}(N - 2t)$ be the winning probability for Game C with $\delta = \alpha$, $\varepsilon = \alpha$, $M = N - 2t$. It holds that*

$$w^{\text{UE}}(N, t) \leq 2^{N-n} w^{\text{full}}(N - 2t). \quad (6.34)$$

Proof. In Game B, getting a bit c_i correct is equivalent to getting u_i correct within a distance α . This corresponds to setting $\delta = \alpha$ and $\varepsilon = \alpha$ in Game C.

Next, in Game B it is required to get at least $N - t$ bits of c correct. Let $e \in \{0, 1\}^N$ denote an error pattern. Let w_e^{UE} be the probability of winning with precisely error pattern e . The bit errors can be arbitrarily distributed over the q part and the p part, which does not nicely fit the symmetric structure of the monogamy game. In order to obtain the symmetric structure we loosen our requirements a little, and allow a surplus of bit errors in one block (if any) to be balanced by additionally allowed bit errors in the other block. In the worst case, all bit errors are located in one block; this leads to an allowed $2t$ errors. We write

$$\begin{aligned} w^{\text{UE}}(N, t) &= \sum_{e \in \{0,1\}^N : |e| \leq t} w_e^{\text{UE}} \leq \sum_{e \in \{0,1\}^N : |e| \leq t} w^{\text{full}}(N - 2t) \\ &= w^{\text{full}}(N - 2t) \left| \{e \in \{0,1\}^N : |e| \leq t\} \right|, \end{aligned} \quad (6.35)$$

where $|e|$ denotes the Hamming weight. Finally we apply the Hamming bound for binary codes. $\square$

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6.3.7. Results

Theorem 6.3.3. *Our QECM scheme is unclonable secure according to Def. 6.1.3, with*

$$\tau(\lambda) = \frac{N}{2} + \left(\frac{N}{2} - t \right) \log(1 + 2\alpha) + t + \frac{1}{2} \log e. \quad (6.36)$$

Proof. The cloning attack in Def. 6.1.3 has a success probability equal to the winning probability in Game A. By Lemma 6.3.1, this equals the winning probability of Game B. Next, by Lemma 6.3.2 this is upper bounded by $2^{N-n} w^{\text{full}}(N - 2t)$. We use the fact that the full monogamy game C is harder (or equally hard) to win than the partial monogamy game 6.1.4. Thus we have $w^{\text{full}}(N - 2t) \leq \sqrt{e}(\frac{1}{2} + \alpha)^{\frac{N}{2} - t}$. $\square$

Remark: For $N \gg 1, \alpha \ll 1$, it holds that $\tau(\lambda) \approx \frac{N}{2}$. This is a large number; however, it still results in a winning probability that is exponentially small in n , which is the main objective for a QECM scheme. Note that for security to hold, care must be taken that $n > N/2$.

We see from the result (6.36) that α needs to be smaller than approximately $\frac{1}{2}$, otherwise the attackers' winning probability is not exponentially small in n . This is unsurprising, for the following reason. One obvious cloning attack on our scheme is to perform a *heterodyne* measurement on each cipherstate mode individually. This corresponds to mixing the squeezed state with the vacuum state using a 50/50 beam splitter. The effect is a

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deterioration of the signal-to-noise ratio: On the one hand, the signal power α^2 is reduced to $\alpha^2/2$. On the other hand, the noise power goes from $\frac{1}{2 \cosh r}$ to $\frac{1}{2}(\frac{1}{2} + \frac{1}{2 \cosh r})$ due to the averaging with the vacuum's shot noise, which has power $\frac{1}{2}$. The resulting signal-to-noise ratio in the heterodyne measurement is $2\alpha^2/(1 + \frac{1}{\cosh r}) \approx 2\alpha^2$. Hence at $\alpha = \mathcal{O}(1)$ there is significant leakage about the codeword bits c_i , putting the unclonability at risk.

We consider the asymptotic case $N \gg 1$. We assume that the code rate approaches capacity and we write $n \approx N[1 - h(\beta)]$. From the security definition we have the requirement that $2^{-n+\tau}$ should be asymptotically small. Substituting $n \rightarrow N[1 - h(\beta)]$ into (6.36), the requirement $-n + \tau < 0$ can be rewritten as

$$h(\beta) - \left(\frac{1}{2} - \beta\right)[1 - \log(1 + 2\alpha)] < 0. \quad (6.37)$$

Where we assumed $t/N \rightarrow \beta$ in the large N limit. As β is a function of α and r (6.11), Eq. (6.37) specifies a region in the (α, r) -plane where it is possible to achieve security asymptotically.

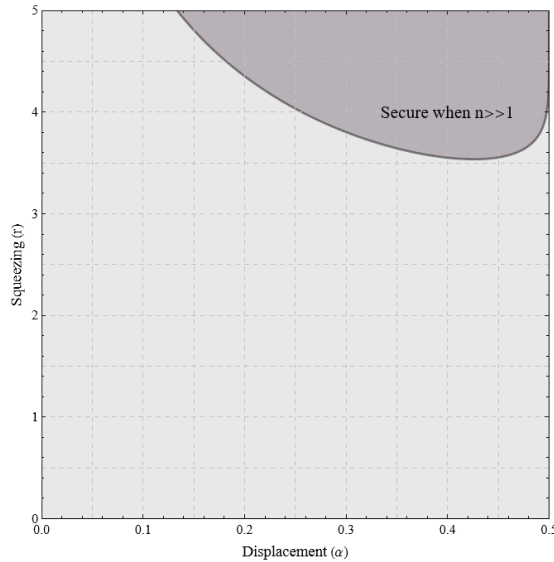


Figure 6.2: A plot of (6.37) as a function of the squeezing r and displacement α . As illustrated, asymptotically the scheme is secure only for $r > 3.5$ for some values of α . Here, we assume that the channel is ideal.

6.4. Practical considerations

Since our main motivation for constructing CV-UE is to have a practically implementable scheme, we examine how the protocol behaves under realistic parameter choices and

noise.

Generating highly squeezed light is experimentally challenging. We therefore restrict r to values that are reasonably close to the current state of the art. A 15 dB squeezed vacuum has been demonstrated [180], and 10 dB is common.¹

The bit error rate (BER) β is given by (6.11). We aim for a small BER, which in turn yields a small decryption–failure probability ε_{DF} (6.12). Fig. 6.3a displays β as a function of r , at $\alpha = 0.4$.

As an example, consider $\alpha = 0.40$ and $r = 3.6$, corresponding to approximately 12.63 dB of squeezing and yielding $\beta \approx 7.74 \cdot 10^{-3}$. For a ciphertext of size $N = 1000$, the expected number of bit errors is $N\beta \approx 7.74$. Choosing $t = 24$, Eq. (6.12) gives $\varepsilon_{\text{DF}} \leq 5.1 \cdot 10^{-6}$. The corresponding asymptotic binary-channel capacity benchmark is $N[1 - h(\beta)] \approx 935$ bits². We will continue to use this relationship between N and n in the remainder of this section.

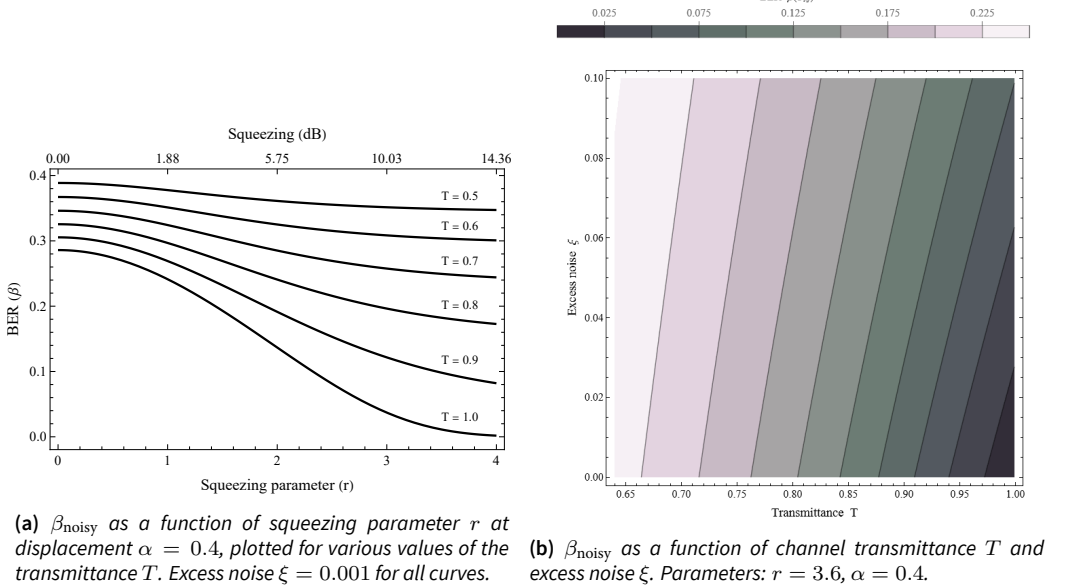


Figure 6.3: Bit error rate (6.39) for different parameters (excess noise in SNU).

¹10 dB corresponds to $\cosh r = 10$, $r \approx 3$; 15 dB corresponds to $\cosh r = 10^{1.5} \approx 31.6$, $r \approx 4.15$.

²The message length achievable by a finite-length code will generally be smaller and depends on the particular code and decoding rule.

6.4.1. Robustness under noise

We model the optical channel by its transmittance $T \in [0, 1]$ and (input-referred) excess-noise power ξ in SNU³ (cf. Section 2.4.1). In this model, the receiver's quadrature variance (SNU) and displacement transform as

$$\frac{1}{\cosh r} \longrightarrow \frac{T}{\cosh r} + (1 - T) + T\xi, \quad \alpha \longrightarrow \sqrt{T}\alpha, \quad (6.38)$$

which follows from the standard covariance-matrix transformation of a thermal-loss (attenuator) Gaussian channel and linear detection as detailed in Section 2.4.1 (alternatively see [80]). Detector inefficiency and electronics noise can be absorbed into the overall transmittance/noise budget in this framework [199]. Under the map (6.38), the BER becomes

$$\beta_{\text{noisy}} = \frac{1}{2} \text{Erfc} \left(\frac{\sqrt{T}\alpha}{\sqrt{\frac{T}{\cosh r} + (1 - T) + T\xi}} \right). \quad (6.39)$$

This expression is plotted in Fig. 6.3b as a function of (T, ξ) for fixed (r, α) .

As an example, consider 5 km of standard single-mode optical telecom fibre at 1550 nm wavelength. The loss rate is 0.22 dB/km; at 5 km this corresponds to $T \approx 0.8$. At 12.63 dB squeezing ($r \approx 3.6$), Fig. 6.3a then gives a BER of around 20%, which can be handled with a low-rate code, e.g. an LDPC code, or a higher-rate code concatenated with a repetition code.

Although the adversaries are granted ideal channels, with $T = 1$ and $\xi = 0$, the honest-channel parameters still enter the security analysis indirectly through the honest receiver's decoding statistics. Loss and excess noise increase the honest bit-error rate, thereby increasing both the error-correction radius and the minimum redundancy required for reliable decoding; both effects weaken the present security bound. Figure 6.4 shows the resulting asymptotic security regions under the assumption of capacity-approaching error correction. Within the experimentally demonstrated squeezing regime considered here, the present proof certifies security only for channels with very low loss. However, the protocol and its security analysis were developed as a proof of principle and were not optimised for operation over noisy channels. The restricted certified region should therefore be understood as a limitation of the present proof-of-principle construction and security reduction, rather than as evidence that the protocol is insecure outside this region.

³Here one SNU denotes the vacuum-noise level; in the quadrature convention of Chapter 2, this corresponds to a homodyne variance of $1/2$.

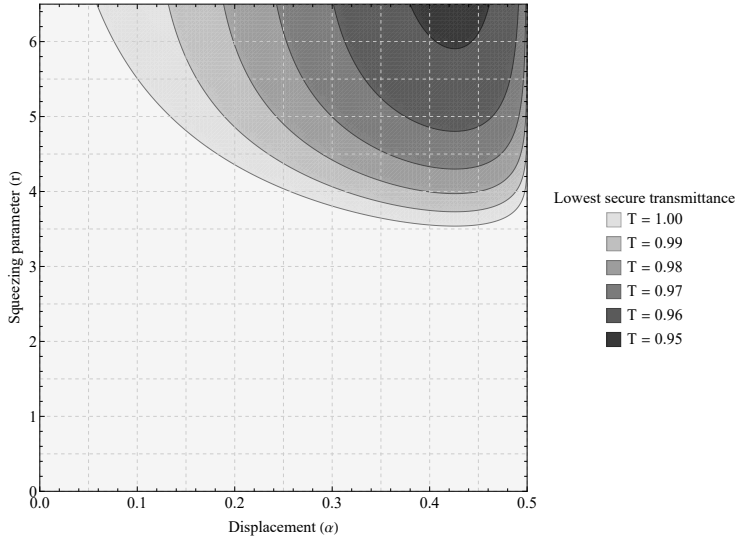


Figure 6.4: Fig.6.2 Eq. (6.37) plotted using β_{noisy} (6.39) for different transmission parameters (transmittance) T . The excess noise was set at $\xi = 0.001$. $T = 0.95$ corresponds to about 1.1 km of standard telecom fibre.

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6.5. Unclonable indistinguishability using CV

Recently it has been shown [30] that information-theoretic unclonability can be achieved in the case of a single-bit message. The security proof makes use of a decoupling theorem [65]: If one starts from a state $\rho^{A_1 E}$, where A_1 and E are potentially entangled, then mixing the A_1 -space into a higher-dimensional Hilbert space $A = A_1 A_2$ using the unitaries from a unitary two-design causes decoupling between A_1 and E . The ‘quality’ of the decoupling (trace distance between actual and ideal situation) is quantified as $\frac{3 \log \log d}{2 \log d}$ [30], where d is the dimension of system A . This result can be generalised to the case of decoupling using ε -approximate unitary two-designs [174], yielding $\frac{3 \log \log d}{2 \log d} \sqrt{1 + \varepsilon \cdot 4d^4}$ (where ε is a bound on the diamond norm).

Below we list the definitions from Section 3.1 again, but modified to facilitate a simpler construction and security proof. All of these come from [30]. The core idea remains unchanged from [42].

Definition 6.5.1 (QECM). A Quantum Encryption of Classical Messages (QECM) is a tuple $Q = (\mathcal{K}, \mathcal{X}, A, \mu, \{\sigma_x^k\}_{k \in \mathcal{K}, x \in \mathcal{X}})$, where $\mathcal{K}$ is the key space; $\mathcal{X}$ is the message space; A is a register that holds the encrypted message; μ is a probability distribution on $\mathcal{K}$; $\sigma_x^k \in \mathcal{D}(\mathcal{H}_A)$ is the encryption of message x with key k . We say that Q is η -correct if there exists a family

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of decryption maps $D^k : \mathcal{D}(\mathcal{H}_A) \rightarrow \mathcal{D}(\mathcal{H}_X)$ such that $\forall_{k \in \mathcal{K}, x \in \mathcal{X}} \langle x | D^k(\sigma_x^k) | x \rangle \geq \eta$. We say that Q is correct if it is 1-correct.

Definition 6.5.2 (Cloning attack and δ -unclonable security). Consider a QECM $Q = (\mathcal{K}, \mathcal{X}, A, \mu, \{\sigma_x^k\}_{k \in \mathcal{K}, x \in \mathcal{X}})$. A **cloning attack** against Q is a tuple $\mathcal{A} = (B, C, \{B_x^k\}_{k \in \mathcal{K}, x \in \mathcal{X}}, \{C_x^k\}_{k \in \mathcal{K}, x \in \mathcal{X}}, \Phi)$, where B and C are registers representing the systems of Bob and Charlie, respectively; $\{B_x^k\}_{x \in \mathcal{X}}$ and $\{C_x^k\}_{x \in \mathcal{X}}$ are POVMs on $\mathcal{H}_B$ and $\mathcal{H}_C$ representing Bob and Charlie's measurements given k ; $\Phi : \mathcal{D}(\mathcal{H}_A) \rightarrow \mathcal{D}(\mathcal{H}_{BC})$ is a CPTP map representing the cloning channel. The success probability of the cloning attack against Q is $c(Q, \mathcal{A}) = \mathbb{E}_k \sum_{x \in \mathcal{X}} \frac{1}{|\mathcal{X}|} \text{Tr}[(B_x^k \otimes C_x^k) \Phi(\sigma_x^k)]$. We say that the QECM Q is **δ -unclonable secure** if

$$\sup_{\mathcal{A}} c(Q, \mathcal{A}) \leq \frac{1}{|\mathcal{X}|} + \delta. \quad (6.40)$$

Definition 6.5.3 (Cloning-distinguishing attack and δ -unclonable-indistinguishable security). Consider a QECM $Q = (\mathcal{K}, \mathcal{X}, A, \mu, \{\sigma_x^k\}_{k \in \mathcal{K}, x \in \mathcal{X}})$. A **cloning-distinguishing attack** against Q is a tuple $\mathcal{A} = (\{x_0, x_1\}, B, C, \{B_b^k\}_{k \in \mathcal{K}, b \in \{0,1\}}, \{C_b^k\}_{k \in \mathcal{K}, b \in \{0,1\}}, \Phi)$, where $x_0, x_1 \in \mathcal{X}$ with $x_1 \neq x_0$; B and C are registers representing the systems of Bob and Charlie, respectively; $\{B_b^k\}_{b \in \{0,1\}}$ and $\{C_b^k\}_{b \in \{0,1\}}$ are POVMs on $\mathcal{H}_B$ and $\mathcal{H}_C$ representing Bob and Charlie's measurements given k ; $\Phi : \mathcal{D}(\mathcal{H}_A) \rightarrow \mathcal{D}(\mathcal{H}_{BC})$ is a CPTP map representing the cloning channel. The success probability of the cloning-distinguishing attack against Q is $\text{cd}(Q, \mathcal{A}) = \mathbb{E}_k \sum_{b \in \{0,1\}} \frac{1}{2} \text{Tr}[(B_b^k \otimes C_b^k) \Phi(\sigma_{x_b}^k)]$. We say that Q is **δ -unclonable-indistinguishable secure** if

$$\sup_{\mathcal{A}} \text{cd}(Q, \mathcal{A}) \leq \frac{1}{2} + \delta. \quad (6.41)$$

Unclonable-indistinguishable security implies unclonable security [42]. The two notions are identical if the message space is $\{0, 1\}$. Furthermore, unclonable-indistinguishable security implies indistinguishable security [42].

Lemma 6.5.4. (Theorem 5.2 in [30]). Let $\mathcal{V} \subset \mathcal{U}(\mathcal{H}_A)$ be a 2-design. Let $d = \dim(\mathcal{H}_A)$. The QECM

$(\mathcal{V}, \{0, 1\}, A, \mu_{\mathcal{V}}, \{U(|x\rangle\langle x| \otimes \tau)U^\dagger\}_{U \in \mathcal{V}, x \in \{0,1\}})$ is correct and δ -unclonable secure with $\delta = \frac{3 \log \log d}{2 \log d}$.

6.5.1. Unclonable Encryption scheme based on position and momentum operators

We propose a QECM for one-bit messages, based on the two-design unitaries from Section 5.3. This results in a new Unclonable Encryption scheme that is based on position and momentum operators. It can be used (at large d) to encrypt a classical bit into a single-mode CV state. Below we detail the scheme and prove unclonable security.

Let $\mathcal{H}_d$ be factorised as $\mathcal{H}_d = \mathcal{H}_{\text{sign}} \otimes \mathcal{H}_{\text{dist}}$, with $\dim(\mathcal{H}_{\text{sign}}) = 2$ and $\dim(\mathcal{H}_{\text{dist}}) = \frac{d}{2}$. A q -boxed state $|j\rangle$ with $j \geq 0$ is decomposed as $|j\rangle = |0\rangle_{\text{sign}} \otimes |j\rangle_{\text{dist}}$ and for $j < 0$ as $|j\rangle = |1\rangle_{\text{sign}} \otimes | |j| - 1 \rangle_{\text{dist}}$. Conversely, a sign bit $s \in \{0, 1\}$ and a distance $r \in \{0, \dots, \frac{d}{2} - 1\}$ represent the q -eigenstate $|(-1)^s(r + s)\rangle$.

The message space is $\{0, 1\}$. The key space is $\mathcal{K} = [0, d)^{6\ell}$, where $\ell \in \mathbb{N}$ is a parameter for the encryption. The ciphertext space is $\mathcal{D}(\mathcal{H}_d)$. The register containing the cipherstate is called A. Let $x \in \{0, 1\}$ be the plaintext, and $k = (\alpha_1 \cdots \alpha_\ell, \alpha'_1 \cdots \alpha'_\ell, \alpha''_1 \cdots \alpha''_\ell, \beta_1 \cdots \beta_\ell, \beta'_1 \cdots \beta'_\ell, \beta''_1 \cdots \beta''_\ell)$ the key. We define

$$\mathcal{V}_\ell(d) = \{U_k\}_{k \in \mathcal{K}}, \quad U_k = (V_{\alpha''_\ell \beta''_\ell} \tilde{V}_{\alpha'_\ell \beta'_\ell} V_{\alpha_\ell \beta_\ell}) \cdots (V_{\alpha''_1 \beta''_1} \tilde{V}_{\alpha'_1 \beta'_1} V_{\alpha_1 \beta_1}) \quad (6.42)$$

with $V_{\alpha\beta} = \omega^{\alpha\hat{Q} + \beta\hat{Q}^2}$, $\tilde{V}_{\alpha\beta} = \omega^{\alpha\hat{P} + \beta\hat{P}^2}$. The steps of the key generation, encryption and decryption are as follows. The parameter ℓ is the number of times the unitary R (see Section 5.3) is applied.

Algorithm 4 Key Generation (Key)

Input: parameters $d \in 2\mathbb{N}$ and $\ell \in \mathbb{N}$.

Output: $k = (\alpha_1, \dots, \alpha_\ell, \alpha'_1, \dots, \alpha'_\ell, \alpha''_1, \dots, \alpha''_\ell, \beta_1, \dots, \beta_\ell, \beta'_1, \dots, \beta'_\ell, \beta''_1, \dots, \beta''_\ell) \in [0, d)^{6\ell}$, a key.

- 1: **for** $i = 1$ to ℓ **do**
 - 2: Sample $\alpha_i, \alpha'_i, \alpha''_i, \beta_i, \beta'_i, \beta''_i$ independently and uniformly from $[0, d)$.
 - 3: **end for**
 - 4: Output k .
-

Algorithm 5 Encryption (Enc)

Input: key $k \in [0, d)^{6\ell}$; message $x \in \{0, 1\}$.

Output: Cipherstate $\sigma^A \in \mathcal{D}(\mathcal{H}_d)$.

- 1: Compute

$$U_k = (V_{\alpha''_\ell \beta''_\ell} \tilde{V}_{\alpha'_\ell \beta'_\ell} V_{\alpha_\ell \beta_\ell}) \cdots (V_{\alpha''_1 \beta''_1} \tilde{V}_{\alpha'_1 \beta'_1} V_{\alpha_1 \beta_1}).$$

- 2: Sample r_0 uniformly from $\{0, \dots, \frac{d}{2} - 1\}$.
- 3: Prepare the state $|x\rangle_{\text{sign}} \otimes |r\rangle_{\text{dist}}$.
- 4: Forget r_0 , obtaining

$$\rho_x^A = |x\rangle\langle x|_{\text{sign}} \otimes \tau^{\text{dist}}.$$

- 5: Apply U_k to ρ_x^A to obtain

$$\sigma^A = U_k \rho_x^A U_k^\dagger.$$

- 6: Output σ^A .
-

Algorithm 6 Decryption (Dec)

Input: key $k \in [0, d)^{6\ell}$; cipherstate $\sigma^A \in \mathcal{D}(\mathcal{H}_d)$.

Output: Message $\hat{x} \in \{0, 1\}$.

1: Compute

$$U_k = (V_{\alpha'_\ell \beta'_\ell} \tilde{V}_{\alpha'_\ell \beta'_\ell} V_{\alpha_\ell \beta_\ell}) \cdots (V_{\alpha'_1 \beta'_1} \tilde{V}_{\alpha'_1 \beta'_1} V_{\alpha_1 \beta_1}).$$

2: Apply $U_k^\dagger$ to σ^A .

3: Measure the $\mathcal{H}_{\text{sign}}$ subsystem in the computational basis $\{|0\rangle, |1\rangle\}$, obtaining outcome $\hat{x}$.

4: Output $\hat{x}$.

Our scheme has the usual structure [136, 30] of mixing a plaintext state into a higher-dimensional space. Formally, we write our QECM as

$$S_{d\ell} = \left(\mathcal{V}_\ell(d), \{0, 1\}, A, \mu_{\text{unif}}, \{U(|x\rangle\langle x|_{\text{sign}} \otimes \tau^{\text{dist}})U^\dagger\}_{U \in \mathcal{V}_\ell(d), x \in \{0, 1\}} \right) \quad (6.43)$$

where μ_{unif} is the uniform distribution. We note that drawing uniformly from $[0, d)^{6\ell}$ is practical. Before proving security, we note the following property of the diamond norm in question.

Lemma 6.5.5 (Diamond vs $2 \rightarrow 2$ norm [132]). *Let $\Phi : \mathcal{B}(\mathcal{H}_d) \rightarrow \mathcal{B}(\mathcal{H}_d)$ be a superoperator. Then $\|\Phi\|_\diamond \leq d\|\Phi\|_{2 \rightarrow 2}$.*

This result helps establish a relationship between the diamond norm used for definitions of 2-designs in the literature and the $2 \rightarrow 2$ norm used here. Now we state the main result.

Theorem 6.5.6. *The QECM $S_{d\ell}$ is correct and δ -unclonable secure, with*

$$\delta = \frac{3 \log \log d}{2 \log d} \sqrt{1 + 4d^{6-\ell}}. \quad (6.44)$$

Proof. This follows exactly the same steps as the proof of Lemma 6.5.4 (see [30]), but with the additional factor $\sqrt{1 + \varepsilon \cdot 4d^4}$ from the generalised decoupling theorem of [174]. In our η -approximate unitary 2-design we have $\eta = d^{-\ell}$, using the $2 \rightarrow 2$ norm in the definition. As [174] uses the diamond norm, we have to take into account $\varepsilon \leq \eta d^2$ (Lemma 6.5.5). $\square$

Note that we need $\ell \geq 6$ in order to get a reasonably small δ .

6.6. Discussion

This chapter developed two complementary routes to unclonable encryption with continuous-variable states. First, a concrete CV protocol can already be shown to satisfy unclonable

security by exploiting monogamy of entanglement; second, by moving to a more structural construction based on decoupling and approximate unitary 2-designs, one can also reach the stronger notion of unclonable indistinguishability. This is also the first analysis of an unclonable encryption protocol in the presence of realistic channel noise, marking a significant step towards practical quantum cryptography.

For the monogamy-based protocol involving realistic noise, making a meaningful comparison with prior work is difficult, since DV–CV performance cannot be fairly assessed by directly comparing the winning probabilities in the cloning game. The most informative benchmark is probably the DV analogue of our scheme, which can be realised via Wiesner’s conjugate coding [192]. Fig. 6.5 plots the winning probability of our UE game as a function of message size, including also the UE game for conjugate coding into qubit states [42]. The conjugate coding scheme first one-time pads a message m and then bitwise encodes the ciphertext into the standard basis or the Hadamard basis. For experimentally feasible squeezing of $r = 3.6 (\approx 15.6 \text{ dB})$, we observe exponentially decreasing probability of the adversaries winning for longer message sizes of the order of 10^2 or greater. We emphasise that Fig. 6.5 is just an illustration for a single choice of parameters, and is not intended as a comprehensive performance comparison.

6

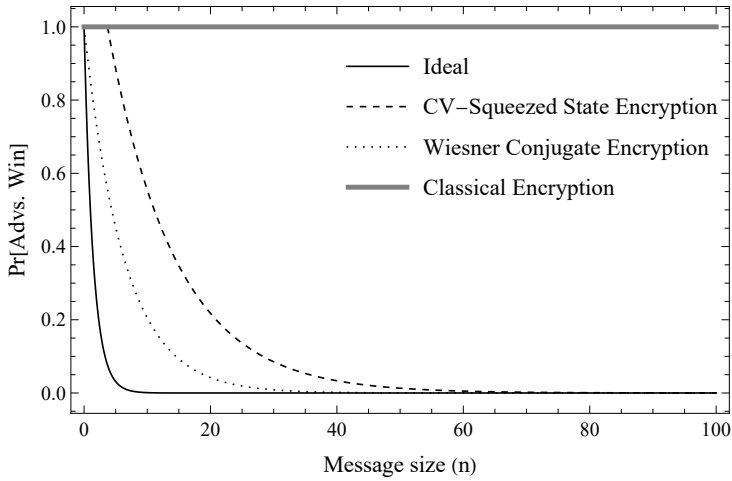


Figure 6.5: Winning probabilities in the unclonability game for various schemes. The ideal curve is the simple guessing probability $(\frac{1}{2})^n$. The winning probability of the conjugate coding scheme is at most $(\frac{1}{2} + \frac{1}{2\sqrt{2}})^n$. For our CV scheme we plotted $2^{-n+\tau(\lambda)}$ (6.36) with parameters $r = 3.6, \alpha = 0.4$ and allowed bit flips $t = 3.5\%$ of the ciphertext size.

As evident from above, arguments based on monogamy of entanglement do not appear to yield guarantees beyond search security; establishing unclonable indistinguishability

for CV protocols requires new proof techniques. This observation motivates the second part of the chapter, where the focus shifts from a concrete protocol to a more structural encryption paradigm based on decoupling.

The second route shows that decoupling provides a natural conceptual framework for CV unclonable encryption once one has access to a suitable notion of approximate unitary 2-design. In finite dimensions, this connection is by now well understood: Haar-random unitaries, and more generally unitary 2-designs, provide the randomisation needed to wash out correlations and thereby yield unclonable indistinguishability [136, 30, 174]. What was missing in the CV setting was an ensemble of unitaries that could play the same role. The contribution made in Chapter 5 shows that such a route becomes possible after a physically motivated regularisation. This leads to a CV encryption scheme with unclonable-indistinguishable security, thereby extending the scope of CV UE beyond what the monogamy-based approach alone can deliver.

Comparison in this case is limited to [30], since it is the only available unclonable-indistinguishable protocol without computational assumptions and also the motivation for the current work. Fig. 6.6 plots the winning probability of the CV UE scheme as a function of the dimension d of the discretised Hilbert space $\mathcal{H}_d$. In contrast to the previous plot, this is the winning probability for one-bit messages. For n -bit messages, each bit can be encrypted independently. We observe that even for $\ell = 7$, the performance is indistinguishable from the discrete scheme. In the discrete case, $n = \log_2 d$ qubits are required to encrypt a single bit. In the CV case, however, d functions as a discretisation parameter and its relationship to the quantum resources required is not straightforward.

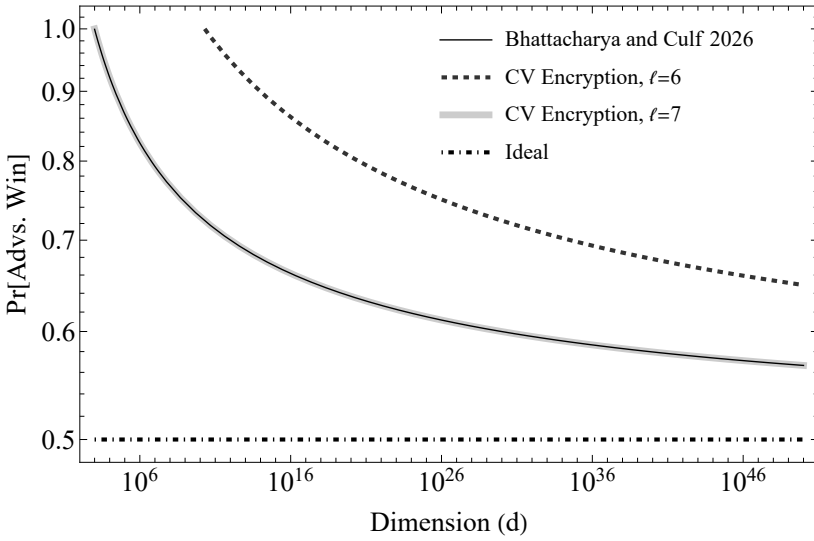


Figure 6.6: Winning probabilities in the unclonable indistinguishability game for the discrete and CV schemes. The ideal line is the guessing probability $\frac{1}{2}$. The winning probabilities of the discrete and CV cases differ by the factor $\sqrt{1 + 4d^{6-\ell}}$, where ℓ is the number of random unitaries (R) applied during encryption.

7

Conclusion

*Whatever knowledge is attainable, must be attained by
scientific methods; and what science cannot discover,
mankind cannot know.*

— Bertrand Russell

The overarching theme of this thesis was to study whether the field of continuous-variable quantum cryptography can be advanced further and, in particular, whether CV systems can robustly support cryptography beyond standard CVQKD. The chapters of the thesis addressed this question by considering four relevant problems: long-distance reconciliation for CVQKD, continuous-variable quantum position verification, approximate unitary two-designs for regularised CV systems, and continuous-variable unclonable encryption.

In this concluding chapter, we return to the research questions stated in Chapter 1. For each question, we summarise the relevant construction, the primary result, and future work.

7.1. Random-codebook reconciliation for long-distance CVQKD

We first recall the research question associated with Chapter 3:

Can we design a reconciliation system for long-distance CVQKD that is parallelisable and produces secret-key rates closer to the Devetak–Winter value than existing methods?

The answer obtained in this thesis is affirmative. The random-codebook reconciliation scheme reaches several percent of the Devetak–Winter value in the low-SNR regime, while keeping the decoding task massively parallelisable. The construction should not be understood as a replacement for highly optimised structured codes in all regimes. Rather, it provides an alternative reconciliation paradigm tailored to the long-distance regime, where the SNR is extremely small and where decoding complexity becomes a central practical constraint.

The main idea is to treat reconciliation as a statistical identification problem. Bob takes his measurement vector $\vec{y}$ and hides it in a table of q candidate vectors. One row is the true vector $\vec{y}$, while the remaining rows are independently drawn fake vectors with the same marginal Gaussian distribution. Alice, who has the correlated vector $\vec{x}$, computes a score for every row and accepts the block only if exactly one row is statistically compatible with her data.

The mathematical reason this works is that the true row and the fake rows have different score statistics. For a suitable score function $J(\vec{x}, \vec{m}_\ell)$, the fake rows are approximately centred, while the true row has a positive shift. The decoding rule is a likelihood-ratio test in which Alice accepts the row ℓ if $J(\vec{x}, \vec{m}_\ell) > \theta$, subject to the condition that exactly one row lies above the threshold. Thus, decoding does not require navigating a structured code. It consists of many independent score computations, which can be distributed over many processors. This is the main operational advantage of the construction: at fixed q , the workload scales linearly with the block length n , and n itself scales favourably with the inverse SNR.

The accept/reject pattern itself may leak information. For this reason, the scheme uses pre-shared secret key material to hide the acceptance information. This reduces the final key ratio, but it allows the security analysis to remain within the standard Gaussian framework. This point is important because, at a non-zero block rejection rate, Eve’s state is in general conditioned on the public accept event. If this conditioning is ignored, the claimed reconciliation performance can look better than what is justified.

Our asymptotic key ratio has the general form

$$\text{SKR}_\infty = \text{raw key} - \text{Eve's quantum info.} - \text{syndrome leak} - \text{accept/reject hiding cost.} \quad (7.1)$$

This expression makes clear where the rate is diminished. The construction creates useful correlated symbols even at extremely low SNR, but the conservative treatment of public acceptance decisions consumes a significant amount of key material.

For a transmission of $T = 10^{-6}$, corresponding to roughly 60 dB of loss or about 300 km of standard fibre, the scheme achieves the following fractions of the low- T Devetak–

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Winter value:

$$\frac{q}{\text{SKR}_{\infty}/\text{DW}} \left| \begin{array}{ccccc} 2^5 & 2^{10} & 2^{15} & 2^{20} & 2^{30} \\ 0.019 & 0.053 & 0.082 & 0.11 & 0.14 \end{array} \right. \quad (7.2)$$

Thus, for $q = 2^{15}$, the protocol reaches about 8.2% of the Devetak–Winter value, and for $q = 2^{30}$ it reaches about 14% in the asymptotic analysis. As shown in Table 3.1, this is a significant improvement over the existing long-distance reconciliation values reported in that table, especially given that the analysis here subtracts the cost of hiding the accept/reject decisions.

The limitations of the result also point to the natural future directions. The first limitation is intrinsic to the random-codebook thresholding mechanism. If one tries to operate too close to Shannon capacity, the separation between the true-score distribution and the largest of the fake-score distributions is not large enough at practical values of q . In the notation of Chapter 3, the relevant separation is governed by a term of order $\sqrt{2 \ln q}$. At reasonable table sizes, this is not sufficient to separate the true entry from all $q - 1$ fake entries with high probability, so the protocol must operate away from the capacity point. This lowers the reconciliation efficiency, and the only direct way to improve the separation is to increase q . However, increasing q is limited by real-time decoding constraints. This motivates further work on optimised parallel implementations of the pseudorandom scheme, including aggressive pruning and FPGA/GPU implementations.

The second limitation is the conservative treatment of acceptance information. Encrypting all accept/reject decisions is safe, but it is also costly. Finding a less wasteful way to account for this conditioning is therefore one of the most important theoretical directions suggested by the chapter and could lead to considerable improvement in the secret-key ratio.

The third limitation is that the cleanest analysis is asymptotic and assumes ideal random fake entries. For an implementable system, the pseudorandom variant should be regarded as the default version, since it avoids storing enormous random tables and more closely resembles practical reconciliation schemes. The statistical quantities entering the key-rate computation, namely the acceptance probability and the symbol error rate, are not expected to change significantly when the fake entries are generated pseudorandomly. Nevertheless, this should be analysed more explicitly, together with a non-asymptotic finite- N treatment of the secret-key ratio.

Finally, several modelling choices should be developed further before the method can be viewed as a complete practical reconciliation system. The construction and security analysis should be adapted to discrete modulation. The quadrature values should also be discretised with a resolution fine enough for the score computation, but not so fine that communication and storage become inefficient; in particular, one should determine how many

bits are required to represent the relevant variables without significantly degrading the reconciliation efficiency. A complete treatment should further include finite-size effects and the statistical uncertainty arising from parameter estimation. In the present analysis, the channel statistics and calibration parameters are assumed to remain stationary throughout the protocol, so that the distributions used to generate the random codebook entries and determine the decoding threshold continue to describe Bob's measurement data. Future work should therefore study the effect of calibration drift and incorporate an explicit channel-monitoring procedure. It should also account for the parameter-estimation pulses in the overall rate: the rates quoted here are per key-generation pulse, since publicly revealed and discarded monitoring pulses are not included in the denominator. The effect of non-negligible excess noise should likewise be studied systematically. Preliminary calculations indicate that $\xi = 10^{-4}$ hardly changes the results, that $\xi = 10^{-3}$ already produces visible degradation, and that $\xi = 10^{-2}$ reduces the $q = 2^{15}$ rate to about 0.6% of the Devetak–Winter value, although the degradation is less severe at larger q .

The present security analysis is restricted to Gaussian collective attacks. An extension to coherent attacks in the asymptotic regime should be possible by supplementing the protocol with an appropriate random symmetrisation and energy test. Establishing finite-size security against coherent attacks would, however, be substantially more involved. One possible route is the Gaussian de Finetti reduction of Ref. [126], which requires the complete protocol to be covariant under the action of the unitary group $U(n)$ and also requires a suitable energy test. The present blockwise reconciliation procedure is not manifestly $U(n)$ -covariant, so applying this reduction would require either an explicit proof that the required covariance is preserved or a corresponding modification of the protocol.

More broadly, the significance of this chapter lies not in the exact key ratios that were obtained, but in the perspective it offers on reconciliation for long-distance CVQKD. The usual emphasis in the literature is on designing highly efficient structured codes and pushing their reconciliation efficiencies as close as possible to the ideal value. The chapter explores what can be achieved when one uses one of the simplest error-correcting codes in the extremely low-SNR regime of QKD. We find that (pseudo)random-codebook error correction can be a practical option for real-time information reconciliation in long-distance CVQKD, even when the security analysis is taken to be highly conservative.

7.2. Continuous-variable quantum position verification

The research question addressed in Chapter 4 is:

Is it possible to achieve secure position verification using continuous-variable

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quantum states in the presence of realistic loss and noise, while also maintaining security against attackers with access to a bounded amount of entanglement?

The answer is again affirmative, but only after distinguishing between two adversarial models. The simplest coherent-state CV-QPV protocol QPV_{coh} (introduced in Section 4.1) is secure against attackers without pre-shared entanglement in an appropriate loss-and-noise regime. However, it is broken by attackers who share an ideal continuous-variable EPR pair. Security against entanglement-assisted attackers therefore requires the modified protocol $\text{QPV}_{\text{coh}}^f$ (introduced in Section 4.4), in which the relevant measurement information is split between the verifiers.

In the basic protocol, a verifier prepares a coherent state whose displacement encodes a random continuous variable. The honest prover receives both the quantum state and the classical information specifying the choice of quadrature, performs the corresponding homodyne measurement, and broadcasts the outcome. Because the channel has attenuation and excess noise, the verifiers do not test for exact equality. Instead, they test whether the reported values have the correct noisy Gaussian statistics.

The security proof against unentangled attackers is based on the entropic uncertainty principle (Lemma 4.3.1). The honest prover's uncertainty about the verifier's displacement is set by the physical channel, whereas at least one of the two separated attackers must have additional uncertainty because the attackers cannot both possess sharp information about conjugate quadratures. In the large-modulation regime, this gives a separation between honest and adversarial response distributions whenever

$$\frac{t}{1+2u} > \frac{e}{4}, \quad (7.3)$$

where t is the transmission and u is the excess-noise power.

The QPV_{coh} protocol also shows the limitation of coherent-state CV-QPV. If the attackers share a single ideal CV-EPR pair, they can use teleportation to reproduce the honest prover's statistics. In that case, the distribution seen by the verifiers is indistinguishable from the honest distribution:

$$\Pr[\text{verifiers accept adversaries}] = \Pr[\text{verifiers accept honest prover}] \quad (7.4)$$

in the ideal EPR limit, with the attackers able to simulate the loss and noise. Therefore, the simplest protocol is not secure against entanglement-assisted adversaries.

The second protocol resolves this by adopting the central idea of $\text{QPV}_{\text{BB84}}^f$ -type constructions. Instead of revealing the measurement basis directly, the two verifiers choose strings x and y , and the prover must determine the correct quadrature from a function value $f(x, y)$. An honest prover at the claimed position receives both pieces of information at

the same time and can compute $f(x, y)$ before measuring. The separated attackers must therefore coordinate under relativistic constraints.

For random functions f , the chapter shows that the continuous-variable version inherits the bounded-entanglement security property known from finite-dimensional QPV. Attackers cannot reproduce the honest response statistics with high probability unless the number of pre-shared qubits grows at least linearly with the size of the classical input. The proof also accounts for attenuation and excess noise.

Beyond the specific protocols analysed here, the chapter also raises a broader question: to what extent do the known structural results on discrete-variable QPV carry over to the continuous-variable setting? The answer is clearly not straightforward. Some ideas do transfer in spirit, but the technical content of the proofs often changes substantially once one passes from discrete alphabets to continuous distributions. It is therefore natural to ask how far the existing attack theory can be generalised. For example, can the recent formulation of CV port-based teleportation [154] be used to reformulate the general attack on discrete-variable QPV from [24] directly in the CV setting? Likewise, known attacks whose cost scales with structural properties of the prover's unitary [170, 64] suggest the question of whether there are natural continuous-variable analogues of notions such as T -count or T -depth that would govern the complexity of attacking CV-QPV protocols. These questions suggest that CV-QPV may require its own attack complexity theory.

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A further open direction concerns loss tolerance. In the discrete-variable case, Ref. [4] has recently shown how to circumvent the usual obstruction caused by transmission loss. Whether a related idea can be made to work in the continuous-variable setting remains open. This is a particularly important question for the long-term practical relevance of CV-QPV. The chapter shows that security can be maintained for non-zero loss and excess noise within certain parameter ranges, but it is also clear that attenuation imposes a fundamental constraint on the current protocol family. A method for overcoming this limitation in the CV setting, such as one based on squeezed states, would substantially increase the practical appeal of continuous-variable position verification.

Thus, the answer to the research question is threefold. First, continuous-variable states can be used for position verification in a meaningful cryptographic sense. Second, security is preserved for imperfect channels, although only within a limited loss-and-noise regime; in particular, the security guarantee breaks down when the loss exceeds 3 dB. Third, security against entanglement-assisted attackers is not achieved by the simplest coherent-state protocol, but it is obtained after splitting the classical measurement information between the verifiers. More broadly, these results support the central perspective of this thesis: the continuous-variable setting is both a rich domain for cryptographic ideas and a practically relevant one, closely aligned with the capabilities of current quantum technology.

7.3. Approximate two-designs for continuous-variable systems

The research question addressed in Chapter 5 was:

Can we construct a physically motivated ε -approximate 2-design for CV systems?

The answer is yes, provided that the statement is understood in a regularised sense. An exact unitary two-design on the full infinite-dimensional CV Hilbert space is not the object constructed in this thesis. Instead, the construction works on a finite-dimensional Hilbert space obtained by imposing a finite quadrature window and a finite resolution. Although it is a technical requirement, this regularisation may be understood as more than that: it reflects the fact that any physical CV implementation necessarily has finite resolution and a finite operational range.

The regularisation is done by dividing the quadrature axis into bins and associating a normalised box state with each bin. This produces a finite-dimensional space $\mathcal{H}_d$, which should be understood as a controlled finite-dimensional approximation to the CV system. On this space, one has a boxed q basis and a Fourier-conjugate boxed p basis. Thus, the construction preserves the basic conjugate-quadrature structure of the CV system, while making it possible to use finite-dimensional design techniques.

The design construction then uses random quadrature phase unitaries. One family is diagonal in the boxed q basis, and the other is diagonal in the boxed p basis. Physically, these operations can be modelled as nonlinear quadrature phase gates with staircase-like phase profiles on a finite quadrature window. Since these are non-Gaussian operations, they are not as immediately accessible as Gaussian operations. Nevertheless, suitably smoothed versions of these profiles fit within the existing framework of measurement-induced nonlinear CV operations and polynomial approximations to nonlinear phase gates.

Mathematically, we show that by alternating the corresponding twirls of these quadrature phase unitaries, one obtains an approximate unitary two-design on the boxed Hilbert space. The main quantitative result is simple: after ℓ repetitions, the construction gives (Corollary 5.3.12)

$$\varepsilon = d^{-\ell}. \quad (7.5)$$

The discretisation dimension d then plays two related roles: it controls both the resolution of the CV approximation and the quality of the approximate two-design. Increasing d improves both approximations, but also produces more, narrower quadrature bins and enlarges the range over which the staircase phase profile must be realised. For the measurement-induced schemes discussed in Section 5.4, this would place stricter

demands on the preparation of non-Gaussian ancillary states, the resolution of the homodyne measurements, and the accuracy of the adaptive feedforward. The challenge is therefore to reproduce increasingly fine phase profiles accurately over an expanding quadrature range. Without a platform-specific error model, however, it is difficult to quantify precisely how the required experimental precision scales with d .

These observations identify concrete steps towards implementing continuous-variable designs and, eventually, approaching the approximately perfect scrambling predicted by the theory in continuous-variable quantum systems.

There is also an intriguing connection to mutually unbiased bases (MUBs). The operators $\omega^{\alpha\hat{Q}+\beta\hat{Q}^2}$, for integers α, β , appear in the construction of MUB states in prime dimension. This suggests a further strengthening of the known relationship between 2-designs and MUBs. In the present thesis, this connection is not developed further, but it points to a possible structural explanation for why alternating random phase operations in conjugate bases are sufficient to produce design-like behaviour.

For the CV setting, several natural extensions remain open. One obvious direction is to investigate whether the q -quadrature cutoff can be carried out without the discretisation. Such an approach could help clarify to what extent the present discretised model is merely a technical device, and to what extent it captures a more intrinsic CV design structure.

It is not known whether the bound $\varepsilon = d^{-\ell}$ is optimal. A second direction is to study, through a sharper analysis, whether tighter bounds on the approximation error ε are possible for the present construction. One may even devise modified phase ensembles that could achieve the same design quality with fewer repetitions or weaker regularisation assumptions. A third direction is to explore whether the MUB connection can be used to construct a broader class of regularised CV approximate designs.

7

7.4. Continuous-variable unclonable encryption

The fourth research question was addressed in the first part of Chapter 6:

Is it possible to achieve unclonable encryption using continuous-variable quantum states in the presence of realistic loss and noise?

The thesis answers this question through a direct CV unclonable-encryption protocol based on displaced squeezed states. A classical message is first encrypted classically and then encoded into a binary error-correcting codeword. Each codeword bit is represented by a displaced squeezed state whose squeezing direction is part of the secret key. The honest receiver, who knows the key, measures the correct quadrature by homodyne detection and decodes the resulting noisy bit string.

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The reason this construction gives unclonability is that an adversary must split the quantum ciphertext before the key is revealed. After the key is revealed, both separated adversarial systems must allow the message to be recovered for the adversaries to succeed. This is where monogamy of entanglement enters.

The proof relates the unclonable-encryption cloning game to a continuous-variable monogamy-of-entanglement game. We obtain an upper bound on the winning probability of the adversaries given by

$$2^{-n + \frac{N}{2} + (\frac{N}{2} - t) \log(1 + 2\alpha) + t + \frac{1}{2} \log e}. \quad (7.6)$$

Here, n is the length of the message, N is the length of the ciphertext, and the remaining quantities are tunable parameters. Finite squeezing is also important because even the honest receiver has a non-zero decoding-failure probability, so correctness must be treated probabilistically.

Loss and excess noise enter both the correctness and security analyses through the honest receiver's decoding statistics. They increase the honest bit-error rate and therefore determine the error-correcting capability and redundancy required of the code, which in turn affect the quantitative security bound. The physical channel thus influences not only whether the honest receiver can decode the ciphertext, but also the parameter regime in which the present proof certifies security, even though the adversaries are granted ideal channels with $T = 1$ and $\xi = 0$. Moreover, error correction is necessary even for an ideal honest channel: at finite squeezing, homodyne outcomes retain intrinsic quantum fluctuations (shot noise), so the corresponding outcome distributions overlap and produce a non-zero bit-error rate. This differs from an idealised discrete-variable protocol, in which orthogonal signal states transmitted through a noiseless channel can produce deterministic honest outcomes.

The protocol and the accompanying security analysis were developed as a proof of principle and were not optimised for noisy channels. In particular, the error-correcting construction introduced here serves only as a simple illustrative example, while the quantitative security bound is not expected to be tight. A practical extension to noisier channels would require an error-correction scheme tailored to the relevant channel statistics, together with a refined security analysis that accounts more efficiently for the redundancy required for decoding. Merely replacing the illustrative code would not, by itself, remove the limitation shown in Figure 6.4, since that plot already assumes capacity-approaching error correction. A complementary direction would be to derive a tighter upper bound on the winning probability in Game C. These improvements could enlarge the parameter regime in which security can be established without changing the underlying squeezed-state construction.

The final research question in the chapter was:

Is it possible to design a continuous-variable unclonable encryption protocol that is also unclonable-indistinguishable secure?

The thesis answers this question using the approximate CV unitary two-design constructed in Chapter 5. The plaintext bit is embedded in a binary degree of freedom inside the regularised boxed Hilbert space. The remaining part of the finite-dimensional space acts as an auxiliary system into which the plaintext is mixed. Encryption consists of applying a randomly chosen unitary from the approximate two-design ensemble, built from alternating random phase operations in the discretised position and momentum bases.

The proof now follows the decoupling route rather than the monogamy route. A unitary two-design reproduces the second-moment behaviour of Haar-random unitaries, which is the randomness needed in the decoupling argument. Since the construction is only approximate, the decoupling statement carries an approximation error controlled by the design error $\varepsilon = d^{-\ell}$. By choosing d and ℓ sufficiently large, the encrypted system becomes sufficiently decoupled from any adversarial reference system. This gives unclonable-indistinguishable security for the corresponding regularised CV scheme, where the probability that the adversaries win is given by

$$\frac{3 \log \log d}{2 \log d} \sqrt{1 + 4d^{6-\ell}}. \quad (7.7)$$

7

Here, as above, d represents the discretisation parameter introduced in Chapter 5, and ℓ is the number of times the designs are applied. This second construction also provides a concrete application for the two-design result presented in Chapter 5.

The main limitation of the decoupling-based construction is that the security proof is not yet fully tailored to the specific CV structure of the protocol. A more specialised decoupling theorem could lead to better bounds. Another point concerns the instantiation of the QECM $S_{d\ell}$ (6.43) using the approximate unitary 2-design of Section 5.3.2, where d is prime. The main advantage of this version is that it yields a discrete key space. However, because the QECM requires an even dimension, the plaintext state $|x\rangle\langle x| \otimes \tau^{\text{dist}}$ must then be prepared in a $(d - 1)$ -dimensional subspace of $\mathcal{H}_d$, with the origin excluded. The subsequent encryption mixes this state into the full d -dimensional space, so the construction deviates slightly from the standard QECM structure. Understanding how this modification affects the security proof is left for future work. Investigating whether a general repetition theorem for n -bit messages can be derived is another important direction for future research.

More generally, the scheme $S_{d\ell}$ admits several possible embellishments and variants. For example, one could investigate encodings in which multiple bits are stored in a single CV mode, or replace the sign degree of freedom by a different binary degree of freedom for

representing the plaintext bit. Another practically relevant direction would be to devise alternative ways of preparing the state τ^{dist} that avoid the need to generate a narrow approximate eigenstate of $\hat{Q}$.

Finally, the results of this chapter suggest that unclonability may become a useful organising principle for further continuous-variable cryptographic tasks. In discrete variables, unclonability has already been connected to primitives such as single-decryptor encryption and revocable commitment [81]. The constructions developed here suggest that analogous CV variants may be possible as well. Another lesson from these results is that different security notions may require completely different proof mechanisms. More generally, the chapter supports a broader thesis theme: many quantum-cryptographic ideas that were first formulated in discrete variables do admit meaningful continuous-variable counterparts, but only after one identifies the right CV-native proof techniques and the right physically motivated regularisations. For unclonable encryption, monogamy and decoupling play exactly these two roles.

7.5. Final words

The results of this thesis show that long-distance CVQKD can be further improved and that continuous-variable quantum systems can support several cryptographic primitives beyond standard key distribution. However, the mathematical treatment of security for CV systems is necessarily more delicate than that for their finite-dimensional analogues. Because a primary motivation for using continuous variables is implementability, one must ensure that the resulting protocols work in realistic regimes with finite channel parameters and must also account for relevant discretisation errors.

For CVQKD, random codebooks show that long-distance reconciliation can be made parallelisable and can recover a non-trivial fraction of the Devetak–Winter value. For CV-QPV, similar security guarantees can be obtained for CV systems using the same proof technique. For CV two-designs, exact Haar-like randomness on the full infinite-dimensional Hilbert space is replaced by an approximate design on a finite-resolution boxed space. For unclonable encryption, both monogamy-based and decoupling-based routes remain available in the continuous-variable setting.

Thus, the overarching conclusion is that continuous variables provide a rich and physically natural setting in which cryptographic security can be recovered once the intrinsic structure of the CV system is treated carefully. The thesis shows that this can be done in several cases and that the resulting protocols retain a close connection to experimentally meaningful CV operations such as coherent-state preparation, squeezing, homodyne detection, and quadrature phase gates, while remaining robust against loss and noise.

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Curriculum Vitae

Arpan Akash Ray was born in Ranchi, Jharkhand, India. He completed his secondary education in the science stream in 2017. In 2022, he received a BS–MS dual degree in science from the Indian Institute of Science Education and Research Mohali, majoring in physics and minoring in mathematics. His master’s thesis, titled *The Kirkwood–Dirac Quasiprobability: Characterising ‘Quantumness’*, focused on quantum information theory. From 2022 to 2026, he pursued his doctoral studies at Eindhoven University of Technology in the Netherlands. Under the supervision of Dr Boris Škorić, his research focused on continuous-variable quantum information theory and its applications to quantum cryptography.



Quantum cryptography uses the laws of quantum physics to protect information. This thesis explores this possibility using continuous-variable systems, in which information is encoded in continuously varying properties of light. Such systems connect naturally with established optical communication technologies.

At its heart, the thesis asks whether continuous-variable quantum systems can support cryptographic tasks beyond quantum key distribution. It develops a new approach to information reconciliation for long-distance quantum key distribution and extends the continuous-variable framework to quantum position verification and unclonable encryption, including the stronger notion of unclonable indistinguishability. It also introduces a new mathematical randomisation tool for continuous-variable quantum information.

Throughout, theoretical security is considered alongside practical limitations such as loss, noise, and restricted quantum resources. Together, the results demonstrate that continuous-variable systems provide a versatile platform for emerging forms of quantum-secure communication.

The artwork on the cover was conceptualised by the author and executed by watercolour artist Jony Akram from Sylhet, Bangladesh. The boundless ocean represents the infiniteness of continuous-variable spaces. The sun, a source of photons shines upon a key. Only when illuminated by the rays of light, the shadow of the key reveals an open doorway. These elements together encapsulate the theme of this thesis into a single artwork: cryptography shaped by the quantum nature of light.